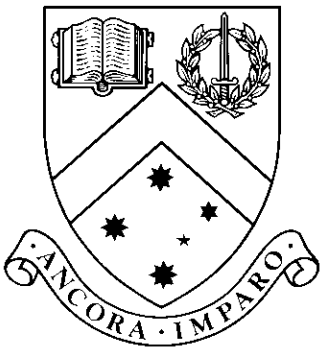
Minimum Message Length Inference of Autoregressive Moving Average Models

by

Daniel Francis Schmidt, BDigSys (Hons)



Thesis

Submitted by Daniel Francis Schmidt

for fulfillment of the Requirements for the Degree of

Doctor of Philosophy (0190)

Supervisor: Assoc. Prof. Andrew P. Paplinski

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School of Computer Science and Software Engineering
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Dedicated to my family: Mum, Dad, Matty, Sarah and Jasmin

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Minimum Message Length Inference of Autoregressive Moving Average Models

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Monash University, 2008

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Abstract

This thesis examines inference of autoregressive-moving average (ARMA) models within the Minimum Message Length (MML) framework. In particular, the first main contribution of this thesis derives message length formulas for ARMA class models expressed in terms of root-space parameters using the Wallace-Freeman (MML87) approximation and presents a modification to overcome difficulties in parameter estimation. The MML87 estimators are empirically compared against well known parameter estimation methods and the results indicate that the MML87 estimators are often superior in terms of prediction capabilities. The second contribution of this thesis introduces a new message length approximation based on the concept of expected uncertainty regions that offers a new way of resolving the issue of point estimation within Dowe's MMLD approximation. The MMLD-S algorithm to approximately evaluate the high dimensional integrals in MMLD and MMLD-like formulas is derived. This algorithm, which handles asymmetric uncertainty regions, is based on systematic sampling of a small number of models and exploits the uncertainty region boundary rule to attain a time complexity that is linear in the number of parameters. The MML87 and MMLD-S message length criteria are then empirically tested against commonly used model selection criteria such as Akaike's Information Criterion (AIC) and Schwarz's Bayesian Information Criterion (BIC) on both synthetic and real data. The results indicate that the MML based methods are competitive with and in many cases superior to the conventional methods in terms of order selection and prediction.

Minimum Message Length Inference of Autoregressive Moving Average Models

Declaration

I declare that this thesis is my own work and has not been submitted in any form for another degree or diploma at any university or other institute of tertiary education. Information derived from the published and unpublished work of others has been acknowledged in the text and a list of references is given.

Daniel Francis Schmidt
December 22, 2008

Acknowledgments

In the spirit of the MML principle, I shall keep this brief and to the point. I would like to thank my fellow postgrads Adrian Bickerstaffe and Enes Makalic for their great friendship, black humour and love of metal; without this, I am not sure I would have managed to stick the whole process out. I must also thank my supervisors for reading my thesis draft and for their comments and suggestions, and Lloyd Allison for many interesting conversations and useful advice. I would like to thank Professor Chris Wallace, for even though I never had the opportunity to speak to him personally, he has indirectly served as a source of inspiration. Finally, I must give big thanks to my mum, dad, my brother Matty and sister-in-law Sarah, and my lovely niece Jasmin for their endless love and support during what has been a difficult period. It is finally over.

Daniel Francis Schmidt

Monash University

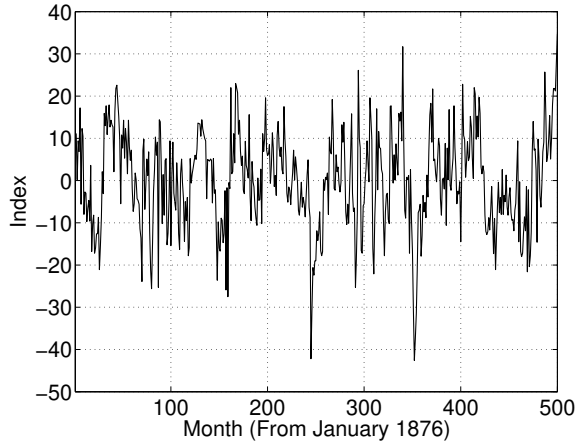
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Chapter 1

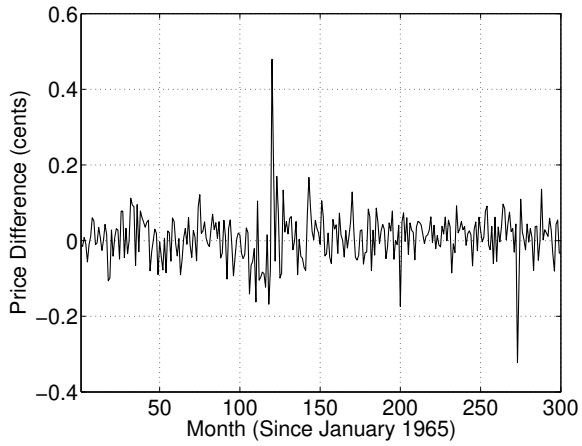
Introduction

1.1 Introduction

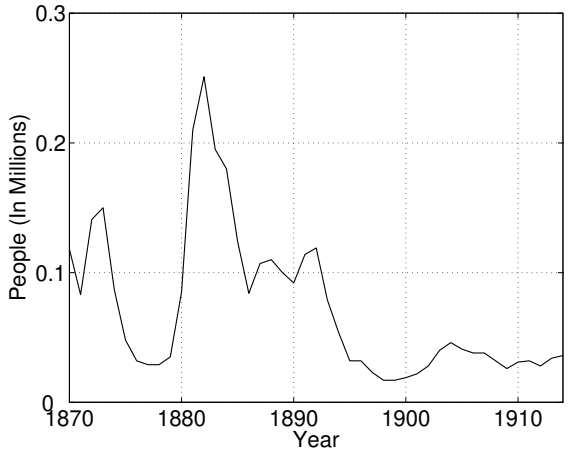
This thesis examines the problem of identification of models from measured sequences of data, more commonly referred to as *time series*, within the Minimum Message Length framework [1] of inductive inference. In particular, it presents new solutions to the problem of parameter estimation and order selection for three common and important dynamic processes: the Autoregressive (AR), Moving Average (MA), and mixed Autoregressive Moving Average (ARMA) models. Time ordered data has been in use since man first began to record statistics of his existence, but the last century has seen extensive study and development on the process undertaken, and linear dynamic systems now assume a very important position in many areas, including engineering, econometrics, the natural sciences, signal processing and the social sciences (several example time series may be observed in Figure 1.1, overleaf); given such a wide ranging applicability of time series data, it is not surprising that an extensive amount of research has been undertaken on estimating, validating and selecting dynamic models from time ordered data. The main contribution of this thesis is to present and explore new estimators for both parameters and model structure for a range of linear dynamic systems based on the principle of Minimum Message Length (MML). The Minimum Message Length criteria provides a unified framework for parameter estimation and model selection that is grounded in information theory, and possesses a slew of desirable properties.



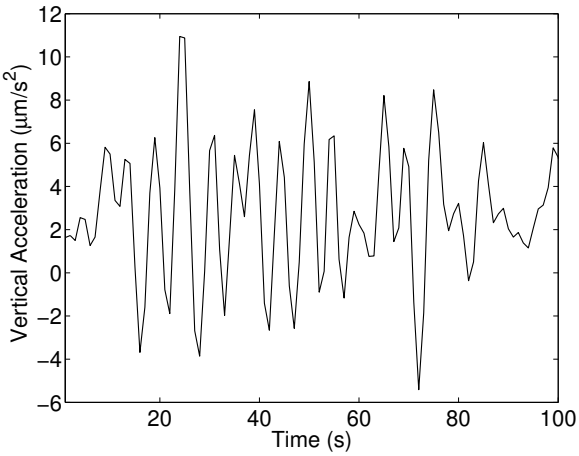
(a) Southern Oscillation Index



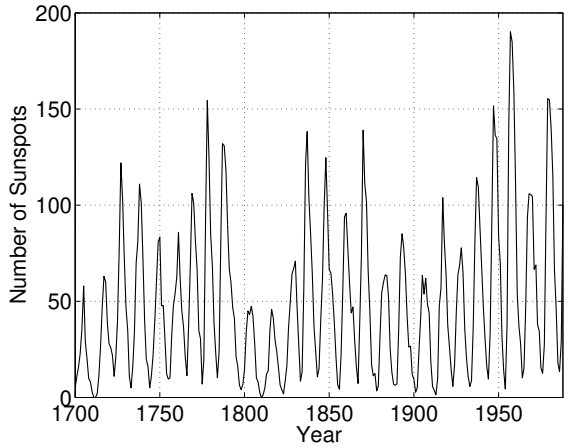
(b) FTA All Share Price Index



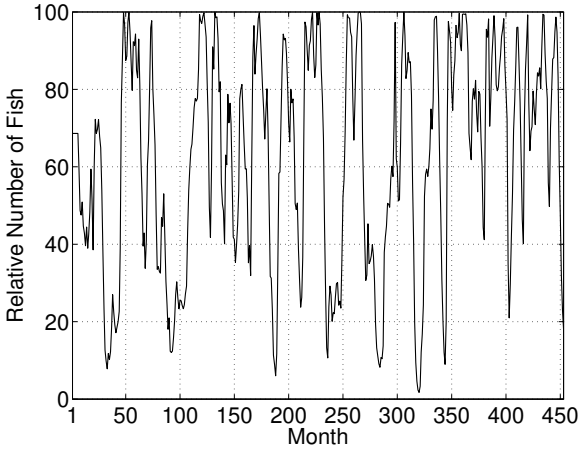
(c) German Immigration to the United States



(d) Seismograph of the Kobe Earthquake



(e) Annual Sunspot Series



(f) Monthly Fish Population in the Pacific Ocean

Figure 1.1: Examples of Time Series

1.2 Problem Statement

Given an observed data sequence $\mathbf{y} \in \mathbb{R}^N$ and an unobserved *innovation* sequence $\mathbf{v} \in \mathbb{R}^N$, the $(P \in \mathbb{Z}_+, Q \in \mathbb{Z}_+)$ order mixed Autoregressive Moving Average (ARMA) model explains measurement y_n as

$$y_n + \sum_{i=1}^{P} a_i y_{n-i} = \sum_{i=1}^{Q} c_i v_{n-i} + v_n \tag{1.1}$$

where $\mathbf{a} \in \mathbb{R}^P$ are the autoregressive coefficients, $\mathbf{c} \in \mathbb{R}^Q$ are the moving average coefficients and the $\mathbf{v}$ are independently and identically distributed Normal variates distributed as per $v_n \sim \mathcal{N}(0, \sigma^2)$, where $\sigma^2 \in \mathbb{R}_+$ is termed the innovation variance. The ARMA model explains the next measurement in a sequence as a weighted linear combination of previous observed measurements and unobserved innovations. By setting $Q = 0$ the ARMA model becomes the pure autoregression (AR), and by setting $P = 0$ it becomes the pure moving average (MA). Well known results [2] show that (1.1) may be manipulated to be represented exactly by an infinite order moving average or an infinite order autoregression, so the strength of the ARMA model is its ability to more parsiminiously model time series than either of its pure derivatives. Given an observed sequence of data $\mathbf{y}$, the most common task is to attempt to *fit* a model of the form (1.1) to the data, such that it approximates the underlying process that generated $\mathbf{y}$ to some degree of accuracy. This entails the estimation of

- The ARMA model orders P and Q
- The autoregressive coefficients, $\mathbf{a}$
- The moving average coefficients, $\mathbf{c}$
- The innovation variance σ^2

These problems have been heavily examined, and a large range of procedures for estimation of both the model orders and the coefficients have been proposed (e.g., [3, 4, 5, 6, 7, 8] to cite just a few); in general, most techniques decompose the problem such that they are either attempting to estimate the order, or estimate the parameters. In contrast, this thesis aims to explore the application of the Minimum Message Length which presents a unified approach to the problem of model fitting by selecting *fully specified models*.

1.3 Inference by Minimum Message Length

The Minimum Message Length (MML) principle is a subjective Bayesian criterion that unifies the traditional problems of parameter estimation and model selection within a single framework. The basic premise is to imagine a sender who wishes to communicate some observed data $\mathbf{y}$ to a receiver through a noiseless communication channel. The process involves the sender stating which model will be used to encode the data, and then sends the data using this model. The first part of this two-part message scheme is called the *assertion* and the second part the *detail*. The Minimum Message Length principle states that the model that renders the complete message length (the sum of the assertion and details) minimum is the best explanation of the observed data. This process may essentially be viewed as a formalism of Occam's Razor based on information theory [9]. The Minimum Message Length principle was birthed with the publication of [10] by Wallace and Boulton; since then it has been refined considerably and now encompasses a range of approximations and approaches that are all based on the same underlying philosophy of short two-part messages; the three main MML criteria are Strict Minimum Message Length, the MMLD approximation and the Wallace and Freeman (MML87) approximation. These are now briefly introduced as they play a particularly important role in the development of the sequel.

1.3.1 Strict Minimum Message Length

The Strict Minimum Message Length (SMML) [11] criterion forms the basis of the two main MML approximations and is the theoretical backbone of the entire principle. It differs fundamentally from the majority of inference procedures by viewing the dataset as the basic unit in inference, as opposed to the model. Unlike traditional inference schemes the process does not involve solving an expression for a particular observed data set but constructing inferences for *all* possible datasets. Assume there is a countable set of datasets $\mathcal{Y}$, a parameterised likelihood function $f(\mathbf{y}|\boldsymbol{\theta})$ and and prior density $h(\boldsymbol{\theta})$. The SMML principle involves finding an allocation of datasets to a set of *groups* $\mathbf{T} = \{\mathbf{t}_1, \ldots, \mathbf{t}_M\}$ with associated point estimates $\hat{\boldsymbol{\theta}}(\mathbf{t}_i)$ that minimises

$$I = -\sum_{i=1}^{M} \left(\log \sum_{\mathbf{y} \in \mathbf{t}_i} r(\mathbf{y}) + \left(\sum_{\mathbf{y} \in \mathbf{t}_i} r(\mathbf{y}) \log f(\mathbf{y}|\hat{\boldsymbol{\theta}}(\mathbf{t}_i)) \right) \right) \tag{1.2}$$

where $r(\mathbf{y})$ is the marginal probability of datum $\mathbf{y}$, and each dataset $\mathbf{y}$ is allocated to one and only one group. Inference for a particular observed dataset then simply involves finding the region that the dataset belongs to, which yields an associated point estimate; this process is examined more thoroughly in Section 2.2. Although this process has strong theoretical properties, the minimisation of (1.2) is an NP problem for all but the simplest of models [12], and is thus untenable in practice. Efforts to overcome this stumbling block and to apply the MML principle in practise have lead to two key approximations to SMML that are significantly more computationally tractable.

1.3.2 Inference by I1F

The I1D or MMLD [13, 1] approximation makes two key modifications to the SMML criteria: it assumes that the assertion codebook is based on the prior distribution rather than the marginal distribution (and thus works in the more conventional model space, as opposed to data space), and it concerns itself only with a subset of model space (termed the *uncertainty region*) that contains the model under consideration. In this thesis a modified form of MMLD, dubbed I1F, is proposed that allows for invariant Bayes point estimation within the MMLD framework. In particular, it makes use of the directed Kullback-Leibler divergence to find the *expected* uncertainty region for a given point estimate. Inference within this new framework involves solving

$$\hat{\boldsymbol{\theta}}_{\mathrm{1F}}(\mathbf{y}) = \arg\min_{\boldsymbol{\theta}\in\boldsymbol{\Theta}} \left\{ L(\mathbf{y}|\boldsymbol{\theta}) + \arg\min_{\boldsymbol{\Omega}\subset\boldsymbol{\Theta}} \left\{ -\log q(\boldsymbol{\Omega}) + \frac{1}{q(\boldsymbol{\Omega})} \int_{\boldsymbol{\theta}\in\boldsymbol{\Omega}} h(\boldsymbol{\theta}) \Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*) d\boldsymbol{\theta} \right\} \right\} \tag{1.3}$$

where

$$q(\boldsymbol{\Omega}) = \int_{\boldsymbol{\theta}\in\boldsymbol{\Omega}} h(\boldsymbol{\theta}) d\boldsymbol{\theta} \tag{1.4}$$

and $L(\mathbf{y}|\boldsymbol{\theta})$ is the negative log-likelihood of the data given the candidate point estimate $\boldsymbol{\theta}$, $\boldsymbol{\Theta}$ is the space of permissible models, $\boldsymbol{\Omega} \subset \boldsymbol{\Theta}$ is the expected *uncertainty region*, $h(\cdot)$ is a prior density over the model space and $\Delta(\boldsymbol{\theta}_2||\boldsymbol{\theta}_1)$ is the Kullback-Leibler divergence [14] between a true distribution indexed by $\boldsymbol{\theta}_1$ and an approximating distribution indexed by $\boldsymbol{\theta}_2$. Sections 2.3 and 2.4 discuss the MMLD and I1F approximations in greater detail.

1.3.3 The MML87 Estimator

By assuming that the prior density is uniform and the Kullback-Leibler divergence is quadratric within the locale of the Uncertainty Region, the Wallace and Freeman (MML87) estimator [15] may be derived from (1.3). Due to the simplifying nature of the assumptions, the MML87 uncertainty region may be found explicitly and inference becomes the problem of solving

$$\hat{\boldsymbol{\theta}}_{87}(\mathbf{y}) = \arg\min_{\boldsymbol{\theta}\in\boldsymbol{\Theta}} \left\{ L(\mathbf{y}|\boldsymbol{\theta}) - \log h(\boldsymbol{\theta}) + \frac{1}{2}\log|\mathbf{J}(\boldsymbol{\theta})| + \frac{P}{2}(\log\kappa_P + 1) \right\} \tag{1.5}$$

where $\mathbf{J}(\cdot)$ is the *Fisher Information Matrix*, and κ_P is the normalised second moment of a cell of a second order optimal P dimensional quantising lattice. This equation is the most well known and widely used implementation of the MML principle. The derivation of MML87 is presented and discussed in Section 2.5.

1.3.4 MML and the ARMA Model

This thesis is focussed on exploring the application of the Minimum Message Length principle to the estimation of ARMA models, i.e. estimating the parameters of the model class described by (1.1) by the MML principle. To this end, the preliminary work of Fitzgibbon [16] is built upon and heavily extended; in the process the weaknesses of the MML87 approximation (1.5) become apparent and are addressed by the introduction of a new technique that is also derived from the stronger MML approximation (1.3), (1.4).

1.4 Thesis Achievements

The main achievements of this thesis are:

- New parameter estimation and order selection criteria for AR, MA and ARMA models based on the MML87 approximation.

- Development of a new MML approximation, MMLD-S, specifically to address problems of the MML87 approximation when applied to the ARMA class of models for order selection.

- Examination of the performance of the MML criteria as compared to traditional model selection criteria, including several order selection criteria combining MML and decision theory.

These achievements are now elaborated upon. Chapters 4, 5 and 6 presents the formulation of the mixed Autoregressive, Moving Average and mixed Autoregressive Moving Average model within the MML87 framework. This approximation is the most easily applicable of all the MML approximations developed thus far and is a suitable starting point for any Message Length analysis. In particular, these chapters examine and derive message length expressions for the Autoregressive and Moving Average processes individually, and then integrate the components for the mixed ARMA process. Several key problems with the ARMA model class that cause the MML87 criteria to breakdown are identified, and some modifications are proposed to help overcome these difficulties. The parameter estimates are shown to more adequately capture the longer term behaviour of the generating process than the convential parameter estimators employed in much of time series analysis. The order selection capabilities of the MML87 criteria are shown to be good when the Maximum Likelihood estimates are employed, but suffer from violations of the MML87 approximation when the MML estimates are used.

Chapter 7 proposes a new approximation to the MMLD criteria, dubbed MMLD-S, that, unlike the existing MMLD techniques, is not based on sampling. In particular, it specifically addresses the issue of non-symmetric coding regions and non-quadratic expected likelihood surfaces that are a property of the ARMA class of models, and that necessitate the fixes proposed in Chapters 4 to 6. By exploiting the boundary rule property of the MMLD uncertainty region, the new approximation achieves a time complexity that is linearly proportional to the number of model parameters. For models with unrestricted parameter spaces and under some basic assumptions the MMLD-S parameter estimates are shown to be invariant to the choice of region geometry. The approximation is then refined to handle non-uniform priors and the more difficult issue of parameter boundaries. A fast algorithm based on rejection sampling is used to overcome the problem posed by parameter boundaries that are not inherent restrictions of the model class that substantially improves the message length estimates produced by the MMLD-S criteria. Finally, the MMLD-S criteria is tested on two well studied inference problems and demonstrates superior results as compared to the MML87 estimator.

In Chapter 8 the new criteria is applied to the Autoregressive, Moving Average and mixed ARMA model classes, and certain issues and problems with its application are identified and solved. In particular, the redundancy of the root-space representation is taken into consideration and solved via the sampling procedure introduced in the previous chapter. Additionally, the difficulty that the hard parameter boundaries pose is also addressed by a suitable transformation of parameters. The message lengths estimated by the MMLD-S and MML87 estimator for the ARMA models are then compared and the results demonstrate that the MMLD-S approximation does not break down in the same fashion as the MML87 estimator.

Finally, Chapter 9 examines the performance of the MML87 and MMLD-S criteria at selecting AR, MA and ARMA models from data. This is undertaken with the aim of determining where the MML-based criteria fit in the in regards to the behaviour of a range of alternative ARMA model order selection criteria. The results demonstrate that the MML schemes offer improvements over the competing criteria in terms of capturing and estimating *long term* structure correctly, at the expense of some performance at short term estimation.

1.4.1 Thesis Summary

This thesis explores the application of the Minimum Message Length principle to the estimation of ARMA models from data. A summary of the content of the chapters is as follows:

- Chapters 2 and 3 are reviews of the current state of Minimum Message Length model selection, and automatic model selection in general. Additionally, Chapter 2 includes a new MML approximation based on the concept of expected uncertainty regions, and details a modified derivation of the MML87 criteria that follows directly from this new approximation. The author believes this more clearly demonstrates the link between the approximation families than existing derivations. With the exception of this new approximation detailed in Section 2.4, which is novel, the reader may skip these two chapters if they are sufficiently versed in the field of inference by compact coding and general model selection.

- Chapters 4 through 10 detail the original contributions of the thesis; in particular, Chapters 4 through 6 detail the inference of ARMA models using MML87, Chapters 7 and 8 introduce the new MMLD-S algorithm and apply it to ARMA model selection, Chapter 9 presents experimental results and Chapter 10 provides a conclusion.

- Appendix A summarises the formulae required to compute the autoregressive Fisher Information matrix in pole space, and is included for completeness. Appendix B details the derivation of the formulas to find the exact uncertainty regions for the univariate Normal Distribution, and Appendix C summarises useful formulae for computing autocovariances, log-likelihoods and other aspects of ARMA models in a convenient fashion.

Chapter 2

Model Selection by Minimum Message Length

2.1 Introduction

The aim of this chapter is to introduce the Minimum Message Length principle and discuss the various criteria and approximations that are used in practice. This is included primarily for completeness as most of the material has been previously published. Some of the content is of note however:

- The new I1F message length approximation derived from the FSMML criterion in Section 2.4 that is applied extensively in the later chapters of this thesis
- The Random Coding algorithm introduced in Section 2.3.1
- A new deriviation of the MML87 criteria directly from the I1F message length equation found in Section 2.5

Readers with background in the field of inference by minimum encoding may elect to skip all but Section 2.4.

2.1.1 The Model Estimation Problem

Let $\mathbf{y} \sim f^*$ be a set of N observed data points generated by some underlying process. The basic problem of estimating parameters for a model may be summarised as seeking a mapping

$$m(\mathbf{y}, \mathcal{M}_P) : \mathcal{Y} \times \boldsymbol{\mathcal{M}} \rightarrow \boldsymbol{\Theta}_{\mathcal{M}_P} \tag{2.1}$$

where $\mathcal{Y}$ is the space of all possible data, N is the number of observations, $\mathcal{M}_P$ denotes a *model class*, $\boldsymbol{\mathcal{M}}$ is the space of all permissible model classes, P are the number of estimated parameters and $\boldsymbol{\Theta}_{\mathcal{M}_P}$ is the space of all permissible realisations of model class $\mathcal{M}_P$. The function m aims to estimate a parameterised density $f_{\boldsymbol{\theta}}$ from the observed data that will best approximate f^*. Generally this is found by minimising some form of cost or loss function; this task is also often referred to as *point estimation*. The *model selection* problem generally involves seeking a mapping

$$s(\mathbf{y}) : \mathcal{Y} \rightarrow \mathcal{M}_P \tag{2.2}$$

that is, the selection of a model class from observations. The combined model selection and parameter estimation problem may be summarised informally as answering the question: **which model, from a (possibly infinite) set of candidate models, best approximates the underlying process that generated the data?** A large number of the existing methods used for model selection are based on attempts to directly answer this question, under various conditions, assumptions and choices of measures of similarity. The bulk of the existing statistical techniques decompose the problem such that first parameters are estimated for a set of competing model classes via $m(\cdot)$, and from these a single model is selected via $s(\cdot)$.

2.1.2 Model Selection by Minimum Encoding

In contrast to the traditional view of model selection and parameter estimation, the Minimum Message Length criterion approaches the problem by solving another question. Rather than trying directly to find the model that best approximates f^*, it attempts to answer the alternative question: **which model, from a (possibly infinite) set of candidate models, best summarises a given dataset?** This may be specified formally as solving

$$\hat{\boldsymbol{\theta}}(\mathbf{y}) = \arg\min_{\boldsymbol{\theta} \in \boldsymbol{\Theta}} \{\Xi(\mathbf{y}|\boldsymbol{\theta})\} \tag{2.3}$$

where $\Xi(\cdot)$ is a measure of how well the model $\boldsymbol{\theta}$ 'summarises' or 'describes' the data set $\mathbf{y}$. The premise is that the model which best summarises the dataset will also provide a good explanation of the underlying process that generated the data. It remains to formally quantify how well a model summarises a data set, i.e. to define $\Xi(\cdot)$. The MML approach

appeals to *information theory* and treats the problem as one of communicating the data $\mathbf{y}$ from a sender to a receiver, over a noiseless communication channel; this may be done using important results in coding theory. If a model $\hat{\boldsymbol{\theta}}(\mathbf{y})$ is been chosen from $\boldsymbol{\Theta}$, there will be resulting distribution over $\mathcal{Y}$ indexed by this model, i.e. $f(\mathbf{y}|\hat{\boldsymbol{\theta}}(\mathbf{y}))$. It is known that given a probability mass function, a suitable prefix code can be created to encode any event over which $f(\cdot)$ is defined. The data will be stated to some level of accuracy ϵ: as long as this accuracy is small compared to any peaks or features of $f(\cdot)$, a suitable probability mass function over $\mathcal{Y}$ may defined. The decodable message that exactly communicates $\mathbf{y}$ may be constructed from $f_{\hat{\boldsymbol{\theta}}}$ with codelength

$$I(\mathbf{y}|\hat{\boldsymbol{\theta}}(\mathbf{y})) = -\log f(\mathbf{y}|\hat{\boldsymbol{\theta}}(\mathbf{y})) - N \log \epsilon \tag{2.4}$$

where $I(\cdot)$ is measured in *nits* (a nit being a base-e digit). If this were taken as the measure of how well the data is summarised by the model, the optimal model would be the one that merely maximised $f(\mathbf{y}|\hat{\boldsymbol{\theta}}(\mathbf{y}))$, i.e. the Maximum Likelihood estimate. However, for the receiver to be able to decode the $\mathbf{y}$ sent using $f_{\hat{\boldsymbol{\theta}}}$, the receiver must know which model the sender selected to encode $\mathbf{y}$. Thus there must be a codeword sent first to instruct the receiver which model is being used to encode the data; the construction of this codeword can pose difficulties. The Minimum Message Length criteria assumes both sender and reciever are aware of a probability distribution over all possible members of $\boldsymbol{\Theta}$, and this may be used as the basis for the construction of a probability mass function $c(\cdot)$ over a *discretised* set of possible models $\boldsymbol{\Theta}'$. This results in a combined *message length* of the data and model given by

$$\begin{aligned} I(\mathbf{y}, \hat{\boldsymbol{\theta}}(\mathbf{y})) &= I(\boldsymbol{\theta}') + I(\mathbf{y}|\boldsymbol{\theta}') && (2.5) \\ &= \underbrace{-\log c(\boldsymbol{\theta}')}_{\text{Assertion}} \underbrace{-\log f(\mathbf{y}|\boldsymbol{\theta}') + e(\mathbf{y}|\boldsymbol{\theta}') - N \log \epsilon}_{\text{Detail}} && (2.6) \end{aligned}$$

where $\boldsymbol{\theta}' \in \boldsymbol{\Theta}'$ is a member of the discretised model space used in place of $\hat{\boldsymbol{\theta}}(\mathbf{y})$, $e(\cdot)$ denotes the excess number of nits required to transmit $\mathbf{y}$ using this discretised estimate (i.e. excess coding length due to rounding-off of $\boldsymbol{\theta}$ to $\boldsymbol{\theta}'$), and $I(\mathbf{y}, \boldsymbol{\theta})$ denotes the total message length. This is the basic idea of *inference by compact coding* [15]. In MML terminology, $I(\boldsymbol{\theta}')$ is often referred to as the *assertion length*, and $I(\mathbf{y}|\boldsymbol{\theta}')$ as the *detail length*. The MML

criteria presented in the sequel refine the process beyond this crude introduction through methods of constructing codes and discretisations of the model space that are optimal in some sense. The model selection process then becomes one of solving

$$\hat{\boldsymbol{\theta}}(\mathbf{y}) = \arg\min_{\boldsymbol{\theta}\in\boldsymbol{\Theta}} \{I(\mathbf{y}, \boldsymbol{\theta})\} \tag{2.7}$$

This selected model is also assumed to be the model that will in some sense best approximate f^*. The MML principle is a formalism of Occam's Razor based on Information Theory. The minimisation of the message length provides a formal way of weighing the complexity of a model against its ability to describe the data; the more complex the model becomes, the shorter the description of the data will be, but the longer than description of the model, and vice versa. It is not the only such method that follows this idea; the closely related Minimum Description Length (MDL) criteria, which postdates the introduction of MML by approximately a decade, also expounds that the model that yields the minimum length message required to describe data will be the best representation of the underyling process. Although related, the MDL criterion does differ significantly in philosophical and technical details [17]. MML is also related to the idea of Kolmogorov complexity [18]. Some of the important properties of the MML approach as compared to other model selection schemes include

- The MML principle does not decompose the model estimation problem into one of parameter estimation and model class selection; instead, it selects *fully specified models*, that is models with all requisite parameters specified. This is in contrast to other schemes that select only model classes (MDL, Bayesian evidence framework). As an example, the MDL approach treats all realisations of second order polynomials as a 'model', and selects between differing orders of polynomial on the basis of observed data. In contrast, the MML criterion treats all fully specified members of all classes on an equal footing, and selects one of the parameterisations of polynomial from one of the orders. It unifies what are traditionally considered seperate issues: that of parameter estimation given a model structure, and discrimination between competing model structures.

- The MML principle makes no assumptions about the underlying generating process f^*. This is contrast to many classical methods that assume the source is a member

of the approximating family of models. As the form of the underlying model is irrelevant, the MML criteria even admits the possibility that the underlying model may not even be expressible by any formally specified model.

- The MML principle provides an invariant method for choosing parameters (point estimation) in a Bayesian setting; this is in contrast to traditional Maximum a Posteriori or mean posterior estimators that are in general not parametrisation invariant [1]

- The MML principle extracts what information is justified from the data and no more. In addition to provide a means to avoid the traditional problems of *overfitting* and *underfitting*, it has been shown to provide statistically consistent estimators for problems where the number of parameters grows with the amount of data

With these theoretical advantages in mind, the technical details of the MML family of estimators is examined next. The final point of import to note before proceeding is that although the basis of the concept is communicating messages, the messages themselves are never actually constructed (though if desired, they may be via arithmetic coding [19]); all that is required is the *length* of the message if it were to be constructed.

2.1.3 Previous Applications of MML

The MML principle, and the various approximations that comprise the practical forms of the principle have been applied to a large range of statistical inference problems with high degrees of success. These include:

- Linear Regression Models and Function Approximation [20, 21, 22]
- Mixture Modelling and Classification [10, 23]
- Bayesian Belief Networks [24, 25, 26]
- Decision Trees and Graphs [27, 28, 29, 30]
- Factor Analysis [31, 32]
- Artifical Neural Networks [33, 34, 35]

In all these cases, the performance of the MML criterion has been comparable, if not superior, to many commonly used and well known model selection schemes.

2.1.4 Chapter Overview

There is no one single Minimum Message Length approximation that is uniformly applicable in all situations. The MML principle covers a range of approximations and ideas, each with different strengths and weaknesses, but all derived from the same central concept. This chapter presents the main approximations and ideas prevalent in the MML community, and is organised as follows: Section (2.2) examines the Strict Minimum Message Length (SMML) criterion, which effectively forms the theoretical backbone of the entire MML concept, and Section (2.3) presents the MMLD approximation to SMML, a significantly more practical method than SMML, Section (2.4) introduces the new I1F approximation, and Section (2.5) derives and discusses the MML87 (or Wallace-Freeman) message length estimator, which is currently the most widely applied MML approximation.

2.2 Strict Minimum Message Length (SMML) Inference

2.2.1 Overview of SMML

The Strict Minimum Message Length criterion offers a radically different approach to the problem of model selection and parameter estimation as compared to traditional methods. In particular, it differs primarily by viewing the basic unit of the inference problem as possible datasets rather than possible models, and by assuming the possible data realisations and estimated models belong to *discrete* sets. This is in contrast to most traditional methods that view the model space and in general the data space as *continua*. The basic concept behind the SMML criterion is to group together datasets that are similar and assign each group a representative point estimate; this induces a discretisation of the parameter space, and a probability mass function is then built over the discrete set of parameter estimates, which is then used as the basis for the assertion codeword. The grouping is done on the basis of minimising the expected message length of a random dataset drawn from the marginal distribution of all data. The next subsections develop this procedure in detail.

2.2.2 Genesis

The following are assumed to be available before induction via SMML may be performed:

- A model set, represented by a parameterised distribution $f(\mathbf{y}|\boldsymbol{\theta})$ indexed by a parameter vector $\boldsymbol{\theta}$, with $\boldsymbol{\theta} \in \Theta$, where $\boldsymbol{\Theta}$ is the complete model space.

- A data space $\mathcal{Y}$ such that $\mathbf{y} \in \mathcal{Y}$; unlike other methods which assume the data space is a continuum, SMML requires that it be a countable set, i.e. $|\mathcal{Y}| \leq |\mathbb{Z}|$. The requirement for the dataspace to be countable poses no practical limitations on the SMML estimator, as all data being analysed on a digital computer must necessarily be countable.

- A prior density $h(\boldsymbol{\theta})$ on the support $\boldsymbol{\theta} \in \boldsymbol{\Theta}$, expressing the *a priori* belief of unseen data being generated by a particular model.

From this a proper posterior can be found, and a usual Bayesian analysis of the data performed. The MML principle however, is to *make sense* of the posterior and provide an method for invariant point-estimation of model parameters. Assume there exists a set of groups $\mathbf{T} = \{\mathbf{t}_i, \ldots, \mathbf{t}_M\}$ where each group $\mathbf{t}_i$ is a subset of the dataspace $\mathcal{Y}$. Each group is defined by the set of data values it contains and has a single fully specified model parameter estimate $\hat{\boldsymbol{\theta}}(\mathbf{t}_i)$ associated with it. The groups are arranged such that they cover the data space completely and disjointly, i.e.

$$\mathbf{t}_1 \cup \mathbf{t}_2 \ldots \cup \mathbf{t}_M = \mathcal{Y} \tag{2.8}$$

$$\mathbf{t}_i \cap \mathbf{t}_j = \emptyset,\ i, j \in [1, M],\ i \neq j \tag{2.9}$$

so that any given data value $\mathbf{y}_i$ belongs to exactly one group. Given some observed dataset $\mathbf{y}$, the SMML parameter estimate is given by

$$\hat{\boldsymbol{\theta}}_{\text{SMML}}(\mathbf{y}) = \hat{\boldsymbol{\theta}}\left(\{\mathbf{t} \in \mathbf{T} : \mathbf{y} \in \mathbf{t}\}\right) \tag{2.10}$$

that is, a surjective mapping between $\mathcal{Y}$ and $\hat{\boldsymbol{\Theta}}$. If all the groups $\mathbf{t}_i$ are taken to be singleton sets, the SMML estimator function (2.10) is similar to most other estimators: for every data point $\mathbf{y} \in \mathcal{Y}$ there is a corresponding one-to-one mapping to a model $\hat{\boldsymbol{\theta}} \in \boldsymbol{\Theta}$. If several data values are merged together into a single group, multiple data values will map to a single estimate. As the data space is mapped to the parameter space, the grouping of multiple data points together amounts to a quantisation of the parameter space. The cost function used to determine the 'optimal' quantisation, or assignment of data values to regions, is the heart of the SMML criterion and is detailed in the next subsection.

2.2.3 Construction of the SMML Estimator

Before a partitioning of the dataspace into groups may commence the marginal probabilities of the datasets in the dataspace must be found. The marginal probability of a dataset $\mathbf{y} \in \mathcal{Y}$ is given by

$$r(\mathbf{y}) = \int_{\boldsymbol{\theta} \in \boldsymbol{\Theta}} h(\boldsymbol{\theta}) f(\mathbf{y}|\boldsymbol{\theta}) d\boldsymbol{\theta} \tag{2.11}$$

The SMML estimator requires that the dataspace consist of a countable set of datasets. If the dataspace is truly discrete (such as that of the binomial distribution) then $r(\mathbf{y})$ is already a probability mass function and construction of the partitions may proceed. If the dataspace is a continuum (say for example, the dataspace of a Normal distribution) then it must be discretised to ensure that $r(\mathbf{y})$ is a probability mass function. This may be done by quantising the dataspace to a suitable accuracy; usually the inherent accuracy to which the real data are expected to be measured is a suitable accuracy to which to quantise the dataspace.

As the point estimate function $\hat{\boldsymbol{\theta}}(\cdot)$ varies only between different groups, it is useful to begin by examining the message lengths of individual groups. Assume that a partitioning $\mathbf{T}$ of M regions $\mathbf{T} = \{\mathbf{t}_1, \ldots, \mathbf{t}_M\}$ that completely and disjointly covers the dataspace has been found. The distribution of datasets within a group $\mathbf{t} \in \mathbf{T}$ is given by the probability mass function

$$r(\mathbf{y}|\mathbf{t}) = \frac{r(\mathbf{y})}{\sum_{\mathbf{x} \in \mathbf{t}} r(\mathbf{x})}, \ \mathbf{y} \in \mathbf{t} \tag{2.12}$$

Next, consider the expected message length of a dataset drawn randomly from a group $\mathbf{t}$

$$\begin{aligned} I(\mathbf{t}) &= \sum_{\mathbf{y} \in \mathbf{t}} r(\mathbf{y}|\mathbf{t}) I(\mathbf{y}) \\ &= -\log c(\mathbf{t}) + \sum_{\mathbf{y} \in \mathbf{t}} r(\mathbf{y}|\mathbf{t}) L\left(\mathbf{y}|\hat{\boldsymbol{\theta}}(\mathbf{t})\right) \end{aligned} \tag{2.13}$$

where $c(\mathbf{t})$ is the probability mass assigned to the codeword for region $\mathbf{t}$ and $L(\cdot)$ is the negative log-likelihood of data given the point estimate. Now that the expected message length for each group has been determined, the expected message length across the entire dataspace may be found. This may be found as the expected message length of a dataset randomly generated by first drawing a group from the set of groups, and subsequently drawing a dataset from the chosen group. The probability mass function over the set of

groups $\mathbf{T}$ is given by

$$r(\mathbf{t}) = \sum_{\mathbf{y} \in \mathbf{t}} r(\mathbf{y}),\ \mathbf{t} \in \mathbf{T} \tag{2.14}$$

The probability mass assigned to each group is exactly equal to the sum of marginal probabilities of all the datasets that the group contains. The expected message length of a random datum drawn from $r(\mathbf{y})$ is then given by

$$I_1 = \underbrace{-\sum_{\mathbf{t} \in \mathbf{T}} r(\mathbf{t}) \log c(\mathbf{t})}_{\text{Expected Assertion}} + \underbrace{\sum_{\mathbf{t} \in \mathbf{T}} r(\mathbf{t}) \left(\sum_{\mathbf{y} \in \mathbf{t}} r(\mathbf{y}|\mathbf{t}) L\left(\mathbf{y}|\hat{\boldsymbol{\theta}}(\mathbf{t})\right) \right)}_{\text{Expected Detail}} \tag{2.15}$$

It remains to select $c(\mathbf{t})$ and $\hat{\boldsymbol{\theta}}(\mathbf{t})$. These two functions are independent which greatly simplifies the problem. Given any partitioning, it is clear that the lower bound on the first term in (2.15) is the entropy [36] of $r(\mathbf{t})$. This may be exactly reached by choosing $c(\mathbf{t})$ as

$$c(\mathbf{t}) = \sum_{\mathbf{y} \in \mathbf{t}} r(\mathbf{y}),\ \forall \mathbf{t} \in \mathbf{T} \tag{2.16}$$

The choice of $\hat{\boldsymbol{\theta}}(\cdot)$ involves only the second term in (2.15); furthermore, each term in the summation depends on one and only one $\hat{\boldsymbol{\theta}}(\cdot)$. Thus, the optimum parameter estimate is given by

$$\hat{\boldsymbol{\theta}}(\mathbf{t}) = \arg\min_{\boldsymbol{\theta} \in \boldsymbol{\Theta}} \left\{ \sum_{\mathbf{y} \in \mathbf{t}} r(\mathbf{y}|\mathbf{t}) L(\mathbf{y}|\boldsymbol{\theta}) \right\}, \forall \mathbf{t} \in \mathbf{T} \tag{2.17}$$

The point estimate is chosen to minimise the average cost of encoding a randomly chosen datum from $\mathbf{t}$. For a singleton region the SMML estimate would coincide with the Maximum Likelihood estimate. For anything larger than a singleton region, the SMML estimator will be 'smeared' by the effects of all the datums within the region, and unlike the Maximum Likelihood estimator, it will in general pick models that may not characterise any single datum as accurately, but will generalise better over the range of datums in the region. The final task remaining is to select a partition $\mathbf{T}$ that minimises (2.15), given (2.16) and (2.17). Unfortunately, at this point it turns out that allocating datums to groups in a greedy fashion will not generally minimise the overall expected message length ([1], pages 156-157). For anything but one-dimensional problems [12] the allocation of datasets to groups is an extremely difficult optimisation problem.

2.2.4 The Ideal Group

Although the minimisation of the message length of single group $\mathbf{t}_i$ will usually not lead to an optimum solution of (2.15), it is quite illuminating. The expected message length of a single group, using the coding probability function (2.16) is given by

$$I(\mathbf{t}_i) = \underbrace{-\log \sum_{\mathbf{y} \in \mathbf{t}_i} r(\mathbf{y})}_{\text{Assertion}} + \underbrace{\sum_{\mathbf{y} \in \mathbf{t}_i} r(\mathbf{y}|\mathbf{t}_i) L\left(\mathbf{y}|\hat{\boldsymbol{\theta}}(\mathbf{t}_i)\right)}_{\text{Detail}} \tag{2.18}$$

This expression offers a large insight into the MML approach in general. The larger the region $\mathbf{t}$, the smaller the assertion becomes. However, at the same time, the detail will grow as a larger number of datasets are being encoded by the same point estimate. At one extreme, $\mathbf{t}$ will cover the entire dataspace and the assertion will cost nothing to state (as $c(\mathbf{t})$ will be equal to one); however, the detail in this case would be expected to be very large given that only a single point estimate is being used to encode all possible datasets in the dataspace. At the other extreme, $\mathbf{t}$ is chosen to contain a single dataset; the detail will be as small as is possible, as the SMML point estimate will be the model that exactly minimises the negative log-likelihood of the dataset. However, the assertion will be very large (unless there is a disproportionate amount of marginal probability on this one dataset). In general, an intermediate group size will be optimal, and this is essentially a trade-off between how accurately an estimate is stated and how well it describes the data. It has been noted that attempting to minimise the expected message length of any random dataset by minimising the expected message lengths within groups individually (i.e. for example, by a greedy search) will rarely lead to the optimum partitioning. However, given that the search for the optimal partitioning is an NP-hard problem [12], the *Ideal Group* [1] (pages 197-202) estimator has been proposed by Dowe as a possible tractable approximation. For a given observed dataset $\mathbf{y}'$, the ideal group estimator allocates similar datasets to a group until the within group expected message length is minimised, without regards to any other groups (hence *ideal*). However, Wallace [1] (page 199, also discussed in [37]) introduces a correction for the Ideal Group estimator; the corrected Ideal Group $\mathbf{t}(\mathbf{y}')$ for observed dataset $\mathbf{y}'$

$$\mathbf{t}(\mathbf{y}') = \arg\min_{\mathbf{t} \subset \mathcal{Y}} \left\{ -\log \sum_{\mathbf{y} \in \mathbf{t}} r(\mathbf{y}) + \sum_{\mathbf{y} \in \mathbf{t}} r(\mathbf{y}|\mathbf{t}) \log\left(r(y)/f(\mathbf{y}|\hat{\boldsymbol{\theta}}(\mathbf{t}))\right) \right\} \tag{2.19}$$

subject to

$$\mathbf{y}' \in \mathbf{t}(\mathbf{y}') \tag{2.20}$$

and $\hat{\boldsymbol{\theta}}(\cdot)$ is used as the point estimate. The Ideal Group estimator is simpler to construct than the SMML estimator, but is still significantly more complex than the other families of SMML approximations that will be introduced in the sequel. It has been applied to the Neyman-Scott problem, successfully yielding consistent estimates of scale (see [1], pages 201-202).

2.2.5 Strengths and Weaknesses of the SMML Estimator

The SMML estimator possesses many strong theoretical properties that make it very desirable. These are briefly summarised below

- **Parameter Invariance**: The SMML estimator is invariant under one-to-one reparameterisations of the log-likelihood function. This is a very desirable property, which is not possessed by the Maximum a Posterior (MAP) estimator or posterior mean.

- **Efficiency**: The SMML estimator is *efficient*, in the sense that it extracts all possible structure from a set of data (to within a bit), and no more. This property, and its impact is discussed in [1] (pages 230-231).

- **Immunity to Parameter Redundancy**: Some model classes possess multiple parameterisations that yield the same likelihood. Such models can cause problems for estimators as there is no longer a single unique model in parameter space that minimises the negative log-likelihood, and indeed may cause troubles for coding schemes that are based on partitioning the parameter space (such as some presented in the next sections). The SMML estimator views the problem from data space, in which the redundancy is not present, and thus avoids the issue completely.

The SMML estimator also possesses several weaknesses

- **Non-smooth Estimator Function**: The SMML estimator is not a smooth or even continuous function of data. The discontinuous nature of the SMML estimator may be a concern for some applications.

- **Time Complexity**: The single biggest weakness the SMML estimator possesses is the fact that construction of the regions is, in general, NP-hard. This renders it completely impractical for almost all problems, and is thus a severe drawback. So far, the only polynomial time construction of the SMML estimator has been for the binomial problem, via Farr's algorithm [12].

The last of these weaknesses, that of the time complexity of the region construction, all but renders the SMML estimator a theoretical concept that is in practice completely unrealisable. However, a considerable amount of further work has been undertaken in the years since the SMML estimator was proposed to produce approximations to the SMML scheme that retain many of its benefits while being computationally tractable. The two most important of these are discussed in the sequel.

2.3 The MMLD Estimator

The previous section covered the Strict MML estimator in some detail, and detailed both its strengths and weaknesses. Foremost amongst these weaknesses is the issue of time complexity; for all but the simplest of problems, the SMML estimator is unacceptably slow to construct. The Ideal Group estimator was previously introduced as an approximation to the SMML estimator [1, 37] that focuses on only constructing the group containing the dataset of interest. Wallace has shown that optimisation of a single group in the SMML scheme will not lead to optimal overall solutions, but the inefficiencies are small enough to be discounted for most practical purposes. While the estimator was derived with the SMML grouping in mind, a more generally useful version is Dowe's MMLD approximation [37] that uses codes based on the *prior* rather than the marginals. If this process is done in direct analogy of the SMML estimator, it amounts to a direct partition of the model space. There do exist strong arguments against a complete partitioning of the parameter space ([1], pages 214-215); however, the focus on a single group in the parameter space has shown to be a powerful technique of inference and it is this method that is the subject of this section. The I1D (or MMLD) approximation to the message length begins by assuming there is some data $\mathbf{y}$ that needs to be transmitted. The approximation then seeks to find a set of models in parameter space, $\boldsymbol{\Omega} \subseteq \boldsymbol{\Theta}$, that are 'suitable' to encode the data. This region is deemed the *uncertainty region*, due to the fact that it contains a

range of models that are deemed impossible to discriminate between on the basis of the observed data. Given this region, and using the prior distribution $h(\boldsymbol{\theta})$ as the basis for a codebook, the probability mass assigned to the codeword for $\boldsymbol{\Omega}$ is given by

$$q(\boldsymbol{\Omega}) = \int_{\boldsymbol{\theta} \in \boldsymbol{\Omega}} h(\boldsymbol{\theta}) d\boldsymbol{\theta} \tag{2.21}$$

Given this, the complete expected message length for data $\mathbf{y}$ is given by

$$I_{1\mathrm{D}}(\mathbf{y}) = -\log q(\boldsymbol{\Omega}) - \frac{1}{q(\boldsymbol{\Omega})} \int_{\boldsymbol{\theta} \in \boldsymbol{\Omega}} h(\boldsymbol{\theta}) \log f(\mathbf{y}|\boldsymbol{\theta}) d\boldsymbol{\theta} \tag{2.22}$$

where $f(\cdot)$ is likelihood function and $\boldsymbol{\Omega}$ is not necessarily contiguous, and may consist of several disjoint regions in space. Comparing this expression to (2.18) there are parallels and differences. The SMML expression uses the marginal probabilities to construct the assertion codeword, and finds the expected negative log-likelihoods for the data grouped within the region, given a fixed point estimate. In contrast (2.22) constructs its codeword using the prior probabilities of point estimates contained with the group, and finds the negative log-likelihoods of the different models within the group given a fixed datum $\mathbf{y}$. It turns out that (2.22) is minimised by finding curves, $\boldsymbol{\Omega}_\phi$, through model space, the interior of which is the region $\boldsymbol{\Omega}$, such that

$$-\frac{1}{q(\boldsymbol{\Omega})} \int_{\boldsymbol{\theta} \in \boldsymbol{\Omega}} h(\boldsymbol{\theta}) \log f(\mathbf{y}|\boldsymbol{\theta}) d\boldsymbol{\theta} - 1 < -\log f(\mathbf{y}|\boldsymbol{\phi}),\ \forall \boldsymbol{\phi} \in \boldsymbol{\Omega}_\phi \tag{2.23}$$

This is deemed the *boundary rule*. The curve $\boldsymbol{\Omega}_\phi$, and thus the boundary of the region $\boldsymbol{\Omega}$, must be a level-set of $f(\mathbf{y}|\boldsymbol{\theta})$. The approximation given by (2.22) is very close to the message length given by the full SMML construction, and indeed the uncertainty region $\boldsymbol{\Omega}$ contains models that could plausibly be used to encode $\mathbf{y}$ within a full SMML construction. The approximation however, while being computationally more tractable than the full SMML construction, still requires evaluation of integrals in potentially very high dimensions. There has been only one real attempt at finding a general algorithm to evaluate (2.22), and this is the MMC algorithm detailed in the sequel. One problem with the MMLD approximation is that while it yields a message length, it gives no indication of how to select a point estimate, beyond the restriction that it must lie inside $\boldsymbol{\Omega}$. Several proposals have been made, one being to simply randomly sample a $\boldsymbol{\theta}$ from prior over the

uncertainty region [1] (page 212), which at least is reasonably straightforward to implement. However, results in [38] suggest that this scheme does not perform particular well, at least for the Poisson and geometric distributions. Furthermore, due to the well known Rao-Blackwell theorem [39], in terms of any convex loss function, the risk of a random coding estimate could always be equalled (and often bettered) by a non-randomised estimator. Another proposal is to appeal to Wald's decision theory [40] and seek the Minimum Expected Kullback-Leibler divergence model within the uncertainty region, i.e.

$$\hat{\boldsymbol{\theta}}(\mathbf{y}) = \arg\min_{\boldsymbol{\theta}^* \in \boldsymbol{\Omega}} \left\{ \int_{\boldsymbol{\theta} \in \boldsymbol{\Omega}} \xi(\boldsymbol{\theta}) \Delta(\boldsymbol{\theta} || \boldsymbol{\theta}^*) d\boldsymbol{\theta} \right\} \tag{2.24}$$

where $\xi(\cdot)$ is a suitable distribution; endorsed choices include $\xi(\boldsymbol{\theta}) = h(\boldsymbol{\theta})$ which is claimed to be better for *induction* (by Wallace, [41], page 72) and $\xi(\boldsymbol{\theta}) = h(\boldsymbol{\theta}) f(\mathbf{y}|\boldsymbol{\theta})$ which is more tailored for *prediction* (by Dowe, [41], page 71). Note there is no need for normalisation as such a constant may simply be taken outside the integral. These estimators have the drawback that they are often difficult to compute (and often must be found numerically).

2.3.1 Random Coding

The MMLD approximation does not construct a complete codebook to transmit the data $\mathbf{y}$, and so it may appear at first that it is not possible to transmit MMLD message to a receiver. However, the message may actually be transmitted by the very clever mechanism of *random coding*. The random coding requires that both transmitter and receiver have a pseudo-random number generator designed to sample randomly from the prior, $h(\cdot)$. Both generators are seeded with the same starting value, and the transmitter begins by randomly drawing models from the random number generator until one lies within the uncertainty region $\boldsymbol{\Omega}$; denote this model $\boldsymbol{\theta}_k$. At this point, the transmitter sends the number of draws k to the receiver, who can then repeat the process and arrive at the model $\boldsymbol{\theta}_k$ with which the transmitter subsequently transmits the data $\mathbf{y}$. Let $p(k)$ denote a probability density over k; this is used to construct the code to transmit k to the receiver. Once the transmitter has arrived at a suitable model, the model and data may be transmitted using a code of length

$$I_{1\mathrm{R}}(\mathbf{y}, \boldsymbol{\theta}_k) = -\log p(k) - \log f(\mathbf{y}|\boldsymbol{\theta}_k) \tag{2.25}$$

It is clear that upon each draw from the random number generator is essentially a Bernoulli trial; accepted if $\boldsymbol{\theta}_k \in \boldsymbol{\Omega}$, and rejected if $\boldsymbol{\theta}_k \ni \boldsymbol{\Omega}$. The uncertainty region contains $q(\boldsymbol{\Omega})$ of the prior probability, and so the chance that a randomly drawn model will lie in $\boldsymbol{\Omega}$ is simply $q(\boldsymbol{\Omega})$. Thus, the number of draws required to arrive at a model inside the uncertainty region is distributed as per a geometric distribution, i.e. $k \sim \text{Geo}(q(\boldsymbol{\Omega}))$. However, it is claimed that by exploiting properties of the likelihood function, a code can be constructed such that $-\log p(k) \approx \log k$ [1] (page 210). Furthermore, it is not difficult to see that $\boldsymbol{\theta}_k$ is distributed independently of k according to the distribution

$$p(\boldsymbol{\theta}_k) = \frac{h(\boldsymbol{\theta}_k)}{\int_{\boldsymbol{\Omega}} h(\boldsymbol{\theta}) d\boldsymbol{\theta}}, \ \boldsymbol{\theta} \in \boldsymbol{\Omega} \tag{2.26}$$

Thus, the message length $I_{1\text{R}}(\mathbf{y}, \boldsymbol{\theta}_k)$ is a random variable; taking expectations with respect to k and $\boldsymbol{\theta}_k$ yields

$$\text{E}_{(k,\boldsymbol{\theta}_k)} \left[I_{1\text{R}}(\mathbf{y}, \boldsymbol{\theta}_k) \right] \approx -\log q(\boldsymbol{\Omega}) - \frac{1}{q(\boldsymbol{\Omega})} \int_{\boldsymbol{\theta} \in \boldsymbol{\Omega}} h(\boldsymbol{\theta}) \log f(\mathbf{y}|\boldsymbol{\theta}) d\boldsymbol{\theta} \tag{2.27}$$

which is clearly the MMLD message length expression (2.22). Thus the random coding process is able to transmit $\mathbf{y}$ with message lengths that are on average as short as the MMLD message lengths.

2.3.2 A Simple Monte-Carlo Approximation Based on Random Coding

Under the random coding scheme the message length $I_{1\text{R}}(\cdot)$ is a random variate; the mean of this variate, given by (2.27) is taken as the MMLD message length. A simple approximation of (2.27) may be constructed by Monte-Carlo simulation. The process is straightforward: let $\boldsymbol{\Theta}' \subset \boldsymbol{\Theta}$ denote a set of N_M models drawn randomly from the prior density. Let $\mathbf{j}$ denote a vector of indices of $\boldsymbol{\Theta}'$ that are deemed part of the uncertainty region, and let the vector $\mathbf{k} \in \mathbb{Z}^{N_K}$ be the vector of 'distances' between each randomly drawn model in the uncertainty region, i.e. the number of draws from the prior between accepted models given by

$$k_n = \begin{cases} j_n - j_{(n-1)} & \text{for } 1 < n \leq N_K \\ j_n & \text{for } n = 1 \end{cases} \tag{2.28}$$

From this an estimate of $I(\mathbf{y})$ may be found numerically as

$$I(\mathbf{y}) \approx \log \sum_{i=1}^{N_K} k_i - \log N_K - \frac{1}{N_K} \sum_{i=1}^{N_K} \log f\left(\mathbf{y}|\boldsymbol{\theta}_{k_i}\right) \tag{2.29}$$

A further simplification is to note that the average number of draws between accepted models may be estimated by (N_K/N_M), i.e. the ratio of accepted models to total number of sampled models. Assume that the models now are sorted in ascending order of their negative log-likelihood, i.e. so that $L(\mathbf{y}|\boldsymbol{\theta}_i) \leq L(\mathbf{y}|\boldsymbol{\theta}_{(i+1)})$. If the first N_K models are are assumed to be in the uncertainty region the message length estimate becomes

$$I(\mathbf{y}) \approx \log N_M - \log N_K - \frac{1}{N_K} \sum_{i=1}^{N_K} \log f\left(\mathbf{y}|\boldsymbol{\theta}_i\right) \tag{2.30}$$

If is straightforward to show that this expression has only a single minimum, and that it may be found by initialising $N_K = 1$ and then incrementing N_K until (2.30) no longer decreases. The assertion term, $(\log N_M - \log N_K)$ is strictly decreasing as N_K increases, and the summation in (2.30) is strictly nondecreasing as N_K increases, given that the models are sorted in ascending order of negative log-likelihood. Thus, the message length must initially be nonincreasing with N_K until some point, and then become strictly increasing. This method of random coding has several advantages and disadvantages. Its main advantage is simplicity: the only requirements are a likelihood function and a suitable method for sampling from the prior. Given that priors are often chosen from amongst well known distributions, very fast samplers are usually available. The second advantage it possesses is that asymptotically, as $N_K \to \infty$, the estimated message length given by (2.29) converges to the exact MMLD message length (2.27),though the rate of convergence may be very slow. The primary, and serious disadvantage, is that in computational (as opposed to statistical) terms the method is very inefficient. The smaller the uncertainty region the more draws are required for the approximation to work well, as most sampled models will lie outside the region and the detail length estimate will be poor. If N_Q models are roughly desired to be used to approximate the uncertainty region, and the prior mass over the uncertainty region is $q(T)$, then the number of required draws is roughly

$$N_M \approx N_Q q^{-1}(T) \tag{2.31}$$

From this it is obvious that as $q^{-1}(T) \to 0$, $N_M \to \infty$. Additionally, as likelihoods are evaluated for each candidate model, the smaller the uncertainty region, the likelihood evaluation is performed for large numbers of models that are will be rejected. The inefficiency becomes greater as N increases, as the volume of the uncertainty region is in general inversely proportional to the number of data points. This is also the case as the number of model parameters P increases. Even assuming the region is roughly hyper-elliptical in shape, the ratio of cube volume to ellipse volume rapidly diminishes as $D \to \infty$. This extreme inefficiency means that for all but low dimension models with small amounts of data, the method is in practice not tenable. The inefficiency may be circumvented by a different Monte Carlo approach, which is covered next.

2.3.3 Message from Monte Carlo (MMC)

The mechanism of Random Coding provides a method by which MMLD messages may be transmitted, and leads to a simple algorithm by which the MMLD message length may be estimated. However, the naive implementation of random coding has several drawbacks, most prominent being the increasing computational inefficiency as N increases or the region prior probability mass $q(\mathbf{\Omega})$ decreases. The Message from Monte Carlo (MMC) algorithm described in [13, 21, 41] attempts to circumvent this inefficiency by using the posterior distribution as an importance density from which to sample. The hope is then that samples will be densely populated near the uncertainty region, and significantly less samples will be required than the random coding approach drawing directly for the prior. Let $\mathbf{\Theta}' \subset \mathbf{\Theta}$ be a finite set of N_M samples drawn from the posterior, i.e.

$$\boldsymbol{\theta}'_i \sim \frac{f(\mathbf{y}|\boldsymbol{\theta})h(\boldsymbol{\theta})}{\int_{\mathbf{\Theta}} f(\mathbf{y}|\boldsymbol{\theta})d\boldsymbol{\theta}},\ i = 1 \ldots N_M \tag{2.32}$$

In the simple random coding algorithm, the estimation of the assertion and detail lengths was trivial because the samples were drawn directly from the prior; in the MMC algorithm, the sample are drawn from an importance density (the posterior) and so must be reweighted. Let $w(\boldsymbol{\theta}_i) = (f(\mathbf{y}|\boldsymbol{\theta}_i))^{-1}$ denote the weight of each sample. Given a set $\mathbf{Q} \subset \mathbf{\Theta}'$ of N_K sampled models, the approximate assertion length under the MMC algorithm is

$$\int_{\boldsymbol{\theta}\in\mathbf{\Omega}} h(\boldsymbol{\theta})d\boldsymbol{\theta} \approx \log\left(\sum_{\boldsymbol{\theta}\in\mathbf{\Theta}'} w(\boldsymbol{\theta})\right) - \log\left(\sum_{\boldsymbol{\theta}\in\mathbf{Q}} w(\boldsymbol{\theta})\right) \tag{2.33}$$

Compare (2.33) to the first two terms of (2.30). If the samples were drawn from the prior, they would have $w(\boldsymbol{\theta}_i) = 1, i = 1...N_M$, and the two expressions would exactly coincide. The estimate of the detail under the MMC algorithm is

$$\left(\int_{\boldsymbol{\theta}\in\boldsymbol{\Omega}} h(\boldsymbol{\theta})d\boldsymbol{\theta}\right)^{-1} \int_{\boldsymbol{\Omega}} h(\boldsymbol{\theta})L(\mathbf{y}|\boldsymbol{\theta})d\boldsymbol{\theta} \approx \left(\sum_{\boldsymbol{\theta}\in\mathbf{Q}} w(\boldsymbol{\theta})\right)^{-1} \left(\sum_{\boldsymbol{\theta}\in\mathbf{Q}} w(\boldsymbol{\theta})L(\mathbf{y}|\boldsymbol{\theta})\right) \tag{2.34}$$

It is clear that if $w(\boldsymbol{\theta}) = 1$ for all samples, then (2.34) simplifies to the arithmetic mean of the negative log-likelihood of samples in $\mathbf{Q}$; this is exactly the same as the last term in (2.30). The MMC algorithm finds the set $\mathbf{Q}$ that minimises the complete estimated message length by successively adding models to $\mathbf{Q}$ in ascending order of their negative log-likelihood, until the boundary rule is violated. Various extensions, including schemes for selecting point estimates and improving inference by various boundary rule conditions are detailed in a successful application of the algorithm to polynomial selection [21]. The main points interest of the MMC algorithm are summarised as

1. The use of the posterior as an importance density means that most of the samples are concentrated near the uncertainty region, yielding a significant improvement in efficiency over the naive random coding algorithm.

2. As the region is implicitly defined by the set of samples $\mathbf{Q}$, the algorithm is very flexible and able to handle very irregular region geometries. This is a strong point of the algorithm, as poorly behaved likelihood functions do not cause the approximation to break down. By the same token, however, the algorithm does not yield a well defined boundary of the region, which may well be very useful given the interesting interpretation of the uncertainty region as a set of possible explanations for $\mathbf{y}$.

3. The algorithm requires the evaluation of the likelihood function for many samples, which yields at least a time complexity of $(N \cdot N_M)$ for many model classes; large numbers of samples may be required in high dimensions to achieve reasonable approximations of the integrals and so the algorithm can be expensive.

4. The requirement to sample from the posterior is the another serious drawback; sampling from the posterior of many distributions is a non-trivial task. For some exponential models with conjugate priors, the problem is reduced significantly in complexity, but for a very large range of models sampling is a computationally expensive and difficult process, especially if the Markov-Chains do not mix rapidly.

The final requirement does lead to the question of whether the posterior density is really the only suitable importance density. The maximum likelihood estimate is perhaps a more useful point around which to sample, as it must always be included in $\mathbf{Q}$, and the likelihood surface is bowled (at least locally) around this point. One alternative possibility is to use a simple normal approximation around the maximum likelihood estimate; this will lead to a much simpler importance density that may have comparable performance to the full posterior. This consideration however, is not pursued further in this thesis as an algorithm is developed later that uses a much smaller set of systematically chosen samples to evaluate the detail integral, by exploiting some properties of the boundary rule.

2.4 I1F: I1D and Expected Uncertainty Regions

This section details a new message length approximation used in this thesis, named I1F; it is included here to ensure a natural flow of topics as it will be shown to provide a link between the MMLD approximation and the MML87 approximation. The I1F approximation is an extension to the I1D approximation that allows for explicit point estimation by application of the message length formula. It is an extension of the Fairly Strict Minimum Message Length estimator, in a similar way that the Ideal Group estimator is itself an extension of the SMML estimator. The Fairly Strict Minimum Message Length (FSMML) [42] estimator is conceptually very similar to the SMML estimator in that it builds a complete codebook before examining any data. The key difference is that the FSMML estimator constructs the codebook in *parameter space* as opposed to data space, using the prior density to construct the codewords rather than the marginal distribution of the data, i.e. there exists a disjoint set of subsets of $\mathbf{\Theta}$ that completely cover the parameter space, each having a representative point estimate. It has been previously noted in Section 2.3 that Wallace [1] felt the complete partitioning of parameter space was not a good idea and would not perform well as a general method for inference; however, the I1F estimator

based on the idea of parameter space partitioning does not seek to perform a complete partitioning, but instead finds a region under *ideal conditions* (i.e. ignoring all other possible regions that would be present in a complete partitioning). This is similar to the MMLD approximation, which has been shown to perform quite well. Furthermore, it is shown in Section (2.5) that under suitable simplifying conditions, the I1F coincides exactly with the MML87 estimator which was strongly endorsed by Wallace, strongly suggesting that for suitably regular problems it will possess the excellent asymptotic properties of MML87.

As a side note, there is some confusion in the literature as to the correct name of the criterion that completely partitions the hypothesis space. Published references to it appear in [21, 43, 44] under the name 'FSMML', while [45] and Wallace's unpublished draft manuscript [42] (which is a draft of what eventually became [1]) refers to it as 'MMLA'. In this thesis, the name FSMML has been used, although it would appear the choice is rather arbitrary. This is also discussed in [37].

2.4.1 The Fairly Strict Ideal Region Estimator

The MMLD approximation has previously been discussed in Section 2.3 and the problem of choosing a suitable point estimate from the uncertainty region was discussed. An alternative approach, proposed in this thesis, is to use the *expected* negative log-likelihoods in place of the negative log-likelihoods in the MMLD formula. This may be interpreted in two ways:

1. Assume that a particular model $\boldsymbol{\theta}^*$ is the true 'generating' process from which the data set $\mathbf{y}$ has been drawn, i.e. $\mathbf{y} \sim f_{\boldsymbol{\theta}^*}$; under this hypothesis, for a given random $\mathbf{y}$ the expected distribution that will minimise the negative log-likelihood will be $f_{\boldsymbol{\theta}^*}$, and that the expected cost of encoding a $\mathbf{y}$ drawn from $\boldsymbol{\theta}^*$ using $\boldsymbol{\theta}$ would be the Kullback-Leibler (KL) divergence between the two. Thus the *expected round-off* cost of encoding a data set drawn from $f_{\boldsymbol{\theta}^*}$ using a sub-optimal model $\boldsymbol{\theta}$ is given by $\Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*)$.

2. The second interpretation is to construct a single region of an FSMML codebook, fixing $\boldsymbol{\theta}^*$ as the point estimate. This is analogous to the case of the Ideal Group estimator for SMML, in that the FSMML region is optimised without regard to the

remaining codebook regions that a complete FSMML construction would require; hence, *ideal*.

The latter interpretion was applied to a degree by Fitzgibbon in [21] to help select a point estimate; however, in that case, models were allocated to the region only if they were initially part of the MMLD uncertainty region. This scheme was therefore another method of selecting a point estimate from an MMLD region, while minimisation of the I1F approximation explicitly yields a point estimate. Given a set $\mathbf{\Omega} \subset \mathbf{\Theta}$, the I1F message length is given by

$$I_{1\mathrm{F}}(\mathbf{y}, \boldsymbol{\theta}^*) = -\log q(\mathbf{\Omega}) - \log f(\mathbf{y}|\boldsymbol{\theta}^*) + \frac{1}{q(\mathbf{\Omega})} \int_{\boldsymbol{\theta} \in \mathbf{\Omega}} h(\boldsymbol{\theta}) \Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*) d\boldsymbol{\theta} \tag{2.35}$$

where $\boldsymbol{\theta}^*$ is the 'point estimate' for the region, $\Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*)$ is the KL divergence between generating model $\boldsymbol{\theta}^*$ and approximating model $\boldsymbol{\theta}$, and $q(\mathbf{\Omega}) = \int_{\mathbf{\Omega}} h(\boldsymbol{\theta}) d\boldsymbol{\theta}$. The assertion length and the expected round-off costs in the detail may be combined into a single *model cost*

$$\mathcal{D}(\boldsymbol{\theta}^*) = -\log q(\mathbf{\Omega}(\boldsymbol{\theta}^*)) + \frac{1}{q(\mathbf{\Omega}(\boldsymbol{\theta}^*))} \int_{\boldsymbol{\theta} \in \mathbf{\Omega}(\boldsymbol{\theta}^*)} h(\boldsymbol{\theta}) \Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*) d\boldsymbol{\theta} \tag{2.36}$$

where the notation $\mathbf{\Omega}(\boldsymbol{\theta}^*)$ refers to the region that minimises (2.36) for model $\boldsymbol{\theta}^*$. This region, $\mathbf{\Omega}(\boldsymbol{\theta}^*)$ is deemed the *expected uncertainty region*. The term 'model cost' is used as (2.36) is the price paid in nits for the privilege of stating a fully specified model. From this a point estimate may be explicitly found by solving

$$\hat{\boldsymbol{\theta}}_{1\mathrm{F}}(\mathbf{y}) = \arg \min_{\boldsymbol{\theta}^* \in \mathbf{\Theta}} \left\{ L(\mathbf{y}|\boldsymbol{\theta}^*) + \min_{\mathbf{\Omega} \subset \mathbf{\Theta}} \left\{ -\log q(\mathbf{\Omega}) + \frac{1}{q(\mathbf{\Omega})} \int_{\boldsymbol{\theta} \in \mathbf{\Omega}} h(\boldsymbol{\theta}) \Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*) d\boldsymbol{\theta} \right\} \right\} \tag{2.37}$$

This formulation provides two advantages: it explicitly describes an invariant scheme for Bayes point estimation based on the MMLD approximation, and thus it allows different point hypotheses to be compared against each other. This has two consequences:

- By seeking the model $\boldsymbol{\theta}^*$ that explicitly minimises (2.37), an invariant Bayesian point estimator may be found that is well behaved even when the model class is badly behaved and the MML87 approximation performs poorly.
- The I1F message length may be used to perform model selection in conjuction with other estimator schemes, i.e. in those cases where the estimates that minimise (2.37) may be to difficult to find or too computationally intensive.

One way to connect (2.22) and (2.35) is to note that the MMLD message length based soley on the likelihood of the data is akin to replacing $\Delta(\cdot||\cdot)$ in (2.35) with the *empirical* Kullback-Leibler divergence

$$\Delta_N(\boldsymbol{\theta}||\boldsymbol{\theta}^*) = L(\mathbf{y}|\boldsymbol{\theta}) - L(\mathbf{y}|\boldsymbol{\theta}^*) \tag{2.38}$$

where $\boldsymbol{\theta}^*$ is now restricted to be the Maximum Likelihood estimate. Under these changes, the MMLD and I1F message length formulas coincide. The I1F message length equation given by (2.35) may be viewed as the basis of the well known MML87 message length estimator, and a new derivation to connect the two is presented in the sequel.

2.4.2 Properties of the Model Cost

This section proves the invariance of the Model Cost under one-to-one transformations, and the existence of the Boundary rule for Model Cost. The main tool for the former proof is the transformation of variables rule, that is if $\boldsymbol{\alpha} = f(\boldsymbol{\theta})$ is a differentiable, one-to-one transformation then the transformed function $g(\boldsymbol{\alpha})$ such that

$$g(\boldsymbol{\theta})|d\boldsymbol{\theta}| = g(\boldsymbol{\alpha})|d\boldsymbol{\alpha}| \tag{2.39}$$

is given by

$$g(\boldsymbol{\alpha}) = |\mathbf{G}(f^{-1})|g(f^{-1}(\boldsymbol{\alpha})) \tag{2.40}$$

where $\mathbf{G}(f^{-1})$ is the Jacobian with entries $G_{(i,j)} = \frac{\partial f^{-1}(\alpha_i)}{\partial \alpha_j}$ [46].

Proposition 2.1: The Model Cost function given by (2.36) is invariant under one-to-one transformation $\boldsymbol{\alpha} = f(\boldsymbol{\theta})$, assuming $h(\boldsymbol{\alpha})$ is constructed from $h(\boldsymbol{\theta})$ to satisfy (2.39).

Proof: Examining (2.36), the invariance of the term

$$q(\boldsymbol{\Omega}) = \int_{\boldsymbol{\theta}\in\boldsymbol{\Omega}} h(\boldsymbol{\theta})d\boldsymbol{\theta} \tag{2.41}$$

follows by the construction of $h(\boldsymbol{\alpha})$ to satisfy (2.39). As this term is invariant, so is the $1/q(\boldsymbol{\Omega})$ outside the integral of the second term. It is known that

$$\Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*) = \Delta(f^{-1}(\boldsymbol{\alpha})||f^{-1}(\boldsymbol{\alpha}^*)) = \Delta(\boldsymbol{\alpha}||\boldsymbol{\alpha}^*)$$

as the transformation does not affect the ratio of the likelihood functions. The integral in the second term may then be written in $\boldsymbol{\alpha}$-space as

$$\int_{\boldsymbol{\alpha}\in f(\boldsymbol{\Omega})} h(\boldsymbol{\alpha})\Delta(\boldsymbol{\alpha}||\boldsymbol{\alpha}^*)d\boldsymbol{\alpha} = \int_{\boldsymbol{\alpha}\in f(\boldsymbol{\Omega})} |\mathbf{G}(f^{-1})|h(f^{-1}(\boldsymbol{\alpha}))\Delta(f^{-1}(\boldsymbol{\alpha})||f^{-1}(\boldsymbol{\alpha}^*))d\boldsymbol{\alpha} \quad (2.42)$$

Now, to prove that this term is invariant, define the new function

$$g(\boldsymbol{\theta}) = h(\boldsymbol{\theta})\Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*) \quad (2.43)$$

and, again assuming $\boldsymbol{\alpha} = f(\boldsymbol{\theta})$, construct $g(\boldsymbol{\alpha})$ to satisfy (2.39); this yields

$$\begin{aligned} \int_{\boldsymbol{\alpha}\in f(\boldsymbol{\Omega})} g(\boldsymbol{\alpha})d\boldsymbol{\alpha} &= \int_{\boldsymbol{\alpha}\in f(\boldsymbol{\Omega})} |\mathbf{G}(f^{-1})|h(f^{-1}(\boldsymbol{\alpha}))\Delta(f^{-1}(\boldsymbol{\alpha})||f^{-1}(\boldsymbol{\alpha}^*))d\boldsymbol{\alpha} \quad (2.44) \\ &= \int_{\boldsymbol{\alpha}\in f(\boldsymbol{\Omega})} h(\boldsymbol{\alpha})\Delta(\boldsymbol{\alpha}||\boldsymbol{\alpha}^*)d\boldsymbol{\alpha} \quad (2.45) \end{aligned}$$

and as $g(\boldsymbol{\alpha})d\boldsymbol{\alpha} = g(\boldsymbol{\theta})d\boldsymbol{\theta}$, this completes the invariance proof. **Q.E.D.**

Proposition 2.2: The Model Cost function (2.36) satisfies the 'Boundary Rule'.

Proof: That the Model Cost form of MMLD satisfies the Boundary Rule is trivial; it has been shown by Wallace (see for example, [41], Appendix A, page 175) that any function of the form

$$g(\boldsymbol{\Omega}) = -\log \int_{\boldsymbol{\theta}\in\boldsymbol{\Omega}} p(\boldsymbol{\theta})d\boldsymbol{\theta} + \frac{1}{\int_{\boldsymbol{\theta}\in\boldsymbol{\Omega}} p(\boldsymbol{\theta})d\boldsymbol{\theta}} \int_{\boldsymbol{\theta}\in\boldsymbol{\Omega}} p(\boldsymbol{\theta})q(\boldsymbol{\theta})d\boldsymbol{\theta} \quad (2.46)$$

satisfies the Boundary Rules, as long as $q(\boldsymbol{\theta})$ is a continuous function of $\boldsymbol{\theta}$. Choosing $p(\boldsymbol{\theta}) = h(\boldsymbol{\theta})$ and $q(\boldsymbol{\theta}) = \Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*)$ renders (2.46) to coincide with (2.36); as $\Delta(\cdot)$ satisfies the continuity restriction, the Model Cost must satisfy the boundary rule. **Q.E.D.**

2.4.3 Random Coding and I1F

The simple random coding algorithm can be applied to estimating the I1F message length in a very similar fashion to the estimation of the MMLD message length. The algorithm is very similar, with the new message length being found by Random Coding given by

$$I_{1\mathrm{F}}(\mathbf{y}) = L(\mathbf{y}|\boldsymbol{\theta}^*) + \mathrm{E}_{(k,\hat{\boldsymbol{\theta}}_k)}\left[-\log f(k) + \Delta(\hat{\boldsymbol{\theta}}_k||\boldsymbol{\theta}^*)\right] \quad (2.47)$$

where k and $\hat{\boldsymbol{\theta}}_k$ are generated by the same process as in Section 2.3.1, and $\boldsymbol{\theta}^*$ is the point estimate under consideration. If the expectations are taken with respect to the prior over the expected uncertainty region, this expression reduces to (2.35). Application of the algorithm in Section 2.3.2 to estimate (2.35) is possible by a straightforward modification; instead of finding the negative log-likelihoods of the data given all N_M sampled models, the KL divergences between the sample models and the point estimate $\boldsymbol{\theta}^*$ are found. The models are then sorted in ascending order of their KL divergences, so that $\Delta(\boldsymbol{\theta}_i||\boldsymbol{\theta}^*) \leq \Delta(\boldsymbol{\theta}_{(i+1)}||\boldsymbol{\theta}^*)$. The algorithm then involves finding the N_K that minimises

$$I(\mathbf{y}) \approx L(\mathbf{y}|\boldsymbol{\theta}^*) - \log N_K + \log N_M - \frac{1}{N_K} \sum_{i=1}^{N_K} \Delta\left(\boldsymbol{\theta}_i||\boldsymbol{\theta}^*\right) \tag{2.48}$$

Again, it is obvious that there is only a single minimum, so that the algorithm may proceed by simply incrementing N_K until the message length no longer decreases. Clearly, this algorithm suffers from the same problems as the Random Coding algorithm for the conventional MMLD estimator, and is not considered viable for most practical applications.

2.5 The MML87 Estimator

The previous section examined the MMLD approximation, and an extension of it based on expected likelihoods called the I1F approximation. These differed from SMML in two key points: they were only concerned with a single coding region and both built the codebook for the assertion using the prior distribution over the model space rather than the marginal distribution of the data. This section now examines the most practical approximation yet devised for Minimum Message Length inference, the Wallace and Freeman estimator [15], from here-on termed the MML87 estimator (due to the year of its introduction). Although the MML87 estimator was devised approximately a decade prior to the introduction of the MMLD estimator, this thesis shows a new result that the MML87 estimator is the exact solution to the I1F message length equation under certain simplifying assumptions. This differs from previous derivations which rely on slightly different arguments. The MML87 approximation makes several assumptions about the priors and the likelihood function that are required for the resulting message length to closely approximate the I1F message length. These are:

- That the priors are roughly flat within the locale of the uncertainty region
- That the KL divergence is roughly quadratic around the parameter estimate $\boldsymbol{\theta}^*$, at least within the locale of the uncertainty region
- That the KL divergence surface has only one minimum which is located at $\boldsymbol{\theta}^*$

The first assumption is, in most cases, reasonably innocuous. In general, it only causes problems if the priors are very sharply curved and the amount of data is small. The second restriction is much more serious: as the uncertainty region cannot be known prior to application of MML87 (for if it were, there would be no point in applying MML87), it is in effect a 'leap of faith' that the uncertainty region will encompass only that area of the KL divergence that is roughly quadratic. The third assumption only causes problems if the model is parametricly redundant; most of the KL divergences of regular distributions are generally unimodal. With these assumptions in mind, a new derivation of the MML87 estimator is made next.

2.5.1 Derivation

The derivation of the MML87 estimator presented in this thesis is new and shows that the MML87 message length equation is an exact solution to the newly proposed I1F message length equation, under the conditions of a locally uniform prior and and locally quadratic KL divergence. It is similar to the derivation given in [47]; however, in that case the MML87 estimator is derived from the MMLA (or FSMML) approximation rather than from the I1F approximation. Begin by considering the I1F message length equation, given a set $\boldsymbol{\Omega} \subset \boldsymbol{\Theta}$

$$I_{1\mathrm{F}}(\mathbf{y}, \boldsymbol{\theta}^*) = L(\mathbf{y}|\boldsymbol{\theta}^*) - \log q(\boldsymbol{\theta}^*) + \frac{1}{q(\boldsymbol{\theta}^*)} \int_{\boldsymbol{\theta} \in \boldsymbol{\Omega}} h(\boldsymbol{\theta}) \Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*) d\boldsymbol{\theta} \tag{2.49}$$

where

$$q(\boldsymbol{\theta}^*) = \int_{\boldsymbol{\theta} \in \boldsymbol{\Omega}} h(\boldsymbol{\theta}) d\boldsymbol{\theta} \tag{2.50}$$

is the prior probability assigned to the codeword, $\boldsymbol{\theta}^*$ is a point estimate that lives within the uncertainty region and $\Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*)$ is the Kullback-Leibler divergence between the 'true' model $\boldsymbol{\theta}^*$ and the model $\boldsymbol{\theta}$. The first simplifying assumption made by the MML87 approximation is that the prior is approximately uniform over the locale of the uncertainty

region. Using this assumption, the message length may be approximated as

$$I_{1\mathrm{F}}(\mathbf{y}, \boldsymbol{\theta}^*) \approx -\log h(\boldsymbol{\theta}^*) - \log w(\boldsymbol{\theta}^*) + \frac{1}{w(\boldsymbol{\theta}^*)} \int_{\boldsymbol{\theta} \in \boldsymbol{\Omega}} \Delta(\boldsymbol{\theta} || \boldsymbol{\theta}^*) d\boldsymbol{\theta} \tag{2.51}$$

where $w(\cdot) = \mathrm{vol}(\boldsymbol{\Omega})$ is the *volume* of uncertainty region. This still leaves the integral in (2.51) to be evaluated. The MML87 approximation now makes a further assumption that the KL divergence curve may be adequately captured by a second order Taylor series expansion around $\boldsymbol{\theta}^*$, at least in the locale of the uncertainty region. Making this assumption leads to

$$I_{1\mathrm{F}}(\mathbf{y}, \boldsymbol{\theta}^*) \approx -\log h(\boldsymbol{\theta}^*) - \log w(\boldsymbol{\theta}^*) + \frac{1}{w(\boldsymbol{\theta}^*)} \left(\int_{\boldsymbol{\theta} \in \boldsymbol{\Omega}} \mathbf{k}(\boldsymbol{\theta}^*) \boldsymbol{\delta}^T + \frac{1}{2} \boldsymbol{\delta} \mathbf{J}(\boldsymbol{\theta}^*) \boldsymbol{\delta}^T d\boldsymbol{\theta} \right) \tag{2.52}$$

where $\boldsymbol{\delta} = (\boldsymbol{\theta}^* - \boldsymbol{\theta})$, the elements of $\mathbf{k}(\boldsymbol{\theta}^*)$ are

$$k_i(\boldsymbol{\theta}^*) = \frac{\partial \Delta(\boldsymbol{\theta} || \boldsymbol{\theta}^*)}{\partial \theta_i}, \; i = 1 \ldots D \tag{2.53}$$

and the elements of $\mathbf{J}(\boldsymbol{\theta}^*)$ are

$$J_{(i,j)}(\boldsymbol{\theta}^*) = \frac{\partial^2 \Delta(\boldsymbol{\theta} || \boldsymbol{\theta}^*)}{\partial \theta_i \partial \theta_j}, \; i, j = 1 \ldots D \tag{2.54}$$

It is clear that the matrix $\mathbf{J}(\cdot)$ is Fisher's Information Matrix, i.e. the matrix of expected second derivatives of the negative log-likelihood [14]. The integral in (2.52) is a prior-weighted average over the region $\boldsymbol{\Omega}$, i.e. an expectation w.r.t. the prior normalised over the region. The third assumption is now made: as nothing is known about the region geometry beyond its requirement for symmetry, the uncertainty region is chosen to be congruent to a cell of suitably scaled and translated optimal second moment quantising lattice. Under this assumption, the first moment of the quantisation error within the region is

$$\int_{\boldsymbol{\theta} \in \boldsymbol{\Omega}} \boldsymbol{\delta} d\boldsymbol{\theta} = \mathbf{0} \tag{2.55}$$

This may also be determined by the fact that $\mathbf{k}$ is the expectation of the score with respect to the data, which is always zero. Only the quadratic term now remains to be evaluated. At this point, Wallace and Freeman introduced a clever device to simplify the message length expression. Through some transformations (see [1], pages 234-235), the round-off

integral may be solved to yield

$$\int_{\boldsymbol{\theta}\in\boldsymbol{\Omega}} \frac{1}{2}\boldsymbol{\delta}\mathbf{J}(\boldsymbol{\theta}^*)\boldsymbol{\delta}^T d\boldsymbol{\theta} = \sqrt[D]{|\mathbf{J}(\boldsymbol{\theta}^*)|} D\kappa_D w(\boldsymbol{\phi}^*)^{(1+2/D)} \tag{2.56}$$

where κ_D is the normalised second moment of the quantising lattice structure [48] from which $\boldsymbol{\Omega}$ is taken. The I1F message length equation then becomes

$$I_{1\mathrm{F}}(\mathbf{y}, \boldsymbol{\theta}^*) \approx -\log h(\boldsymbol{\theta}^*) - \log w(\boldsymbol{\theta}^*) + \frac{1}{2}\sqrt[D]{|\mathbf{J}(\boldsymbol{\theta}^*)|} D\kappa_D w(\boldsymbol{\phi}^*)^{(2/D)} \tag{2.57}$$

Minimising this expression with respect to $w(\boldsymbol{\theta}^*)$ yields

$$w(\boldsymbol{\theta}^*) = \frac{1}{\sqrt{|\mathbf{J}(\boldsymbol{\theta}^*)|}\kappa_D^{(D/2)}} \tag{2.58}$$

Substitution of (2.58) into (2.57) yields the MML87 message length equation

$$I_{1F}(\mathbf{y}, \boldsymbol{\theta}^*) \approx I_{87}(\mathbf{y}, \boldsymbol{\theta}^*) = L(\mathbf{y}|\boldsymbol{\theta}^*) - \log h(\boldsymbol{\theta}^*) + \frac{1}{2}\log|\mathbf{J}(\boldsymbol{\theta}^*)| + \frac{D}{2}(\log\kappa_D + 1) \tag{2.59}$$

Finally, this expression can be related back to the components of the I1F message length as

$$q(\boldsymbol{\theta}^*) \approx \frac{h(\boldsymbol{\theta}^*)}{\sqrt{|\mathbf{J}(\boldsymbol{\theta}^*)|}\kappa_D^{(D/2)}} \tag{2.60}$$

$$\frac{1}{q(\boldsymbol{\theta}^*)}\int_{\boldsymbol{\theta}\in\boldsymbol{\Omega}} h(\boldsymbol{\theta})\Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*)d\boldsymbol{\theta} \approx \frac{D}{2} \tag{2.61}$$

where the equalities are exact if the assumptions under which the MML87 estimator is derived are met. It is clear to see that under the assumptions from which the MML87 estimator is derived the round-off cost is dependent only on the dimensionality of the model class, and only the assertion length is dependent on the particular point estimate $\boldsymbol{\theta}^*$. This is not generally the case with the exact I1F estimator, at least in the non-asymptotic case. Inference by MML87 is then performed by seeking the estimates that minimise (2.59).

It is clear to see that $\sqrt{|\mathbf{J}(\boldsymbol{\theta}^*)|}$ is proportional to how sensitive the expected likelihood function is to changes in $\boldsymbol{\theta}^*$; assuming reasonably flat priors, an MML87 estimator is dominated by a trade-off between a fitting the data well (the log-likelihood term) and selecting an estimate from a region with low parameter sensitivity (the Fisher Information

term). This is the salient feature of the MML87 estimator, and it results in an estimator which is significantly less 'confident' than the ML estimator (or MAP estimator, assuming flattish priors). If two estimates yield very similar log-likelihoods, but one comes from a region where the parameter sensitivity is lower, the MML87 estimator will select this estimate at the expense of a slight increase in log-likelihood. A region of the likelihood function with lower sensitivity usually indicates a more 'vague' model of the data, and one which is less likely to be 'surprised' by unobserved data.

2.5.2 Asymptotic Properties of MML87

Consider the MML87 message length equation. For all but a few densities, the log-determinant of the Information Matrix may be written as

$$\frac{1}{2}\log|\mathbf{J}(\boldsymbol{\theta}^*)| = \frac{P}{2}\log N + \frac{1}{2}\log|\mathbf{J}_1(\boldsymbol{\theta}^*)| \tag{2.62}$$

where $\mathbf{J}_1(\boldsymbol{\theta}^*)$ is the Information Matrix for a single datum. Holding the number of parameters, P, fixed, the asymptotic behaviour of the message length as $N \to \infty$ may be found by disregarding those terms dominated by N, leaving

$$I_{87}(\mathbf{y}, \boldsymbol{\theta}^*) = L(\mathbf{y}|\boldsymbol{\theta}^*) + \frac{P}{2}\log N \tag{2.63}$$

This indicates that, asymptotically, each parameter can be stated to a suitable accuracy in $\left(\frac{1}{2}\log N\right)$ nits. This corroborates with similar results by alternative criteria and is discussed in the sequel in more detail. Additionally, examination of (2.63) reveals that asymptotically the MML estimates do not depend on the coding volume, and coincide with the Maximum Likelihood estimates. This is comforting, as the ML estimates are well known to have strong asymptotic properties, and these will be shared by the MML87 estimates. An important exception occurs when P is implicitly dependent on N; that is, when the number of model parameters P grows as some function of the number of observations. In this case there is no clear 'asymptotic' behaviour as the data will never swamp the Fisher Information term or the priors. This disagreement with Maximum Likelihood turns out to be advantageous, as MML87 estimates have been shown to be statistically consistent for such troublesome problems as the Neyman-Scott problem [49].

2.6 MML Inference of the Binomial Distribution

To complete this discussion of the MML87 approximation we present a worked derivation an MML87 estimator for a simple and well studied model: the Binomial distribution. The MML87 estimator for the Binomial was derived quite some time ago [1] (pages 246-247), but is presented here as it is one of the few models for which a complete SMML estimator solution may be obtained, and is illustrative of several important aspects of MML estimators.

2.6.1 The MML87 Estimator

The Binomial distribution is used to model the outcome of a series of unordered independent Bernoulli trials. Given N trials $\mathbf{y} \in \{0,1\}^N$, let $\bar{y} = \sum_{i=1}^{N} y_i$ be the number of observed '1's. For probability $\theta \in [0,1]$ of a '1', the Binomial likelihood is

$$f(\mathbf{y}|\theta) = \binom{N}{\bar{y}} \theta^{\bar{y}} (1-\theta)^{(N-\bar{y})} \tag{2.64}$$

where the first term accounts for the fact the order of the trials is considered irrelevant. From this the negative log-likelihood is found as

$$L(\mathbf{y}|\theta) = -\log \binom{N}{\bar{y}} - \bar{y} \log \theta - (N-\bar{y}) \log (1-\theta) \tag{2.65}$$

It is then easy to derive the Maximum Likelihood estimator by minimising (2.65)

$$\hat{\theta}_{\mathrm{ML}}(\mathbf{y}) = \frac{\bar{y}}{N} \tag{2.66}$$

To find the MML87 estimator, a prior over θ must first be chosen; a simple uniform prior $h(\theta) = 1, \theta \in [0,1]$ is appropriate if nothing is known about θ. The next step is to find the Fisher Information $J(\theta)$. Differentiating (2.65) with respect to θ yields

$$\frac{\partial^2}{\partial \theta^2} \{L(\mathbf{y}|\theta)\} = \frac{\bar{y}}{\theta^2} + \frac{N-\bar{y}}{(1-\theta)^2} \tag{2.67}$$

Taking expectations yields

$$J(\theta) = \mathrm{E}\left[\frac{\bar{y}}{\theta^2} + \frac{N-\bar{y}}{(1-\theta)^2}\right] = \frac{N}{\theta(1-\theta)} \tag{2.68}$$

Substitution of the prior and (2.68) into the MML87 estimator expression (2.59) and minimisation with respect to θ yields the MML87 estimate

$$\hat{\theta}_{87}(\mathbf{y}) = \frac{\bar{y} + \frac{1}{2}}{N + 1} \tag{2.69}$$

2.6.2 Comparison of the MML87 and ML Estimators

Comparing (2.66) to (2.69) it may be observed that the MML87 estimator augments the data set with half a head and half a tail. This has the effect of making the MML87 estimator less 'confident' than the ML estimator, particularly when for the extreme cases when $y = 0$ or $y = N$. In these situations, the ML estimator will select $\theta = 0$ or $\theta = 1$, completely ruling out the chance of a success (or failure) in any future throws; in contrast, the MML estimator will select a small chance of success (or failure) even when it has seen no evidence of such an occurence. If a conjugate Beta prior (of which the uniform is a special case) with parameters (α, β) is chosen, the MAP estimator for θ is given by

$$\hat{\theta}_{MAP}(\mathbf{y}) = \frac{y + \alpha - 1}{N + \alpha + \beta - 2} \tag{2.70}$$

It is immediately obvious that the MML87 estimator is a special case of the MAP estimator (with $\alpha = \beta = \frac{3}{2}$); in fact, if a conjugate Beta prior is chosen for the MML87 Binomial estimator, the two can always be made to coincide for suitable choices of (α, β). This is due to the fact that for this model (and for other exponential models), the Fisher Information is 'conjugate' with the likelihood, that is, the effect of the Fisher Information may be absorbed completely into the conjugate prior. The difference between the MAP and MML87 estimator is that if particular values of α and β are chosen to match any real prior beliefs we may have, the two estimators will differ. This is clear for the case of $\alpha = \beta = 1$ (i.e. the uniform prior), in which case the MAP estimator exactly coincides with the ML estimator. Let θ_* denote the 'true' value of θ from which the data is generated. The bias, variance, and squared error risk (a good introduction to these topics can be found in [39]) of the ML and MML87 estimators of θ_* are summarised in Table 2.1. It is obvious to see (and well known) that the ML estimator is the uniformly minimum variance unbiased estimator of θ_*, and thus its squared-error risk is simply its variance. The MML87 estimator is biased, but possesses a uniformly smaller variance, and it is easy

Table 2.1: Statistics of the ML and MML87 estimators for the Binomial Distribution

Estimator	Bias	Variance	Squared-Error Risk	Bayes Risk
ML	0	$\frac{\theta_*(1-\theta_*)}{N}$	$\frac{\theta_*(1-\theta_*)}{N}$	$\frac{1}{6(N+1)}$
MML87	$\frac{\frac{1}{2}-\theta_*}{N+1}$	$\frac{N\theta_*(1-\theta_*)}{(N+1)^2}$	$\frac{(N-1)\theta_*(1-\theta_*)+\frac{1}{4}}{(N+1)^2}$	$\frac{N+\frac{1}{2}}{6(N+1)^2}$

to be show that the MML87 estimator variance achieves the Cramer-Rao lower bounds for biased estimators [50]. The squared error risk is interesting, as the ML risk can be shown to be smaller than the MML87 risk only on $[0, \varepsilon_1) \cup (\varepsilon_2, 1]$, where $(\varepsilon_1 \leq \varepsilon_2)$ are the two solutions of $\left(-\theta_*^2 + \theta_* - \frac{N}{4(3N+1)}\right)$. The Bayes risk [51], taken with respect to the uniform prior is meaningless for the ML estimator, but is included as the MAP and ML estimators are identical in this case. It is interesting to observe that despite the bias of the MML87 estimator, the Bayes risk is smaller for all sample sizes.

Of further interest is the Kullback-Leibler divergence risk of the estimators. It is difficult to derive exact expressions for this, but a very simple analysis reveals a severe limitation of the ML estimate $\hat{\theta}_{\mathrm{ML}}$. If $\bar{y} = 0$, the ML estimator will select $\hat{\theta}_{\mathrm{ML}} = 0$; the Kullback-Leibler divergence of this estimate to any other θ_* is infinite, as the model puts no probability mass on the '1' event. A similar result occurs if $\bar{y} = N$. For $\theta_* \in (0, 1)$, any finite N samples drawn from a Binomial distribution will always have a non-zero chance of $\bar{y} = 0$ or $\bar{y} = N$ occuring. The KL risk of the ML estimator is therefore always infinite. In contrast, the MML87 estimator will never select an estimate of $\hat{\theta}_{87} = 0$ or $\hat{\theta}_{87} = 1$, irrespective of N, and will therefore always choose a model that puts some probability mass on both '0' and '1' events. Thus, the KL risk will always be finite, and therefore uniformly lower than the KL risk of the ML estimator for all N and all $\theta_* \in (0, 1)$.

This example of trading an increase in bias for a decrease in risk, in particular the KL risk, is a common feature of many MML87 estimators (it can be readily shown for the Poisson and Geometric distributions, as further examples), and arises because the MML87 estimator is willing to only state an estimate to a precision warranted by the data. Further studies of the MML87 estimator may be found in [52] (wherein it is termed, perhaps a little controversially, the 'MDL' estimator).

2.6.3 MML87 and SMML

The previously derived MML87 estimator for the Binomial is now compared to the SMML estimator. For this simple problem, the exact SMML partitioning can be performed in

linear time by application of Farr's algorithm [12], and thus it is possible to the approximation to the true 'optimal' partitioning. This is made even easier, as data y in the binomial model is the sufficient statistic, there is a direct mapping between the data space and the parameter space; if $\hat{\theta}(y)$ is an estimator for θ given y, it satisfies

$$y_1 \leq y_2 \Longleftrightarrow \hat{\theta}(y_1) \leq \hat{\theta}(y_2) \tag{2.71}$$

Thus, for the Binomial model, partitionings in data space (SMML) are comparable with partitionings in parameter space (MML87). In particular, is becomes possible to examine the region size selected by the SMML criterion and compare it to the region size chosen by the MML87 estimator. The MML87 estimator determines the uncertainty region volume from the spacing function $w(\cdot)$, so given an estimate $\hat{\theta}$ the uncertainty region is given by $\Omega = \left\{\theta : \theta_{87}^{-} \leq \theta \leq \theta_{87}^{+}\right\}$, with the boundaries found as

$$\theta_{87}^{-} = \hat{\theta} - \frac{1}{2}\left(\frac{12}{J(\theta)}\right)^{\frac{1}{2}},\ \theta_{87}^{+} = \hat{\theta} + \frac{1}{2}\left(\frac{12}{J(\theta)}\right)^{\frac{1}{2}} \tag{2.72}$$

These may be converted into data counts simply by scaling by N, i.e. $\left(\bar{y}_{87}^{-} = N\theta_{87}^{-}\right)$. The table below summarises the partitions chosen by the SMML criterion, and the estimated partitions found by the MML87 approximation when $\hat{\theta}_{87} = \hat{\theta}_S$ (i.e. the partition point estimate). This is done as the MML87 estimator always estimates the uncertainty region centered on the model under consideration, and the partition estimate will usually lie in a roughly central location in the partition. The $\hat{\theta}_S$ column lists the point esti-

Table 2.2: Binomial Partitions for $N = 20$ using SMML and MML87

$\hat{\theta}_S$	$\bar{y}_S^-$	$\bar{y}_S^+$	$\bar{y}_{87}^-$	$\bar{y}_{87}^+$
0	0	0	-	-
0.15	1	5	0.23	5.7659
0.475	6	13	5.6319	13.3681
0.825	14	19	13.55	19.4432
1	20	20	-	-

mates associated with each partition, the SMML partitions themselves being given by $\left\{\theta : \bar{y}_S^- \leq \theta \leq \bar{y}_S^+\right\}$ and the estimated MML87 partitions given by $\left\{\theta : \bar{y}_{87}^- \leq \theta \leq \bar{y}_{87}^+\right\}$. For this sample size and problem, it can be seen that the MML87 partitions are quite close to the SMML partitions, though there is a clear difference for the extreme cases. The SMML

criterion places both extreme cases $\bar{y} = 0$ and $\bar{y} = N$ into singleton groups, whereas the MML87 estimator is unable to produce an estimate of a partition for $\theta = 0$ or $\theta = 1$ as the Fisher Information term diverges. This is an artefact of the approximation, albeit a useful one as the expected KL divergences of the MML87 estimates will be lower (and indeed, finite) than the corresponding expected KL divergences of the SMML estimates. This comparison of SMML data group partitions versus MML87 parameter space uncertainty region estimates is only valid because of the sufficient statistic nature of the number of head counts H upon which the Binomial likelihood function depends, but demonstrates that for sufficiently regular problems the MML87 approximation is a useful tool for estimating uncertainty regions, and thus, message lengths.

2.7 Conclusion

This chapter has reviewed an Information Theoretic solution to the problem of inductive inference. In particular, it examined the Strict Minimum Message Length criterion which forms the theoretical backbone of the entire MML family of approximations. The more practical MMLD approximation to SMML was then introduced, along with a new approximation based on random coding. Following this, a new derivation of the well known MML87 estimator was made by viewing it as an *estimator* the MMLD message length, and the chapter concluded with a brief application of MML to a simple problem and a summary of model classes to which MML has been successfully applied. As a final note, the most important notation introduced in the chapter will now be summarised:

- $\mathbf{y} \in \mathcal{Y}$: A set of N data points from the space of all possible data $\mathcal{Y}$
- $\boldsymbol{\theta}$: A fully specified model, from the space of all permissible models $\boldsymbol{\Theta}$
- $h(\boldsymbol{\theta})$: the prior distribution over $\boldsymbol{\theta}$
- $I(\boldsymbol{\theta})$, $I(\mathbf{y}|\boldsymbol{\theta})$, $I(\mathbf{y}, \boldsymbol{\theta})$: the assertion, detail length and combined message length of model $\boldsymbol{\theta}$ given data $\mathbf{y}$, i.e. $I(\mathbf{y}, \boldsymbol{\theta}) = I(\boldsymbol{\theta}) + I(\mathbf{y}|\boldsymbol{\theta})$
- $q(\boldsymbol{\theta})$: the prior mass of the assertion
- $\mathcal{D}(\boldsymbol{\theta})$: the penalty cost of a model, i.e. $I(\mathbf{y}, \boldsymbol{\theta}) = L(\mathbf{y}|\boldsymbol{\theta}) + \mathcal{D}(\boldsymbol{\theta})$
- $\mathbf{J}(\boldsymbol{\theta})$, $\mathbf{J}(\boldsymbol{\theta}, \mathbf{y})$: the exact and observed Fisher Information Matrix, respectively

- $\Delta(f||g)$: the Kullback-Leibler divergence between source distribution g and approximating distribution f

The next chapter reviews some of the many existing methods found in the literature for the estimation of Autoregressive Moving Average parameters and order selection from observed data.

Chapter 3

Estimation of ARMA Models

This chapter introduces the most common and important methods used in the literature for identifying the structure of ARMA processes and estimating the parameters from data. These two tasks are usually decomposed; the *parameter estimation* step estimates

- $\mathbf{a}$, the autoregressive coefficients
- $\mathbf{c}$, the moving average coefficients
- σ^2, the innovation variance

while the *model selection* step usually estimates

- P, the number of autoregressive coefficients
- Q, the number of moving average coefficients

Neither of these problems are trivial and a lot of work has been dedicated to their solution. This chapter examines what are the most important parameter estimation and model selection techniques for estimating ARMA processes. The methods examined are chosen because of their importance in the literature, either due to their classic status or because they are modern methods that attempt to correct issues with the classic methods. The chapter begins by examining the parameter estimation aspect and covers the most important methods currently used to estimate ARMA parameters, before proceeding to review the available methods for order selection. Beginning with the traditional methods based on the correlograms, the chapter proceeds to review the very important area of

Minimum Discrepancy Estimators, before examining the Finite Sample Information Criteria, Predictive Least Squares, the Minimum Description Length principle and Bayesian methods.

3.1 ARMA Models and Autocovariances

Before the various model selection and parameter estimation techniques are examined, a few basic properties of the ARMA model shall be examined. This serves to introduce some quantities and terminology that is used extensively in the sequel. Recall the definition of the ARMA model: given an observed data sequence $\mathbf{y} \in \mathbb{R}^N$ and an unobserved *innovation* sequence $\mathbf{v} \in \mathbb{R}^N$, the $(P \in \mathbb{Z}_+, Q \in \mathbb{Z}_+)$ order mixed Autoregressive Moving Average (ARMA) model explains measurement y_n as

$$y_n + \sum_{i=1}^{P} a_i y_{n-i} = \sum_{i=1}^{Q} c_i v_{n-i} + v_n \tag{3.1}$$

where $\mathbf{a} \in \mathbb{R}^P$ are the autoregressive coefficients, $\mathbf{c} \in \mathbb{R}^Q$ are the moving average coefficients and the v_n are identically and independently distributed Normal variates distributed as per $v_n \sim \mathcal{N}(0, \sigma^2)$, where $\sigma^2 \in \mathbb{R}_+$ is termed the innovation variance. Assume that the data $\mathbf{y}$ was generated by an ARMA(P,Q) process with parameters $\boldsymbol{\theta}^* = \left\{\mathbf{a}^*, \mathbf{c}^*, \sigma^{*2}\right\}$. Furthermore, assume that all the *poles* and *zeros* of the ARMA process lie within the unit circle; the ARMA model is then *stationary* and *invertible* [2]. If $\mathbf{y}$ were was generated by this model, the autocovariances are:

$$\mathrm{E}\left[y_n y_{(n+k)}\right] = \mathrm{E}\left[y_n y_{(n-k)}\right] = \gamma_k\left(\boldsymbol{\theta}^*\right) \tag{3.2}$$

where $\gamma_k(\cdot)$ is the *autocovariance function* for the ARMA model; it is usual to drop the dependence on the parameters and use the symbol γ_k when the generating model is clear from the context. Due to the assumption of normally distributed noise, the stationary ARMA model becomes *weakly stationary* [53], and the theoretical second moments (i.e. the autocovariances) completely capture all the behaviour of process. Thus, the likelihood of a dataset $\mathbf{y}$ being generated by true ARMA process with parameters $\boldsymbol{\theta}^*$ is

$$f(\mathbf{y}|\boldsymbol{\theta}^*) = \frac{1}{(2\pi)^{\frac{N}{2}}\sqrt{|\Gamma(\boldsymbol{\theta}^*)|}} \exp\left(-\frac{1}{2}\mathbf{y}\boldsymbol{\Gamma}^{-1}(\boldsymbol{\theta}^*)\mathbf{y}^T\right) \tag{3.3}$$

where $\Gamma(\cdot)$ is a Toeplitz structure autocovariance matrix of the ARMA model, defined as

$$\mathbf{\Gamma}(\boldsymbol{\theta}^*) = \begin{bmatrix} \gamma_0(\boldsymbol{\theta}^*) & \gamma_1(\boldsymbol{\theta}^*) & \gamma_2(\boldsymbol{\theta}^*) & \dots & \gamma_{(N-1)}(\boldsymbol{\theta}^*) \\ \gamma_1(\boldsymbol{\theta}^*) & \gamma_0(\boldsymbol{\theta}^*) & & & \vdots \\ \gamma_2(\boldsymbol{\theta}^*) & & \ddots & & \gamma_2(\boldsymbol{\theta}^*) \\ \vdots & & & \gamma_0(\boldsymbol{\theta}^*) & \gamma_1(\boldsymbol{\theta}^*) \\ \gamma_{(N-1)}(\boldsymbol{\theta}^*) & \dots & \gamma_2(\boldsymbol{\theta}^*) & \gamma_1(\boldsymbol{\theta}^*) & \gamma_0(\boldsymbol{\theta}^*) \end{bmatrix} \tag{3.4}$$

which may be more succinctly described by

$$\Gamma_{(i,j)}(\boldsymbol{\theta}^*) = \gamma_{|i-j|}(\boldsymbol{\theta}^*) \tag{3.5}$$

The ARMA model is a parsimonious description of a time series model because the number of parameters $(P + Q + 1)$ is usually significantly less than the number of entries in the autocovariance matrix (N^2). The ARMA autocovariance function provides a template for the autocovariance matrix that is tunable by selecting different values of $\mathbf{a}$, $\mathbf{c}$ and σ^2. The autocovariance function and methods for efficient evaluation of the likelihood function (3.3) may be found in Appendix C.

3.2 Parameter Estimation

The task of estimating the parameters of a model class from observed data, also known as *point estimation* [39], is one of the central tasks in statistical analysis. Let $\psi \in \Psi \in \mathbb{Z}$ denote a model class indicator, and $\mathbf{\Theta}^{(\psi)}$ the parameter space for model class ψ; a point estimator m is a mapping such that

$$m : \mathbf{y} \in \mathcal{Y}, \psi \in \Psi \rightarrow \boldsymbol{\theta} \in \mathbf{\Theta}^{(\psi)} \tag{3.6}$$

that is, m maps a dataset $\mathbf{y}$ and a chosen model class ψ to a member of the model class parameter space $\mathbf{\Theta}^{(\psi)}$. Despite the underlying linearity of the ARMA model family, point estimation for AR, MA and mixed ARMA models is considered a difficult task and a range of point estimation schemes have been proposed over the last century. The problems arise due to the fact that the data is not independently and identically distributed, and because

the innovation sequence cannot be directly observed. Several of the most important and commonly used parameter estimators in the literature are now discussed.

3.2.1 Least Squares Estimation

Perhaps the most straightforward method for estimating parameters of autoregressive models is to treat them as conventional linear regressions and estimate the parameters by the method of least squares. The design matrix $\mathbf{X}$ is constructed as a matrix of lagged values of the sequence $\mathbf{y}$ as $\mathbf{X} = \left[\mathbf{x}_P^T, \ldots, \mathbf{x}_N^T\right]$, where $\mathbf{x}_n = x_{(n-P:n)}$ is the state vector at time n. The least squares estimates $\hat{\mathbf{a}}_{\mathrm{LS}}$ minimise the fitted sum of squares, i.e. they solve

$$\hat{\mathbf{a}}_{\mathrm{LS}} = \arg\min_{\mathbf{a}\in\mathbb{R}^P} \left\{ (\mathbf{y} - \mathbf{a}\mathbf{X}) (\mathbf{y} - \mathbf{a}\mathbf{X})^T \right\} \tag{3.7}$$

The least squares estimates have the advantage of making very few assumptions about the nature of data or the generating model, and will produce estimates regardless of the stationarity of the underlying generating source. Additionally, the solution to (3.7) is well known and simple to compute and the least squares estimates are known to be consistent [2]. Unfortunately, they also suffer from several drawbacks. The most obvious of these is that there is a difficulty in handling the first P points for which a state vector cannot be formed. There are several solutions to this problem, such as treating them as zero, backcasting or simply ignoring them, though all these methods throw away useful information if the underlying model is considered stationary. The least squares estimates clearly ignore the correlated nature of the time series data, and this can introduce bias' in the parameter estimates for small sample sizes. Perhaps the most serious problem is that for any finite sample size there is a non-zero chance the least squares estimates will be nonstationary, even if the underyling source is stationary. Although this chance decreases to zero as $N \to \infty$, it can cause problems when the amount of data is small. The simple expedient of restricting the estimates to lie in the stationarity region is less than satisfactory as this generally yields estimates that move as close to the boundary as allowed. However, if the sample size is large, the least squares estimates are generally a good choice. The least squares estimates can be readily extended to estimation of MA and ARMA models by filtering the innovations $\hat{\mathbf{v}}$ from the data by inversion of the ARMA

model, i.e.

$$\hat{v}_n = -\sum_{i=1}^{Q} c_i \hat{v}_{n-i} + \sum_{i=1}^{P} y_{n-i} + y_n \tag{3.8}$$

where $\hat{v}_i = 0$ for $i = (1-Q)\ldots 0$ and $y_i = 0$ for $i = (1-P)\ldots 0$. The least squares estimates $\hat{\mathbf{a}}_{\mathrm{LS}}$ and $\hat{\mathbf{c}}_{\mathrm{LS}}$ are then found by minimising the sum of innovations $\hat{\mathbf{v}}\hat{\mathbf{v}}^T$. Although relatively simple to compute the least squares estimates can degrade if the generating source model has zeros that are close to the unit circle, as the transients introduced by setting the initial conditions to zero may then take a considerable number of samples to 'die away' [53]. The least squares estimates have also been successfully been extended to the estimation of nonlinear ARMA models [54, 55].

3.2.2 Maximum Likelihood Estimation

The method of Maximum Likelihood (ML) [56] is a very general estimation method pioneered by R. A. Fisher. The idea behind the ML estimator is quite simple. Given a parameterised likelihood function $f(\mathbf{y}|\boldsymbol{\theta}, \psi)$ where $\mathbf{y}$ is data, $\boldsymbol{\theta}$ is a parameter vector and ψ a model class indicator, the Maximum Likelihood estimates are found by solving

$$\boldsymbol{\theta}_{\mathrm{ML}}^{(\psi)} = \arg \max_{\boldsymbol{\theta} \in \boldsymbol{\Theta}^{(\psi)}} \{f(\mathbf{y}|\boldsymbol{\theta}, \psi)\} \tag{3.9}$$

The ML estimator selects the parameter estimate that renders the likelihood of the observed data $\mathbf{y}$ to be greatest. The Maximum Likelihood estimates have in general several properties that make them good choices for point estimation: they are invariant under transformations of the model and data, if the model admits sufficient statistics then they depend entirely on the sufficient statistics and no more, and for many model classes, including ARMA models [57], they are asymptotically efficient. That is, as $N \to \infty$

$$\hat{\boldsymbol{\theta}}_{\mathrm{ML}}^{(\psi)} \sim \mathcal{N}\left(\boldsymbol{\theta}^*, \mathbf{J}^{-1}(\boldsymbol{\theta}^*)\right) \tag{3.10}$$

where $\boldsymbol{\theta}^*$ are the parameters of the generating model and $\mathbf{J}(\cdot)$ is the Fisher Information Matrix. It is usual to work with negative log-likelihoods rather than directly with likelihoods to help with numerical issues; for ARMA models the maximum likelihood estimator

involves solving

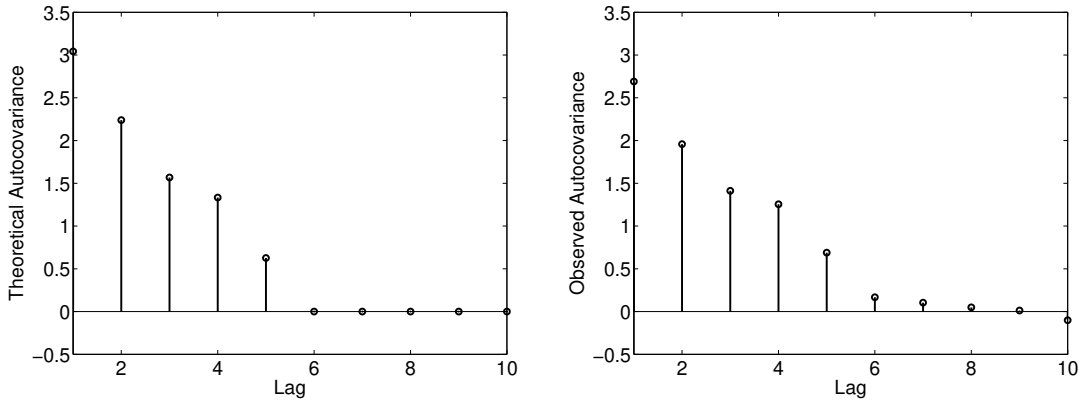
$$\hat{\boldsymbol{\theta}}_{\mathrm{ML}}^{(P,Q)} = \arg \min_{\boldsymbol{\theta} \in \boldsymbol{\Theta}^{(P,Q)}} \left\{ \frac{N}{2} \log 2\pi + \frac{1}{2} \log |\boldsymbol{\Gamma}(\boldsymbol{\theta})| + \frac{1}{2} \mathbf{y} \boldsymbol{\Gamma}^{-1}(\boldsymbol{\theta}) \mathbf{y}^T \right\} \tag{3.11}$$

where the negative log-likelihood may be efficiently evaluated via the Kalman Filter [58, 59, 60]. It can be shown that the least squares estimates and the exact Maximum Likelihood estimates are very similar, differing mainly in the way in which the first samples are handled. In the case of pure AR models, the likelihood can be broken into a conditional component that is the same as the form of the conditional least squares when the first P samples are disregarded, and an unconditional component that explicitly models the first P samples. For ARMA model and sufficiently large sample sizes the Kalman Filter essentially coincides with the inversion (3.8) after a sufficient number of samples have been processed; thus the ML and LS approach here differ in the way the initial values of the innovation sequence and first P data points are handled. For ARMA models the ML estimates have been proven to be consistent [61, 62] and given that they explicitly take into account the correlated nature of the data, are often considered a good choice for small sample sizes. The largest drawbacks with the ML estimates is that for anything but AR(1) models they must be found by a numerical search, and for MA and ARMA models zeros near the invertibility boundary can lead to strange behaviour for small sample sizes (the so called 'piling up' effect) [63, 64].

3.2.3 Other AR/ARMA Estimation Methods

Several other estimation schemes for AR and ARMA models are now briefly examined. An early method that explicitly takes the correlated nature of the data into account when estimating parameters is Yule-Walker estimator. This is based on the method-of-moments, and works by equating sample autocovariances with the theoretical autocovariances of the AR model and solving for the parameters. The Yule-Walker estimates are simple to find and are guaranteed to be stationary, but unfortunately, the empirical autocovariance estimates on which they are based require a large amount of data for any reasonable level of accuracy and this can lead to bias and poor estimates for small sample sizes [65].

The popular Burg estimator [66] works by sequentially estimating partial autocorrelations, and transforming these back to coefficients. The Burg estimator is fast, guarantees



(a) Exact Autocovariances (b) Observed Autocovariances

Figure 3.1: Exact and Observed Autocovariances for an MA(4) Model

stationarity of the estimated models and yields good estimates even for small numbers of data points [65].

For estimating MA and ARMA models, Durbin's methods [67, 68] estimate MA models by exploiting the duality between AR and MA structures. The basic approach is to estimate a high order autoregressive process from the data, which is subsequently used to arrive at a suitable moving average model. Recently, these procedures have been re-examined and modernised [69, 70, 71, 72] and along with the Burg estimator, form the basis of the ARMAsel procedure [73].

3.3 Order Selection Based on Observed Correlations

Amongst the earliest methods used for order selection of Autoregressive Moving Average processes are those based on inspection of the correlogram. These are motivated by the properties of the theoretical autocorrelation sequences of autoregressive and moving average processes. Given a data set $\mathbf{y}$, the first step in these traditional methods is to estimate the process autocorrelations; these are often referred to as the *observed* autocorrelations (as opposed to theoretical), and may be estimated as

$$\hat{\gamma}_k = \frac{\mathbf{y}_{(1:N-k)}\mathbf{y}^T_{(k+1:N)}}{N-k} \tag{3.12}$$

$$\hat{\rho}_k = \frac{\hat{\gamma}_k}{\hat{\gamma}_0} \tag{3.13}$$

where $\hat{\gamma}_k$ and $\hat{\rho}_k$ denote the k-th observed autocovariance and autocorrelation, respectively. Once these have been estimated, they can be used to estimate the order of a moving average

process by recalling that for an MA(Q) process only the autocovariances γ_0 through γ_Q are non-zero. Due to the finite memory of an Moving Average process, the rest are necessarily zero. Examination of the observed autocorrelations is then undertaken to see if such a pattern is present in the data; if the correlogram rapidly drops to near zero after the Q-th autocorrelation, a Q-th order moving average process is assumed to be a suitable fit. Figures 3.1(a) and 3.1(b) show the theoretical autocovariance sequence of an MA(4) process and the observed autocoviariance sequence from $N = 500$ realisations of this process. With this amount of data the first five autocovariances are readily identified and the correct order would be conjectured. This process is not directly applicable to Autoregressive processes as their theoretical autocovariance functions decay to zero only as $k \to \infty$; however, it is well known that the theoretical partial autocorrelations of an AR(P) process are non-zero for only $(P + 1)$ samples. The observed k-th observed partial autocorrelations may be estimated from the data by first solving

$$\hat{\boldsymbol{\phi}}_k = \text{toep}\left(\hat{\boldsymbol{\rho}}_{(0:k-1)}\right)^{-1} \hat{\boldsymbol{\rho}}^T_{(1:k)} \tag{3.14}$$

The entry $\hat{\phi}_{(k,k)}$ is then the k-th observed partial autocorrelation. The same process may then be undertaken as for the Moving Average model by examining the observed partial autocorrelations, and determining when they drop rapidly to zero. These methods of pattern identification in observed partial autocorrelations and autocovariances have progressed over the years ([74] examines several) but do suffer from several drawbacks: most obviously, as the amount of data grows smaller, the estimated autocorrelations have large variances and become worse and worse and it may be extremely difficult to determine a suitable 'cut-off' point; extreme care must be exercised when identifying models via this method [75]. A second problem is that it involves extensive human interaction and is not suitable when large numbers of models must be estimated from large numbers of time series or when the process must be done in an online fashion. A third major drawback is that the mixed autoregressive moving average process 'blurs' both autocorrelation and partial autocorrelation sequences (as it is a convolution of the two processes) and is thus particularly difficult to estimate from observed correlation sequences.

3.4 Minimum Discrepancy Estimators

A very important class of model selection criteria are the so called *minimum discrepancy estimators* (MDE) (an excellent introduction to these approaches may be found in [76]). Assume there is a countable set of parameterised distributions $F = \left\{ f^{(\psi)} | \psi \in \Psi \right\}$, $f^{(\psi)} = \left\{ f_{\boldsymbol{\theta}}^{(\psi)} | \boldsymbol{\theta} \in \boldsymbol{\Theta}^{(\psi)} \right\}$, where $\psi \in \Psi \subset \mathbb{N}$ is the set of model class indicators, and $\boldsymbol{\theta}^{(\psi)} \in \boldsymbol{\Theta}^{(\psi)}$ are the model parameters for model class $\psi \in \Psi$. Assuming that some data has been generated by the 'true' distribution f^*; the Minimum Discrepancy approach aims to solve

$$f(\boldsymbol{\theta}, \psi) = \arg \min_{\psi \in \Psi, \theta \in \boldsymbol{\Theta}^{(\psi)}} \left\{ \Phi \left(f(\boldsymbol{\theta}, \psi), f^* \right) \right\} \tag{3.15}$$

where $\Phi(\cdot)$ is a *discrepancy* function that measures how well one distribution approximates another, i.e. the MDE approach aims to find the model that minimises the discrepancy to the true generating source f^*. Of course, the true model is not known (otherwise there would be no need to perform induction) but is observed through a realisation $\mathbf{y} \sim f^*$ of the generating source. The aim is to find a solution (or approximate solution) to (3.15) using only the observed data and a chosen $\Phi(\cdot)$. Several important discrepancy functions include Squared Prediction Errors and the directed and symmetric Kullback-Leibler divergence [14]. While other discrepancy functions such as the Gauss discrepancy, Kolmogorov discrepancy and Pearson discrepancy [76] do exist, the Kullback-Leibler discrepancies have assumed a particularly important role in the development of MDE techniques. As the true generating source is not known, the discrepancy between any estimated model and the true model also cannot be known. The general method employed to circumvent this problem is to *estimate* the discrepancy from the observed data $\mathbf{y}$. The most naive approach would be to use the *empirical* discrepancy function, $\Phi_N \left(f_{\left(\boldsymbol{\theta}^{(\psi)}, \psi\right)}, \mathbf{y} \right)$, which uses the discrepancy of a distribution $f_{(\boldsymbol{\theta}, \psi)}$ from the empirical distribution defined by the data $\mathbf{y}$ as an estimate of the exact discrepancy from f^*. Solving

$$f_{(\hat{\boldsymbol{\theta}}_{\Phi}^{(\psi)}, \psi)} = \arg \min_{\psi \in \Psi} \left\{ \Phi_N \left(f_{\left(\hat{\boldsymbol{\theta}}^{(\psi)}, \psi\right)}, \mathbf{y} \right) \right\} \tag{3.16}$$

yields the minimum empirical discrepancy estimator, where

$$\hat{\boldsymbol{\theta}}_{\Phi}^{(\psi)} = \arg \min_{\boldsymbol{\theta} \in \boldsymbol{\Theta}_{\psi}} \left\{ \Phi_N \left(f_{\left(\boldsymbol{\theta}^{(\psi)}, \psi\right)}, \mathbf{y} \right) \right\}$$

is the point estimate within model class ψ that minimises the empirical discrepancy. Unfortunately, it is well known that the empirical discrepancy is usually a *biased* estimate [77, 78] of the true discrepancy, and if the model class under consideration is *nested* minimising the empirical discrepancy will simply result in the most complex candidate model being selected. The bias corrected estimate of the discrepancy takes the form

$$MDE(\mathbf{y}, \psi) = \underbrace{\Phi_N \left(f_{\left(\hat{\boldsymbol{\theta}}_{\Phi}^{(\psi)}, \psi\right)}, \mathbf{y} \right)}_{\text{empirical fit}} + \underbrace{\alpha(\Phi, P, N)}_{\text{bias}} \tag{3.17}$$

where $P = \dim\left(\hat{\boldsymbol{\theta}}_{\Phi}^{(\psi)}\right)$ and N is the number of data points. The first term in (3.17) measures how well the model fits the data (which is a realisation of the true process) and will decrease as the number of parameters grows. The second term, $\alpha(\cdot)$ is the expected discrepancy due to the variance in the parameter estimates $\hat{\boldsymbol{\theta}}_{\Phi}^{(\psi)}$, and this will increase as the number of parameters grows (a straightforward result of the Cramer-Rao inequality [79, 80]). The MDE is chosen such that it satisfies

$$\mathrm{E}_{f^*}\left[MDE(\mathbf{y}, \psi)\right] \approx \mathrm{E}_{f^*}\left[\Phi\left(\hat{\boldsymbol{\theta}}_{\Phi}^{(\psi)}, f^*\right)\right] \tag{3.18}$$

under certain conditions. How accurate the estimate $MDE(\cdot)$ is depends on various regularity conditions, the class of models considered and the choice of discrepancy function $\Phi(\cdot)$. The next section introduces one of the most important MDE's, the Akaike Information Criteria (AIC).

3.4.1 The Akaike Information Criterion (AIC)

The Akaike Information Criterion (AIC) [3] was one the very first MDE type criteria introduced, and is an important landmark in the automatic selection of models in general. The AIC is based on the directed Kullback-Leibler (KL) divergence; given an approximating distribution $\hat{f}$ and a 'true' distribution f^*, the KL divergence is

$$\begin{aligned} \Delta(\hat{f}||f^*) &= -\mathrm{E}_{f^*}\left[\log \hat{f}\right] + \mathrm{E}_{f^*}\left[\log f^*\right] && (3.19) \\ &= H(\hat{f}||f^*) - H(f^*||f^*) && (3.20) \end{aligned}$$

where $H(f^*)$ is the *entropy* of f^*, and $H(\hat{f}||f^*)$ is the expected negative log-likelihood of data generated by f^* and encoded using $\hat{f}$. The KL divergence therefore has the information theoretic interpretation of the number of nits in excess of the entropy required to transmit data generated by f^* using a code based on $\hat{f}$. The AIC is derived under several key assumptions (for a rigorous derivation, see [76] Appendix A, pages 237-243):

- The model family under consideration is *nested*, that is $f^{(P+1)} \subset f^{(P)}$, where $P \in \Psi \subset \mathbb{N}_+$ now simply corresponds the number of parameters of the model class.
- That the 'true' generating source is a model in the model family, i.e. $f^* \in f^{(P^*)}$, where $P^* \in \Psi$ is the 'true model' class.
- That the model class $f^{(P)}$ under consideration has at least as many parameters as the model class to which f^* belongs, i.e. $P^* \leq P$

The parameter estimate that minimises the empirical KL divergence for a given model class ψ is simply the Maximum Likelihood estimate $\hat{\boldsymbol{\theta}}_{\mathrm{ML}}^{(P)}$; by exploiting the fact that asymptotically in N the ML estimate is normally distributed for many classes, i.e.

$$\hat{\boldsymbol{\theta}}_{\mathrm{ML}}^{(P)} \sim N\left(\mathbf{0}, \mathbf{J}^{-1}(\hat{\boldsymbol{\theta}}_{\mathrm{ML}}^{(P)})\right)$$

the required expectations may be computed yielding the AIC criterion

$$AIC(P) = 2L(\mathbf{y}|\hat{\boldsymbol{\theta}}_{\mathrm{ML}}^{(P)}) + 2(P+1) \tag{3.21}$$

where $L(\cdot)$ is the negative log-likelihood of the data. The number of parameters is augmented by one to take into account the estimate of the noise variance parameter. The AIC is an asymptotically unbiased estimator of KL divergence and satisfies

$$\mathrm{E}_{f^*}\left[AIC(P)\right] = \mathrm{E}_{f^*}\left[2H\left(f_{\hat{\boldsymbol{\theta}}_{\mathrm{ML}}^{(P)}}||f^*\right)\right] + \mathrm{o}(1) \tag{3.22}$$

if the conditions of derivation are met. The AIC criterion is not statistically consistent, and thus will not be guaranteed to select the correct model order P^* as $N \to \infty$. It is however, statistically efficient in the sense that as $N \to \infty$ it will select the model class that yields the minimum KL divergence from the true model.

3.4.2 The Corrected Akaike Information Criterion (AICc)

In the development of the AIC, asymptotic arguments were employed and the final criterion neglects an o(1) term. For small to moderate N, this o(1) term can be quite large and simply ignoring it can lead to significant overfitting. The corrected Akaike Information Criterion (AICc) [81, 82] takes this term into account and is given by

$$AICc(P) = 2L(\mathbf{y}|\hat{\boldsymbol{\theta}}_{\mathrm{ML}}^{(P)}) + 2\frac{(P+1)N}{N-P-2} \tag{3.23}$$

Examination of (3.23) shows that as $N \to \infty$, $AICc(P) \to AIC(P)$, that is they asymptotically converge. Given that the AICc criterion is designed to take into account an o(1) term this is expected. The AICc criterion satisfies:

$$\mathrm{E}_{f^*}\left[AICc(P)\right] = \mathrm{E}_{f^*}\left[2H\left(f_{\hat{\boldsymbol{\theta}}_{\mathrm{ML}}^{(P)}}||f^*\right)\right] \tag{3.24}$$

that is, the AICc is an exactly unbiased estimator of the KL divergence, under the conditions of derivation. The AICc penalty term is larger than the AIC penalty term for moderate or small N, and the *per parameter* penalty $AICc(P)/(P+1)$ increases as P grows rather than remaining constant as in the case of AIC. The AICc has been extensively tested and applied in many experiments in the literature, and in almost all situations is reported to offer much improved performance in terms of correct order selections and prediction errors when compared to AIC [81, 82, 83].

3.4.3 The Symmetric KL Divergence Criterion (KIC)

A more recent development has been the MDE based on the symmetric Kullback-Leibler divergence, or J divergence. The J divergence between 'true' distribution f^* and approximating distribution is $\hat{f}$ is given by

$$\begin{aligned} \mathcal{J}(\hat{f}, f^*) &= \Delta(\hat{f}||f^*) + \Delta(f^*||\hat{f}) \\ &= H(\hat{f}||f^*) + H(f^*||\hat{f}) - H(f^*) - H(\hat{f}) \end{aligned} \tag{3.25}$$

that is, the J divergence is the sum of the KL divergence between the two distributions, in both directions. It is symmetric but does not satisfy the triangle inequality and is therefore not a true distance measure [46]. The directed KL divergence tends to favour

models with larger estimates of variance parameters (in the Gaussian noise case); these models have lower negative log-likelihoods and are therefore lower order models. The KL divergence is therefore seen as preferring lower order models. The motivation behind the choice of the J divergence as a discrepancy is that its symmetric nature punishes over and under estimations of variance parameters more equitably and would be expected to underfit less than criteria based on the KL divergence. Using the Maximum Likelihood estimates $\hat{\boldsymbol{\theta}}_{\mathrm{ML}}^{(P)}$ the asymptotic J divergence criterion, or KIC [5], is given by

$$KIC(P) = 2L\left(\mathbf{y}|\hat{\boldsymbol{\theta}}_{\mathrm{ML}}^{(P)}\right) + 3(P+1) \tag{3.26}$$

Clearly there are striking similarities between (3.21) and (3.26); the AIC estimates the KL divergence due to estimation by P, and the KIC estimates the J divergence due to estimation as $\left(\frac{3}{2}P\right)$. That the KIC penalty is greater than the AIC penalty is straightforward when one considers that the KIC estimates a quantity that is *at least* as large as the asymmetric KL divergence estimated by AIC. The KIC is therefore *more* conservative than the AIC, which is interesting given the motivation for its development. It can be noted however, that if one was to divide the J-divergence by two to arrive at the average KL divergence in both directions, the KIC would be less conservative than AIC. The KIC satisfies

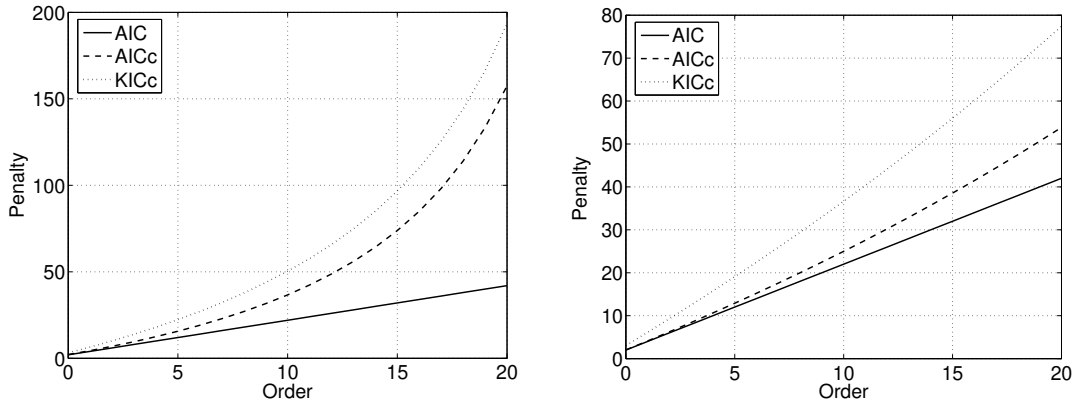
$$\mathrm{E}_{f^*}\left[KIC(P)\right] = 2\mathrm{E}_{f^*}\left[H\left(f_{\hat{\boldsymbol{\theta}}_{\mathrm{ML}}^{(P)}}||f^*\right) + H\left(f^*||f_{\hat{\boldsymbol{\theta}}_{\mathrm{ML}}^{(P)}}\right)\right] + \mathrm{o}(1) \tag{3.27}$$

where in a similar fashion to AIC, the o(1) term is ignored by asymptotic arguments. Experiments indicate that it performs better in terms of correct order selections and prediction errors for moderate and smaller N. Like the AIC and AICc, the KIC criterion is not asymptotically consistent but is asymptotically efficient.

3.4.4 The Corrected Symmetric KL Divergence Criterion (KICc)

The final MDE Information Criterion considered is the corrected KIC criterion (KICc) [6]. In a similar fashion to AICc, this explicitly takes into account the o(1) term in the KIC, yielding the criterion

$$KICc(P) = 2L\left(\mathbf{y}|\hat{\boldsymbol{\theta}}_{\mathrm{ML}}^{(P)}\right) + 2\frac{(P+1)N}{(N-P-2)} - N\psi\left(\frac{N-P}{2}\right) + N\log\frac{N}{2} \tag{3.28}$$



(a) $N = 30$ (b) $N = 100$

Figure 3.2: Penalty Terms for MDE Criterion

where $\psi(\cdot)$ is the digamma function. Given the asymptotic behaviour of $\psi(\cdot)$, as $N \to \infty$, $KICc(P) \to KIC(P)$, as would be expected given the nature of the derivation of KICc. It can be seen that the AICc penalty term appears in (3.28); this is not unsurprising as the J divergence is the sum of two directed KL divergences, and the AICc is based on one of these directed KL divergences. The KICc satisfies

$$\mathrm{E}_{f^*}\left[KICc(P)\right] = 2\mathrm{E}_{f^*}\left[H\left(f_{\hat{\boldsymbol{\theta}}_{\mathrm{ML}}^{(P)}}||f^*\right) + H\left(f^*||f_{\hat{\boldsymbol{\theta}}_{\mathrm{ML}}^{(P)}}\right)\right] \tag{3.29}$$

that is, the KICc is an exactly unbiased estimator of the J divergence, under the conditions of its derivation. Although the KICc is relatively new and has not been tested extensively in empirical investigations, several pieces of work have shown it to be highly competitive, outperforming AIC and AICc in terms of order selection and prediction errors for small N and performing reasonably well for moderate N [84]. Figures 3.2(a) and 3.2(b) show the penalty curves for AIC, AICc and KICc for a 'small' sample size, and a 'large' sample size. The plots clearly demonstrate that the KICc is more conservative than AICc, which in turn is more conservative than AIC, and clearly show that for $N = 100$ the AICc curve is much closer to the AIC curve than for $N = 30$; for $N = 100$, the KICc curve is also quite close to linear, which is expected as asymptotically KICc coincides with KIC.

3.5 Sampling Approaches to MDE

The previous MDE criteria used the distribution of the Maximum Likelihood estimates to take expectations and derive closed form estimates of the discrepancies. The following section examines MDE approaches that are based on *sampling*, both systematic and random; these methods enable the use of arbitrary parameter estimators and discrepancy functions, but at the cost of greatly increased computational complexity. All the subsequent methods work by creating new datasets from the observed dataset $\mathbf{y}$; let $\mathbf{y}^{(k)} \subset \mathbf{y}$ denote a new dataset formed by taking datapoints from the complete dataset $\mathbf{y}$.

3.5.1 Cross Validation

The Cross-Validation (CV) [85, 86] procedure was one of the earliest sampling based approaches to find an MDE, and was originally introduced for application to univariate and multivariate linear models. The idea is to estimate the prediction error of a model class from its prediction errors against carefully chosen subsets of the data. The procedure requires the choice of a suitable loss function, denoted $\varphi(\mathbf{y}, \boldsymbol{\theta}^{(\psi)})$ that measures the ability of model $\boldsymbol{\theta}^{(\psi)}$ to make predictions about data $\mathbf{y}$. Once a suitable loss function has been chosen, the procedure is reasonably straightforward: the observed data $\mathbf{y}$ is divided into M pairs of training datasets $\mathbf{y}^{(k)} \in \mathbb{R}^{N_T}$ and testing or validation datasets $\mathbf{y}^{(k)*} \in \mathbb{R}^{N^*}$, chosen such that $(N_T + N^* \leq N)$ and $\mathbf{y}^{(k)} \bigcap \mathbf{y}^{(k)*} = \emptyset$, i.e. that each pair of testing and training sets are disjoint and do not share data. For $\psi \in \Psi$, and all $k = 1 \dots M$ training/testing pairs, the following is performed:

1. Estimate a model from the training set $\mathbf{y}^{(k)}$, denoted $\hat{\boldsymbol{\theta}}_{(k)}^{(\psi)}$, using a suitable estimator (such as the minimum empirical discrepancy estimator).

2. Compute the loss $\varepsilon_{(k)}^{(\psi)}$ of model $\hat{\boldsymbol{\theta}}_{(k)}^{(\psi)}$ against the testing dataset $\mathbf{y}^{(k)*}$

$$\varepsilon_{(k)}^{(\psi)} = \varphi\left(\mathbf{y}^{(k)*}, \hat{\boldsymbol{\theta}}_{(k)}^{(\psi)}\right)$$

Once this has been done, model class with the minimum total loss is selected as the best model class. This involves solving

$$\hat{\psi} = \arg\min_{\psi \in \Psi} \left\{ \sum_{k=1}^{M} \varepsilon_{(k)}^{(\psi)} \right\} \tag{3.30}$$

There exists a very large range of possible loss functions and ways to divide the data set $\mathbf{y}$ into testing and training sets. Amongst the most common form of CV used is the Leave-One-Out (LOO) scheme using the squared-prediction error loss. Under this scheme the training and testing sets are formed as

$$y_i^{(k)} = y_i, \ (i = 1 \dots N, \ i \neq k) \tag{3.31}$$

$$y^{(k)*} = y_k \tag{3.32}$$

The LOO simply does as the name suggests: the training sets are the complete data set, sans one point which is then used as the testing point. This means the number of training and testing sets is simply equal to N, which leads to a more manageable time complexity. The CV approach has been applied to a large range of practical problems [55] and has shown to perform well if the datasets are large. It does however suffer from several drawbacks: for many model classes it is extremely time consuming to estimate a new model for each training dataset, and under certain conditions the criterion may be statistically inconsistent. Furthermore, to easily apply cross validation to autoregressive models the unconditional components of the likelihood are usually ignored and the models are treated as linear regressions, which for small sample sizes involves throwing away information. An interesting property of CV using the squared prediction error loss is that for large N and under certain conditions it is essentially the same as the AIC [87] discussed in Section (3.4.1); this is perhaps not as surprising as it first seems if one considers that for linear regression models the KL divergence is simply the scaled squared prediction error. The related *bootstrap* [88] technique also has similar problems to cross validation when applied to selection of time series models; the next section presents a sampling approach to MDE that is more suited to time series.

3.5.2 Accumulated Prediction Error

A subtle modification of the CV approach leads to a new method for model selection called the Accumulated Prediction Error (APE) procedure. The APE advocates the following

special choice of $(N-1)$ training and testing datasets

$$y_i^{(k)} = y_i,\ (i = 1 \dots k) \tag{3.33}$$

$$y^{(k)*} = y_{(k+1)} \tag{3.34}$$

that is, the training sets are chosen so that each set contains only the first k datapoints, and the testing set always contains the subsequent datapoint. Under this scheme, the losses $\varepsilon_{(k)}^{(\psi)}$ become known as *true prediction errors*. The APE procedure is known to be statistically consistent even in cases when CV is not [89], and this is due to the fact that unlike the CV LOO criterion the predictions are honest in the sense that no future data is ever used to compute them. The one key drawback with the APE criterion is that it requires an ordering of the data, and this ordering may have a large impact on the resulting prediction errors. In the specific case of time series models, the data is inherently ordered and this does not pose a problem.

When combined with the squared prediction error loss, the APE procedure is also known as the Predictive Least Squares (PLS) [90, 4] algorithm. An important advantage of this process is that it makes no explicit assumptions about the nature of the data or the noise, unlike other procedures that make assumptions of normally distributed innovation noise. Additionally, due to the fact that the training sets are always ordered subsets of the full time series, an application of APE to AR models (and ARMA models) may use estimates that minimise exact negative log-likelihood rather than simply treating the AR models as linear regressions and throwing away important information. Unfortunately, for small and moderate data sizes the PLS procedure has been shown to perform quite poorly [91, 84]. The PLS procedure also suffers from the need to estimate N models to evaluate each model class, although for AR models this can be alleviated by clever use of lattice filters [92].

3.6 Bayesian Inference

The Bayesian school of inference [93, 94] advocates making inferences based not on likelihoods but on *posterior* distributions. The Bayesian process involves choosing a distribution $h(\cdot)$ over the parameter space for each model class under consideration. This is the belief that a particular model will turn out to be the true model, before any data has been

observed. Once some data $\mathbf{y}$ has been observed, the posterior distribution $p(\boldsymbol{\theta}^{(\psi)}, \psi|\mathbf{y})$ over parameter space of model class $\psi \in \Psi$ satisfies

$$p(\boldsymbol{\theta}^{(\psi)}, \psi|\mathbf{y}) \propto f(\mathbf{y}|\boldsymbol{\theta}^{(\psi)}, \psi)h(\boldsymbol{\theta}^{(\psi)}|\psi)h(\psi),\ \boldsymbol{\theta}^{(\psi)} \in \boldsymbol{\Theta}^{(\psi)} \tag{3.35}$$

The expression in (3.35) is not necessarily a proper distribution because it will in general not integrate to unity. The normalising factor $r(\mathbf{y}, \psi)$ is given by

$$r(\mathbf{y}, \psi) = \int_{\boldsymbol{\Theta}^{(\psi)}} h(\boldsymbol{\theta}|\psi)f(\mathbf{y}|\boldsymbol{\theta}, \psi)d\boldsymbol{\theta} \tag{3.36}$$

This is also known as the Bayesian *evidence*. While Bayesian inference has been widely applied, there are still several contentious issues surrounding its use. The first is how to choose a prior; subjective Bayesians advocate that the prior should be chosen to seriously reflect one's prior beliefs about the state of the world (or at least that part of the world that the inference is concerned with). On the other hand, objective Bayesians attempt to formulate prior distributions that are uninformative and express the notion of ignorance in some fashion. The most famous of these include the *Jeffrey's prior* [95] and the *reference priors* [96], but other uninformative priors do exist [97]. An interesting discussion of priors can be found in [1], pages 45-54. The second contentious issue involves selecting a point estimate (what has been referred to as a model, or fully specified model in this thesis) from the posterior. The obvious solution is to use some suitable summary statistic of the posterior distribution such as the mode (the so called Maximum a Priori (MAP) estimator), or the mean

$$\hat{\boldsymbol{\theta}}^{(\psi)}_{\text{MAP}} = \arg\min_{\boldsymbol{\theta} \in \boldsymbol{\Theta}^{(\psi)}} \{p(\boldsymbol{\theta}, \psi|\mathbf{y})\} \tag{3.37}$$

$$\hat{\boldsymbol{\theta}}^{(\psi)}_{\text{MEAN}} = \int_{\boldsymbol{\Theta}^{(\psi)}} p(\boldsymbol{\theta}, \psi|\mathbf{y})\boldsymbol{\theta}d\boldsymbol{\theta} \tag{3.38}$$

The problem with these estimators is that for general prior distributions they are not invariant under one-to-one reparameterisations of the likelihood function. The previous Chapter discussed the Minimum Message Length principle and noted that it solves this very problem by providing a Bayes estimator that is invariant. Bayesian model selection generally involves the Bayes factor; given two competing classes ψ_1 and ψ_2, the Bayes

Factor is given by

$$\frac{r(\mathbf{y}, \psi_1)}{r(\mathbf{y}, \psi_2)} \tag{3.39}$$

A Bayes Factor less than one is in favour of model class ψ_2, and a Bayes factor greater than one is in favour of model class ψ_1. This procedure may be used to compare multiple model classes; evaluation of the Bayesian evidence $r(\cdot)$ needed to compute the Bayes factor can be extremely difficult for arbitrary likelihood functions and prior distributions. Monte-Carlo Markov Chain methods [98, 99] are one commonly used approach to evaluate these difficult integrals and to perform Bayesian model selection in general. The Bayesian approach in general, and the requirement and meaning of prior distributions has caused a great deal of controversy and argument amongst the general statistics community (for example, see [100] and the discussions in [101]). However, this matter is far beyond the scope of this thesis and will no new arguments will presented to justify subjective Bayes procedures. There have been a large number of applications of Bayesian methods to inference of AR and ARMA models with various choices of priors, including [102] which performs inferences in partial autocorrelation space and [103, 104] which perform inferences in root-space. While the MML principle is also a Bayesian approach to model selection, it differs in many aspects from 'orthodox' Bayesian inference; this is discussed in more detail in [105].

3.6.1 The Bayesian Information Criteria (BIC)

One method of simplifying the posterior is to apply the Laplace approximation [106]. This method approximates an arbitrary posterior density by a multivariate normal distribution centred on the posterior mode. Under certain conditions, it is known that

$$\hat{\boldsymbol{\theta}}^{(k)}_{\mathrm{MAP}} \rightarrow \hat{\boldsymbol{\theta}}^{(k)}_{\mathrm{ML}}, \text{ as } N \rightarrow \infty \tag{3.40}$$

that is, asymptotically the MAP and ML estimators coincide. This is not unexpected as for large N the effects of the prior distribution will be swamped by the effects of the data. The Bayesian Information Criteria [107] exploits this fact to offer an approximation to the Bayesian evidence. By applying the Laplace expansion to the posterior and taking limits, the BIC is given by

$$BIC(\psi) = L(\mathbf{y}|\hat{\boldsymbol{\theta}}^{(\psi)}_{\mathrm{ML}}) + \frac{P}{2}\log N \tag{3.41}$$

where $P = \dim\left(\hat{\boldsymbol{\theta}}_{\mathrm{ML}}^{(\psi)}\right)$ is the number of model parameters. The BIC satisfies

$$-\log \int_{\boldsymbol{\Theta}^{(\psi)}} h(\boldsymbol{\theta}|\psi) f(\mathbf{y}|\boldsymbol{\theta}, \psi) d\boldsymbol{\theta} = BIC(\psi) + \mathrm{o}(1) \tag{3.42}$$

The BIC given by (3.41) is very similar in form to the AIC/AICc and KIC/KICc criteria introduced earlier, consisting of a negative log-likelihood term plus a penalty term that is a function of the number of parameters and data. In this case, however, as N increases, the penalty term also increases unboundedly. The result of this is that the BIC criterion is consistent in the sense that if the true model is in the model classes considered, as $N \to \infty$ the probability of selecting it via minimising the BIC approaches one. This is a powerful result and is one of the reasons the BIC is so widely applied. The AMAP criterion [108] is derived in a similar fashion BIC, but the way in which the limits are taken leads to slightly different penalty terms for some model classes; this has been shown improve model selection results for some model classes [109]. For AR models however, it yields the same criterion.

3.7 Finite Sample Information Criteria

The Finite Sample Information Criteria developed by Broersen and his collaborators were designed to take into consideration the properties of autoregressive parameter estimators for 'finite' sample sizes. A sample size N is roughly described as finite if $(N/P < 10)$, i.e. there are less than 10 data points per parameter to be estimated. The key components of the finite sample information criteria are the finite sample parameter variances for several of the popular autoregressive model estimators. These are the variances of the P-th estimated reflection coefficient (for the Burg and Yule-Walker estimators), and the variances of the P-th estimated coefficient (for the least-squares methods), for an AR(P) model, and are given by [110]

$$v_j^{(YW)} = \frac{N-j}{N(N+2)} \tag{3.43}$$

$$v_j^{(BURG)} = \frac{1}{N+1-j} \tag{3.44}$$

$$v_j^{(LSFB)} = \frac{1}{N+\frac{3}{2}(1-j)} \tag{3.45}$$

$$v_j^{(LSF)} = \frac{1}{N+2-2j} \tag{3.46}$$

where (YW) refers to the Yule-Walker estimates, $(LSFB)$ the least-squares forwards-backwards method and (LSF) the least squares forwards only method. It is clear that when N is small relative to j the variances may differ considerably from the asymptotic variance $(1/N)$. These finite sample variances may be used to find two important expectations; the first

$$\mathrm{E}\left[\hat{\tau}_{(\cdot)}^{(P)}\right] = \tau^* \prod_{i=1}^{P} \left(1 - v_i^{(\cdot)}\right) \tag{3.47}$$

is the expected estimated variance of an AR(P) model fitted to N data points, where $(\cdot)$ is replaced by the appropriate estimator. Clearly, as P increases, $\mathrm{E}\left[\hat{\sigma}^2\right]$ decreases; thus, using $\hat{\sigma}^2$ as a measure of the models fit to *unseen* data is impractical. The second expectation is the expected prediction error

$$\mathrm{E}\left[\hat{y}_{(N+1)} - y^*_{(N+1)}\right] = \tau^* \prod_{i=1}^{P} \left(1 + v_i^{(\cdot)}\right) \tag{3.48}$$

where $\hat{y}_{(N+1)}$ is the prediction of the next value in the observed sequence, and $y^*_{(N+1)}$ is the true (and unknown) value. In contrast to (3.47), the expected prediction error (3.48) increases as P increases, and as P approaches N the error begins to rise quite rapidly. These expectations may be used to derive a third expectation

$$\mathrm{E}\left[\frac{\sum_{i=1}^{N}(\hat{y}_i - y_i^*)}{\hat{\tau}_{(\cdot)}^{(P)}}\right] \approx N \prod_{i=1}^{P} \frac{1 + v_i^{(\cdot)}}{1 - v_i^{(\cdot)}} \tag{3.49}$$

where the approximation $\mathrm{E}\left[x/y\right] \approx \mathrm{E}\left[x\right]/\mathrm{E}\left[y\right]$ is employed. It can be seen that the true (unknown) variance is not present in (3.49) as it has been cancelled out. This expectation may be used to derive the Finite Sample Information Criteria [111] from the negative log-likelihood as

$$FSIC(P) = \log \hat{\tau}_{(\cdot)}^{(P)} + \prod_{i=1}^{P} \frac{1 + v_i^{(\cdot)}}{1 - v_i^{(\cdot)}} - 1 \tag{3.50}$$

The expectation for $\left(\log \hat{\tau}_{(\cdot)}^{(P)}\right)$ relies on the unknown true variance σ^{*2} and thus is estimated by the observed variance. Another criterion that may be derived from the expectations given by (3.47) and (3.48) is the Finite Information Criteria [110]. This is based on a finite sample modification to Akaike's Final Prediction Error criteria [112] and is given

by

$$FIC(P) = \log \hat{\tau}_{(\cdot)}^{(P)} + \alpha \prod_{i=1} v_i^{(\cdot)} \tag{3.51}$$

where α is a penalty term. Investigations have suggested $\alpha = 3$ yields good order selection results [113]. The final criterion of note is the Combined Information Criterion (CIC) [65] which essentially combines the FIC and FSIC criteria and is given by

$$CIC(P) = \log \hat{\tau}_{(\cdot)}^{(P)} + \max \left\{ \prod_{i=1}^{P} \frac{1 + v_i^{(\cdot)}}{1 - v_i^{(\cdot)}} - 1, \alpha \prod_{i=1} v_i^{(\cdot)} \right\} \tag{3.52}$$

For higher values of N/P the CIC is the same as the FIC, and for smaller N/P the CIC is equal to the FSIC. This criteria combines the strong asymptotic properties of FIC with the good small sample properties of FSIC, and has been shown to yield good performance in autoregressive order selection on a range of examples.

3.8 The Minimum Description Length (MDL) Principle

The Minimum Description Length (MDL) principle [114, 115, 116, 117, 118] is the last model selection method to be examined. The introduction of the MDL principle postdates the Minimum Message Length (MML) principle by approximately 10 years and the two have many aspects in common, particularly the philosophy that the model that most compresses the observed data is the best explanation of said data. There are however many areas in which the two principles disagree, both philosophically and technically, the largest difference being on the topic of prior information. In its original and 'crude' form (the terminology being taken from [119]) the MDL principle was based around universal integer codes and codelengths of $(\frac{1}{2} \log N)$ per continuous parameter. This lead to the unfortunate (and incorrect) conclusion that 'MDL is BIC'. Modern MDL is instead concerned with the concept of *universal models*. A central concept is that of the coding *regret*; this is defined as the excess number of nits (though it may be negative) required by a distribution $\tilde{f}(\cdot)$ to code data $\mathbf{y}$ over the 'best' possible codelength based on $f^{(\psi)}$, i.e.

$$R\left(\mathbf{y}, \tilde{f}, f^{(\psi)}\right) = -\log \tilde{f}(\mathbf{y}) + \log f_{\hat{\boldsymbol{\theta}}_{\mathrm{ML}}^{(\psi)}}^{(\psi)}(\mathbf{y}) \tag{3.53}$$

A model $\tilde{f}(\cdot)$ is universal relative to a model class ψ if, for any ε there exists an N such that

$$\frac{1}{N}\max_{\mathbf{y}\in\mathcal{Y}}\left\{R\left(\mathbf{y},\tilde{f},f^{(\psi)}\right)\right\}\leq\varepsilon \tag{3.54}$$

that is, for any suitably large datasize the difference between coding the data using the best model in the mode class ψ or the universal model becomes arbitrarily small. In the modern MDL approach, the so called Normalised Maximum Likelihood universal model has assumed a position of pre-eminence; this is due to the fact that it achieves minimax coding regret. The NML distribution $\tilde{f}^{(\psi)}_{\text{NML}}$ relative to model class ψ is the solution of

$$\tilde{f}^{(\psi)}_{\text{NML}}=\arg\min_{\tilde{f}\in\mathcal{F}}\left\{\sup_{\mathbf{y}\in\mathcal{Y}}\left\{R\left(\mathbf{y},\tilde{f},f^{(\psi)}\right)\right\}\right\} \tag{3.55}$$

where $\mathcal{F}$ is the space of potential probability distributions. The NML distribution is minimax optimal in that it minimises the maximum coding regret. The MDL approach attempts to avoid the requirement for specification of subjective priors (as opposed to MML) and so must appeal to optimality other than minimisation of Bayes loss; in decision theory a well known alternative if there exists no prior knowledge is the minimax approach [40]. The NML distribution is given by

$$\tilde{f}^{(\psi)}_{\text{NML}}(\mathbf{y})=\frac{f\left(\mathbf{y}|\hat{\boldsymbol{\theta}}^{(\psi)}_{\text{ML}}(\mathbf{y}),\psi\right)}{\int_{\mathbf{y}'\in\mathcal{Y}}f\left(\mathbf{y}'|\hat{\boldsymbol{\theta}}^{(\psi)}_{\text{ML}}(\mathbf{y}),\psi\right)d\mathbf{y}'} \tag{3.56}$$

where $\hat{\boldsymbol{\theta}}^{(\psi)}_{\text{ML}}(\mathbf{y})$ is the ML estimate for model class ψ given data $\mathbf{y}$. The NML distribution is termed a *one-part* code as no point estimate is explicitly named before the data is sent; instead a codebook is defined for all possible data values based around the chosen model class ψ. This removes the redundancy present in two-part codes [84] but at the same time sacrifices the ability to make a statement about a fully specified model. Model selection in the NML framework involves solving

$$\hat{\psi}_{\text{NML}}=\arg\min_{\psi\in\Psi}\left\{-\log f\left(\mathbf{y}|\hat{\boldsymbol{\theta}}^{(\psi)}_{\text{ML}}(\mathbf{y}),\psi\right)+\log\int_{\mathbf{y}'\in\mathcal{Y}}f\left(\mathbf{y}'|\hat{\boldsymbol{\theta}}^{(\psi)}_{\text{ML}}(\mathbf{y}'),\psi\right)d\mathbf{y}'\right\} \tag{3.57}$$

where the negative log of the normalising constant in (3.56) is termed the *parametric complexity*, denoted $\text{COMP}(\psi)$, and the complete negative log-likelihood of (3.56) is termed the *stochastic complexity*, denoted $\text{SC}(\mathbf{y},\psi)$. It is clear to see that model classes which

are flexible and allow for short code lengths for a large number of possible datasets will have larger parametric complexities than models which allow for short code lengths on only a small number of datasets; this forms a measure of their complexity in terms of how well they can fit the possible datasets. The NML codelength then consists of a sum of a goodness of fit term (the negative log-likelihood of the data given the ML estimate) and the parametric complexity term, which measures how complex the model class is. It is also interesting to note that the regret of any dataset coded using $\tilde{f}^{(\psi)}_{\mathrm{NML}}$ will be uniformly equal to $\mathrm{COMP}(\psi)$. Evaluation of the normalising integral in (3.56) is far from trivial and for many models no closed forms have been found. Rissanen introduced an asymptotic approximation for the stochastic complexity of the NML distribution that is valid under certain regularity conditions (such as the asymptotic normality of the ML parameter estimates); this approximation [118] is given by

$$\mathrm{SC}(\mathbf{y},\psi) = -\log f\left(\mathbf{y}|\hat{\boldsymbol{\theta}}^{(\psi)}_{\mathrm{ML}},\psi\right) + \frac{P}{2}\log\frac{N}{2\pi} + \int_{\boldsymbol{\Theta}^{(\psi)}} \sqrt{|\mathbf{J}_1(\boldsymbol{\theta})|}d\boldsymbol{\theta} + \mathrm{o}(1) \tag{3.58}$$

where $\mathbf{J}_1(\boldsymbol{\theta})$ is the *per data* Fisher Information Matrix and P is the number of model parameters. Although the NML universal model appears to offer a good basis upon which to make inferences (see [119], Section 14.6 for an interesting discussion of minimax coding), for certain model classes $\mathrm{COMP}(\psi) = \infty$ (see for example [120]); the model class is deemed to have infinite parametric complexity. There are several methods proposed to solve this problem, including renormalisation [121], hybrid two-part and NML codes and luckiness functions [120, 119], although in general the issue of infinite parametric complexity is problematic. The NML distribution is not the only universal model considered in the modern MDL literature; an alternative is the Bayes mixture distribution given by

$$\tilde{f}^{(\psi)}_{\mathrm{BAYES}}(\mathbf{y}) = \int_{\boldsymbol{\Theta}^{(\psi)}} h(\boldsymbol{\theta}) f(\mathbf{y}|\boldsymbol{\theta},\psi)d\boldsymbol{\theta} \tag{3.59}$$

This distribution is literally a Bayesian averaging of all possible distributions within model class ψ, taken with respect to some prior $h(\cdot)$. The problem of selecting a suitable prior remains; choosing the uninformative Jeffrey's prior $h(\boldsymbol{\theta}) \propto \sqrt{|\mathbf{J}_1(\boldsymbol{\theta})|}$ leads to a special Bayesian mixture that like the NML distribution, is minimax with respect to coding regret. The other two types of universal model that are often considered in the MDL literature are the two-part models and the prequential models [117]. The former have similarities

to the MML codes in that they first name a model, and then transmit the data using the named model. The prequential codes have close ties to Dawid's principle of prequential inference [122] and to the method of Accumulated Prediction Error discussed in Section 3.5.2. A very thorough discussion of the MDL principle is given by [119], and a discussion of differences and similarities between MML and MDL may be found in [17]. The key differences between MDL and MML are now summarised:

- The MML principle is a subjective Bayes procedure that embraces the use of prior distributions to specify and utilise prior information. This is in contrast with the MDL principle which attempts to circumvent the issue of priors altogether by appealing to alternate ideas of 'optimal' coding. Both philosophies have their merits and it is impossible to discredit either; the important issue to note is that they differ significantly on this point.

- While the MDL principle encompasses a range of universal models, the MML principle strictly employs two-part codes. The two-part MML codes differ from the MDL two-part codes in that they are especially designed to minimise *expected* code lengths, with respect to the chosen priors. Given that the MDL approach is to avoid explicitly specifying priors, the two-part (and one-part) universal codes are based on other metrics of optimality and may differ substantially from the MML codes, at least in the non-asymptotic case.

- The MML principle involves selecting a *fully specified* model, that is, it chooses a point estimate from within a model class (and therefore enables various competing point estimates to be compared). The MDL principle instead chooses only a model class and does not endorse any particular point estimate within the class (although some very recent work does examine the issue of MDL parameter estimation [119], Section 15.4). Once again, it may be argued that there are merits to both approaches. The MDL approach means that once a class has been selected, an estimator known to have certain properties may be employed to find a suitable point estimate. On the other hand, the MML point estimator has itself been shown to have advantageous properties for many model families (and has remained consistent even for such problems as Neyman-Scott [49]).

The MDL criterion has been applied to order selection of autoregressive and autoregressive moving average models in a large number of papers, ranging from early work on crude MDL [117], to predictive least squares and prequential methods [90] to the very latest work based on the NML universal models [123]. Empirical results in this latest work suggests that the NML selection criterion performs very well for small and moderate sample sizes, at least in terms of correct order identification. The MDL principle has been successfully applied to a range of real world problems including oil spill segmentation [124] and ultrasonic measurement systems [125] to name a few.

3.9 Evaluation of Model Selection Criteria

The final section of this chapter examines some loss functions which may be used to assess how well an model estimated from data approximates the true model that generated the data. Begin by defining $\boldsymbol{\theta}^* = \left\{\mathbf{a}^*, \mathbf{c}^*, {\sigma^2}^*\right\}$ as the 'true' model, and $\hat{\boldsymbol{\theta}} = \left\{\hat{\mathbf{a}}, \hat{\mathbf{c}}, \hat{\sigma}^2\right\}$ as some approximating model. For convenience, assume that the autoregressive and moving average parameter vectors of both models are of the same length, with zeros padding out the vectors if required. Let P and Q denote the lengths of the autoregressive and moving average parameter vectors respectively, and let $R = \max(P, Q)$.

3.9.1 Squared Prediction Errors

Given a *known* sequence $\mathbf{y}_{(1:n-1)}$ and innovation sequence $\mathbf{v}_{(1:n-1)}$, the one-step ahead squared prediction error for y_n between $\boldsymbol{\theta}^*$ and $\boldsymbol{\theta}$ is given by

$$\left(\sum_{i=1}^{P} a_i^* y_{(n-i)} - \sum_{i=1}^{Q} c_i^* v_{(n-i)} - \sum_{i=1}^{P} \hat{a}_i y_{(n-i)} + \sum_{i=1}^{Q} \hat{c}_i v_{(n-i)}\right)^2 \tag{3.60}$$

If $\mathbf{y}$ is distributed per the ARMA process defined by $\boldsymbol{\theta}^*$, and $\mathbf{v}$ distributed as per an i.i.d. normal, the expectation of (3.60) is given by

$$\mathrm{SPE}_1(\boldsymbol{\theta}, \boldsymbol{\theta}^*) = \left([-\mathbf{a}^*, \hat{\mathbf{c}}^*] - [-\hat{\mathbf{a}}, \mathbf{c}]\right) \boldsymbol{\Psi}(\boldsymbol{\theta}^*) \left([-\mathbf{a}^*, \mathbf{c}^*] - [-\hat{\mathbf{a}}, \hat{\mathbf{c}}]\right)^T \tag{3.61}$$

where $\Psi(\cdot)$ is the $(P + Q \times P + Q)$ auto- and cross-covariance matrix of the true model, i.e.

$$\boldsymbol{\Psi}(\boldsymbol{\theta}) = \mathrm{E}\left[\left[\mathbf{y}_{(1:P)}, \mathbf{v}_{(1:Q)}\right]^T \left[\mathbf{y}_{(1:P)}, \mathbf{v}_{(1:Q)}\right]\right] \tag{3.62}$$

In the case of pure autoregressive models, scaling (3.61) by σ^{-2} yields the *Model Error* defined in [110]. The multiple-step ahead prediction error is a measure of how much two sequences $\mathbf{y}^*$ and $\hat{\mathbf{y}}$ differ over some number of samples, the two sequences being generated by feeding the true model and the estimated model with the *same* innovation sequence. Given an initial observation sequence $\mathbf{y}^0 \in \mathbb{R}^R$ and a known innovation sequence $\mathbf{v} \in \mathbb{R}^{(N^P+R)}$, the future sequences are found by

$$y_n^* = -\sum_{i=1}^{P} a_i^* y_{(n-i)}^* + \sum_{i=1}^{Q} c_i^* v_{(n-i)}, \ (n = R+1 \ldots N_P) \tag{3.63}$$

$$\hat{y}_n = -\sum_{i=1}^{P} \hat{a}_i \hat{y}_{(n-i)} + \sum_{i=1}^{Q} \hat{c}_i v_{(n-i)} \tag{3.64}$$

where $\hat{y}_n = y_n^* = y_n^0$ for $1 \leq n \leq R$. The N_P multi-step squared prediction error is then given by

$$\mathrm{SPE}_{N_P} = \frac{1}{(N_P - R)} \sum_{i=R+1}^{N_P} \mathrm{E}\left[(y_i^* - \hat{y}_i)^2\right] \tag{3.65}$$

The larger the N_P the more the ability of the model to capture overall dynamic behaviour of the process is tested. As both $\mathbf{y}^*$ and $\hat{\mathbf{y}}$ are zero-meaned and stationary, the difference of the two will also be a zero-mean stationary sequence. Thus, one choice is to let $N_P \to \infty$; this is simply denoted SPE_∞ in the sequel. The final issue to address is that of making the squared prediction errors commensurate between different true models. This is a concern if Monte Carlo experiments are undertaken to assess the behaviour of an estimator. As the signal-to-noise ratio of ARMA models is inherently tied to the values of the model parameters and is not affected by the value of σ^2, any experiment in which many different models are used will necessary mean that models with different signal powers will be compared. A model with poles close to the origin will have a smaller signal power than a model with poles close to the stationarity boundary; to put prediction errors for the two different models on an equal footing they may be normalised to produce *normalised* errors by

$$\mathrm{SPE}_1'(\hat{\boldsymbol{\theta}}, \boldsymbol{\theta}^*) = \frac{\mathrm{SPE}_1(\hat{\boldsymbol{\theta}}, \boldsymbol{\theta}^*)}{\gamma_0(\boldsymbol{\theta}^*)}. \tag{3.66}$$

where $\gamma_0(\cdot)$ is the zero-order autocorrelation of the true process; the same arguments apply to the multi-step SPE scores. This normalisation by the variance of the process has been

performed for many problems to ensure that loss functions are easily compared across the parameter space (see for example [126], for comparing estimators of Poisson means).

3.9.2 Kullback-Leibler Divergences

The squared prediction error loss measures how well the estimated model captures the time behaviour of the true process; the directed Kullback-Leibler (KL) [14] divergence also measures this, but at the same time takes into account the estimates of the innovation noise variance by measuring the difference between the probability density functions of the true and estimated models. The KL divergence is literally the excess number of nits required to transmit N data points generated by the true process $\boldsymbol{\theta}^*$ using the estimated model $\hat{\boldsymbol{\theta}}$, and for ARMA models is given by

$$\Delta(\hat{\boldsymbol{\theta}}||\boldsymbol{\theta}^*) = \frac{1}{2}\left(\log\frac{|\boldsymbol{\Gamma}(\hat{\boldsymbol{\theta}})|}{|\boldsymbol{\Gamma}(\boldsymbol{\theta}^*)|} + \mathrm{Tr}\left(\boldsymbol{\Gamma}^{-1}(\hat{\boldsymbol{\theta}})\cdot\boldsymbol{\Gamma}(\boldsymbol{\theta}^*)\right) - N\right) \tag{3.67}$$

where $\boldsymbol{\Gamma}(\cdot)$ are the autocovariance matrices for the true and estimated models. This exploits the fact that an ARMA model can be expressed in the form of a multivariate Normal distribution. While the directed KL divergence does take all parameters of the model into account, it punishes an *underestimation* of the magnitude of autocovariances more heavily than an overestimation, and is therefore particularly sensitive to underestimation of the innovation variance parameter. The symmetric KL divergence, known as the J divergence is given by

$$\mathrm{J}(\hat{\boldsymbol{\theta}}, \boldsymbol{\theta}^*) = \Delta(\hat{\boldsymbol{\theta}}||\boldsymbol{\theta}^*) + \Delta(\boldsymbol{\theta}^*||\hat{\boldsymbol{\theta}}) \tag{3.68}$$

By adding the directed KL divergences in both directions the J divergence punishes overestimations and underestimations of the magnitude of the autocovariances more equitably, and will not naturally tend to favour an overestimation of the innovation variance parameter.

3.9.3 Conditional Prediction Errors

The previous metrics were *unconditional* in the sense that the they were found by taking expectations of different losses. They summarise the expected loss for data randomly generated from the true model; it is also of interest to examine the expected losses conditioned on exactly the data that was observed. Let $\mathbf{y} \in \mathbb{R}^N$ be the sequence of data generated by

$\boldsymbol{\theta}^*$ from which the estimate $\hat{\boldsymbol{\theta}}$ was found. A predictive distribution $f_{(N+1)}(\cdot|\mathbf{y}, \boldsymbol{\theta}^*)$ may be found given the true ARMA model $\boldsymbol{\theta}^*$. If the innovations are Normally distributed the true predictive distribution will be a univariate normal distribution with mean $\mu^*_{(N+1)}$ and variance $\tau^*_{(N+1)}$. There will also exist an estimated predictive distribution $f_{(N+1)}(\cdot|\mathbf{y}, \hat{\boldsymbol{\theta}})$ with mean $\hat{\mu}_{(N+1)}$ and variance $\hat{\tau}_{(N+1)}$. From this, the conditional one-step ahead prediction error is given by

$$\mathrm{SPE}_{1c}(\boldsymbol{\theta}, \boldsymbol{\theta}^*, \mathbf{y}) = \left(\mu^*_{(N+1)} - \hat{\mu}_{(N+1)}\right)^2 \tag{3.69}$$

which may be normalised by dividing by $\gamma_0(\cdot)$ if required, as discussed in the previous section. The conditional one-step ahead directed KL divergence is given by

$$\Delta_{1c}(\hat{\boldsymbol{\theta}}||\boldsymbol{\theta}^*; \mathbf{y}) = \frac{1}{2}\left(\log\frac{\hat{\tau}_{(N+1)}}{\tau^*_{(N+1)}} + \frac{\tau^*_{(N+1)} + \left(\mu^*_{(N+1)} - \hat{\mu}_{(N+1)}\right)^2}{\hat{\tau}_{(N+1)}} - 1\right) \tag{3.70}$$

and from this the one-step ahead J divergence may be found by

$$\mathrm{J}_{1c}(\hat{\boldsymbol{\theta}}, \boldsymbol{\theta}^*, \mathbf{y}) = \Delta_{1c}(\hat{\boldsymbol{\theta}}||\boldsymbol{\theta}^*; \mathbf{y}) + \Delta_{1c}(\boldsymbol{\theta}^*||\hat{\boldsymbol{\theta}}; \mathbf{y}) \tag{3.71}$$

The unconditional losses measure the discrepancy between an estimated model and a true model for all possible datasets; the conditional losses literally measure the predictive ability of the model for the next data point in an observed sequence. As such, the conditional losses are perhaps less truly indicative of how close an estimated model is to the true generating source than the unconditional losses. By the same token, if prediction of the next sample in a sequence is the ultimate aim, these losses are obviously very useful measures. The topic of the differences between the aims of induction and prediction in inference are deep and lie beyond the scope of this thesis; in the empirical experiments presented in the sequel a range of loss functions will be employed to describe a picture of the performance of the model selection criteria that is not biased to single point of view.

3.9.4 Correct Order Selection

This criterion is often used to evaluate the performance of a model selection criterion's ability to correctly identify structure of the underlying model, i.e. to correctly estimate P and Q. There are several arguments against the validity of this criterion as a gauge of an estimator's performance: first, due to the parameterisation of the model, or the

mathematical form of the likelihood, it is very possible that the estimates $\hat{\mathbf{a}}$, $\hat{\mathbf{c}}$ and $\hat{\sigma}^2$ yield very high large negative log-likelihoods and prediction errors when the correct model order is used, while a less complex model may yield much better log likelihood/prediction errors. This is particular true when the amount of data from which the inferences are drawn is very small in comparison to the number of parameters in the model [127]. What then is the point of knowing the true structure of the underlying model if the ability of this estimated model to predict the behaviour of the true process is poor?

The second problem with this criterion is that it is only applicable in those cases where the structure of the underlying model is both known, and known to be of the same class as the model being inferred from the data. For real data, where the underlying generating model can never really be known, the concept of correct order identification becomes somewhat meaningless.

3.10 Conclusion

This chapter has examined a range of the most important and frequently used methods for estimating ARMA models from observed data in the literature. In particular, the following methods were reviewed:

- The Yule-Walker, Least Squares and Burg Estimators for AR parameters
- The Maximum Likelihood and long AR estimators for ARMA parameters
- Minimum Discrepancy Estimation techniques, including the AIC and KIC criteria
- Sampling approaches to MDE through Cross-Validation and Accumulated Prediction Errors
- Bayesian Inference
- Broersen's Finite Sample criteria
- The Minimum Description Length principle and universal models

From these methods, it was determined that the AIC/AICc, KIC/KICc, BIC, CIC and SC criteria appear to offer a convenient mix of ease of computation and performance, and will be examined against the MML based criteria in Chapter 9.

Chapter 4

MML87 Inference of Autoregressive Models

4.1 Chapter Overview

This chapter presents the first major contribution of this thesis: a new order and parameter estimator for autoregressive models using the MML87 message length approximation. Autoregressive models have been previously examined within an MML framework [16] with success, the inferences being made in coefficient space. This chapter examines both model selection (i.e. order selection) and parameter estimation capabilities of the MML estimators for AR model within *pole space*, and the results demonstrate the extra effort is rewarded with inferences that outperform those made by conventional estimators. The chapter begins by formalising the problem statement, and then considers suitable priors for the model parameters. The likelihood function and Fisher Information matrix are then presented. Following this several modifications to the basic MML87 approximation are proposed to correct several issues that arise with autoregressive models. The task of parameter estimation is then examined, and some properties of the resulting estimator discussed. Finally, results of parameter estimation and order selection on both synthetic and real datasets within the MML87 framework are presented.

4.2 Autoregressive (AR) Models

Given a sequence of observed data $\mathbf{y}$ the AR(P) explanation for observation y_n is given by

$$y_n + \sum_{i=1}^{P} a_i y_{n-i} = v_n \tag{4.1}$$

where $v_i \sim \mathcal{N}(0, \sigma^2)$ are the i.i.d. normally distributed innovations. The most commonly prescribed method for fitting ARMA models to data is to estimate the coefficients $\mathbf{a}$ for range of candidate model orders, and select the one that minimises some selection criterion. The MML principle is a Bayesian approach to inference, and thus requires the specification of prior densities over the model parameters. The drawback with performing inferences on the coefficients $\mathbf{a}$ is that they have no clear and direct interpretation on the behaviour of the model. Instead, this thesis adopts the approach taken in [103] and [104], and works in the *root*-space. By defining the *delay* operator q^{-d} such that $y_n q^{-d} = y_{n-d}$ it is possible to express the ARMA equation (1.1) as

$$A(q) y_n = v_n \tag{4.2}$$

where

$$A(q) = \left(1 + \sum_{i=1}^{P} a_i q^{-i}\right) \tag{4.3}$$

The roots of $A(q)$ are the system *poles* and it is well known that their location determines the dynamic response of the model. An AR process is stable, and consequently stationary, if and only if all the roots lie within the unit circle [2].Let $\mathbf{p} \in \mathbb{C}^P$ denote a vector of the roots of $A(q)$; as the coefficients $\mathbf{a}$ are real, the poles may come in only two variants: wholey real poles, denoted $p_i = \alpha$, and complex conjugate pole pairs, denoted $p_{(i,j)} = \{r \exp(\mathrm{j}\omega), r \exp(-\mathrm{j}\omega)\}$. Let $\boldsymbol{\alpha} \in (-1, 1)^{N_R}$ denote the vector of real pole positions, and $\mathbf{r} \in (-1, 1)^{N_C}$ and $\boldsymbol{\omega} \in (0, \pi)^{N_C}$ denote the radial and angular components of the complex pole pairs. The root-space parameter vector $\boldsymbol{\beta}$ is given by

$$\boldsymbol{\beta} = [\alpha_1, \ldots, \alpha_{N_R}, r_1, \omega_1, \ldots, r_{N_C}, \omega_{N_C}] \tag{4.4}$$

and the corresponding vector of poles

$$\mathbf{p} \equiv \mathbf{p}(\boldsymbol{\beta}) = \left[\alpha_1, \ldots, \alpha_{N_R}, r_1 \mathrm{e}^{(\mathbf{j}\omega_1)}, r_1 \mathrm{e}^{(-\mathbf{j}\omega_1)}, \ldots, r_{N_C} \mathrm{e}^{(\mathbf{j}\omega_{N_C})}, r_{N_C} \mathrm{e}^{(-\mathbf{j}\omega_{N_C})}\right] \tag{4.5}$$

The AR model may then be expressed as

$$\prod_{k=1}^{P} \left(1 - p_k(\boldsymbol{\beta}) q^{-1}\right) y_n = v_n \tag{4.6}$$

The root-space parameterisation admits a more natural interpretation of the AR model's behaviour and makes the process of specifying prior distributions for the parameters significantly more intuitive. This of course has the consequence of complicating the resulting model's Fisher Information Matrix significantly; there are however, several key advantages to using this representation of the model that justify the extra work:

1. The AR model's time response is determined largely by the poles of the model. The autoregressive coefficient parameters do not have an intuitive time-response interpretation and thus it is difficult to select suitable priors for these parameters. In contrast, the root-parameters have a direct effect on the time response of the model.

2. This structure allows AR models to be examined based on their underlying structural components. In coefficient space, an AR(P) process is a single model class; a root-space representation allows an AR(P) process to be broken into $(\lfloor P/2 \rfloor + 1)$ (where $\lfloor \cdot \rfloor$ denotes the integer component) different processes by examining the different combinations of root structure allowed with P delays.

3. For regime switching models comprised of multiple AR processes, shared structural behaviours of models in different regimes may be easily identified; in coefficient space the coefficients are formed by combinations of structure behaviour parameters and this is extremely difficult.

4. The 'uninformative' uniform prior over the autoregressive coefficients induces priors over the poles that vary with the order of autoregressive model under consideration. There is usually no reason to believe that the distribution of pole magnitudes and angular components will be different depending on the order of model, and even if this is desired, it may be easily implemented within pole space. In contrast, selecting a prior in coefficient space that induces priors that do not vary with model order over the poles is very difficult.

The MML principle is a Bayesian technique and thus requires that prior distributions are placed over all parameters that are to be inferred. Suitable priors for our AR model parameters are now examined.

4.3 Priors for Autoregressive Models

The full prior density over a complete AR model is examined first, i.e. $\boldsymbol{\theta} = \left\{P, \boldsymbol{\beta}, \sigma^2\right\}$. By assuming that the model parameters are *a priori* independent, the total prior density $h(\boldsymbol{\theta})$ can be factorised into the following products

$$h(\boldsymbol{\theta}) = h(P) \cdot h(\boldsymbol{\beta}) \cdot h(\sigma^2) \tag{4.7}$$

Suitable priors for all parameters in $\boldsymbol{\theta}$ are examined in the next subsections.

4.3.1 Priors for $\boldsymbol{\beta}$

Priors for the autoregressive root-space parameters $\boldsymbol{\beta}$ are examined first. Begin by recalling that for an AR model to be stationary all of its poles $\mathbf{p}$ must lie within the perimeter of the unit circle centred at $p = 0$. A simple prior would then be to make $h(p) \propto 1$ on the unit circle. However, it should be recalled that poles only come in two flavours: wholly real poles, and complex conjugate pole pairs, and the moduli and radii of these poles help to determine the time response of the AR model [128]. Priors for both types of parameters are now discussed.

Priors on Real Poles

For a purely real pole, $p_i = \alpha_i$ and thus $|p_i| = |\alpha_i|$ and $\angle p_i = 0$; thus a prior density is required only on the real component of p_i. The effect of a particular prior density upon our inferences can be roughly determined by examining the time response of AR(1) model, i.e. an autoregressive model with a single real pole. The impulse response of an unforced AR(1) model with real pole at α is given exactly by

$$y_n = \alpha^n, \;\; n \in \mathbb{Z}, \; n \geq 0 \tag{4.8}$$

This sequence may be exactly produced by sampling the continuous time process

$$y_t = \exp\left(-\frac{t}{\tau}\right) \tag{4.9}$$

at uniform intervals $t = 0, 1, 2, ..., \infty$, where τ is the time constant of the process. The relation between α and τ is given by

$$\tau = -\frac{1}{\log \alpha} \tag{4.10}$$

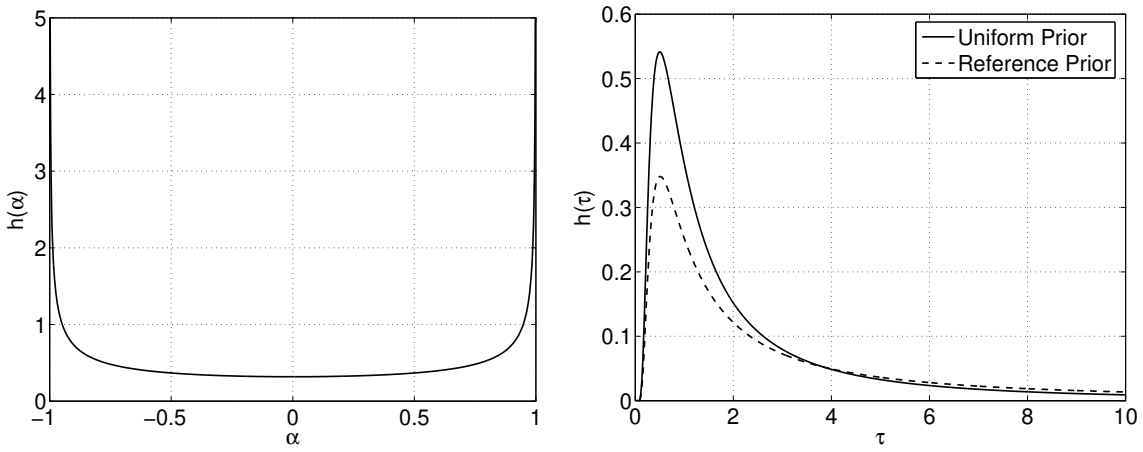
This nonlinear relationship indicates that as $\alpha \to 1$ the range of different time responses as characterised by their time constants becomes denser. As the AR model is designed to model time series data, and thus time responses, it is helpful to view the model parameters in terms of this time constant. For some prior distribution $h(\alpha)$, the resulting induced density on τ can be found by

$$\begin{aligned} h(\tau) &= \frac{\mathrm{d}}{\mathrm{d}\tau}\left\{\exp\left(-\frac{1}{\tau}\right)\right\} \cdot h\left(\alpha|\alpha = \exp\left(-\frac{1}{\tau}\right)\right) && (4.11)\\ &= \frac{\exp\left(-\frac{1}{\tau}\right)}{\tau^2} \cdot h\left(\alpha|\alpha = \exp\left(-\frac{1}{\tau}\right)\right) && (4.12) \end{aligned}$$

Several candidate priors for α are now considered, and the resultant induced prior density on τ observed using (4.12). Over the course of many years of Bayesian analysis of autoregressive processes a wide range of possible priors have been proposed; a subset of popular choices is examined next:

$$\begin{aligned} h_u(\alpha) &= \frac{1}{2} && (4.13)\\ h_r(\alpha) &= \frac{1}{\pi\sqrt{(1-\alpha^2)}} && (4.14)\\ h_b(\alpha|a,b) &= \frac{1}{2}B(a,b)|\alpha|^{a-1}(1-|\alpha|)^{b-1} && (4.15)\\ h_n(\alpha|\sigma_\alpha^2) &= \sqrt{2}\exp\left(-\frac{1}{2}\log\left(-\frac{\alpha+1}{\alpha-1}\right)^2\sigma_\alpha^{-2}\right)\left(\sqrt{\pi}\sigma_\alpha|\alpha^2-1|\right)^{-1} && (4.16)\\ h_p(\alpha|T) &= \left((1-\alpha^2)^{-1}\left(T-\frac{1-\alpha^{2T}}{1-\alpha^2}\right)\right)^{\frac{1}{2}} \cdot \Omega(T)^{-1} = \frac{\mathcal{P}(\alpha|T)}{\Omega(T)} && (4.17) \end{aligned}$$

with all distributions defined on the support $\alpha \in (-1, 1)$, i.e. the stationarity region. A discussion of these priors follows:



(a) Reference prior in α-space

(b) Uniform and Reference priors in τ-space

Figure 4.1: Reference prior in α-space, and Uniform and Reference priors in τ-space

1. **Uniform Prior**: The prior in (4.13) is clearly the uniform prior. While this is a traditionally 'uninformative' prior, it can be observed from its induced prior on τ that it puts very little probability mass on $\tau > 4$, which can lead it to favour models with faster poles.

2. **Reference Prior**: The prior given by (4.14) is the Yang and Berger [129, 130] reference prior and is shown in α-space in Figure 4.1(a); for AR(1) models it is also a Jeffrey's prior. Figure 4.1(b) shows that it places more mass for $\tau > 4$ and is thus significantly less biased towards faster models than the uniform prior. This is clear from the plot of the ratio of uniform prior on reference prior in τ-space given in Figure 4.2. For values of $\tau > 4$ the reference prior assigns significantly higher proportion of probability mass than does the uniform prior.

3. **Beta Prior**: The prior specified by (4.15) is a Beta prior on the modulus of α; the scaling by $\frac{1}{2}$ takes into account the sign of α, giving positive and negative values equal weighting. This prior is tunable, and by selecting different values of the hyper-parameters a and b it may be shaped to emphasise slower or faster poles and thus take into account any actual prior information that may be available. For an AR(1) model, a suitably uninformative is given by $a = 5$ and $b = \frac{1}{2}$ [131]. In general, as $b \to 1$ the prior favours faster models, and as $b \to 0$ the prior favours slower models.

4. **Transformed Normal Prior**: The prior specified by (4.16) is built by placing a Normal distribution over a continuous variable $x \in (-\infty, \infty)$ and transforming this

variable to α via

$$\alpha = \frac{2\exp(x)}{1+\exp(x)} - 1 \tag{4.18}$$

which is simply an offset $\tanh(\cdot)$ function. This prior, used in [104], allows for a flexible specification of priors in τ-space by choosing suitable values for σ_α^2. Its main drawback is that the light tail of the Normal distribution can lead to priors that punish large values of τ too heavily.

5. **Phillips' Prior**: The prior given by (4.17) was proposed by Phillips [132] as the stationary part of a complete prior over $\alpha \in (-\infty, \infty)$. The tunable parameter T allows it to emphasise larger or smaller τ. The term $\Omega(T)$ is a normalisation constant and is given by

$$\Omega(T) = \int_{-1}^{1} \mathcal{P}(\alpha|T)d\alpha \tag{4.19}$$

where $\mathcal{P}(\cdot)$ is the unnormalised 'Phillips' prior' function.

In this work, the uniform and reference priors have been chosen for study, as they do not depend on further hyperparameters. The uniform prior is suitable for models with faster poles, and the reference prior is a reasonable choice if the data is believed to have been generated by a slower process (i.e. one with poles closer to the unit circle).

Priors on Complex Pole Pairs

To determine suitable priors for complex conjugate pole pairs, we begin by recalling that complex poles only appear in conjugate pairs of the form $p_{(i,j)} = r_i \exp(\mathbf{j}\omega_i)$, $r_i \exp(-\mathbf{j}\omega_i)$. Thus, r_i is a modulus parameter and ω_i the angle parameter. The impulse response of an AR(2) model with two complex conjugate pairs given by

$$y_n = Kr^n cos(\omega n), \;\; n \in \mathbb{Z},\; n \geq 0 \tag{4.20}$$

It can be observed then that the r parameter has a similar effect on the time response to the α parameter in a real pole, and thus a similar prior may be appropriate. One choice of prior on ω is to make it uniform on $\omega \in [0, \pi]$, i.e. the range of normalised frequencies a discrete system may assume [104], that is

$$h(r, \omega) = \frac{2h(|r|)}{\pi} \tag{4.21}$$

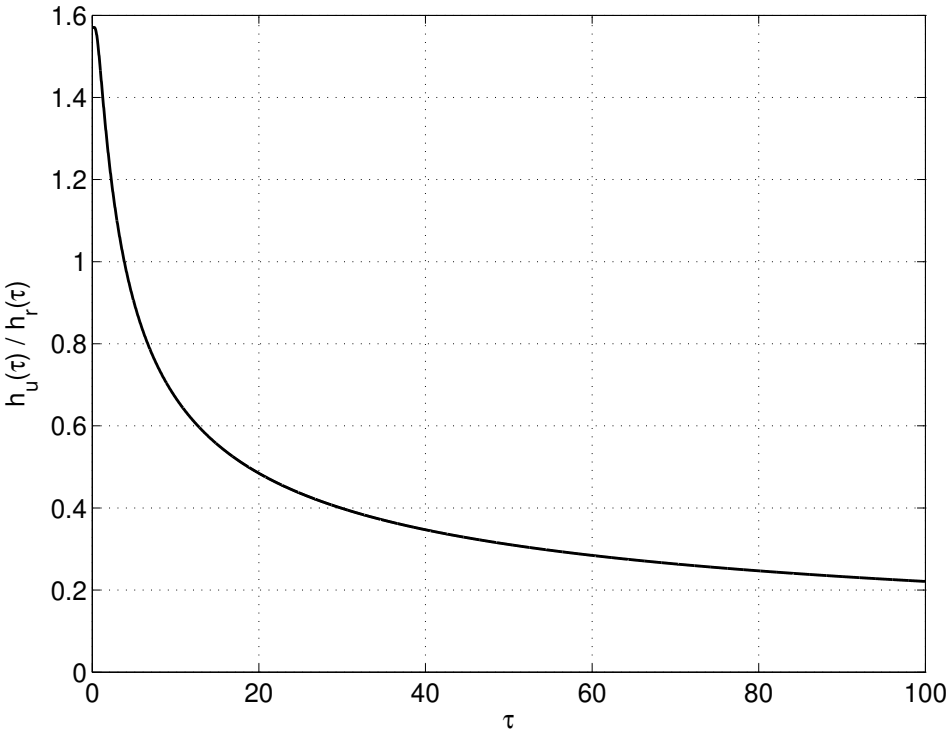


Figure 4.2: Plot of $h_u(\alpha)/h_r(\alpha)$ in τ-space

Another option is the 'component reference prior', which is formed by taking a uniform prior over the coefficients of an AR(2) model with complex poles [103]. This prior, which biases away from models with extremely high or low frequency components leads to the prior for a complete complex pair given by

$$h(r, \omega) = 2h(|r|) \cdot \left(\frac{1}{2}\sin\omega\right) = h(|r|)\sin\omega \tag{4.22}$$

where $h(|r|)$ is selected from the previous section, i.e. from priors (4.13)-(4.17). The prior is scaled by a factor of two to account for the fact that $r \in (0, 1)$, whereas the priors on real poles are defined on the support $(-1, 1)$. For the experiments presented later in this thesis, the uniform prior given by (4.21) was used, with the uniform prior taken over $h(|r|)$.

4.3.2 Parameter Redundancy

An issue arises when applying the MML87 approximation to the autoregressive model specified in root-parameter space: such a model is a redundant representation of an equivalent order model in coefficient space. This redundancy arises because the ordering of the

poles has no effect on the resulting coefficient space model, i.e.

$$(q + 0.3)(q - 0.5) = (q - 0.5)(q + 0.3) = q^2 - 0.2q - 0.15 \tag{4.23}$$

The resulting model in coefficient space is thus identical regardless of an assignment $\boldsymbol{\beta} = [-0.3, 0.5]$ or $\boldsymbol{\beta} = [0.5, -0.3]$. To avoid this parameter redundancy, the roots are always assumed to be ordered by their radial components (as per [103]). In particular, the real poles are ordered as

$$\alpha_1 \geq \alpha_2 \geq \ldots \geq \alpha_{P_R} \tag{4.24}$$

and the complex poles as

$$r_1 \geq r_2 \geq \ldots \geq r_{P_C} \tag{4.25}$$

The final parameter vector is always constructed as

$$\boldsymbol{\beta} = [\alpha_1, \ldots, \alpha_{P_R}, r_1, \omega_1, \ldots, r_{P_C}, \omega_{P_C}] \tag{4.26}$$

While these removes the issue of parameter redundancy, it obviously cuts out a large portion of parameter combinations. The prior densities are taken over each α_i individually on the support $(-1, 1)$ and on each r_i over $(0, 1)$. Thus, the combined prior density for these parameters puts probabilty mass over events that are not possible, e.g. for $\alpha_2 > \alpha_1$. To resolve this issue, the resulting joint prior probability density over all real poles is given by

$$h(\alpha_1, \ldots, \alpha_{P_R}) = \frac{\prod_{i=1}^{P_R} h(\alpha_i)}{\int_{\mathbf{x} \in \Lambda_\alpha} \prod_{i=1}^{P_R} h(\alpha_i = x_i) d\mathbf{x}} \tag{4.27}$$

where Λ_α is a region such that

$$\Lambda_\alpha = \left\{ \mathbf{x} \in (-1, 1)^{P_R} : x_1 \geq x_2 \geq \ldots \geq x_{P_R} \right\} \tag{4.28}$$

that is, the region of valid parameter combinations in parameter space. The same construction is required for the radii components of complex pole pairs. The joint probability density over all radii components is then

$$h(r_1, \ldots, r_{P_R}) = \frac{\prod_{i=1}^{P_C} h(r_i)}{\int_{\mathbf{x} \in \Lambda_r} \prod_{i=1}^{P_C} h(r_i = x_i) d\mathbf{x}} \tag{4.29}$$

where Λ_r is a region such that

$$\Lambda_r = \left\{ \mathbf{x} \in (0,1)^{P_C} : x_1 \geq x_2 \geq \ldots \geq x_{P_C} \right\} \tag{4.30}$$

In the case of the uniform prior, the integrals simplify to the volume of a simplex

$$\int_{\mathbf{x} \in \Lambda_\alpha} \prod_{i=1}^{P_R} h(\alpha_i = x_i) d\mathbf{x} = \frac{1}{P_R!} \tag{4.31}$$

$$\int_{\mathbf{x} \in \Lambda_r} \prod_{i=1}^{P_C} h(r_i = x_i) d\mathbf{x} = \frac{1}{P_C!} \tag{4.32}$$

This indeed holds for any size region as long as all parameters are defined on the same support, and the priors are uniform. If the uniform prior is not chosen, the integrals are more difficult to evaluate. However, as the integrals are only dependent on the number of parameters P_R and P_C, and that the range of model orders considered is always small (i.e., usually much less than a hundred) they can be pre-computed once by Monte-Carlo integration [46] and used to generate a simple look-up table. The final prior probability density over the $\boldsymbol{\beta}$-parameters is then

$$h(\boldsymbol{\beta}) = h(\alpha_1, \ldots, \alpha_{P_R}) \cdot h(r_1, \ldots, r_{P_C}) \cdot \prod_{i=1}^{P_C} h(\omega_i) \tag{4.33}$$

with priors densities for individual parameters chosen accordingly.

4.3.3 Priors for σ^2

A very common choice of prior that has been widely used for model variance σ^2 is the *scale invariant* prior [1]. This prior is designed by placing a uniform prior on $\log \sigma^2$; appropriate transformation of $h(\log \sigma^2)$ to $h(\sigma^2)$ yields

$$\frac{\mathrm{d}}{\mathrm{d}\sigma^2} \left\{ \log \sigma^2 \right\} \cdot 1 = \frac{1}{\sigma^2} \tag{4.34}$$

This prior is clearly improper, and can be rendered proper by using a suitable normalisation over an appropriate support. The final scale-invariant prior on σ^2 is then given by

$$h(\sigma^2) = \frac{1}{\sigma^2} \cdot \frac{1}{\log(\sigma_U^2) - \log(\sigma_L^2)}, \ \sigma^2 \in [\sigma_L^2, \sigma_U^2] \tag{4.35}$$

Table 4.1: Enumerations of Pole Structures for the AR(P) Models for $P = 1 \ldots 4$

S	P	P_R	P_C	$\boldsymbol{\beta}$
1	1	1	0	$\{\alpha_1\}$
2	2	2	0	$\{\alpha_1, \alpha_2\}$
3	2	0	2	$\{r_1, \omega_1\}$
4	3	3	0	$\{\alpha_1, \alpha_2, \alpha_3\}$
5	3	1	2	$\{\alpha_1, r_1, \omega_1\}$
6	4	4	0	$\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$
7	4	2	2	$\{\alpha_1, \alpha_2, r_1, \omega_1\}$
8	4	0	4	$\{r_1, \omega_1, r_2, \omega_2\}$

This prior is used for all experiments in this technical report. The inverse-gamma [104] is another possible prior for σ^2, though using this does not lead to as simple MML estimators as the above scale invariant prior.

4.3.4 Priors for Model Structure

One possible prior for the autoregressive structure is to select a maximum number of poles that is considered reasonable (the maximum 'order' of the AR model) and enumerate all possible $\boldsymbol{\beta}$-structures in this range. Placing a uniform distribution on all enumerations allows for an equal *a priori* belief on any individual structure being chosen. For an AR(P) model, there are $(\lfloor P/2 \rfloor + 1)$ (where $(\lfloor \cdot \rfloor)$ denotes the integer component of an expression) different possible $\boldsymbol{\beta}$-structures the model may assume. If it is assumed $P \leq P_{MAX}$, all possible model $\boldsymbol{\beta}$-structures may be enumerated for all model orders less than or equal to P_{MAX}. Let S select which of these enumerated structures is to be used. A suitable prior for S is then given by

$$h(S) = \left(\sum_{i=1}^{P_{MAX}} \lfloor i/2 \rfloor + 1 \right)^{-1} \tag{4.36}$$

which gives a uniform prior on all possible structures for orders $P = 1, \ldots, P_{MAX}$. Table 4.1 shows the enumerations for AR(1) to AR(4) models, where P_C is the number of complex poles and P_R the number of real poles, and clearly $P = P_C + P_R$ for all S. An alternative prior is to state the order, P, of the process using a uniform distribution, and then state the number of complex poles, P_C, from a uniform distribution based on P, i.e.

$$h(P) = (P_{MAX})^{-1} \tag{4.37}$$

$$h(P_C|P) = \lfloor P/2 \rfloor^{-1} \tag{4.38}$$

This prior has the effect of punishing higher order models slightly, with the relative punishment decreasing as $P \to \infty$. In this thesis, the uniform over structures prior given by (4.36) was used in all experiments.

4.4 Likelihood Function

The AR(P) explanation of measurement y_n is then given by

$$\prod_{i=1}^{P}(1 - q^{-1}p_i)y_n = v_n \tag{4.39}$$

where $\mathbf{p}$ are the model poles, and v_n is a random disturbance assumed to be i.i.d. as $v_n \sim \mathcal{N}(0, \sigma^2)$. The most straightforward way to model (4.39) is in the unconditional likelihood form [2]

$$L(\mathbf{y}|\boldsymbol{\theta}) = \frac{N}{2}\log(2\pi) + \frac{N}{2}\log\boldsymbol{\Gamma}(\boldsymbol{\theta}) + \frac{1}{2}\mathbf{y}\boldsymbol{\Gamma}^{-1}(\boldsymbol{\theta})\mathbf{y}^T \tag{4.40}$$

where $\boldsymbol{\Gamma}(\boldsymbol{\theta})$ is the $(N \times N)$ process autocovariance matrix. However, as N grows, evaluation of the likelihood in this form becomes extremely inefficient. Given the linearity of (4.39) and the normality of the innovations, it is clear that the mean of the distribution over measurement y_n may be computed from the previous P measurements, for $n > P$. Let $\mathbf{x}_n = \left[y_{(n-1)}, \ldots, y_{(n-P)}\right]$ denote the state vector at time n, and build the design matrix $\mathbf{X} = \left[\mathbf{x}_{(P+1)}^T, \ldots, \mathbf{x}_N^T\right]$. The negative log-likelihood is then given by

$$\begin{aligned} L(\mathbf{y}|\boldsymbol{\theta}) \quad = \quad & \frac{N}{2}\log(2\pi) + \frac{1}{2}\log\left(|\boldsymbol{\Gamma}(\boldsymbol{\theta})|\right) + \frac{1}{2}\mathbf{y}_{(1:P)}\boldsymbol{\Gamma}^{-1}(\boldsymbol{\theta})\mathbf{y}_{(1:P)}^T + \frac{(N-P)}{2}\log(\sigma^2) \\ & + \frac{1}{2\sigma^2}\left(\mathbf{y}_{(P+1:N)} + \mathbf{aX}\right)\left(\mathbf{y}_{(P+1:N)} + \mathbf{aX}\right)^T - N\log\epsilon \end{aligned} \tag{4.41}$$

where $\mathbf{a}$ are the autoregressive coefficients, ϵ is the measurement accuracy of the data and $\boldsymbol{\Gamma}(\cdot)$ is now the $(P \times P)$ autocovariance matrix of the autoregression used to model the first P measurements for which no conditional mean can be computed.

4.5 The Fisher Information Matrix

These next subsections are concerned with the computation of the Fisher Information Matrix for the AR model described in root parameter-space, which is presented for the

sake of completeness. Ignoring the measurement accuracy, begin by dividing the complete negative log-likelihood (4.41) into two components

$$L^{(\Gamma)}(\mathbf{y}_{(1:P)}|\boldsymbol{\theta}) = \frac{P}{2}\log(2\pi) + \frac{1}{2}\log\left(|\boldsymbol{\Gamma}(\boldsymbol{\theta})|\right) + \frac{1}{2}\mathbf{y}_{(1:P)}\boldsymbol{\Gamma}^{-1}(\boldsymbol{\theta})\mathbf{y}_{(1:P)}^{T} \quad (4.42)$$

$$L^{(X)}(\mathbf{y}_{(P+1:N)}|\boldsymbol{\theta}) = \frac{N-P}{2}\log(2\pi) + \frac{N-P}{2}\log(\sigma^2) + \frac{1}{2\sigma^2}\left(\mathbf{y}_{(P+1:N)} + \mathbf{aX}\right)\left(\mathbf{y}_{(P+1:N)} + \mathbf{aX}\right)^{T} \quad (4.43)$$

where $L^{(\Gamma)}(\cdot)$ is the negative log-likelihood of the unconditional component, and $L^{(X)}(\cdot)$ is the negative log-likelihood of the conditional component. Given the linearity of the differentiation and expectation operators, the Fisher Information Matrix for the full AR log-likelihood given by (4.41) may be broken into a summation of two sub Information Matrices, one for each of the two likelihoods

$$\mathbf{J}(\boldsymbol{\theta}) = \mathbf{J}^{(\Gamma)}(\boldsymbol{\theta}) + \mathbf{J}^{(X)}(\boldsymbol{\theta}) \quad (4.44)$$

These two Information Matrices are covered in the next two subsections.

4.5.1 Unconditional Likelihood Information Matrix

The unconditional likelihood is in effect the modelling of the first P data values as a zero meaned multivariate Normal distribution, characterised by the covariance matrix $\boldsymbol{\Gamma}$. The Fisher Information Matrix for this term may then be found exactly by the algorithm given in [133] using the autocovariance computations for root space parameters given in [134]. However, this is unnecessarily complex and a suitable approximation may be found via differencing. Using the simple approximation

$$\frac{\partial\gamma_k(\boldsymbol{\beta})}{\partial\beta_i} \approx \frac{\gamma_k(\boldsymbol{\beta} + \delta\mathbf{e}_i) - \gamma_k(\boldsymbol{\beta})}{\delta} \quad (4.45)$$

where $\gamma_k(\boldsymbol{\beta})$ denotes the k-th autocovariance of an AR process with root-space parameters $\boldsymbol{\beta}$, $\mathbf{e}_i$ is the indicator vector of all zeros and a single one at the i-th element, and δ is a suitably small perturbation value. Using this approximation, the entry $\mathbf{J}^{(\Gamma)}_{(\beta_i,\beta_j)}(\cdot)$ may be computed as

$$\mathbf{J}^{(\Gamma)}_{(\beta_i,\beta_j)}(\boldsymbol{\theta}) = \frac{1}{2}\mathrm{Tr}\left(\boldsymbol{\Gamma}^{-1}(\boldsymbol{\theta})\frac{\partial\boldsymbol{\Gamma}(\boldsymbol{\theta})}{\partial\beta_i}\boldsymbol{\Gamma}^{-1}(\boldsymbol{\theta})\frac{\partial\boldsymbol{\Gamma}(\boldsymbol{\theta})}{\partial\beta_j}\right) \quad (4.46)$$

where Tr denotes the trace operator. The $\mathbf{J}_{(\beta_i,\sigma^2)}(\cdot)$ entries are simply zero, and the variance entry is found as

$$\begin{aligned} J^{(\Gamma)}_{(\sigma^2,\sigma^2)}(\boldsymbol{\theta}) &= \frac{1}{2}\mathrm{Tr}\left(\boldsymbol{\Gamma}^{-1}(\boldsymbol{\theta})\frac{\boldsymbol{\Gamma}(\boldsymbol{\theta})}{\sigma^2}\boldsymbol{\Gamma}^{-1}(\boldsymbol{\theta})\frac{\boldsymbol{\Gamma}(\boldsymbol{\theta})}{\sigma^2}\right) \\ &= \frac{P}{2(\sigma^2)^2} \end{aligned} \tag{4.47}$$

The previous work by Fitzgibbon et al [16] on AR models did not compute the FIM for the unconditional component, instead approximating it by absorbing it into the FIM for the conditional component, i.e. the asymptotic Information Matrix was employed [2]. For large sample sizes this makes little difference, but when the sample size is small and the estimated poles are near to the unit circle boundary this can have a significant effect, tending penalise such models more heavily.

4.5.2 Conditional Likelihood Information Matrix

It now remains to compute $\mathbf{J}^{(X)}(\cdot)$ for the conditional component. Begin by defining the modelling 'error' as

$$z_n = y_n + \mathbf{a}\mathbf{x}_n^T,\ n > P \tag{4.48}$$

By recalling the definition of $\mathbf{a}$, the 'error' polynomial can be defined as

$$z_n \equiv e_n(q) \equiv e_n(\mathbf{p},\mathbf{y},q) = \prod_{i=1}^{P}(1-p_i q)y_n \tag{4.49}$$

where q is a formal polynomial variable. Now, consider the expression

$$\frac{N-P}{2}\log(2\pi) + \frac{N-P}{2}\log(\sigma^2) + \frac{1}{2\sigma^2}\sum_{n=P+1}^{N} e_n(\mathbf{p},\mathbf{y},q_1)e_n(\mathbf{p},\mathbf{y},q_2) \tag{4.50}$$

Expanding the polynomials in the right hand term and then making the substitution $q_1^k = q_2^k = y_{n-k}$ yields the the conditional part of negative log-likelihood as in (4.43). The second derivatives of (4.50) with respect to the root parameters (β_i, β_j) are then found by expanding

$$\frac{\partial^2 L^{(X)}(\mathbf{y}|\boldsymbol{\theta})}{\partial\beta_i\partial\beta_j} = \frac{1}{\sigma^2}\sum_{n=P+1}^{N}\left(\frac{\partial^2 e_n(q_1)}{\partial\beta_i\partial\beta_j}e_n(q_2) + \frac{\partial e_n(q_1)}{\partial\beta_i}\frac{\partial e_n(q_2)}{\partial\beta_2}\right) \tag{4.51}$$

and subsequently making the substitutions $q_1^k = q_2^k = y_{n-k}$. The Fisher Information Matrix is then found by taking expectation with respect to the data. Recalling that $e_n(\cdot) = z_n$, the first term in (4.51) vanishes under expectations and we are left with

$$J^{(X)}_{(\beta_i,\beta_j)}(\boldsymbol{\theta}) = \frac{(N-P)}{\sigma^2}\mathrm{E}\left[\frac{\partial e_n(q_1)}{\partial \beta_i}\frac{\partial e_n(q_2)}{\partial \beta_2}\right] \tag{4.52}$$

The expectation of (4.52) is simply found by further substituting

$$y_n y_{(n-k)} = \gamma_k \tag{4.53}$$

where γ_k is the theoretical k-th autocovariance of the sequence, i.e. $\gamma_k = \mathrm{E}\left[y_n y_{n-k}\right] = \mathrm{E}\left[y_{n-k} y_n\right]$ [2]. The (β_i, β_j) entries of the Information Matrix have the following general form

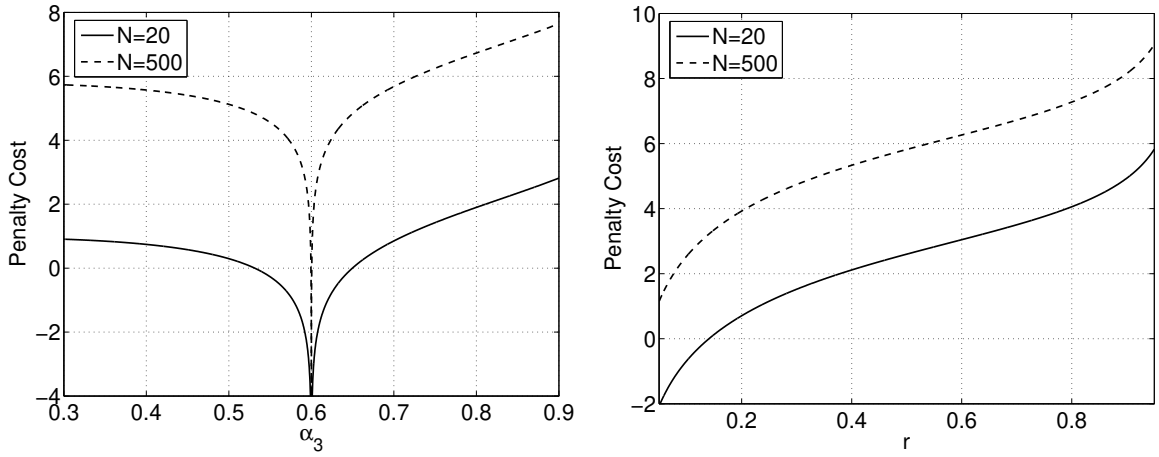
$$J^{(X)}_{(\beta_i,\beta_j)}(\boldsymbol{\theta}) = \frac{(N-P)}{\sigma^2}\sum_{k=1}^{P}\eta_k^{(i,j)}\gamma_{(k-1)} \tag{4.54}$$

that is, the entries are linear combinations of the first P theoretical model autocovariances. An important point to note is that the autocovariances contain the innovation variance σ^2 as a factor; this is cancelled by the scaling of (4.54) by $1/\sigma^2$. The Fisher Information entries for the $\boldsymbol{\beta}$ parameters are therefore independent of the innovation variance. Efficient closed form expressions for the entries of the Fisher Information Matrix are given in Appendix A (taken from [135] and [123]) which do not depend explicitly on the model autocovariances and are thus significantly easier (and more numerically stable) to compute. The FIM also contains entries for the variance parameter σ^2. The FIM entry for the (β_i, σ^2) and (σ^2, σ^2) terms are given by

$$J^{(X)}_{(\beta_i,\sigma^2)}(\boldsymbol{\theta}) = -\frac{1}{2\sigma^2}\sum_{n=P+1}^{N}\mathrm{E}_y\left[v_n\frac{\partial z_n}{\partial \beta_i}\right] = 0 \tag{4.55}$$

$$J^{(X)}_{(\sigma^2,\sigma^2)}(\boldsymbol{\theta}) = \frac{1}{(\sigma^2)^3}\sum_{n=P+1}^{N}\mathrm{E}_y\left[z_n^2\right] - \frac{N-P}{2(\sigma^2)^2} = \frac{N-P}{2(\sigma^2)^2} \tag{4.56}$$

The expectation in (4.56) is simply the innovation variance.



(a) Model Cost for AR(3) Model with Real Poles (b) Model Cost for AR(2) Model with Complex Poles

Figure 4.3: Potential Problems with MML87 Approximation

4.6 Potential Issues with MML87 Inference of AR Models

There are several potential problems with the inference of autoregressions in root-space in an MML87 framework. These are primarily due to the use of the Fisher Information Matrix in determining the optimal precision to which the autoregressive parameters must be stated. Begin by recalling that in the MML87 framework the 'model cost' of a model (i.e. assertion cost and excess round-off) is given by

$$\mathcal{D}_{87}(\boldsymbol{\theta}) = -\log h(\boldsymbol{\theta}) + \frac{1}{2}\log|\mathbf{J}_1(\boldsymbol{\theta})| + \frac{P}{2}\log N + \frac{D}{2}(\log \kappa_P + 1) \tag{4.57}$$

where $\mathbf{J}_1(\cdot)$ is the 'per case' Information Matrix. There are two situations which potentially violate the MML87 approximation assumptions of quadraticity within the uncertainty region

- Singularities in the Information Matrix due to parameter correlation
- Singularities in the Information Matrix due to flat spots in the KL-divergence surface

The first of these issues arise due to the redundancy in parameterisation of AR models in root-space, and the second arises due to very shallow curvatures of the KL-divergence for some particular parameter combinations. Both problems essentially manifest themselves as values of $|\mathbf{J}(\cdot)|$ that are very small or near zero, resulting in a gross underestimation of the uncertainty region, and thus highly optimistic (or even nonsensical) message lengths.

The first type of singularity in the FIM may arise when two or more real roots, or two or more complex roots coincide (or nearly coincide). Under these conditions the effects of these parameters on the expected negative log-likelihood are indistinguishable. This arises because the order of the root space parameters is irrelevant to the coefficient space polynomial they map to, as discussed previously in Section 4.3.2. This is a serious concern as, regardless of the number of data points N, the FIM will remain singular if two parameters coincide. If the parameters *nearly* co-incide, the approximation becomes increasingly better as N increases but for small sample sizes still poses a potential concern. Figure 4.3(a) shows the log-determinant of the FIM of an AR(3) model with two roots fixed to $(\alpha_1 = -0.3, \alpha_2 = 0.6)$, and a third root varied on $\alpha_3 \in [0.3, 0.9]$. The plots clearly demonstrate the large negative peak (undefined when $\alpha_2 = \alpha_3$) as $\alpha_3 \to \alpha_2$ from both sides. Furthermore, for $N = 20$, the model cost is nonpositive (and thus, nonsensical) on approximately the region $\alpha_3 \in [0.55, 0.65]$. The plot for $N = 500$ shows that the region of nonpositive model costs is reduced dramatically to a quite narrow locality around $\alpha = 3$; however, the model costs in the region (roughly $\alpha \in [0.575, 0.625]$) drop very rapidly and form quite poor approximations to the true model cost. A larger concern is that regardless of N

$$\forall \alpha_2 \in (-1, 1), \forall K < 0, \exists \epsilon \in \mathbb{R} : \frac{1}{2} \log |\mathbf{J}_1([\alpha_1, \alpha_1 + \epsilon, \boldsymbol{\theta}_{(3:P)}])| < K \tag{4.58}$$

i.e. no matter what the sample size, for any AR model containing at least two roots, the model cost may always be rendered nonsensical by making the difference between the two roots as small as required.

The second form of singularity arises for models with complex conjugate pole pairs. For suitably small values of the radial component r_i, the curvature of the KL-divergence along the r_i and ω_i axis will be extremely small; the result of this is that the quadratic approximation of the KL-divergence no longer holds over the entire uncertainty region, leading to gross overestimations of the uncertainty region. The curvature terms approach zero as $r \to 0$, and it is easy to see that at $r = 0$ the complex conjuguate pole pair is equivalent to model with two real poles $\alpha_1 = \alpha_2 = 0$, which is itself is clearly singular. Figure 4.3(b) shows the model cost for an AR(2) model with a single complex conjugate pole pair as the radial component r is varied; it is obvious that as $r \to 0$ the model cost begins to rapidly drop off, and for $N = 20$ and $r < 0.1$ the model cost becomes negative.

4.6.1 Impact on Inference

The previous section has detailed several cases in which the MML87 approximation yields poor estimates (and in some cases, completely nonsensical) of the model cost. It may be argued that these cases represent pathalogical instances that will rarely be observed in reality and should have little practical effect on the performance of the MML87 criteria; such a claim will be investigated in the sequel through empirical studies. There is, however, one task in which these concerns are guaranteed to cause serious problems: that of *parameter estimation* by minimising the message length. This amounts to solving

$$\hat{\boldsymbol{\theta}}_{87}(\mathbf{y}) = \arg\min_{\boldsymbol{\theta}\in\boldsymbol{\Theta}} \{L(\mathbf{y}|\boldsymbol{\theta}) + \mathcal{D}_{87}(\boldsymbol{\theta})\} \tag{4.59}$$

It is easy to see that if the parameter values may be freely selected from the permissable space, the model cost can be made as small as is desired by judicious selection of root parameters, for all AR models of order two or greater. Thus, the effects of the likelihood may be completely swamped by the effects of the model cost by moving the parameters into the regions which violate the MML87 approximation assumptions; this leads to poor and heavily biased parameter estimates. There are several possibilities to handling this problem:

- Application of the 'Small Sample' Approximation ([1], Page 235)
- Addition of terms to the diagonals of the FIM ([31] and [1], Pages 270-275, and O'Hagan's comments in the discussion of [15])
- Diagonalisation of the FIM [35]

The first of these works by bounding the prior mass such that $\text{vol}(\mathcal{R})h(\boldsymbol{\theta}) \in (0, 1]$; this has the advantage of preserving invariance, but suffers from two drawbacks. The model cost under this regime is given by

$$-\log q(\boldsymbol{\theta}) = \frac{1}{2}\left(1 + \frac{|\mathbf{J}(\boldsymbol{\theta})|\kappa_P^P}{h(\boldsymbol{\theta})^2}\right) + \frac{P}{2} \tag{4.60}$$

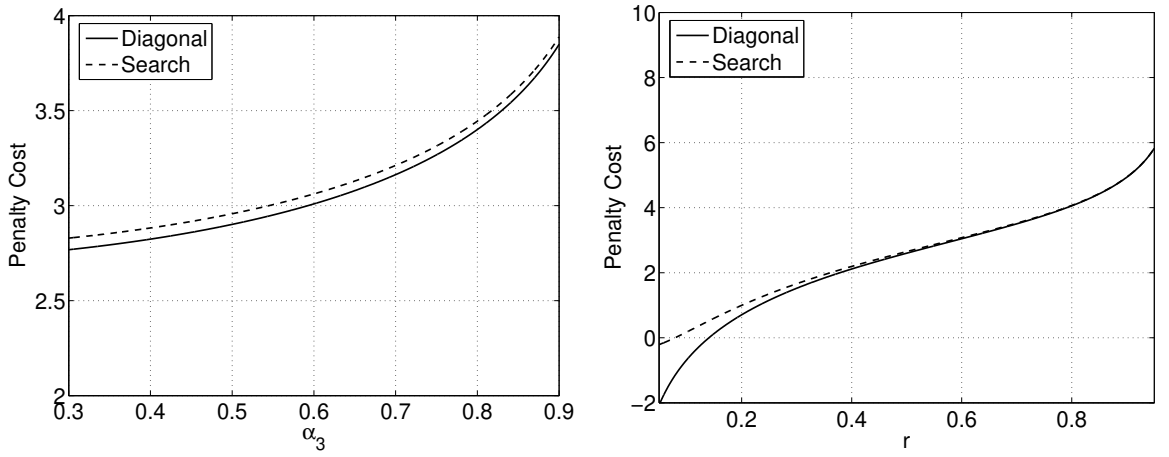
The first difficulty arises from the fact that the addition of one to the prior-coding volume product inside the log term means that the expression no longer simply expands to the more well known form (4.57); this means that differentiating the expression becomes

difficult. A second, more fundamental concern is that while the model cost is now clearly bounded on $\left(\frac{P}{2}, \infty\right)$, the model cost becomes constant for all parameter configurations that render $\mathbf{J}(\cdot)$ singular. This means that if a parameter estimation scheme chooses a singular configuration for $\mathbf{J}(\cdot)$, the estimator reduces to the Maximum Likelihood estimator. More seriously, if the model costs are then used in an order selection procedure, a large order model can reduce its model cost to $\frac{P}{2}$ by simply selecting two real roots, or one of the complex roots radius to a singular configuration; this may lead to serious overfitting.

The work on single factor analysis and linear regression overcomes possible singularities in the Information Matrix by the addition of diagonal terms to the FIM; however, these diagonal terms are derived by encoding the model parameters as 'extra data' which leads to serious restrictions on the form of the priors that may be chosen, and is not a general solution. The last option is to adopt a more pragmatic approach, at least for the purposes of parameter estimation, and make heuristic modifications that yield adequate results. The simplest of these is to diagonalise the Information Matrix by simply disregarding the off-diagonal terms. This procedure is akin to assuming that only the curvature along the component axis of the parameter space has any effect on the uncertainty region (this is in fact similar to earlier flavours of MML, such as [10]). The immediate effect of this modification is to remove the singularities caused by the coincidence of real poles/radial components and 'smooth' the model cost surface, as the resulting matrix will be of full rank. The problem of small radial components yielding small curvature remains; a simple modification that can be applied to bound the $\log|\mathbf{J}_1(\cdot)|$ term (and thus upperbound the volume of the uncertainty region) is to simply add the identity matrix to the diagonalised FIM. For areas with small curvature, the log-determinant term will smoothly approach the lower bound of zero (as each diagonal element is bounded on $[1, \infty)$), and for areas of the parameter space with large curvatures, the addition of the identity matrix will make little difference to the magnitude of the log-determinant term. The final model cost under these modifications is given by

$$\mathcal{D}_{87S}(\boldsymbol{\theta}) = -\log h(\boldsymbol{\theta}) + \frac{1}{2}\sum_{i=1}^{P} \log\left(J_{(i,i)}(\boldsymbol{\theta}) + 1\right) + \frac{P}{2}\left(\log \kappa_P + 1\right) \tag{4.61}$$

which is clearly lowerbounded by $\left(-\log h(\boldsymbol{\theta}) + \frac{P}{2}(\log \kappa_P + 1)\right)$. This is dubbed the 'Search' approximation as it has been derived specifically as an estimate of the model



(a) Model Cost for AR(3) Model with Real Poles (b) Model Cost for AR(2) Model with Complex Poles

Figure 4.4: 'Corrected' Model Cost Estimates

cost useful for finding parameter estimates close to the true MML estimates. The approximation has the advantage of preserving the general behaviour of the model cost surface: areas with larger curvatures cost more to state than those with lower curvatures. However, at the same time, the possibility of arbitrarily small model costs has been removed. It also has the practical advantage of being significantly less expensive to compute as only the diagonal terms of the FIM are required; these can significantly improve the speed of numerical searches when fitting high order models. While it may seem ad-hoc to add the identity matrix to the Fisher Information, it can be noted that the enforced choice of $\mathcal{N}(0, 1)$ priors in the factor-analysis work [31] produces the same effect, and is essentially also motivated by a desire to render the FIM full rank. Figure 4.4 shows the model cost curves of the example models from the previous section for the diagonalised FIM, and the Search modification (diagonalised plus identity matrix). Two things of note may be observed from these figures: the first is that the differences between the purely diagonalised criteria and the 'Search' criteria are negligible in those regions of parameter space where the curvature is large. The second is that in those areas of the parameter space where the curvature becomes signficiantly small, the 'Search' criteria clearly prevents the model cost from becoming unboundedly small and produces a curve that could be reasonably extrapolated based on the curve from $r \in [0.4, 0.95]$. The perfomance of this Search approximation is tested empirically in a later section of this Chapter, and Chapter 7 presents a new message length approximation algorithm that circumvents the

problems of the MML87 approximation by approximating the KL divergence surface in a more robust manner, avoiding the requirement for ad-hoc modifications.

4.7 Parameter Estimation

The issue of estimation of model parameters $(\boldsymbol{\beta}, \sigma^2)$ from a measurement sequence $\mathbf{y}$ is considered in this subsection. Closed form solutions for the MML estimator of $\boldsymbol{\beta}$ are difficult to obtain, so a numerical search is employed. The estimates for σ^2, assuming a scale invariant prior, can be found explicitly, so at each evaluation of the cost function a suitable σ^2 is estimated, given an estimate of $\boldsymbol{\beta}$.

4.7.1 Maximum Likelihood Estimate of σ^2

The Maximum Likelihood estimator for σ^2 is derived first. Begin by noting that σ^2 can be factorised out of the autocovariance matrix $\boldsymbol{\Gamma}(\boldsymbol{\theta})$ in (4.41), and the likelihood may be rewritten as

$$\begin{aligned} L(\mathbf{y}|\boldsymbol{\theta}) &= \frac{N}{2}\log(2\pi) + \frac{1}{2}\log\left(\left(\sigma^2\right)^P \cdot |\boldsymbol{\Gamma}^*(\boldsymbol{\theta})|\right) + \frac{1}{2\sigma^2}\mathbf{y}_{(1:P)}\boldsymbol{\Gamma}^{*-1}(\boldsymbol{\theta})\mathbf{y}_{(1:P)}^T \\ &\quad + \frac{N-P}{2}\log(\sigma^2) + \frac{1}{2\sigma^2}\left(\mathbf{y}_{(P+1:N)} + \mathbf{aX}\right)\left(\mathbf{y}_{(P+1:N)} + \mathbf{aX}\right)^T \end{aligned} \tag{4.62}$$

where

$$\boldsymbol{\Gamma}^*(\boldsymbol{\theta}) = \frac{1}{\sigma^2}\boldsymbol{\Gamma}(\boldsymbol{\theta}) \tag{4.63}$$

is the process autocovariance matrix with an innovation variance of one. The likelihood may be rewritten as

$$L(\mathbf{y}|\boldsymbol{\theta}) = \frac{N}{2}\log(\sigma^2) + \frac{1}{2\sigma^2}\left(\mathbf{y}_{(1:P)}\boldsymbol{\Gamma}^{*-1}(\boldsymbol{\theta})\mathbf{y}_{(1:P)}^T + \left(\mathbf{y}_{(P+1:N)} + \mathbf{aX}\right)\left(\mathbf{y}_{(P+1:N)} + \mathbf{aX}\right)^T\right) + K_L(\boldsymbol{\beta}) \tag{4.64}$$

where $K_L(\boldsymbol{\beta})$ are the terms that do not depend on σ^2. Differentiating and solving for σ^2 yields the maximum likelihood

$$\hat{\sigma}^2_{\mathrm{ML}}(\mathbf{y}) = \frac{\mathbf{y}_{(1:P)}\boldsymbol{\Gamma}^{*-1}(\boldsymbol{\theta})\mathbf{y}_{(1:P)}^T + \left(\mathbf{y}_{(P+1:N)} + \mathbf{aX}\right)\left(\mathbf{y}_{(P+1:N)} + \mathbf{aX}\right)^T}{N} \tag{4.65}$$

4.7.2 Minimum Message Length Estimate of σ^2

The Minimum Message Length estimate of σ^2 is considered next. The procedure is similar to that of the Maximum Likelihood estimate, but must now also consider the effect of the FIM term and priors. Examination of the elements of the Fisher Information Matrix reveal that all entries for the $\boldsymbol{\beta}$ parameters are independent σ^2. and noting that the $(\partial\boldsymbol{\beta}_i, \partial\sigma^2)$ entries of the FIM are zero, the determinant can be rewritten as

$$|\mathbf{J}(\boldsymbol{\theta})| = |\mathbf{J}_{(\beta_1:\beta_P,\beta_1:\beta_P)}(\boldsymbol{\theta})| \cdot J_{(\sigma^2,\sigma^2)}(\boldsymbol{\theta}) \tag{4.66}$$

The message length may then be rewritten as

$$\begin{aligned} I_{87}(\mathbf{y},\boldsymbol{\theta}) &= L(\mathbf{y}|\boldsymbol{\theta}) - \log h(\sigma^2) + \frac{1}{2}\log J_{(\sigma^2,\sigma^2)}(\boldsymbol{\theta}) + K_I(\boldsymbol{\beta}) && (4.67)\\ &= L(\mathbf{y}|\boldsymbol{\theta}) + \log(\sigma^2) - \log(\sigma^2) + K(\boldsymbol{\beta}) && (4.68) \end{aligned}$$

where $K(\boldsymbol{\beta})$ are all the terms not dependent on σ^2. Clearly the effect of the Fisher and the prior terms cancel, and render the MML estimate for σ^2 to be

$$\hat{\sigma}^2_{87}(\mathbf{y}) = \hat{\sigma}^2_{\mathrm{ML}}(\mathbf{y}) \tag{4.69}$$

where $\hat{\sigma}^2_{\mathrm{ML}}(\mathbf{y})$ is given by (4.65).

4.7.3 Estimation of the AR(1) Model

For the case of $P = 1$, i.e. the AR(1) model, the estimation problem is reduced significantly in complexity and more exact expressions for the estimation of the single real pole α may be derived. The autocovariance function for an AR(1) model is

$$\gamma_k = \frac{\sigma^2\alpha^k}{1-\alpha^2} \tag{4.70}$$

and using this in the negative log-likelihood (4.41) yields

$$L(\mathbf{y}|\alpha,\sigma^2) = \frac{N}{2}\log(2\pi) + \frac{N}{2}\log\sigma^2 - \frac{1}{2}\log\left(1-\alpha^2\right) + \frac{(1-\alpha^2)y_0^2 + S_1 + \alpha^2 S_2 + 2\alpha S_3}{2\sigma^2} \tag{4.71}$$

where $S_1 = \left(\sum_{i=2}^{N} y_i^2\right)$, $S_2 = \left(\sum_{i=1}^{N-1} y_i^2\right)$, $S_3 = \left(\sum_{i=2}^{N} y_i y_{i-1}\right)$ are summaries of the data. The Maximum Likelihood estimate of σ^2 is then given by

$$\hat{\sigma}^2_{\mathrm{ML}}(\mathbf{y}) = \frac{(1-\alpha^2)y_0^2 + S_1 + \alpha^2 S_2 + 2\alpha S_3}{N} \tag{4.72}$$

which as per Section 4.7.2 is seen to coincide with the MML estimate, assuming a scale-invariant prior.

4.7.4 Maximum Likelihood Estimate of α

The Maximum Likelihood estimate may be found by differentiating (4.71) with respect to α, substituting $\hat{\sigma}^2_{\mathrm{ML}}(\mathbf{y})$ for σ^2 and solving for the stationary point. This involves solving the cubic

$$((n-1)S_2 + (1-n)y_1^2)\alpha^3 + (N-2)\alpha^2 + ((n-1)y_1^2 - S_1 - NS_2)\alpha - S_3 N = 0 \tag{4.73}$$

and selecting the permissible root as the estimate of α.

4.7.5 Minimum Message Length Estimate of α

Two MML estimates for α are derived; the first uses only the conditional Fisher Information term, and the second uses the exact Fisher Information. Ignoring those terms irrelevant to the estimation of α, the message length for the AR(1) model is

$$L(\mathbf{y}|\alpha, \sigma^2) - \frac{1}{2}\log\left(1-\alpha^2\right) \tag{4.74}$$

Differentiating this expression with respect to α and substituting $\sigma^2 = \hat{\sigma}^2_{87}(\mathbf{y}) = \hat{\sigma}^2_{\mathrm{ML}}(\mathbf{y})$ yields

$$((n-2)S_2 + (2-n)y_1^2)\alpha^3 + (n-4)\alpha^2 + ((n-2)y_1^2 - 2S_1 - nS_2)\alpha - NS_3 = 0 \tag{4.75}$$

The permissible root of this cubic yields the conditional Fisher MML estimate of α. Comparison of (4.73) to (4.75) shows that they are largely similar, the only difference being several terms scaled by two. This is unsurprising since that in the case of the AR(1) model, the addition of the log-Fisher Information simply results in the $\left(\frac{1}{2}\log\gamma_0\right)$ term

being scaled by two. The MML estimates when the exact Fisher Information is used are significantly more complex; this is primarily due to the fact that the Information term is the sum of two separate terms, which does not lead to simplifications when viewed through a log; the Fisher Information term $J_{(\alpha,\alpha)}(\cdot)$ for this model is

$$J_{(\alpha,\alpha)}(\alpha) = \frac{1}{2\gamma_0^2}\left(\frac{\partial\gamma_0}{\partial\alpha}\right)^2 + \frac{(N-1)\gamma_0}{\sigma^2} \tag{4.76}$$

Substiting (4.70) into the above expression and using it in the message length equation yields

$$L(\mathbf{y}|\alpha,\sigma^2) + \frac{1}{2}\log\left(\frac{2\alpha^2}{(1-\alpha^2)^2} + \frac{(N-1)}{(1-\alpha^2)}\right) \tag{4.77}$$

Differentiating this with respect to α and substituting $\sigma^2 = \hat{\sigma}^2_{\mathrm{ML}}(\mathbf{y})$ yields

$$\begin{aligned}\frac{\partial I_{87}(\mathbf{y}|\boldsymbol{\theta})}{\partial\alpha} &= \left[(n^2+6-5n)S_2 + (5n-6-n^2)y_1^2\right]\alpha^5 + (12+n^2-7n)\alpha^4 \\ &\quad +((6+2n^2-8n)y_1^2 + (6n-2n^2)S_2 + (6-2n)S_1)\alpha^3 \\ &\quad +(-2n^2+8n)S_3\alpha^2 + \left[(3n-n^2)y_1^2 + (n^2-n)S_2 + 2S_1 n\right]\alpha \\ &\quad +(n^2-n)S_3 \end{aligned} \tag{4.78}$$

Solving this quintic for the only valid solution yields the MML87 estimator using the exact Fisher Information. It is clear to see this expression differs significantly from the ML and previous MML estimators; this is hardly surprising considering the extra complexity of the exact Fisher Information; however, solving a quintic is a considerably more straightforward task than optimising the message length directly.

4.7.6 Minimum Message Length Estimate of $\boldsymbol{\beta}$ for $P > 1$

The previous subsection detailed estimator expressions for the case that $P = 1$; to find the estimates $\boldsymbol{\beta}$ for $P > 1$ a numerical search is employed. This process can be prone to some instabilities due to the FIM becoming near singular for certain parameter combinations (see Section 4.6), so the diagonal search stabilisation is applied. The procedure for finding the MML estimates for an AR(P) model is then given by

1. Find initial coefficient estimates, $\hat{\mathbf{a}}_0$, for an AR(P) model using a suitable estimation scheme; the Burg method is recommended as it is quick and provides quite good estimates.

2. Extract the root structure, $\hat{\boldsymbol{\beta}}_0$, of our initial estimates $\hat{\mathbf{a}}_0$.

3. Numerically search for the MML estimates, $\hat{\boldsymbol{\beta}}_{87}$, within this given root structure, using the diagonal search modification of the conditional FIM matrix, with the estimates $\hat{\boldsymbol{\beta}}_0$ as a starting point. The diagonal search modification is a simple robust approximation of the coding quantum, and is given by

$$\frac{1}{2}\log|\mathbf{J}(\boldsymbol{\theta})| \approx \frac{1}{2}\sum_{i=1}^{P+1}\log\left(J_{(i,i)}(\boldsymbol{\theta})+1\right) \tag{4.79}$$

   A search performed under this regime will minimise the trade-off between the coding volume and the negative log-likelihood, under the assumption that the parameters are uncorrelated. While this is a false assumption in the case of AR models, and may lead to estimates that yield slightly longer message lengths than if the full FIM was used, the fact that the diagonal elements of the FIM generally dominate the off-diagonals, and the increased robustness in the search makes it a useful approximation. This approximation was discussed in more detail in Section 4.6.1.

It also becomes clear at this point that parameterisation of the AR model in terms of root-parameters has another advantage: for a given structure, the stationarity constraints reduce to a search within a hypercube, with parameters bounded according to their type, i.e. $\alpha_i \in (-1,1)$, $r_i \in (0,1)$ and $\omega_i \in (0,\pi)$ [128]. This is in contrast to a search in coefficient-space where the stationarity region is bounded by a much more complex polytope.

4.8 Remarks on the MML87 AR Estimator

This section examines the MML87 estimator equation for the AR model.

4.8.1 The Effect of σ^2 on the Parameter Accuracy

Excluding the influence of the priors, the main extra term in the MML87 estimator is the coding quantum (as estimated from the Fisher Information). The first thing of note when examining the Fisher Information Matrix is that for the $\boldsymbol{\beta}$-parameters the variance in the denominator cancels when expectations are taken. This is at odds with a linear regression interpretation of the auto-regressive process, and the explanation is relatively simple: in

the case of a standard linear regression, the value of σ^2 determines the signal-to-noise ratio. The larger the σ^2, the smaller the signal-to-noise ratio of the model. The SNR of an AR process with root parameters $\boldsymbol{\beta}$ is given by

$$\frac{\mathrm{E}\left[y_i^2\right]}{\mathrm{E}\left[v_i^2\right]} = \frac{\gamma_0(\boldsymbol{\beta}, \sigma^2)}{\sigma^2} = \frac{\sigma^2 \gamma_0(\boldsymbol{\beta}, 1)}{\sigma^2} = \gamma_0(\boldsymbol{\beta}, 1) \tag{4.80}$$

where $\gamma_0(\boldsymbol{\theta}, \sigma^2)$ is the zero-th order autocovariance of an AR process with root space parameters $\boldsymbol{\beta}$ and innovation variance σ^2. It is clear from (4.80) that in the case of the AR model, the signal-to-noise ratio is completely determined by the position of the model poles. Indeed, the closer they are to the boundary of the unit circle, the larger the signal-to-noise ratio becomes (as the process autocorrelations increase as the roots approach the boundary) [2]. Thus, the value of σ^2 is irrelevant in determining the precision to which the $\boldsymbol{\beta}$-parameters must be stated.

4.8.2 The Effect of the FIM on the $\boldsymbol{\beta}$-parameter Estimates

For auto-regressive parameter estimation we search using only the diagonals of the Fisher Information. The magnitude of the determinant of this modified Fisher Information matrix is roughly proportional to the magnitude of the model's autocovariances. These autocovariances are in turn proportional to the magnitude of the real poles and radii components of the imaginary poles; the closer these are to the edge of the stationarity region, the larger the autocovariances become. This leads to an estimator that will tend to select pole estimates that are further away from edge of the stationarity region than the corresponding Maximum Likelihood estimates would be.

In this sense the MML87 estimator is a type of (fixed penalty) shrinkage estimator, as it shrinks the parameters towards zero; this type of behaviour has been examined quite extensively since the discovery by Stein that the sample mean is inadmissable in terms of squared error loss for the case of normally distributed data. There have been many shrinkage estimators proposed in the case of linear regressions, including methods such as the James-Stein Estimator, Ridge Regression estimators, the Lasso and various Empirical Bayes estimators [136, 137, 138, 139, 140, 51, 126]. In these cases, the reduction in estimator variance has often lead to significant improvements in prediction errors, despite the

increase in estimator bias the shrinkage introduces. The MML87 estimator is expected to have the following properties:

- **Improved Long Term Estimates**: The MML estimator, by selecting slightly faster models, tends to capture the long term structure of the model more accurately than the ML estimator; given the nonlinear relationship between model pole magnitudes and resulting time constants, a slight underestimation in pole magnitude will lead to lower long term errors than a slight overestimation of model pole positions.

- **Larger Scale Estimates**: The Maximum Likelihood estimator selects the coefficient estimates that minimise the resulting residual variance. The MML87 estimates will necessarily differ from the ML estimates (due to the influence of the Fisher Information term), and thus must result in slightly inflated residual variance.

- **Asymptotic Behaviour**: Ignoring the unconditional component (which becomes negigible as $N \gg P$, the FIM term for the autoregressive model may be rewritten as

$$\frac{1}{2}\log|\mathbf{J}(\boldsymbol{\theta})| = \frac{1}{2}\left(\log|\mathbf{J}_1^{(X)}(\boldsymbol{\theta})| + \log|\mathbf{J}^{(\Gamma)}(\boldsymbol{\theta})|\right) + \frac{P}{2}\log(N-P) \tag{4.81}$$

  where $\mathbf{J}_1^{(X)}(\cdot)$ is the conditional FIM for a single datum. It is then clear that parameter dependent aspect of the FIM does not increase proportional to N. From this it follows that as $N \to \infty$, the FIM term will be dominated by the likelihood term which does increase proportional to N. This also holds for any priors, and thus asymptotically the MML estimates converge to the Maximum Likelihood estimates, and thus possess their desirable properties of asymptotic efficiency and consistency [61].

The improved long term estimation of the MML87 estimator is tested extensively by Monte Carlo experiments in the results section of this chapter, and the experimental evidence suggests the MML87 estimates better capture the long-term behaviour of the underlying process.

4.9 Autoregressive Parameter Estimation Results

This section discusses experimental results designed to test the performance of the MML87 criterion when used for parameter estimation of autoregressive models.

4.9.1 Experimental Design

The parameter estimation experiments undertaken were designed simply to test the performance of the MML87 parameter estimation scheme against several well known autoregressive parameter estimation schemes. The estimators chosen for comparison purposes are the Maximum Likelihood estimator, Burg estimator, the Yule-Walker estimator and the constrained least-squares estimator, as all four of these schemes provide estimates that are guaranteed to be stationary. These standard estimators were to be compared against the estimates found by directly minimising the search-modified message length expression. The loss functions used to judge the performance of the estimators were

- Multiple step expected Squared Prediction Error, SPE_∞
- Single step expected Squared Prediction Error, SPE_1
- One hundred step Kullback-Leibler divergence, KL_{100}
- One hundred step symmetric Kullback-Leibler divergence, J_{100}

Given that the models within a test run will be randomly generated from different areas of the stationarity region, the squared prediction error scores are normalised to ensure they are comparable between models. These loss functions are summarised in more detail in Section 3.9. The test procedure utilised was simple: a number of models I is randomly generated from parameter prior distributions and from each model N data points are generated by simulating the model with $\mathcal{N}(0,1)$ innovation noise, discarding the first 500 samples to remove the effect of initial conditions. Then, parameters are estimated for a fixed model structure by each of the parameter estimation schemes, and the losses is computed. The models were drawn from:

$$\alpha_i \sim U(-1,1) \tag{4.82}$$

$$r_i \sim U(0,1) \tag{4.83}$$

$$w_i \sim U(0,\pi) \tag{4.84}$$

where $U(\cdot)$ is a uniform distribution. One thousand models were generated for each order; the sample sizes chosen were $N = \{k(3P + 1)\}$ with $k = \{1, 2, 4\}$ and P the order of the model being estimated. These sizes were chosen as they represent reasonable range of low data-to-parameter ratio situations, in which all estimators are expected to perform differently. For very large N, the MML estimator is known to coincide with the ML estimator and thus experimental tests would present little new.

4.9.2 Discussion

Tables of Results

The results for autoregressive parameter estimation are presented in Tables 4.7 through to 4.10, with standard deviations shown in parenthesis. A summary of the general findings is:

- The most obvious trend is that for all sample sizes and model orders, the MML87 estimates attain very good squared prediction errors and KL divergences, particularly SPE_∞ losses which indicates the pulling in of the poles (and subsequent downwards bias of the pole magnitudes) leads to better long term prediction errors. The improvements over Maximum Likelihood and the Burg estimator are most marked for situations where the number of data points to parameters is low.

- The large bias in the Yule-Walker estimates leads to quite good performance for models with pole magnitudes drawn from uniform priors, at least for low order models and in terms of SPE losses. For $P = 1$ and $P = 2$ the Yule-Walker estimator marginally outperforms or is very close to the MML87 in terms of SPE loss. However, the KL divergences show that the Yule-Walker estimates of scale can be very poor especially for large P. The very large J-divergence losses for higher order models indicate that the scale is being grossly overestimated for some models, and examination of the results confirmed this to be the case.

- The Burg and Maximum Likelihood estimators exhibit very similar performance in terms of all the losses, particularly as P and N increase. For lower order models the Maximum Likelihood estimator attains slightly lower SPE and KL/J divergence losses, and for higher order models the trend is slightly reversed. This confirms previous findings that the Burg estimator is a very practical substitute for the Maximum

Liklihood estimator, especially in light of the manifold decrease in speed required to compute the Burg estimates.

- For all sample sizes and model orders the least squares estimates perform the worse in terms of SPE losses, particularly the SPE_∞ loss. This is not unexpected as the autocorrelations in the data lead to biased estimates (affecting the SPE_1 loss) and stationarity is ensured by simple constraining the estimates to lie within the unit circle, often leading to models that have poles almost on the boundary. These models will clearly have very large time constants and this can lead to poor long term predictions.

The presented losses must be interpreted as Bayes risks given the manner in which the models were generated and may be argued to favour the MML87 estimates; of course, a simple counter argument is that the other estimates attempt to be *objective* by avoiding the concepts of prior distributions completely and should perform well regardless of the distribution of models to which they are applied. It is of course not clear that 'real life' will generate models that have a uniform distribution of pole magnitudes.

However, given the choice of uniform priors on the root parameters, the traditional Bayesian MAP estimator coincides with the ML estimator. It is therefore of interest to that the MML estimator, which is essentially a *discretised* MAP estimator (in the sense that it selects the model with the largest *a posteriori* probability mass), significantly outperforms the standard MAP estimator. One could argue that the mean posterior estimator may be more competitive, but such an estimator is driven by minimising a specific loss function rather than choosing the most probable model in light of the priors and the data; additionally, it is also very difficult to compute for AR models in high dimensions, requiring MCMC methods [103].

4.9.3 Non-Bayesian Parameter Estimation Experiment

A further test was undertaken to examine the behaviour of the MML87S parameter estimates in a non-Bayesian setting. Rather than randomly generating models from the prior distributions, four AR(7) models were used as generating sources. Each of the models has coefficients of the form $\mathbf{a} = [1, -\xi, (-\xi)^2, (-\xi)^3, (-\xi)^4, (-\xi)^5, (-\xi)^6, (-\xi)^7]$, with $\xi = \{0.2, 0.4, 0.6, 0.8\}$, yielding a mix of real and complex poles all at radius $|\xi|$ (in a

similar fashion to the experiments in [141]). This allows for the behaviour of the different parameter estimators to be tested across a range of different pole strengths. For each model one thousand samples was randomly generated, and the results of the MML87S, Maximum Likelihood and Burg estimators are presented in Table 4.2;

The results are interesting. In all cases, the MML87S estimates attain lower SPE_∞ and KL/J divergence scores in comparison to the ML and Burg estimators. The scores for the ML and Burg estimators increase only slowly as ξ grows larger, while the MML87S scores are significantly smaller for $\xi = 0.2$ than for $\xi = 0.8$. This is not surprising, as the largest gains made by the shrinkage due to the Fisher Information term would be expected to be made when the generating model has poles closer to the origin. As the poles grow in strength, the shrinkage will yield less gains and for models with many very large poles that are at odds with the prior, the shrinkage will actually lead to degraded performance as compared to the Burg/ML estimator. Table 4.3 shows the biases and variances of the autoregressive coefficients estimated by both MML87S and ML for $\xi = 0.6$. It is clear the MML87S estimates have a larger bias than the ML estimates, but possess significantly lower variance, particular for $(a_4, \ldots, a_7)$.

It is interesting that although the shrinkage leads to poorer SPE_1 losses for $\xi = 0.8$, the MML87S estimates still possess better SPE_∞ and KL/J divergence losses, indicating that the shrinkage is still beneficial for capturing the 'long term' behaviour of the model, even if it results in degraded short term performance. For larger ξ the SPE_1 losses can grow quite significantly because the models are strongly at odds with the chosen priors; however, these results show that for a large range of pole strengths the MML87S estimates yield

Table 4.2: Autoregressive Parameter Estimation Results, SPE Scores, $N = 22$

	MML87		ML		Burg	
ξ	SPE_∞	SPE_1	SPE_∞	SPE_1	SPE_∞	SPE_1
0.2	**0.0918**	**0.0770**	0.6017	0.4788	0.5172	0.4351
0.4	**0.1051**	**0.0926**	0.6136	0.4784	0.5278	0.4331
0.6	**0.1385**	**0.1787**	0.6316	0.4759	0.5427	0.4336
0.8	**0.1959**	0.5409	0.6545	0.4260	0.5705	**0.4084**
ξ	KL_{100}	J_{100}	KL_{100}	J_{100}	KL_{100}	J_{100}
0.2	**0.0971**	**0.1634**	0.5768	0.8201	0.5159	0.7343
0.4	**0.1062**	**0.1804**	0.5821	0.8257	0.5178	0.7368
0.6	**0.1349**	**0.2505**	0.5869	0.8305	0.5211	0.7411
0.8	**0.2100**	**0.4985**	0.6041	0.8453	0.5438	0.7651

Table 4.3: Autoregressive Parameter Estimate Biases and Variances

	MML87						
	a_1	a_2	a_3	a_4	a_5	a_6	a_7
Bias	-0.0939	0.2205	-0.1715	0.1136	-0.0743	0.0449	-0.0279
Var	(0.0717)	(0.1032)	(0.0689)	(0.0181)	(0.0041)	(0.0010)	(0.0004)
	Maximum Likelihood						
	a_1	a_2	a_3	a_4	a_5	a_6	a_7
Bias	-0.0123	-0.0331	0.0083	-0.0386	0.0014	-0.0289	-0.0048
Var	(0.0713)	(0.0942)	(0.1106)	(0.1055)	(0.0966)	(0.0948)	(0.0714)

good parameter estimates. Of course, priors should always be chosen using any knowledge that is available; if a real life process is expected to have many large magnitude poles, then perhaps another prior that more accurately reflects the prior beliefs should be used in place of the uniform (for example the reference prior, or a suitably tuned Beta prior).

An Illustrative Example

The way in which the MML87 estimator differs from the Maximum Likelihood, Burg, and to an extent, Yule-Walker estimates can be revealed by examining a single experiment on an AR(10) process, drawn from the larger experiment set. A sequence of 31 data points was generated from the 10-th order AR process, $A_t(q)$, given by

$$\begin{aligned} A_t(q) &= 1 + 0.06465q^{-1} - 0.05855q^{-2} - 0.5921q^{-3} - 0.3985q^{-4} \\ &\quad -0.0157q^{-5} + 0.07233q^{-6} + 0.05173q^{-7} - 0.01658q^{-8} \\ &\quad +0.004312q^{-9} - 0.0004345q^{-10} \end{aligned} \tag{4.85}$$

This process was randomly generated by sampling from the prior, using uniform priors on the model radii. The roots of $A_t(q)$ are a mix of slow and fast real and complex poles. The Maximum Likelihood estimate for an AR(10) model from this data was

$$\begin{aligned} A_{\mathrm{ML}}(q) &= 1 + 0.2915q^{-1} + 0.2798q^{-2} - 0.4843q^{-3} - 0.6584q^{-4} \\ &\quad -0.4458q^{-5} - 0.6106q^{-6} + 0.1736q^{-7} + 0.04555q^{-8} \\ &\quad +0.2382q^{-9} + 0.1903q^{-10} \end{aligned} \tag{4.86}$$

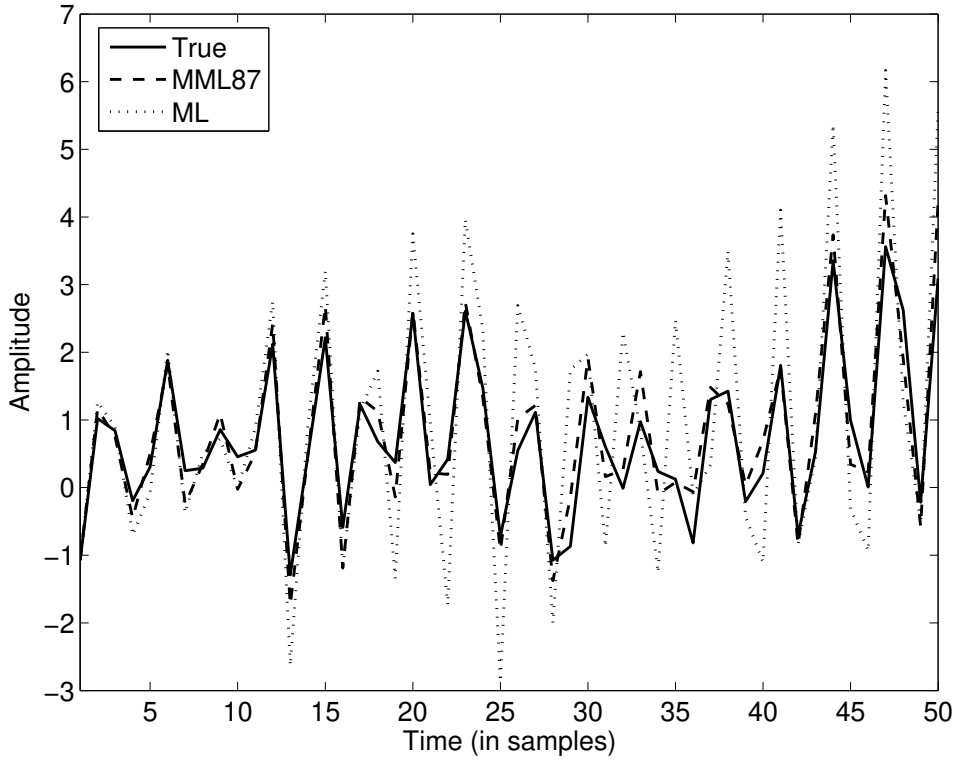


Figure 4.5: AR(10) predictions from ML and MML87 Estimated Models

The model had a negative log-likelihood of 44.34 nits, and a message length of 69.84 nits. In contrast the MML87 estimated model was

$$\begin{aligned} A_{87}(q) &= 1 + 0.196q^{-1} + 0.01273q^{-2} - 0.6743q^{-3} - 0.3962q^{-4} \\ &\quad -0.07755q^{-5} - 0.005284q^{-6} - 6.19 \times 10^{-05}q^{-7} - 4.036 \times 10^{-06}q^{-8} \\ &\quad -1.24 \times 10^{-08}q^{-9} - 7.943 \times 10^{-08}q^{-10} \end{aligned} \tag{4.87}$$

This model had a negative log-likelihood score of 49.58 nits, and a message length of 67.34 nits. The difference in parameter estimates is quite clear: the MML87 estimator has preferred to set many of the parameters to contribute very little, if anything to the explanation. Examination of the poles of $A_{\mathrm{ML}}(q)$ and $A_{87}(q)$ reveals the same. Table 4.4 shows the magnitudes of the poles of the ML and MML87 estimates, sorted in descending

Table 4.4: Magnitude of poles of ML and MML87 estimates for an AR(10) model

	$\lvert p_1\rvert, \ldots, \lvert p_{10}\rvert$									
True	0.9631	0.8246	0.8246	0.6523	0.6196	0.6196	0.4600	0.2050	0.2050	0.1371
ML	0.9943	0.9661	0.9661	0.9053	0.8542	0.8542	0.8277	0.8277	0.8153	0.5558
MML	0.9860	0.8476	0.8476	0.1914	0.1914	0.1913	0.0200	0.0200	0.0200	0.0200

order. It can be seen that all of the ML poles are large in magnitude (at least greater than 0.5), while only six of the MML87 poles are larger in magnitude than 0.02 (which is a lower bound in the search), and of those only three are larger in magnitude than 0.2. The MML87 estimator has eschewed the saving of approximately 5 nits on the negative log-likelihood term in preference of saving 8.9 nits by selecting parameters that require significantly less accuracy to state; it believes that there is no real justification for saving on the likelihood when the parameters required to realise this saving must be stated much more accurately than those which result in a higher negative log-likelihood.

The fundamental difference in performance does not lie in the fact that the MML estimator selects parameters more accurately than ML, but that it only selects estimates that are highly informative (in a 'Fisher' sense) if it believes the data justifies their selection. As only six of the root-space parameters in the MML model are actually contributing to the negative log-likelihood, it has in a sense performed model selection while estimating parameters; the resultant model is effectively an AR(6) model rather than an AR(10) model. This type of behaviour is not uncommon amongst other 'shrinkage' estimators (such as the Lasso [140]). Figure 4.5 shows predictions made by both models when fed with the same innovation sequence as the true model; the MML estimates are clearly much closer in behaviour to the true model than the ML estimates. In particular, while both estimates roughly capture the dynamic behaviour of the model, the overall gain of the model is more correctly captured by the MML estimates.

4.9.4 Autoregressive Order Selection Results

This chapter concludes with a simple examination of the MML87 criteria when used for order selection. This experiment is not intended to study the behaviour of the estimator in exacting detail; rather it is used to illustrate that the potential problems previously outlined in Section 4.6 do have a serious impact on selection between competing model orders. To this end, a simple test was performed: one thousand AR(2) models were randomly generated from the model priors, and from each model a segment of $N = 20$ data points was produced. Models from order $P = \{0, ..., 7\}$ were then estimated from the data, and several model selection criteria were used to select the optimal model from this set. The criteria chosen in this case were the Akaike Information Criterion (AIC), the corrected Akaike Information Criteria (AICc) and the Bayesian Information Criteria

(BIC). The exact MML87 and MML87 Search criteria were also used to select models from the candidate set. The results of order selection when the Maximum Likelihood estimates were used by all criteria are detailed in Table 4.5 below. The number of times each criterion identified the optimal model as being of a particular order is presented, with the highest number of correct identifications in boldface.

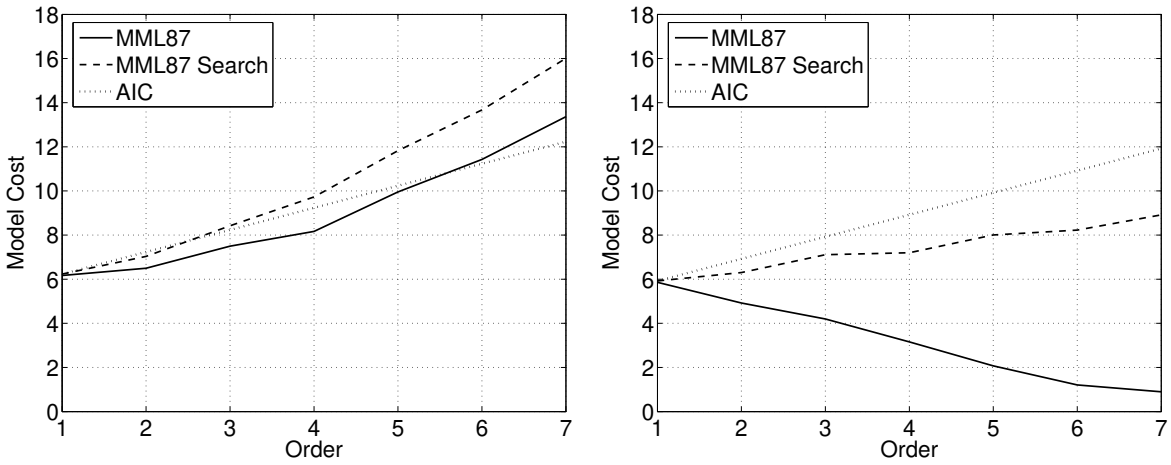
Table 4.5: Autoregressive Order Identification Results (Using ML Estimates)

	Order							
Criterion	0	1	2	3	4	5	6	7
AIC	0	137	584	108	54	39	36	42
AICc	0	194	705	71	20	8	2	0
BIC	0	201	679	69	24	16	4	7
MML87	1	38	390	180	159	86	73	73
MML87$_S$	1	109	**732**	102	52	2	1	1

The results indicate that the exact MML87 criteria performs quite poorly in terms of order selection, scoring the lowest correct order selections and by far the most overestimations of order. The MML87 Search modification brings quite dramatic improvements, with the highest number of correct identifications and a substantially reduced number of overestimations, though still noticable greater than the AICc and BIC criteria. One of the main reasons for the poor performance of the exact MML87 criterion was found to be due to models being estimated with almost coincident real poles; for small sample sizes the Maximum Likelihood estimator often produces models with poles near the unit circle, and these poles have similar magnitude and thus lead to a problems. It is also clear that the more poles being estimated, the greater than chance that several will be almost coincident, and this means that it is usually higher order models that violate the message length approximation, leading to larger numbers of overestimations. The MML87 Search modification avoids this problem because it examines only the diagonalised Information Matrix, in which singularities due to coincident roots can not occur. The results of order selection when the MML approximations used the MML estimates are presented in Table 4.6.

Table 4.6: Autoregressive Order Identification Results (Using MML Estimates)

	Order							
Criterion	0	1	2	3	4	5	6	7
MML87	0	0	1	7	23	42	94	833
MML87$_S$	1	48	**277**	86	204	107	169	108



(a) Model Costs for ML Estimates

(b) Model Costs for MML Estimates

Figure 4.6: Model Costs for MML87 and MML87 Search Approximations

The results show that the order selection performance has degraded even further when the MML estimates are used (the exact MML87 estimates are particularly poor), which is contrary to expectations. The primary reason for this poor performance is due to the very same mechanism that allows the MML parameter estimates to perform well: if there is little structure in the data, the estimates will contain poles that are very uninformative (in a Fisher sense) regions. In particular, the estimator will often select complex conjugate pole pairs with small radii components, as they contribute little to the explanation, and subsequently cost very little to state. These are just the sort of situations that throw doubt on the validity of the MML87 approximation. Figure 4.6 shows the average Model Costs per order for the MML87 and MML87S approximations, using both the ML and MML estimates. For the Maximum Likelihood estimates the MML87S Model Costs are overall higher, which is not unexpected as the Search approximation does not take into account any parameter correlations. For the MML estimates, the curves are clearly significantly shallower than for the ML estimates; this is also to be expected, as the MML estimates for each model order will naturally have a lower model cost. However, the increase in average Model Cost per order is even lower than the AIC criterion which is known to perform badly for small samples sizes, indicating that the models are causing some problems even for the MML87 Search approximation. The exact MML Model Costs when the MML estimates demonstrate reasonably comprehensively that the approximation is breaking down at least for a large number of the estimated models as the Model Cost is now a decreasing function of model complexity!

These examples demonstrate that while the performance of the MML87 parameter estimates are excellent, the order selection results fall somewhat short of expectations. Of course, it has been argued earlier in this thesis that correct order identification is far from the only worthwhile way of evaluating a model selection criteria, and the sequel presents more detailed examinations of the MML87 approximations on various losses. However, this simple experiment illuminates the shortcomings of MML87 when applied to the autoregressive model in pole space, especially as a method of making a justified inductive guess at the generating source.

4.10 Conclusion

This chapter introduced new parameter estimation and order selection criteria for autoregressive models based on the MML87 approximation. The main contributions of this chapter are now summarised.

- This chapter has presented new parameter and order selection estimators for autoregressive processes. In particular, these processes were characterised by their root-space parameters which are a natural setting for the specification of priors.

- Suitable priors for root-space parameters and structural parameters were examined, an examination of the possible problems with an MML87 formulation of autoregressive models was undertaken and a 'search' modification to MML87 dubbed MML87S was introduced to help overcome the difficulties. The key differences between the MML estimator and the ML estimator were then highlighted.

- Experimental results showed the good performance of the MML estimates at capturing long-term behaviour of a process, particular for small sample sizes. Simple order selection experiments subsequently confirmed the MML87S estimator provides reasonable inferences as long as the estimated models do not lie within the parameter ranges that seriously violate the MML87 assumptions.

The next chapter considers an MML87 formulation of a closely related model class: the moving average process.

Table 4.7: Autoregressive Parameter Estimation SPE Results, AR(1-5)

	MML87		ML		Burg		Yule-Walker		Least Squares	
N	SPE_∞	SPE_1	SPE_∞	SPE_1	SPE_∞	SPE_1	SPE_∞	SPE_1	SPE_∞	SPE_1
	AR(1)									
4	0.2691	**0.1616**	0.8581	0.2106	0.9477	0.2001	**0.2577**	0.1657	8.4844	0.2256
	(0.46)	*(0.27)*	*(2.82)*	*(0.37)*	*(3.58)*	*(0.35)*	*(0.29)*	*(0.27)*	*(22.36)*	*(0.39)*
8	**0.1734**	**0.0915**	0.2967	0.1023	0.2883	0.0974	0.1856	0.0926	1.6913	0.1039
	(0.24)	*(0.17)*	*(0.71)*	*(0.20)*	*(0.72)*	*(0.18)*	*(0.23)*	*(0.16)*	*(7.78)*	*(0.18)*
16	**0.1108**	**0.0453**	0.1416	0.0464	0.1427	0.0456	0.1253	0.0458	0.3114	0.0472
	(0.16)	*(0.08)*	*(0.24)*	*(0.08)*	*(0.26)*	*(0.08)*	*(0.19)*	*(0.08)*	*(1.60)*	*(0.08)*
	AR(2)									
7	0.4208	**0.1797**	1.0055	0.2619	0.9141	0.2426	**0.3526**	0.1906	13.533	0.2965
	(1.00)	*(0.30)*	*(2.29)*	*(0.43)*	*(2.56)*	*(0.38)*	*(0.30)*	*(0.27)*	*(201.60)*	*(0.50)*
14	**0.2225**	**0.0871**	0.3744	0.1005	0.3666	0.0984	0.2453	0.0936	1.0854	0.1046
	(0.26)	*(0.13)*	*(0.62)*	*(0.14)*	*(0.61)*	*(0.14)*	*(0.25)*	*(0.12)*	*(4.59)*	*(0.15)*
28	**0.1256**	**0.0404**	0.1634	0.0421	0.1809	0.0422	0.1559	0.0425	0.2478	0.0430
	(0.16)	*(0.06)*	*(0.26)*	*(0.06)*	*(0.40)*	*(0.06)*	*(0.20)*	*(0.05)*	*(0.79)*	*(0.06)*
	AR(3)									
10	**0.3421**	**0.1270**	0.9144	0.2305	0.9134	0.2060	0.3970	0.1750	43.390	0.2771
	(0.57)	*(0.23)*	*(2.00)*	*(0.45)*	*(2.18)*	*(0.35)*	*(0.31)*	*(0.32)*	*(1161)*	*(0.51)*
20	**0.2034**	**0.0628**	0.3667	0.0832	0.3930	0.0826	0.2755	0.0819	1.0946	0.0911
	(0.21)	*(0.11)*	*(0.49)*	*(0.12)*	*(0.65)*	*(0.12)*	*(0.25)*	*(0.11)*	*(3.63)*	*(0.13)*
40	**0.1291**	**0.0294**	0.1925	0.0366	0.2141	0.0364	0.1885	0.0380	0.2933	0.0382
	(0.16)	*(0.05)*	*(0.26)*	*(0.05)*	*(0.35)*	*(0.05)*	*(0.22)*	*(0.05)*	*(0.72)*	*(0.05)*
	AR(4)									
13	**0.3295**	**0.1092**	0.8773	0.1930	0.7543	0.1739	0.4292	0.1426	4.4986	0.2372
	(0.37)	*(0.17)*	*(1.11)*	*(0.30)*	*(0.90)*	*(0.25)*	*(0.30)*	*(0.18)*	*(18.40)*	*(0.34)*
26	**0.2005**	**0.0524**	0.3756	0.0735	0.3885	0.0711	0.3047	0.0700	0.8708	0.0791
	(0.24)	*(0.10)*	*(0.52)*	*(0.13)*	*(0.51)*	*(0.13)*	*(0.26)*	*(0.10)*	*(2.43)*	*(0.14)*
52	**0.1283**	**0.0242**	0.1970	0.0319	0.2091	0.0318	0.2028	0.0337	0.2769	0.0337
	(0.15)	*(0.04)*	*(0.24)*	*(0.04)*	*(0.28)*	*(0.04)*	*(0.22)*	*(0.04)*	*(0.62)*	*(0.05)*
	AR(5)									
16	**0.3439**	**0.0841**	0.9081	0.1593	0.8996	0.1485	0.4372	0.1213	5.7016	0.2015
	(0.61)	*(0.16)*	*(1.39)*	*(0.27)*	*(1.44)*	*(0.23)*	*(0.32)*	*(0.16)*	*(26)*	*(0.30)*
32	**0.2099**	**0.0384**	0.4074	0.0596	0.4142	0.0584	0.3199	0.0596	0.8321	0.0659
	(0.23)	*(0.07)*	*(0.54)*	*(0.10)*	*(0.70)*	*(0.10)*	*(0.27)*	*(0.08)*	*(2.23)*	*(0.11)*
64	**0.1353**	**0.0194**	0.2131	0.0264	0.2301	0.0261	0.2292	0.0278	0.3437	0.0272
	(0.16)	*(0.03)*	*(0.26)*	*(0.04)*	*(0.43)*	*(0.04)*	*(0.24)*	*(0.03)*	*(1.00)*	*(0.04)*

Table 4.8: Autoregressive Parameter Estimation SPE Results, AR(1-5)

	MML87		ML		Burg		Yule-Walker		Least Squares	
N	SPE_∞	SPE_1	SPE_∞	SPE_1	SPE_∞	SPE_1	SPE_∞	SPE_1	SPE_∞	SPE_1
	AR(6)									
19	**0.3193**	0.1222	0.8584	0.1397	0.8200	0.1240	0.4639	**0.1127**	6.6656	0.1719
	(0.37)	*(1.47)*	*(1.07)*	*(0.28)*	*(1.00)*	*(0.23)*	*(0.35)*	*(0.26)*	*(66.06)*	*(0.28)*
38	**0.2003**	**0.0313**	0.3814	0.0507	0.3934	0.0497	0.3365	0.0513	0.8111	0.0555
	(0.22)	*(0.06)*	*(0.43)*	*(0.08)*	*(0.59)*	*(0.08)*	*(0.27)*	*(0.07)*	*(2.07)*	*(0.08)*
76	**0.1370**	**0.0152**	0.2086	0.0216	0.2226	0.0217	0.2403	0.0242	0.2505	0.0225
	(0.16)	*(0.02)*	*(0.25)*	*(0.03)*	*(0.33)*	*(0.03)*	*(0.24)*	*(0.03)*	*(0.36)*	*(0.03)*
	AR(7)									
22	**0.4306**	**0.0565**	0.8672	0.1215	0.8531	0.1134	0.4903	0.0942	3.3481	0.1597
	(2.30)	*(0.11)*	*(1.09)*	*(0.30)*	*(1.11)*	*(0.27)*	*(0.29)*	*(0.11)*	*(9.81)*	*(0.39)*
44	**0.2026**	**0.0259**	0.3970	0.0444	0.4032	0.0438	0.3721	0.0463	0.6662	0.0493
	(0.21)	*(0.05)*	*(0.46)*	*(0.07)*	*(0.46)*	*(0.07)*	*(0.28)*	*(0.06)*	*(1.27)*	*(0.08)*
88	**0.1285**	**0.0131**	0.2092	0.0192	0.2271	0.0192	0.2624	0.0221	0.2942	0.0202
	(0.15)	*(0.02)*	*(0.27)*	*(0.03)*	*(0.37)*	*(0.03)*	*(0.25)*	*(0.03)*	*(0.75)*	*(0.03)*
	AR(8)									
25	**0.2884**	**0.0643**	0.8364	0.1506	0.7060	0.1403	0.4822	0.1179	3.4040	0.1710
50	**0.1950**	**0.0224**	0.4070	0.0396	0.3739	0.0383	0.3660	0.0405	1.5773	0.0434
	(0.21)	*(0.04)*	*(0.45)*	*(0.06)*	*(0.40)*	*(0.06)*	*(0.26)*	*(0.05)*	*(26.37)*	*(0.07)*
100	**0.1225**	**0.0109**	0.1941	0.0169	0.1909	0.0168	0.2603	0.0198	0.2253	0.0175
	(0.18)	*(0.02)*	*(0.24)*	*(0.03)*	*(0.21)*	*(0.02)*	*(0.24)*	*(0.02)*	*(0.30)*	*(0.03)*
	AR(9)									
28	**0.3536**	**0.0403**	0.8888	0.0808	0.7699	0.0761	0.5074	0.0739	6.5016	0.1093
	(1.34)	*(0.09)*	*(0.90)*	*(0.15)*	*(0.77)*	*(0.13)*	*(0.28)*	*(0.10)*	*(82)*	*(0.19)*
56	**0.2103**	**0.0193**	0.4269	0.0307	0.4098	0.0302	0.3848	0.0346	0.7546	0.0348
	(0.29)	*(0.04)*	*(0.51)*	*(0.05)*	*(0.49)*	*(0.05)*	*(0.27)*	*(0.04)*	*(1.95)*	*(0.06)*
112	**0.1237**	**0.0090**	0.2159	0.0132	0.2047	0.0132	0.2795	0.0167	0.2546	0.0140
	(0.15)	*(0.02)*	*(0.28)*	*(0.02)*	*(0.25)*	*(0.02)*	*(0.25)*	*(0.02)*	*(0.50)*	*(0.02)*
	AR(10)									
31	**0.3035**	**0.0385**	0.8656	0.0806	0.7191	0.0746	0.5107	0.0713	2.6345	0.1138
	(0.59)	*(0.09)*	*(0.87)*	*(0.15)*	*(0.63)*	*(0.14)*	*(0.28)*	*(0.10)*	*(7.77)*	*(0.22)*
62	**0.1905**	**0.0165**	0.3973	0.0315	0.3699	0.0307	0.4016	0.0352	0.8022	0.0349
	(0.21)	*(0.04)*	*(0.37)*	*(0.06)*	*(0.37)*	*(0.06)*	*(0.28)*	*(0.05)*	*(2.83)*	*(0.06)*
124	**0.1331**	**0.0077**	0.1932	0.0130	0.1925	0.0130	0.2930	0.0166	0.2230	0.0137
	(0.51)	*(0.02)*	*(0.20)*	*(0.02)*	*(0.19)*	*(0.02)*	*(0.25)*	*(0.02)*	*(0.28)*	*(0.02)*

Table 4.9: Autoregressive Parameter Estimation Results, KL Divergences, AR(1-5)

	MML87		ML		Burg		Yule-Walker		Least Squares	
N	KL_{500}	J_{500}	KL_{500}	J_{500}	KL_{500}	J_{500}	KL_{500}	J_{500}	KL_{500}	J_{500}
	AR(1)									
4	**1.0441**	**1.2800**	1.7308	2.0191	2.7951	3.1133	3.5807	3.9055	1.4784	2.8434
	(3.04)	*(3.19)*	*(11.16)*	*(11.30)*	*(21.27)*	*(21.43)*	*(23.98)*	*(24.18)*	*(4.44)*	*(7.05)*
8	**0.3117**	**0.4306**	0.3232	0.4566	0.3846	0.5209	0.4190	0.5534	0.3328	0.5698
	(0.78)	*(0.87)*	*(0.77)*	*(0.87)*	*(1.15)*	*(1.25)*	*(1.29)*	*(1.38)*	*(0.84)*	*(1.20)*
16	0.1076	0.1666	**0.1042**	**0.1660**	0.1077	0.1708	0.1190	0.1811	0.1069	0.1755
	(0.20)	*(0.25)*	*(0.19)*	*(0.24)*	*(0.20)*	*(0.25)*	*(0.23)*	*(0.27)*	*(0.19)*	*(0.25)*
	AR(2)									
7	1.1785	1.4270	**1.0216**	**1.3110**	1.8468	2.1106	1.7042	2.3203	2.0820	3.9458
	(12.63)	*(12.68)*	*(7.77)*	*(7.83)*	*(15.05)*	*(15.13)*	*(10.47)*	*(11.67)*	*(17.09)*	*(24.55)*
14	**0.2086**	**0.3150**	0.2428	0.3595	0.2615	0.3825	0.3143	0.5743	0.2496	0.4205
	(0.44)	*(0.50)*	*(0.45)*	*(0.52)*	*(0.51)*	*(0.58)*	*(0.58)*	*(2.09)*	*(0.45)*	*(0.68)*
28	**0.0760**	**0.1278**	0.0787	0.1320	0.0833	0.1385	0.1149	0.2082	0.0819	0.1398
	(0.11)	*(0.14)*	*(0.11)*	*(0.14)*	*(0.11)*	*(0.15)*	*(0.20)*	*(0.46)*	*(0.11)*	*(0.15)*
	AR(3)									
10	**0.5812**	**0.7508**	1.2986	1.5542	1.4721	1.7371	1.5444	2.7543	1.4087	3.7864
	(2.50)	*(2.56)*	*(6.53)*	*(6.59)*	*(7.45)*	*(7.53)*	*(9.22)*	*(17.52)*	*(7.04)*	*(32.7)*
20	**0.1657**	**0.2480**	0.2161	0.3233	0.2501	0.3603	0.3913	1.0259	0.2393	0.4182
	(0.29)	*(0.33)*	*(0.35)*	*(0.40)*	*(0.63)*	*(0.67)*	*(1.40)*	*(8.09)*	*(0.38)*	*(0.58)*
40	**0.0699**	**0.1123**	0.0806	0.1313	0.0836	0.1353	0.1490	0.3623	0.0853	0.1399
	(0.14)	*(0.16)*	*(0.15)*	*(0.17)*	*(0.15)*	*(0.17)*	*(0.35)*	*(1.39)*	*(0.14)*	*(0.16)*
	AR(4)									
13	**0.3937**	**0.5663**	0.9487	1.2061	0.9218	1.1737	0.7736	10.079	0.9974	2.4301
	(0.82)	*(0.92)*	*(2.03)*	*(2.14)*	*(2.68)*	*(2.82)*	*(2.10)*	*(137)*	*(2.03)*	*(5.05)*
26	**0.1422**	**0.2231**	0.2059	0.3120	0.2189	0.3265	0.3618	2.7093	0.2183	0.3584
	(0.21)	*(0.27)*	*(0.36)*	*(0.41)*	*(0.39)*	*(0.44)*	*(0.77)*	*(30)*	*(0.36)*	*(0.45)*
52	**0.0581**	**0.0975**	0.0699	0.1186	0.0724	0.1216	0.1605	1.0562	0.0729	0.1252
	(0.07)	*(0.10)*	*(0.07)*	*(0.10)*	*(0.08)*	*(0.11)*	*(0.29)*	*(9.29)*	*(0.08)*	*(0.10)*
	AR(5)									
16	**0.3927**	**0.5373**	0.9914	1.2395	0.9430	1.1881	0.9651	7.3723	1.0248	2.5832
	(1.20)	*(1.25)*	*(3.07)*	*(3.14)*	*(2.49)*	*(2.56)*	*(1.87)*	*(55)*	*(2.89)*	*(9.26)*
32	**0.1468**	**0.2176**	0.2345	0.3382	0.2489	0.3528	0.4325	4.0258	0.2494	0.3865
	(0.52)	*(0.54)*	*(1.18)*	*(1.19)*	*(1.40)*	*(1.41)*	*(1.78)*	*(32)*	*(1.07)*	*(1.10)*
64	**0.0581**	**0.0949**	0.0741	0.1234	0.0768	0.1265	0.2102	2.6534	0.0777	0.1328
	(0.09)	*(0.11)*	*(0.11)*	*(0.13)*	*(0.13)*	*(0.14)*	*(0.53)*	*(22)*	*(0.13)*	*(0.15)*

Table 4.10: Autoregressive Parameter Estimation Results, KL Divergences, AR(6-10)

	MML87		ML		Burg		Yule-Walker		Least Squares	
N	KL_{500}	J_{500}	KL_{500}	J_{500}	KL_{500}	J_{500}	KL_{500}	J_{500}	KL_{500}	J_{500}
	AR(6)									
19	**0.3600**	**0.5101**	0.8416	1.0902	0.8715	1.1123	1.1212	8.8097	0.9134	1.9520
	(1.26)	*(1.33)*	*(1.68)*	*(1.77)*	*(2.23)*	*(2.29)*	*(4.71)*	*(65)*	*(1.89)*	*(3.24)*
38	**0.1226**	**0.1897**	0.1915	0.2916	0.1979	0.2993	0.3825	4.6803	0.2063	0.3406
	(0.25)	*(0.28)*	*(0.28)*	*(0.32)*	*(0.34)*	*(0.38)*	*(0.94)*	*(51)*	*(0.32)*	*(0.38)*
76	**0.0537**	**0.0889**	0.0721	0.1198	0.0736	0.1220	0.2040	2.2550	0.0744	0.1247
	(0.08)	*(0.10)*	*(0.10)*	*(0.12)*	*(0.10)*	*(0.12)*	*(0.34)*	*(17)*	*(0.09)*	*(0.11)*
	AR(7)									
22	**0.3425**	**0.4860**	0.9581	1.1993	0.9576	1.1948	1.0813	18.964	1.0313	2.2445
	(1.10)	*(1.17)*	*(2.47)*	*(2.52)*	*(2.44)*	*(2.49)*	*(2.48)*	*(220)*	*(2.79)*	*(6.38)*
44	**0.1095**	**0.1709**	0.1860	0.2855	0.1948	0.2944	0.4357	5.4466	0.2059	0.3295
	(0.16)	*(0.19)*	*(0.23)*	*(0.26)*	*(0.32)*	*(0.35)*	*(0.66)*	*(37)*	*(0.27)*	*(0.31)*
88	**0.0466**	**0.0792**	0.0642	0.1110	0.0659	0.1132	0.2392	2.9518	0.0681	0.1188
	(0.05)	*(0.07)*	*(0.05)*	*(0.08)*	*(0.06)*	*(0.08)*	*(0.38)*	*(19)*	*(0.06)*	*(0.08)*
	AR(8)									
25	**0.2590**	**0.3938**	0.7937	1.0313	0.6990	0.9256	0.7540	15.220	0.8330	1.7277
	(0.47)	*(0.54)*	*(1.43)*	*(1.49)*	*(1.26)*	*(1.32)*	*(1.41)*	*(89.45)*	*(1.36)*	*(2.37)*
50	**0.0948**	**0.1565**	0.1712	0.2732	0.1679	0.2672	0.3712	7.5955	0.1860	0.3306
	(0.10)	*(0.13)*	*(0.16)*	*(0.20)*	*(0.16)*	*(0.20)*	*(0.45)*	*(53)*	*(0.17)*	*(0.34)*
100	**0.0414**	**0.0735**	0.0620	0.1091	0.0626	0.1096	0.2248	3.4990	0.0648	0.1149
	(0.04)	*(0.06)*	*(0.04)*	*(0.06)*	*(0.05)*	*(0.07)*	*(0.34)*	*(22)*	*(0.05)*	*(0.07)*
	AR(9)									
28	**0.2257**	**0.3531**	0.7777	1.0165	0.7206	0.9435	0.8125	22.4442	0.8358	1.9123
	(0.34)	*(0.44)*	*(1.88)*	*(1.92)*	*(2.03)*	*(2.05)*	*(1.42)*	*(220)*	*(1.92)*	*(4.44)*
56	**0.0990**	**0.1593**	0.1980	0.2993	0.2097	0.3097	0.4625	8.1987	0.2259	0.3603
	(0.16)	*(0.19)*	*(0.69)*	*(0.71)*	*(1.07)*	*(1.08)*	*(1.25)*	*(43)*	*(0.90)*	*(0.92)*
112	**0.0405**	**0.0699**	0.0616	0.1079	0.0622	0.1084	0.2662	3.9216	0.0649	0.1144
	(0.04)	*(0.06)*	*(0.04)*	*(0.06)*	*(0.05)*	*(0.06)*	*(0.40)*	*(20)*	*(0.05)*	*(0.07)*
	AR(10)									
31	**0.2209**	**0.3432**	0.7214	0.9560	0.6442	0.8670	0.8239	26.9150	0.8160	1.6766
	(0.39)	*(0.52)*	*(1.73)*	*(1.76)*	*(1.66)*	*(1.70)*	*(1.41)*	*(242.16)*	*(2.13)*	*(2.90)*
62	**0.0906**	**0.1465**	0.1740	0.2747	0.1714	0.2697	0.4626	8.9691	0.1903	0.3201
	(0.11)	*(0.14)*	*(0.16)*	*(0.19)*	*(0.16)*	*(0.19)*	*(0.55)*	*(46)*	*(0.20)*	*(0.24)*
124	**0.0383**	**0.0671**	0.0615	0.1075	0.0620	0.1079	0.2851	5.8198	0.0656	0.1147
	(0.04)	*(0.06)*	*(0.05)*	*(0.06)*	*(0.05)*	*(0.07)*	*(0.39)*	*(43)*	*(0.06)*	*(0.07)*

Chapter 5

MML87 Inference of Moving Average Models

5.1 Introduction

The second class of model to be examined in this thesis is the moving average (MA) model. An MA model can be formed from the general ARMA model by setting $P = 0$. The MA(Q) explanation of measurements $\mathbf{y} \in \mathbb{R}^N$ is then given by

$$y_n = \sum_{i=1}^{Q} c_i v_{n-i} + v_n \tag{5.1}$$

where $\mathbf{v} \in \mathbb{R}^N$ is a vector of random disturbance, assumed to be i.i.d. as $v_n \sim \mathcal{N}(0, \sigma^2)$, $\mathbf{c} \in \mathbb{R}^Q$ is the vector of moving average coefficients and $Q \in \mathbb{N}$ is the *order* of the moving average. Order selection of MA processes using MML has been studied previously by Sak et al [142]; however, in that case the inferences were based on the *conditional likelihood* in which estimates of the innovations are filtered from the sequence $\mathbf{y}$ through inversion of the moving average process, and a different Fisher Information matrix approximation is employed. In contrast, the work presented here is based on the exact unconditional likelihood of the data $\mathbf{y}$, and has also been extended to estimation of the $\mathbf{c}$ parameters. As per the autoregressive models, it is possible to reparameterise the moving average model in terms of it's zero-space parameters; let $\boldsymbol{\zeta} = [\boldsymbol{\alpha}, \mathbf{r}, \boldsymbol{\omega}]$ denote the vector of components of the model's zeros. However, the bulk of the work in this thesis considers the moving average

models in coefficient space; it is not considered generally useful to work in zero-space for moving average models for several reasons:

- The zero-space parameters have less clearly defined effects on the time response of the MA model than on the AR model and the statistician will generally have less prior knowledge on the zero positions than on the pole positions.

- Moving average models are rarely considered suitable explanations to time series by themselves, more commonly introduced simply to increase the parsimony of autoregressive explanations.

- The Information Matrix in coefficient space does not possess the same issues as the Information Matrix in zero-space.

These three issues, in particular the third, indicate that formulating the message length for the moving average model may be most easily done in coefficient space, and this is the focus of the chapter. However, it should be noted that a zero-space formulation is shown to be easily made based on the results of the previous chapter.

5.2 Priors for c

The starting point for specifying priors over the moving average coefficients is to make the assumption that the model is invertible. From this it follows that the roots of

$$C(q) = 1 + \sum_{i=1}^{Q} c_i q^{-i} \tag{5.2}$$

must be constrained to lie within the unit circle. As the likelihoods of the models presented in the sequel are parameterised in moving average coefficient space it is certainly valid to choose priors over the parameter polynomial $C(q)$ rather than its roots. Given no prior knowledge on the values of the parameters other than that they must define an invertible model, a suitable choice is the uniform prior over the valid region of values the vector $\mathbf{c}$ may assume, that is

$$h(\mathbf{c}) = \frac{1}{\text{vol}\,(\Lambda_Q)},\ \mathbf{c} \in \Lambda_Q \tag{5.3}$$

where Λ_k is the hyper-region of k-ary polynomial parameters for which all roots are within the unit circle, i.e.

$$\Lambda_k = \left\{ C(q) \in \mathbb{R}^{k+1} : C_1(q) = 1,\ ||\text{roots}(C(q))||_\infty < 1 \right\} \tag{5.4}$$

and $\text{vol}(x)$ is the volume of some hyper-region x. A method of computing the volume of the region Λ_k is given by [143] and is summarised below for completeness

$$M_{k+1} = \frac{k}{k+1} M_{k-1},\ \ M_1 = 2 \tag{5.5}$$

$$\text{vol}\left(\Lambda_k\right) = \left(M_1 \cdot M_3 \cdot M_5 \cdot ... M_{k-1}\right)^2,\ \text{if } k \text{ is even} \tag{5.6}$$

$$\text{vol}\left(\Lambda_{k+1}\right) = \text{vol}\left(\Lambda_k\right) M_{k+1} \tag{5.7}$$

This prior, uniform over the coefficient space is the same as previously used by Fitzgibbon [1] and Sak [2] with success.

5.2.1 An Interpretation of the Prior

The pole-space priors that are used over the autoregressive components express the belief that the individual poles will be distributed according to some fashion that does not vary with the order of the model. This has the advantage that estimating a higher order model will not lead to bias towards different pole magnitudes than estimating a lower order model and that the prior beliefs on pole magnitudes and angles may be explicitly stated. However, the disadvantage of this structure is that if uniform priors are placed over the pole magnitudes it leads to coefficient space distributions that favour smaller magnitude coefficients for the higher order terms, i.e. models with a large number of smaller coefficients.

In contrast, a uniform prior over the coefficient space expresses the belief that the model will tend to consist of a set of strongly defined coefficients and will shy away from models with small magnitude coefficients. As moving average components, especially in mixed ARMA models, are often viewed as a way to increase the parsimonity of a model rather than telling us as much about the behaviour of the process as the autoregressive components (as the zero positions are much harder to use to interpret a model than the pole positions which can quickly summarise the entire model's approximate behaviour)

it was decided a prior that favoured including only the strongly defined moving average terms would be suitable.

This leaves the user able to concentrate on specifying the autoregressive pole priors based on their prior belief of time constants and underlying natural frequencies while avoiding the requirement of specifying zeros based on more difficult prior beliefs of phase shifts. Instead, they may simply use the colourless (over coefficient space) uniform prior which will be biased towards including only strong moving average effects which will help decrease complexity and aid in the interpretation of the resulting model.

5.3 Likelihood Function

The negative log-likelihood function for a MA(Q) moving average model is given by

$$L(\mathbf{y}|\boldsymbol{\theta}) = \frac{N}{2}\log(2\pi) + \frac{1}{2}\log|\mathbf{W}(\boldsymbol{\theta})| + \frac{1}{2}\mathbf{y}^T \cdot \mathbf{W}^{-1}(\boldsymbol{\theta}) \cdot \mathbf{y} \tag{5.8}$$

where $\mathbf{W}(\boldsymbol{\theta}) \in \mathbb{R}^{(N\times N)}$ is the autocovariance matrix of the data. This matrix is given by

$$\mathbf{W}(\boldsymbol{\theta}) = \sigma^2 \cdot \text{toep}\left(\left[\text{toepu}\left([1,\mathbf{c}]\right)\cdot[1,\mathbf{c}]^T, \mathbf{0}_{(1\times(N-Q-1))}\right]\right) \tag{5.9}$$

where toepu$(\cdot)$ is an upper-triangular Toeplitz matrix. The expression (5.8) may be efficiently evaluated via the Kalman Filter [59] (also see Appendix C).

5.4 The Fisher Information Matrix

The exact Information Matrix for a zero meaned moving average process with Normal innovations is given by [133]

$$J_{(i,j)}(\boldsymbol{\theta}) = \frac{1}{2}\text{Tr}\left(\mathbf{W}^{-1}(\boldsymbol{\theta})\frac{\partial \mathbf{W}(\boldsymbol{\theta})}{\partial \theta_i}\mathbf{W}^{-1}(\boldsymbol{\theta})\frac{\partial \mathbf{W}(\boldsymbol{\theta})}{\partial \theta_j}\right) \tag{5.10}$$

In particular, it is important to note that the entries are not merely functions of the parameters scaled by the number of data points; this is due to the fact that the data $\mathbf{y}$ is not i.i.d. Beyond the considerable time required to compute (5.10), the largest issue surrounding the use of the exact Information Matrix in an MML87 formulation of the moving average problem is it's instability. There are many combinations of parameters

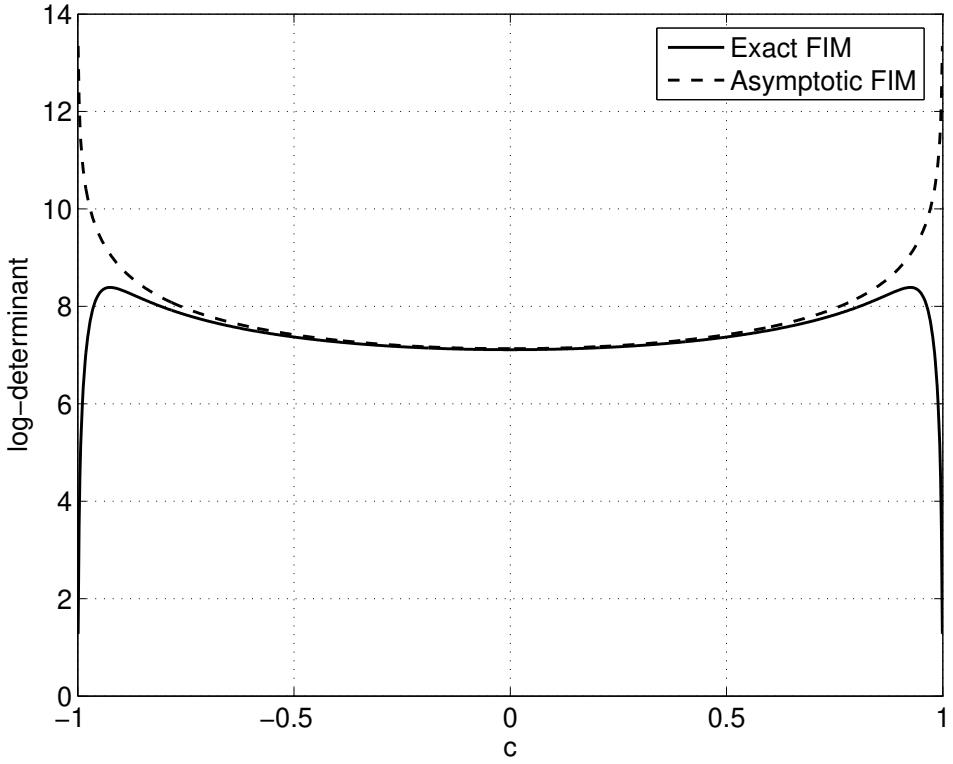


Figure 5.1: Asymptotic and Exact FIM log-determinant plots for the MA(1) model (N=50)

that yield near singular Information Matrices, and this is clearly a source of great concern for an MML87 application. In this work the *asymptotic* Information Matrix is used in place of the exact Matrix. Although for small amounts of data the exact and the asymptotic Information Matrices will differ considerably as the zeroes of the **c** parameters approach the boundary of the unit circle, the increased stability of the asymptotic variant is deemed to outweigh any discrepancies in those regions where the exact matrix is also stable. Figure 5.1 shows a plot of the log-determinant of the exact FIM vs the log-determinant of the asymptotic FIM for the MA(1) model for fifty datapoints. The two coincide closely near $c = 0$, and begin to diverge as $c \to 1$. As N increases, the curves coincide for an increasingly larger region around $c = 0$, and eventually converge as $N \to \infty$, as is to be expected. Using the Whittle's asymptotic Fisher Information Matrix [7], entry (i, j) is approximated by

$$\mathbf{J}_{(i,j)}(\boldsymbol{\theta}) \approx \frac{N}{4\pi} \int_{-\pi}^{\pi} \frac{\frac{\partial \phi(\omega)}{\partial \theta_i} \cdot \frac{\partial \phi(\omega)}{\partial \theta_j}}{\phi^2(\omega)} \mathrm{d}\omega \tag{5.11}$$

where N is the size of the data, and $\phi(\omega)$ is the spectral power density function given by

$$\phi(\omega) = w_0 + 2\sum_{k=1}^{\infty} w_k \cos(k\omega) \tag{5.12}$$

where $w_k = \mathrm{E}\left[y_n y_{n-k}\right]$ are the moving average autocovariances. For ARMA processes the integrals in (5.11) can be solved exactly to yield simple expressions for the Information Matrix [2]. In the case of the moving average process, the asymptotic Information Matrix is given by

$$\mathbf{J}(\boldsymbol{\theta}) = N \begin{bmatrix} \dfrac{\mathrm{E}_{\mathbf{r}}\left[\mathbf{r}_{(1:Q)}^T \mathbf{r}_{(1:Q)}\right]}{\sigma^2} & \mathbf{0} \\ \mathbf{0} & \dfrac{N}{2\left(\sigma^2\right)^2} \end{bmatrix} \tag{5.13}$$

where $\mathbf{r} = [r_1, ..., r_Q]$ is a sequence assumed to be generated from the auxilliary AR(Q) model

$$r_n + \sum_{i=1}^{Q} c_i r_{n-i} = v_n \tag{5.14}$$

and

$$\mathrm{E}_{\mathbf{r}}\left[\mathbf{r}_{(1:Q)}^T \mathbf{r}_{(1:Q)}\right] = \boldsymbol{\Gamma}_r(\boldsymbol{\theta}) = \mathrm{toep}\left(\left[\gamma_0(\boldsymbol{\theta}), \ldots, \gamma_{(Q-1)}(\boldsymbol{\theta})\right]\right) \tag{5.15}$$

is a $(Q \times Q)$ autocovariance matrix generated from the auxilliary AR(Q) process given by (5.14).

5.5 Parameter Estimation

This section examines the issue of estimating the $\mathbf{c}$ and σ^2 parameters for the moving average model.

5.5.1 Maximum Likelihood Estimation of the Noise Variance

It is possible to find closed form maximum likelihood estimates for the innovation variance; begin by factorising the noise variance σ^2 out of the negative log-likelihood function

$$L(\mathbf{y}|\mathbf{c}, \sigma^2) = \frac{N}{2}\log(2\pi) + \frac{1}{2}\log\left(\left(\sigma^2\right)^N \cdot |\mathbf{W}^*(\mathbf{c})|\right) + \frac{1}{2\sigma^2}\mathbf{y}^T \mathbf{W}^{*-1}(\mathbf{c})\mathbf{y} \tag{5.16}$$

where

$$\mathbf{W}^*(\boldsymbol{\theta}) = \frac{1}{\sigma^2}\mathbf{W}(\boldsymbol{\theta}) \tag{5.17}$$

is the process autocovariance matrix for unit innovation variance. Considering only terms involving σ^2, the likelihood reduces to

$$L(\mathbf{y}|\sigma^2) = \frac{N}{2}\log(\sigma^2) + \frac{1}{2\sigma^2}\mathbf{y}^T\mathbf{W}^{*-1}(\boldsymbol{\theta})\mathbf{y} \tag{5.18}$$

Differentiating with respect to to σ^2, and solving yields

$$\hat{\sigma}^2_{\mathrm{ML}}(\mathbf{y}) = \frac{\mathbf{y}^T\mathbf{W}^{*-1}(\boldsymbol{\theta})\mathbf{y}}{N} \tag{5.19}$$

5.5.2 Minimum Message Length Estimation of the Noise Variance

Given the chosen priors, it is also possible to find a closed form MML estimator for the innovation variance. Examination of the asymptotic Fisher Information Matrix given by (5.13) reveals that it depends on the autocovariance matrix of the auxilliary AR process. This autocovariances contain σ^2 as a factor, and thus the log-determinant of the FIM may be expressed as

$$|\mathbf{J}(\boldsymbol{\theta})| = |\boldsymbol{\Gamma}^*_w(\boldsymbol{\theta})| \cdot \frac{N}{2\sigma^2} \tag{5.20}$$

where

$$\boldsymbol{\Gamma}^*_w(\boldsymbol{\theta}) = \frac{1}{\sigma^2}\boldsymbol{\Gamma}_w(\boldsymbol{\theta}) \tag{5.21}$$

is the auxilliary process autocovariance matrix with unit innovation variance. Considering only those terms in the Message Length that include σ^2 yields

$$I_{87}(\mathbf{y}, \sigma^2) = \frac{N}{2}\log(\sigma^2) + \frac{1}{2\sigma^2}\mathbf{y}^T\mathbf{W}^{*-1}(\boldsymbol{\theta})\mathbf{y} + \log(\sigma^2) - \frac{1}{2}\log(\sigma^4) \tag{5.22}$$

The last two terms are due to the FIM and the prior respectively, and as in the AR model, these cancel. Upon simplification, this reduces to (5.18), and thus the MML estimator for the variance is also

$$\hat{\sigma}^2_{87}(\mathbf{y}) = \frac{\mathbf{y}^T\mathbf{W}^{*-1}(\boldsymbol{\theta})\mathbf{y}}{N} = \hat{\sigma}^2_{\mathrm{ML}}(\mathbf{y}) \tag{5.23}$$

5.5.3 The CL_{87} Estimator

This section presents an alternate parameter estimator for moving average models that is based on the conditional least squares estimator. In this scheme, for moving average models, the innovations are estimated by inversion of the moving average model, and the

set of coefficients that minimise the variance of innovations is sought as the estimate. The CL_{87} works in a similar fashion, but instead uses the estimated innovations in an approximation to the negative log-likelihood, and subsequently arrive at an approximate message length of the model. The first step is to estimate the innovations, $\hat{\mathbf{v}}$, via the autoregressive process

$$\hat{v}_n = y_n - \sum_{i=1}^{Q} \hat{c}_i \hat{v}_{n-i} \tag{5.24}$$

with initial conditions

$$\hat{v}_{(1-Q):0} = 0 \tag{5.25}$$

Once the innovations have been estimated, the next step is to approximate the Message Length of the exact model as

$$I_{\text{CL}}(\mathbf{y}, \hat{\boldsymbol{\theta}}) \approx \frac{N}{2}\left(\log(2\pi) + \log\left(\hat{\sigma}^2_{\text{CL}}(\mathbf{y})\right) + 1\right) + \frac{1}{2}\log|\mathbf{J}(\hat{\boldsymbol{\theta}})| - \log h(\hat{\boldsymbol{\theta}}) + \kappa_M(Q+1) \tag{5.26}$$

where

$$\hat{\sigma}^2_{\text{CL}}(\mathbf{y}) = \frac{\hat{\mathbf{v}}^T\hat{\mathbf{v}}}{N} \tag{5.27}$$

is the estimate of the innovation variance. The CL_{87} estimates are then those $\hat{\mathbf{c}}$, $\hat{\sigma}^2$ that minimise (5.26) with $\hat{\mathbf{v}}$ estimated via (5.24). These estimates were found using the gradient free multi-level coordinate search [144]. To ensure estimates are invertible, the search was conducted in the partial autocorrelation space. In this space, the permissible values that characterise an invertible moving average process are constrained to lie within a Q-dimensional hypercube bounded on $(-1, 1)$ in all dimensions. A one-to-one transformation exists between partial autocorrelation space and coefficient space [145, 146], and this was applied to find the message length approximation. For large amounts of data, one would expect the estimators based on the conditional likelihood and the exact likelihood to be virtually indistinguishable.

5.6 Remarks on the MML87 MA Estimator

The MML87 estimator for the moving average model behaves in a similar fashion to the MML estimator for the AR models. Given uniform priors on $\mathbf{c}$ and the scale invariant

prior on σ^2, the only term that differs between the ML and the MML estimator is the log-determinant of the FIM. The first thing to note is that, as in the case of the autoregressive model, the innovation variance σ^2 has no effect on the accuracy to which the $\mathbf{c}$ parameters are stated. This is due to the fact that the signal-to-noise ratio of an MA model is completely defined by the values of the coefficients:

$$\frac{\mathrm{E}\left[y_i^2\right]}{\mathrm{E}\left[v_i^2\right]} = \frac{\sigma^2\left(\sum_{i=1}^{Q} c_i^2 + 1\right)}{\sigma^2} = 1 + \sum_{i=1}^{Q} c_i^2 \tag{5.28}$$

This is clearly independent of σ^2. The log-determinant of the FIM behaves similarily to the autoregressive proces. As the $\mathbf{c}$ parameters move closer to the invertibility boundary, the log-determinant of the FIM becomes larger; the estimates will then tend to be 'shrunk' towards the origin by the effect of this penalty term. This leads to several features of the MML87 moving average parameter estimator as compared to the Maximum Likelihood estimator:

- **Flatter Frequency Responses**: The height and steepness of peaks (or troughs) in the frequency response of moving average model increases as its the zeros approach the boundary of invertibility. As the log-determinant of FIM in the MML87 estimator will drive the zeros further away from the boundary than the Maximum Likelihood estimator, the frequency responses of the MML models will tend to be 'flatter' and more smeared. As a consquence, these models will be less definite on how strongly the frequencies are being affected by the model.

- **No 'piling-up' effect**: As the log-determinant approaches infinity as the roots approach the boundary, the 'piling-up' effect on the unit circle boundary present in the exact Maximum Likelihood estimates [63] for small sample sizes does not occur in the MML87 estimates (see Section 5.8.3).

- **Larger Scale Estimates**: In a similar fashion to the MML AR model, the scale estimates of the MML MA estimator will be larger than the corresponding ML estimates. This follows easily from the observation that the ML estimator will select the estimates $\hat{\mathbf{c}}_{\mathrm{ML}}$ that minimise $\hat{\sigma}^2_{\mathrm{ML}}$ (under the condition that it is estimated as per (5.19)). As $\hat{\mathbf{c}}_{87}$ must necessarily differ from $\hat{\mathbf{c}}_{\mathrm{ML}}$, and as $\hat{\sigma}^2_{87}$ is chosen in the same fashion as $\hat{\sigma}^2_{\mathrm{ML}}$, it follows that $\hat{\sigma}^2_{87} > \hat{\sigma}^2_{\mathrm{ML}}$.

These comments apply equally to the CL_{87} estimator when compared to a scheme based on simply minimising the conditional negative log-likehood function.

5.7 Moving Average Models in Root-space

If there is a desire to parameterise the moving average models in zero-space, this is entirely possible and is now briefly discussed. Let $\boldsymbol{\zeta} \in \mathbb{R}^Q$ denote the vector of zero-space parameters, analogous to the pole-space parameters $\boldsymbol{\beta}$ for autoregressive processes. Once again the Asymptotic Information matrix may be used, and application of Whittle's asymptotic approximation (5.11) yields an Information Matrix that is the scaled conditional component of the autoregressive root-space matrix of the auxiliary autoregression

$$r_n + \sum_{i=1}^{Q} c_i r_{n-i} = v_n \tag{5.29}$$

The Fisher Information Matrix $\mathbf{J}(\cdot)$ for the MA model in zero-space parameters $\boldsymbol{\zeta}$ is then given by

$$\mathbf{J}(\boldsymbol{\zeta}, \sigma^2) = \frac{(N+Q)\mathbf{J}^{(X)}(\boldsymbol{\zeta}, \sigma^2)}{N} \tag{5.30}$$

where $\mathbf{J}^{(X)}(\cdot)$ is the conditional component of the AR model parameterised by the roots of the MA model, and may be computed as per Section 4.5. Priors for the zero-space parameters may be chosen as per Section 4.3. The main concern with this parameterisation is that the Information Matrix now shares the same issues of singularities and flat-spots that cause problems for the autoregressive models. Any parameter estimation within this parameterisation should also employ the diagonal search modification proposed in Section 4.6.1 to stabilise the search procedure.

5.8 Moving Average Experiments

5.8.1 Parameter Estimation Experiments

The MML87 moving average parameter estimation scheme was tested against the Maximum Likelihood estimates, the Prediction Error Method as implemented in the Matlab System Identification Toolbox, the ARMASel method [73] and the CL_{87} estimator. Models were randomly generated by sampling uniformly from the stationarity region as per [147].

For each order, the estimators were tested on datasets of sample size $N = \{k(3Q+1)\}$, generated from one thousand different models, where $k = \{1, 2, 4\}$. These sample sizes were chosen to contain the minimum number of data points the MATLAB implementations required for the PEM estimates to be computed, and they represent a suitably small data per parameter ratio. Experiments were performed for $Q = \{1, \cdots, 10\}$ which captured a large range models and the estimated models were tested in terms of the normalised unconditional SPE_∞ and KL_{100} losses, and the normalised conditional SPE_{Qc} and KL_{Qc} losses, where the predictions were taken over the next Q samples conditioned on the observed n data points. The results for moving average Parameter estimation are presented in Tables 5.4 through to 5.7, with standard deviations shown in parenthesis. A summary of the general findings are

- For all sample sizes, and all model orders, the Maximum Likelihood and PEM estimators perform generally the poorest, in terms of all losses. Part of this may be attributed to the piling up effect present in both of the estimators.

- The KL_{100} scores for the ML and PEM estimators are in some cases extremely large, while the KL_{Qc} scores for the same cases are not significantly greater than those of the MML87 and ARMASel estimators. This would indicate that the problems with the ML and PEM estimators manifest themselves more in long-term forecasting of behaviour than in short-term predictions.

- The ARMASel estimator performs generally worse in terms of prediction errors as compared to the MML87 and CL87 estimators, though remains quite competitive in terms of KL divergences, especially for moderate data-to-parameter ratios.

- The exact MML87 estimator generally outperforms the CL87 estimator for most combinations of sample sizes and model orders, though the CL87 estimator has quite good performance for lower order models. This is possibly due to an increase in the effect of the conditional likelihood 'transients' as the model order increases.

Again, we note the coincidence of the conventional Bayesian MAP estimator with the Maximum Likelihood estimator under the choice of uniform priors over the coefficient space. It is interesting that taking the accuracy to which a Bayes estimate must be stated leads to a variant of the MAP estimator (i.e. the MML87 estimator) that is free of the

'piling-up' effect that affects the standard MAP estimator (see Section 5.8.3). Furthermore, it is interesting to note that in terms of all the loss functions, the MML87 estimator clearly outperforms the MAP estimator equipped with the same priors.

5.8.2 An Example of the MML87 Moving Average Estimator

A good understanding of the general behaviour of the MML87 moving average parameter estimation scheme as compared to the Maximum Likelihood estimator (and those based on maximising approximations of the likelihood) can be gained by examining a single test case. A sequence of 44 data points was drawn from the MA(7) model

$$\begin{aligned} C(q) &= 1 + 0.87q^{-1} + 1.14q^{-2} - 0.41q^{-3} - 0.52q^{-4} \\ &\quad -0.82q^{-5} - 0.11q^{-6} + 0.1q^{-7} \end{aligned} \tag{5.31}$$

This model was generated by sampling uniformly from the invertibility region, and contains a mix of real and complex zeros. The Maximum Likelihood estimate was given by

$$\begin{aligned} C_{\mathrm{ML}}(q) &= 1 + 0.9q^{-1} + 0.9q^{-2} - 0.81q^{-3} - 0.58q^{-4} \\ &\quad -0.9q^{-5} - 0.13q^{-6} - 0.30q^{-7} \end{aligned} \tag{5.32}$$

and the MML87 estimates are given by

$$\begin{aligned} C_{87}(q) &= 1 + 0.74q^{-1} + 0.85q^{-2} - 0.6q^{-3} - 0.55q^{-4} \\ &\quad -0.87q^{-5} - 0.11q^{-6} - 0.07q^{-7} \end{aligned} \tag{5.33}$$

The ML model achieved a training negative log-likelihood of 84.15 nits, with a message length of 103.06 nits. In contrast the MML model achieved a training negative log-likelihood of 84.5 nits, but a message length of 100.31 nits. Thus it trades less than 0.4 nits of training score to reduce the message length by almost 3 nits. The zeros of all three models are presented in Table 5.1. The most obvious thing to note is that the zeros of the ML estimated model are much stronger in magnitude than the corresponding zeros of the MML estimates; in particular, all the ML zeros are greater in magnitude than 0.5 while in the case of the MML estimates two zeros are less than 0.3. The effect of this is shown quite clearly in the Power Spectral Density plots of all three models, presented in

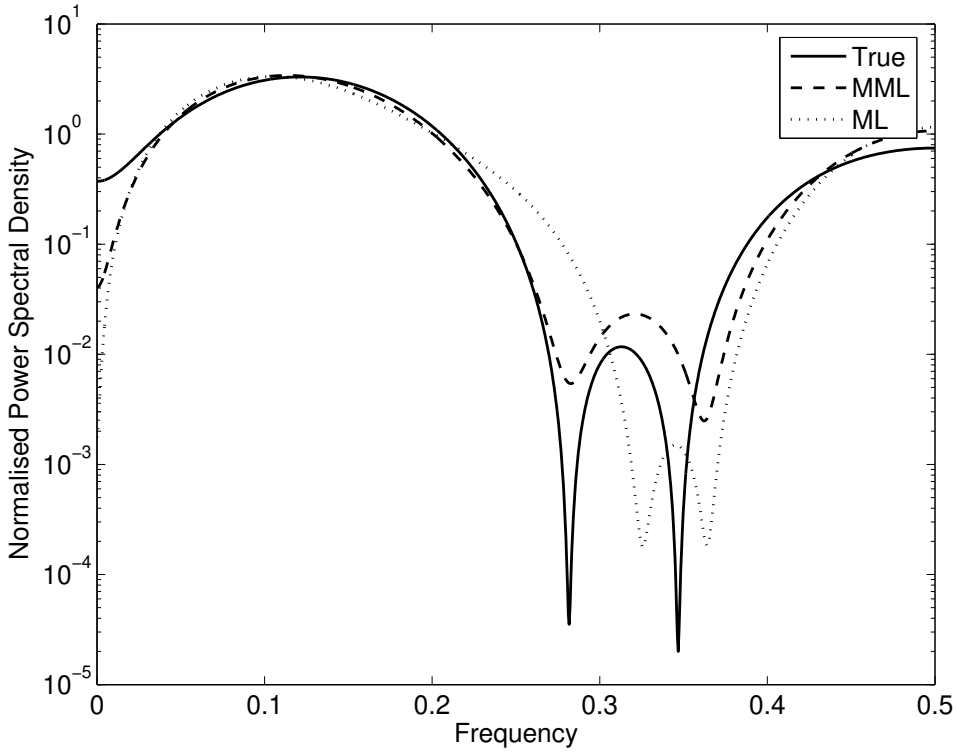
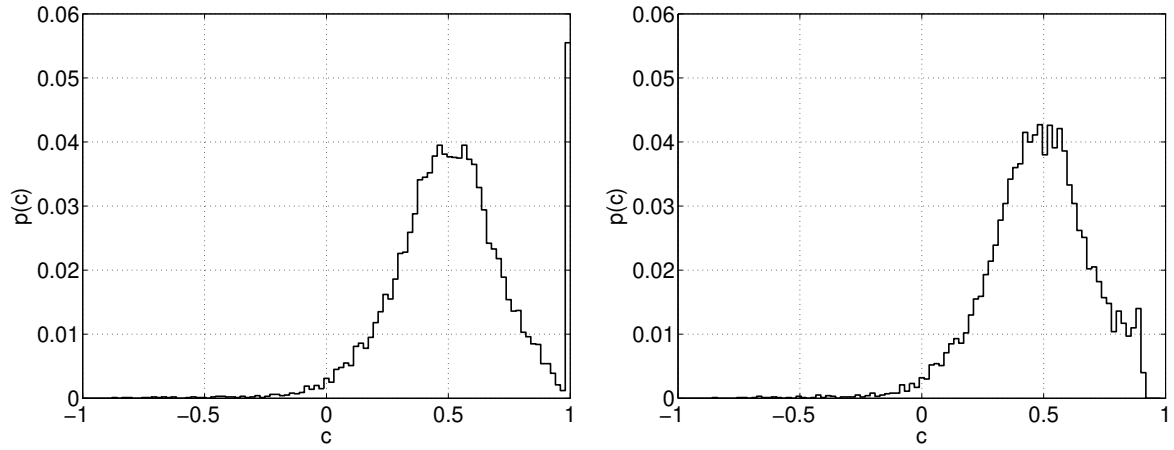


Figure 5.2: Power Spectral Densities of MA(7) ML and MML estimates

Figure 5.2. Although the ML model captures the depths of the troughs in the frequency response more accuately than the MML87 model, the positions of the troughs are incorrect. This leads to a strong attenuation of the wrong frequencies, which clearly has strong effects on the Kullback-Liebler divergence between the models. The MML87 estimator does not correctly capture the depths of the troughs, but more accurately estimates the position of one of them ($f \approx 0.27$) and by underestimating the strength of attenuation of the other bandstop mode, causes the KL-divergence to be reduced. Again, the MML estimates refuse to 'overcommit' to a more precise model, which is in this case one in which frequencies are more heavily attenuated, if it is not warranted by the data.

Table 5.1: Magnitude of zeros of ML and MML87 estimates for an MA(7) model

	$\lvert z_1\rvert, \ldots, \lvert z_7\rvert$						
True	0.9953	0.9953	0.9947	0.9947	0.8424	0.4417	0.2624
ML	0.9908	0.9785	0.9785	0.9775	0.9775	0.5755	0.5755
MML	0.9530	0.9548	0.9548	0.9372	0.9372	0.2931	0.2931



(a) Empirical Distribution of ML Estimates

(b) Empirical Distribution of MML87 Estimates

Figure 5.3: Empirical Distributions of MA(1) Parameter Estimates for $c_1^* = \frac{1}{2}$, $N = 25$

5.8.3 MML and the 'Piling Up' Effect

The exact Maximum Likelihood estimator for moving average coefficients suffers from a well known phenomenon called the 'piling-up effect' [63]. For certain datasets the Maximum Likelihood estimator will estimate parameters on the boundary of invertibility, and for small amounts of data this tendency increases. The result is that for a given source moving average model with coefficients $\mathbf{c}^*$ the distribution of the estimates $\hat{\mathbf{c}}_{\mathrm{ML}}$ will consist of a mixture of a unimodal distribution centered near $\mathbf{c}^*$ and a distribution with all mass on parameters on the invertibility boundary. That the piling up effect is not present in the MML87 estimator is straightforward to see. Consider the MML87 estimator equation:

$$I_{87}(\mathbf{y}, \boldsymbol{\theta}) = L(\mathbf{y}|\boldsymbol{\theta}) + \frac{1}{2}\log \mathbf{J}(\boldsymbol{\theta}) + o(1)$$

The log-determinant of the asymptotic Information Matrix grows to infinity as any one of the roots approaches the invertibility boundary. Any $\mathbf{c}$ with all roots inside the unit circle will lead to a finite sum of negative log-likelihood and $\frac{1}{2}\log \mathbf{J}(\cdot)$, and thus possess a smaller message length than a model with parameters on the boundary (a sum of finite likelihood term and an infinite Fisher term). As a simple example, the empirical distributions of $\hat{\mathbf{c}}_{87}$ and $\hat{\mathbf{c}}_{\mathrm{ML}}$ for the MA(1) model with $N = 10$, $c_1^* = 0.5$ over 2000 test runs are presented in Figure 5.3. It is clear to see that the Maximum Likelihood estimate distribution has definite point mass at $c_1 = 1$, which is absent in the MML87 distribution. If the point masses were removed from the ML distribution, it follows a much more

Table 5.2: Statistics of MA(1) Estimators for $c_1^* = \frac{1}{2}$, $N = 25$

Estimator	Bias	Variance	SE Risk	KL Risk	J Risk
ML	-0.0226	0.0559	0.0564	1.5709	3.0870
MML87	0.0121	0.0437	0.0439	1.5642	3.0765

conventional shape than the MML87 estimate distribution in which the right tail drops off rapidly after approximately $c_1 > 0.9$; this is presumably due to the increased 'shrinkage' effects of the asymptotic Fisher Information as the parameter begins to approach the invertibility boundary. The summary statistics of the two distributions are presented in Table 5.2. The bias' indicate that MML87 generally underestimates c_1^* and ML generally overestimates it, though the differences are quite marginal; this is not unsurprising given the 'shrinkage' effect of the MML87 estimator and the tendency of the ML estimate to selected parameters on the boundary. It is clear that the MML87 estimator has a lower variance, the variance of the ML estimator being 'blown out' by the large number of estimates on the boundary; clearly, as the bias' are very close to zero the variance of the estimators is the dominant contributor to their squared error risk, and thus the MML87 estimator in this case outperforms the ML estimator. The KL and J-divergence risks are very close, though the fact that the MML87 estimator outperforms both is an indication that it is estimating correlations that are on average closer than the ML estimator; again this is probably largely down to the effect of the boundary estimates. From the 10^4 test runs performed, 550 Maximum Likelihood estimates were on the $c_1 = 1$ boundary. This is approximately a probability of 0.055 which is quite close to the probability of 0.076 as estimated in [63].

That the MML87 estimator does not dominate [39] the ML estimator for all values of $c_1^* \in (-1, 1)$ in terms of squared error risk is relatively simple to demonstrate. Consider the case when $c_1^* = 0.99$ and $N = 25$; from the calculations in [63], the probability of the ML estimator selecting $\hat{c}_1 = 1$ is at least greater than 0.5. From this, at least around fifty percent of the estimates will have a squared error loss of 10^{-4}. In contrast, due to the large effects of the Fisher Information as the parameter estimates approach the boundary, the MML87 estimator will be expected to have a much larger squared error risk. It is well known that any biased estimator that systematically selects $\hat{c}_1 = \kappa$ will generally be impossible to dominate as its squared error risk for $c_1^* = \kappa$ will be zero; of course such an estimator is not particularly useful for discovering anything about the data and so the

criterion of dominance is not the only one that should be used to assess an estimator. A slightly larger analysis of the performance of the MML87 estimator is undertaken in the sequel in comparison against a new class of MML estimator.

5.8.4 Moving Average Order Selection Results

A brief test of the order selection capabilities of MML87 for the moving average model is now presented. A set of one thousand MA(2) models was randomly generated from the invertibility region, and from each model $N = 15$ data points was generated. For each of these datasets, models of orders $Q = \{0, \ldots, 5\}$ were estimated via the method of Maximum Likelihood and several model selection criteria were asked to select a model from the candidate set. The number of times each criterion selected each model order, as well as average SPE_∞ and KL_{100} scores are presented in Table 5.3. The results are

Table 5.3: Moving Average Order Identification Results (Using ML Estimates)

	Order						Scores	
Criterion	0	1	2	3	4	5	SPE_∞	KL_{100}
AIC	181	224	**395**	115	65	20	0.3769	11.496
AICc	320	297	363	19	1	0	**0.2369**	4.1275
BIC	295	263	366	54	18	4	0.2861	4.1672
MML87	577	279	97	31	11	5	0.2848	**0.7885**

interesting. In terms of correct order selection, the AIC criterion is the winner, though it is obvious that it is also the worst offender in terms of over estimations of the order. The MML87 criterion is clearly the worst performer in terms of correct order selections, and produces the most under estimations of model order. This is most likely due to the nature of the ML estimates; because of the piling up effect many of the models will have zeros on or extremely close to the invertiblity boundary. The MML87 criterion uses the asymptotic Information Matrix to compute the model cost; the asymptotic MA Information Matrix is the same as the AR asymptotic information matrix, which grows very large as model roots approach the stationarity boundary. Therefore, in many cases the zeros near the unit circle boundary lead to extremely large message lengths. The zero order model has no zeros, and therefore the message length will never be unrealistically large for any estimate. This would seem explain why the zero order model is chosen so frequently.

The squared prediction error and KL divergence scores are also quite interesting. Although the AIC criterion scored the most correct order identifications it is clearly the

worst performer in terms of squared prediction errors and KL divergences. This certainly supports the view that correct order identification is not necessarily the only way to assess model selection criteria. The AICc and BIC criteria have smaller SPE and KL divergence losses than AIC, but still significantly larger KL divergences than the MML87 criterion. This is also due to the nature of the parameter estimates. The AICc and BIC criteria do not vary their penalty terms models depending on the model parameter estimates; thus, models with zeros on or close to the unit circle that will in general lead to large KL divergence losses are treated on the same footing as those with zeros closer to the origin. In contrast, the MML87 criterion (as discussed above) will assign large message lengths to these models and in general, not select them. The interesting thing here is that in many cases a model with no parameters beyond a scale estimate is closer to the true model in terms of KL divergence than a model with one or two parameters as estimated by Maximum Likelihood.

5.9 Conclusion

This chapter has examined the inference of moving average models using the MML87 message length approximation. The main findings and contributions of this chapter are now summarised:

- The issue of singularities in the exact Information Matrix in the formulation of the MML87 message length equation for the MA models was resolved by the application of the asymptotic Information Matrix.
- An examination of the behaviour of the MML estimator showed that the 'piling-up' problem present in the Maximum Likelihood parameter estimator does not occur in the MML estimator.
- Experimental results show that the MML parameter estimates for the MA model perform very well in terms of squared prediction errors, KL divergences and one-step ahead predictive scores, outperforming Maximum Likelihood and Prediction Error methods quite convincingly in many cases.

The next chapter examines the more general mixed ARMA model within the MML87 framework.

Table 5.4: Moving Average Parameter Estimation Results, SPE Scores, MA(1-5)

	MML87		ML		PEM		ARMASel		CL_{87}	
N	SPE_∞	SPE_{1c}	SPE_∞	SPE_{1c}	SPE_∞	SPE_{1c}	SPE_∞	SPE_{1c}	SPE_∞	SPE_{1c}
					MA(1)					
4	0.1891	0.1613	0.4034	0.2126	0.4899	0.3264	0.2072	0.1702	**0.1754**	**0.1567**
	(0.25)	*(0.42)*	*(0.53)*	*(0.56)*	*(0.60)*	*(1.08)*	*(0.26)*	*(0.43)*	*(0.22)*	*(0.39)*
8	0.1199	**0.0964**	0.2139	0.1281	0.2070	0.1505	**0.1111**	0.0993	0.1227	0.0967
	(0.20)	*(0.25)*	*(0.37)*	*(0.35)*	*(0.32)*	*(0.51)*	*(0.16)*	*(0.24)*	*(0.20)*	*(0.25)*
16	**0.0625**	**0.0753**	0.0989	0.1013	0.0830	0.0975	0.0645	0.0820	0.0655	0.0786
	(0.12)	*(0.22)*	*(0.20)*	*(0.33)*	*(0.15)*	*(0.33)*	*(0.10)*	*(0.23)*	*(0.12)*	*(0.22)*
					MA(2)					
7	0.2429	**0.1462**	0.5040	0.1900	0.7706	0.6811	0.2945	0.1689	**0.2411**	0.1503
	(0.29)	*(0.33)*	*(0.68)*	*(0.41)*	*(0.82)*	*(1.65)*	*(0.34)*	*(0.36)*	*(0.27)*	*(0.34)*
14	**0.1150**	**0.0893**	0.1818	0.1172	0.2451	0.1921	0.1279	0.1009	0.1212	0.0922
	(0.17)	*(0.23)*	*(0.24)*	*(0.28)*	*(0.35)*	*(0.45)*	*(0.14)*	*(0.22)*	*(0.15)*	*(0.21)*
28	**0.0498**	**0.0461**	0.0706	0.0558	0.0862	0.0894	0.0629	0.0577	0.0561	0.0509
	(0.08)	*(0.12)*	*(0.11)*	*(0.13)*	*(0.15)*	*(0.34)*	*(0.09)*	*(0.13)*	*(0.07)*	*(0.13)*
					MA(3)					
10	**0.2501**	**0.1433**	0.5406	0.1993	0.7655	0.4441	0.3234	0.1655	0.2654	0.1539
	(0.27)	*(0.29)*	*(0.68)*	*(0.39)*	*(0.90)*	*(0.92)*	*(0.34)*	*(0.31)*	*(0.26)*	*(0.30)*
20	**0.1246**	**0.0788**	0.1979	0.0965	0.2688	0.1818	0.1511	0.0971	0.1328	0.0824
	(0.18)	*(0.16)*	*(0.29)*	*(0.19)*	*(0.39)*	*(0.35)*	*(0.17)*	*(0.18)*	*(0.16)*	*(0.16)*
40	**0.0502**	**0.0409**	0.0657	0.0506	0.0835	0.0759	0.0726	0.0579	0.0595	0.0452
	(0.06)	*(0.08)*	*(0.09)*	*(0.11)*	*(0.10)*	*(0.15)*	*(0.07)*	*(0.09)*	*(0.07)*	*(0.08)*
					MA(4)					
13	**0.2413**	**0.1176**	0.4784	0.1644	0.7727	0.4915	0.3357	0.1515	0.2648	0.1295
	(0.24)	*(0.19)*	*(0.60)*	*(0.29)*	*(0.80)*	*(0.95)*	*(0.33)*	*(0.23)*	*(0.23)*	*(0.20)*
26	**0.1150**	**0.0780**	0.1788	0.0997	0.2722	0.1838	0.1492	0.0898	0.1344	0.0852
	(0.13)	*(0.14)*	*(0.22)*	*(0.17)*	*(0.32)*	*(0.27)*	*(0.14)*	*(0.13)*	*(0.14)*	*(0.14)*
52	**0.0464**	**0.0350**	0.0601	0.0437	0.0833	0.0759	0.0707	0.0515	0.0587	0.0405
	(0.05)	*(0.06)*	*(0.07)*	*(0.08)*	*(0.09)*	*(0.12)*	*(0.07)*	*(0.08)*	*(0.06)*	*(0.06)*
					MA(5)					
16	**0.2692**	**0.1297**	0.5354	0.1718	0.7840	0.4195	0.3398	0.1597	0.2838	0.1368
	(0.33)	*(0.20)*	*(0.61)*	*(0.27)*	*(0.88)*	*(0.65)*	*(0.31)*	*(0.23)*	*(0.26)*	*(0.21)*
32	**0.1222**	**0.0712**	0.1933	0.0902	0.2657	0.1693	0.1522	0.0833	0.1388	0.0795
	(0.13)	*(0.11)*	*(0.22)*	*(0.14)*	*(0.27)*	*(0.23)*	*(0.13)*	*(0.12)*	*(0.13)*	*(0.12)*
64	**0.0525**	**0.0374**	0.0705	0.0480	0.0916	0.0818	0.0759	0.0533	0.0638	0.0429
	(0.06)	*(0.06)*	*(0.09)*	*(0.08)*	*(0.10)*	*(0.14)*	*(0.07)*	*(0.08)*	*(0.06)*	*(0.07)*

Table 5.5: Moving Average Parameter Estimation Results, SPE Scores, MA(6-10)

	MML87		ML		PEM		ARMASel		CL_{87}	
N	SPE_∞	SPE_{1c}	SPE_∞	SPE_{1c}	SPE_∞	SPE_{1c}	SPE_∞	SPE_{1c}	SPE_∞	SPE_{1c}
	MA(6)									
19	**0.2382**	**0.1188**	0.5029	0.1644	0.7533	0.4723	0.3321	0.1520	0.2566	0.1218
	(0.23)	*(0.19)*	*(0.54)*	*(0.29)*	*(0.75)*	*(0.82)*	*(0.29)*	*(0.23)*	*(0.21)*	*(0.18)*
38	**0.1148**	**0.0703**	0.1767	0.0901	0.2541	0.1859	0.1555	0.0895	0.1354	0.0774
	(0.12)	*(0.11)*	*(0.18)*	*(0.13)*	*(0.25)*	*(0.33)*	*(0.13)*	*(0.12)*	*(0.13)*	*(0.11)*
76	**0.0488**	**0.0337**	0.0637	0.0409	0.0885	0.0722	0.0737	0.0484	0.0616	0.0393
	(0.06)	*(0.05)*	*(0.08)*	*(0.06)*	*(0.14)*	*(0.17)*	*(0.07)*	*(0.06)*	*(0.06)*	*(0.06)*
	MA(7)									
22	**0.2309**	**0.1226**	0.4717	0.1670	0.7393	0.5770	0.3298	0.1521	0.2704	0.1366
	(0.21)	*(0.20)*	*(0.48)*	*(0.27)*	*(0.67)*	*(3.93)*	*(0.26)*	*(0.22)*	*(0.22)*	*(0.21)*
44	**0.1088**	**0.0658**	0.1664	0.0851	0.2419	0.1616	0.1463	0.0794	0.1378	0.0754
	(0.11)	*(0.10)*	*(0.17)*	*(0.13)*	*(0.23)*	*(0.22)*	*(0.12)*	*(0.10)*	*(0.12)*	*(0.11)*
88	**0.0420**	**0.0280**	0.0533	0.0337	0.0768	0.0630	0.0682	0.0438	0.0582	0.0334
	(0.04)	*(0.04)*	*(0.05)*	*(0.05)*	*(0.08)*	*(0.10)*	*(0.05)*	*(0.05)*	*(0.05)*	*(0.04)*
	MA(8)									
25	**0.2414**	**0.1102**	0.5064	0.1517	0.7702	0.3840	0.3531	0.1453	0.2695	0.1195
	(0.21)	*(0.14)*	*(0.49)*	*(0.20)*	*(0.65)*	*(0.52)*	*(0.29)*	*(0.18)*	*(0.20)*	*(0.16)*
50	**0.1122**	**0.0640**	0.1744	0.0838	0.2531	0.1703	0.1593	0.0845	0.1414	0.0730
	(0.11)	*(0.10)*	*(0.18)*	*(0.13)*	*(0.26)*	*(0.25)*	*(0.12)*	*(0.11)*	*(0.13)*	*(0.10)*
100	**0.0463**	**0.0290**	0.0606	0.0370	0.0851	0.0693	0.0741	0.0434	0.0619	0.0353
	(0.05)	*(0.04)*	*(0.07)*	*(0.06)*	*(0.09)*	*(0.11)*	*(0.06)*	*(0.05)*	*(0.06)*	*(0.05)*
	MA(9)									
28	**0.2361**	**0.1110**	0.4912	0.1460	0.7864	0.4206	0.3533	0.1420	0.2661	0.1216
	(0.20)	*(0.14)*	*(0.45)*	*(0.18)*	*(0.72)*	*(0.71)*	*(0.30)*	*(0.16)*	*(0.18)*	*(0.15)*
56	**0.1091**	**0.0636**	0.1718	0.0825	0.2611	0.1717	0.1533	0.0812	0.1344	0.0728
	(0.10)	*(0.09)*	*(0.17)*	*(0.12)*	*(0.27)*	*(0.26)*	*(0.11)*	*(0.10)*	*(0.11)*	*(0.10)*
52	**0.1273**	**0.0251**	0.1970	0.0349	0.2091	0.0348	0.2028	0.0385	0.1479	0.0280
	MA(10)									
31	**0.2306**	**0.1053**	0.4755	0.1471	0.7721	0.4294	0.3641	0.1452	0.2683	0.1143
	(0.19)	*(0.13)*	*(0.45)*	*(0.19)*	*(0.72)*	*(0.66)*	*(0.37)*	*(0.21)*	*(0.19)*	*(0.15)*
62	**0.1081**	**0.0616**	0.1688	0.0797	0.2503	0.1763	0.1497	0.0820	0.1386	0.0732
	(0.10)	*(0.08)*	*(0.16)*	*(0.10)*	*(0.23)*	*(0.25)*	*(0.11)*	*(0.10)*	*(0.12)*	*(0.11)*
124	**0.0459**	**0.0295**	0.0584	0.0364	0.0840	0.0666	0.0722	0.0445	0.1323	0.1245
	(0.04)	*(0.04)*	*(0.05)*	*(0.05)*	*(0.08)*	*(0.09)*	*(0.06)*	*(0.05)*	*(0.64)*	*(1.06)*

Table 5.6: Moving Average Parameter Estimation Results, KL Divergences, MA(1-5)

	MML87		ML		PEM		ARMASel		CL_{87}	
N	KL_{100}	KL_{1c}	KL_{100}	KL_{1c}	KL_{100}	KL_{1c}	KL_{100}	KL_{1c}	KL_{100}	KL_{1c}
	MA(1)									
4	1.0115	0.6751	16.0675	0.8783	14.5686	0.8467	1.0606	0.6708	**0.8703**	**0.6332**
	(2.65)	*(1.53)*	*(55)*	*(1.86)*	*(47)*	*(1.79)*	*(2.83)*	*(1.49)*	*(2.19)*	*(1.45)*
8	0.3557	0.2227	3.9862	0.2788	2.4480	0.2703	**0.2938**	**0.2106**	0.3738	0.2221
	(1.01)	*(0.39)*	*(15)*	*(0.49)*	*(7.18)*	*(0.46)*	*(0.93)*	*(0.37)*	*(1.02)*	*(0.39)*
16	0.1280	0.1042	0.8535	0.1276	0.3667	0.1193	**0.1108**	**0.1037**	0.1396	0.1066
	(0.24)	*(0.20)*	*(2.68)*	*(0.26)*	*(1.06)*	*(0.24)*	*(0.19)*	*(0.18)*	*(0.28)*	*(0.20)*
	MA(2)									
7	0.9032	0.5054	34.0341	0.8989	54.693	0.9232	0.9330	0.5015	**0.7186**	**0.4575**
	(4.15)	*(1.09)*	*(427)*	*(2.47)*	*(632)*	*(1.28)*	*(3.23)*	*(1.01)*	*(1.56)*	*(0.78)*
14	0.2732	0.1919	3.7378	0.2755	1.5598	0.3285	**0.2194**	**0.1846**	0.2662	0.1906
	(0.51)	*(0.33)*	*(36.42)*	*(0.47)*	*(7.36)*	*(0.60)*	*(0.30)*	*(0.29)*	*(0.48)*	*(0.32)*
28	0.0989	**0.0808**	0.3249	0.1023	0.3071	0.1284	**0.0946**	0.0903	0.1060	0.0871
	(0.15)	*(0.13)*	*(0.68)*	*(0.15)*	*(0.79)*	*(0.21)*	*(0.13)*	*(0.14)*	*(0.19)*	*(0.14)*
	MA(3)									
10	0.6579	0.4513	13.5478	0.9807	6.4992	0.9408	0.7229	0.4600	**0.6352**	**0.4447**
	(1.83)	*(0.92)*	*(100.19)*	*(2.53)*	*(36.42)*	*(1.79)*	*(1.82)*	*(0.82)*	*(1.36)*	*(0.82)*
20	0.2638	0.1825	1.8618	0.2655	1.7204	0.3254	**0.2234**	**0.1787**	0.2488	0.1826
	(0.50)	*(0.28)*	*(11.93)*	*(0.41)*	*(17.77)*	*(0.45)*	*(0.32)*	*(0.24)*	*(0.44)*	*(0.26)*
40	**0.0925**	**0.0760**	0.2478	0.0993	0.2382	0.1297	0.0957	0.0911	0.1004	0.0845
	(0.13)	*(0.09)*	*(0.76)*	*(0.13)*	*(0.36)*	*(0.18)*	*(0.10)*	*(0.09)*	*(0.12)*	*(0.10)*
	MA(4)									
13	0.6033	**0.3955**	13.4286	0.8007	5.4284	0.9963	0.6576	0.4113	**0.5949**	0.4047
	(1.24)	*(0.56)*	*(99.65)*	*(1.33)*	*(17.01)*	*(2.02)*	*(1.27)*	*(0.54)*	*(0.95)*	*(0.52)*
26	0.2491	**0.1864**	1.3446	0.2873	1.4437	0.3802	**0.2269**	0.1872	0.2632	0.1970
	(0.43)	*(0.27)*	*(5.86)*	*(0.45)*	*(7.97)*	*(0.53)*	*(0.33)*	*(0.22)*	*(0.46)*	*(0.26)*
52	**0.0940**	**0.0738**	0.2390	0.0978	0.2846	0.1430	0.0943	0.0911	0.0979	0.0871
	(0.35)	*(0.08)*	*(1.50)*	*(0.12)*	*(1.62)*	*(0.18)*	*(0.16)*	*(0.09)*	*(0.12)*	*(0.09)*
	MA(5)									
16	**0.5739**	0.4184	8.5290	0.9586	4.0540	0.9583	0.6409	0.4333	0.5740	**0.4081**
	(0.99)	*(0.60)*	*(45.11)*	*(1.91)*	*(13.69)*	*(1.46)*	*(1.87)*	*(0.67)*	*(0.94)*	*(0.49)*
32	0.2339	**0.1776**	1.0716	0.2790	1.0077	0.3468	**0.2041**	0.1776	0.2729	0.1956
	(0.29)	*(0.19)*	*(3.43)*	*(0.34)*	*(2.36)*	*(0.34)*	*(0.20)*	*(0.16)*	*(0.62)*	*(0.20)*
64	**0.0821**	**0.0741**	0.1821	0.1020	0.2159	0.1413	0.0869	0.0910	0.0914	0.0874
	(0.10)	*(0.07)*	*(0.36)*	*(0.11)*	*(0.30)*	*(0.16)*	*(0.06)*	*(0.08)*	*(0.09)*	*(0.08)*

Table 5.7: Moving Average Parameter Estimation Results, KL Divergences, MA(6-10)

	MML87		ML		PEM		ARMASel		CL_{87}	
N	KL_{100}	KL_{1c}	KL_{100}	KL_{1c}	KL_{100}	KL_{1c}	KL_{100}	KL_{1c}	KL_{100}	KL_{1c}
	MA(6)									
19	**0.5325**	0.3805	7.3393	0.9411	5.8614	0.9529	0.5718	0.3909	0.5471	**0.3665**
	(1.16)	*(0.88)*	*(44.94)*	*(2.89)*	*(57.37)*	*(1.63)*	*(1.40)*	*(0.59)*	*(1.79)*	*(0.48)*
38	0.2236	**0.1701**	0.8606	0.2665	1.0261	0.3708	**0.2035**	0.1794	0.2287	0.1868
	(0.45)	*(0.20)*	*(3.58)*	*(0.31)*	*(6.25)*	*(0.47)*	*(0.25)*	*(0.18)*	*(0.38)*	*(0.21)*
76	**0.0776**	**0.0695**	0.1499	0.0923	0.1873	0.1352	0.0851	0.0867	0.0887	0.0850
	(0.09)	*(0.07)*	*(0.29)*	*(0.09)*	*(0.25)*	*(0.17)*	*(0.06)*	*(0.07)*	*(0.09)*	*(0.08)*
	MA(7)									
22	**0.4563**	**0.3483**	4.9791	0.8327	3.2779	0.8902	0.5069	0.3899	0.5254	0.3830
	(0.53)	*(0.38)*	*(28.39)*	*(1.43)*	*(17.06)*	*(1.39)*	*(0.64)*	*(0.47)*	*(0.76)*	*(0.45)*
44	0.1993	**0.1618**	0.8490	0.2674	0.9362	0.3648	**0.1866**	0.1710	0.2268	0.1857
	(0.26)	*(0.15)*	*(5.58)*	*(0.49)*	*(5.49)*	*(0.40)*	*(0.17)*	*(0.14)*	*(0.26)*	*(0.16)*
88	**0.0687**	**0.0652**	0.1159	0.0864	0.1666	0.1338	0.0794	0.0849	0.0842	0.0839
	(0.06)	*(0.06)*	*(0.14)*	*(0.08)*	*(0.18)*	*(0.12)*	*(0.05)*	*(0.07)*	*(0.07)*	*(0.07)*
	MA(8)									
25	**0.4742**	**0.3580**	8.2900	0.8393	3.0058	0.9430	0.5377	0.4010	0.5542	0.3960
	(0.53)	*(0.38)*	*(125.52)*	*(1.35)*	*(8.76)*	*(1.22)*	*(0.62)*	*(0.39)*	*(0.90)*	*(0.49)*
50	0.1912	**0.1599**	0.5924	0.2561	0.6920	0.3600	**0.1888**	0.1749	0.2279	0.1893
	(0.21)	*(0.15)*	*(1.37)*	*(0.25)*	*(2.13)*	*(0.31)*	*(0.16)*	*(0.13)*	*(0.30)*	*(0.17)*
100	**0.0693**	**0.0671**	0.1137	0.0924	0.1725	0.1519	0.0807	0.0890	0.0829	0.0876
	(0.08)	*(0.06)*	*(0.15)*	*(0.09)*	*(0.21)*	*(0.16)*	*(0.06)*	*(0.07)*	*(0.07)*	*(0.08)*
	MA(9)									
28	**0.4686**	**0.3602**	3.4903	0.8185	2.9799	0.9862	0.5025	0.3872	0.4995	0.3774
	(0.58)	*(0.39)*	*(12.86)*	*(1.04)*	*(14.06)*	*(2.60)*	*(0.55)*	*(0.35)*	*(0.56)*	*(0.35)*
56	**0.1851**	**0.1603**	0.5255	0.2654	0.6605	0.3755	0.1855	0.1741	0.2069	0.1837
	(0.19)	*(0.14)*	*(0.83)*	*(0.26)*	*(1.13)*	*(0.37)*	*(0.14)*	*(0.12)*	*(0.19)*	*(0.15)*
52	**0.1273**	**0.0251**	0.1970	0.0349	0.2091	0.0348	0.2028	0.0385	0.1479	0.0280
	MA(10)									
31	**0.4567**	**0.3396**	4.4930	0.8503	8.4753	1.0758	0.5235	0.3952	0.4821	0.3615
	(0.74)	*(0.37)*	*(42.29)*	*(1.90)*	*(187.29)*	*(3.05)*	*(0.80)*	*(0.44)*	*(0.57)*	*(0.31)*
62	0.1826	**0.1559**	0.6602	0.2608	0.6194	0.3784	**0.1786**	0.1718	0.2055	0.1875
	(0.21)	*(0.13)*	*(5.25)*	*(0.34)*	*(0.97)*	*(0.31)*	*(0.13)*	*(0.12)*	*(0.20)*	*(0.15)*
124	**0.0634**	**0.0670**	0.0927	0.0918	0.1392	0.1378	0.0764	0.0876	0.0957	0.1075
	(0.06)	*(0.05)*	*(0.09)*	*(0.08)*	*(0.13)*	*(0.12)*	*(0.05)*	*(0.06)*	*(0.11)*	*(0.15)*

Chapter 6

MML87 Inference of ARMA Models

6.1 Introduction

The previous two chapters have examined inference of autoregressive (AR) and moving average (MA) models within an MML87 framework respectively. This chapter combines the previous work to find suitable message length expressions for the mixed autoregressive moving average (ARMA) model. This model class is popular because of its ability to more parsiminously model stationary time series than either the autoregressive or moving average models on their own. The ARMA(S,Q) explanation of measurement y_n of a time series is given by

$$\prod_{i=1}^{P}(1 - q^{-1}p_i)y_n = (1 + \sum_{i=1}^{Q} c_i q^{-i})v_n \tag{6.1}$$

where $P = \dim(\mathbf{p})$, S is the model structure indicator defined as per Section 4.3.4, and the innovations $\mathbf{v}$ are assumed to be i.i.d. as $v_n \sim \mathcal{N}(0, \sigma^2)$. As the name suggests, this model class includes both an auto-regressive component and a moving average component and generalises both classes. This chapter is organised as follows: ARMA parameter priors are reviewed in Section 6.2, Section 6.3 presents the likelihood function for ARMA models, Section 6.4 discusses a suitable Fisher Information Matrix approximation, Section 6.5 discusses estimation of MML87 parameters, Section 6.6 presents experimental results and Section 6.7 provides concluding remarks.

6.2 Priors

The complete prior density over the ARMA parameters is given by

$$h(\boldsymbol{\theta}) = h(\boldsymbol{\beta}) \cdot h(\mathbf{c}) \cdot h(S) \cdot h(Q) \cdot h(\sigma^2) \tag{6.2}$$

that is, the model contains both sets of parameters from its two sub-models. The parameter priors may then be chosen from those priors discussed in Sections 4.3 and 5.2. For completeness, a suitable set of general priors for the ARMA model, assuming uniform priors for radial and angular pole components are summarised below:

$$\begin{aligned}
h(S) &= \left(\sum_{i=1}^{P_{MAX}} (\lfloor i/2 \rfloor + 1) \right)^{-1} \\
h(Q) &= (Q_{MAX})^{-1} \\
h(\sigma^2) &= \frac{1}{\sigma^2} \left(\log(\sigma_U^2) - \log(\sigma_L^2) \right)^{-1} \\
h(\mathbf{c}) &= \frac{1}{\mathrm{vol}(\Lambda_c)} \\
h(\boldsymbol{\beta}) &= \left(\frac{1}{\pi} \right)^{N_C} \cdot \left(\frac{1}{2} \right)^{N_R} \cdot N_C! \cdot N_R!
\end{aligned}$$

where N_C and N_R are the number of complex and real poles of $\mathbf{a}$, σ_U^2 and σ_L^2 are upper and lower bounds on σ^2, P_{MAX} is the maximum permitted autoregressive order and Λ_c is the Q-parameter stationarity region, and may be found by the equations given in Section 5.2.

6.3 Likelihood Function

The most straightforward way to model a sequence with an ARMA process is through the unconditional likelihood function

$$L(\mathbf{y}|\boldsymbol{\theta}) = \frac{N}{2} \log(2\pi) + \frac{1}{2} \log |\mathbf{R}(\boldsymbol{\theta})| + \frac{1}{2} \mathbf{y} \mathbf{R}^{-1}(\boldsymbol{\theta}) \mathbf{y}^T \tag{6.3}$$

where $\mathbf{R}(\boldsymbol{\theta}) \in \mathbb{R}^{(N \times N)}$ is the ARMA process theoretical autocovariance matrix for N datapoints. This likelihood function may be evaluated efficiently in linear time via the Kalman Filter (see Appendix C).

6.4 The Fisher Information Matrix

The exact Information Matrix for an ARMA process is given by

$$J_{(i,j)}(\boldsymbol{\theta}) = \frac{1}{2}\mathrm{Tr}\left(\mathbf{R}^{-1}(\boldsymbol{\theta})\frac{\partial \mathbf{R}(\boldsymbol{\theta})}{\partial \theta_i}\mathbf{R}^{-1}(\boldsymbol{\theta})\frac{\partial \mathbf{R}(\boldsymbol{\theta})}{\partial \theta_j}\right) \tag{6.4}$$

Although straightforward, evaluation of (6.4) is slow unless special measures are taken (such as the algorithms in [133] and [148]), and more importantly, the resulting matrix is badly behaved for small N (just as in the case of the pure moving average model). This causes significant problems for the MML87 approximation that uses the log-determinant of $\mathbf{J}(\cdot)$ to determine the coding volume of the model. Therefore, in this formulation, the asymptotic Information Matrix is used in place of the exact matrix, just as was done previously for pure moving average models. To find the asymptotic Information matrix, begin by defining two auxilliary autoregressive processes r_n and s_n:

$$\prod_{i=1}^{P}(1 - q^{-1}p_i)r_n = \epsilon_n \tag{6.5}$$

$$s_n + \sum_{i=1}^{P} a_i s_{n-i} = -\epsilon_n \tag{6.6}$$

where $\boldsymbol{\epsilon}$ is the same Normally i.i.d. innovation sequence for both processes. Using Whittle's approximation as a basis, the Information sub-matrix for $\boldsymbol{\beta}$ is given by

$$\mathbf{J}_{(\boldsymbol{\beta},\boldsymbol{\beta})}(\boldsymbol{\theta}) = \begin{bmatrix} \mathrm{E}\left[\frac{\partial L(\mathbf{r}|\boldsymbol{\theta})}{\partial \beta_1}\frac{\partial L(\mathbf{r}|\boldsymbol{\theta})}{\partial \beta_1}\right] & \cdots & \mathrm{E}\left[\frac{\partial L(\mathbf{r}|\boldsymbol{\theta})}{\partial \beta_1}\frac{\partial L(\mathbf{r}|\boldsymbol{\theta})}{\partial \beta_P}\right] \\ \vdots & \ddots & \vdots \\ \mathrm{E}\left[\frac{\partial L(\mathbf{r}|\boldsymbol{\theta})}{\partial \beta_P}\frac{\partial L(\mathbf{r}|\boldsymbol{\theta})}{\partial \beta_1}\right] & \cdots & \mathrm{E}\left[\frac{\partial L(\mathbf{r}|\boldsymbol{\theta})}{\partial \beta_P}\frac{\partial L(\mathbf{r}|\boldsymbol{\theta})}{\partial \beta_P}\right] \end{bmatrix} \tag{6.7}$$

The Information sub-matrix for the moving average coefficients $\mathbf{c}$ is given by

$$\mathbf{J}_{(\mathbf{c},\mathbf{c})}(\boldsymbol{\theta}) = N \cdot \mathrm{E}\left[\mathbf{s}_{(1:Q)}^T \mathbf{s}_{(1:Q)}\right] \tag{6.8}$$

The Information sub-matrix for the cross-terms is given by

$$\mathbf{J}_{(\boldsymbol{\beta},\mathbf{c})}(\boldsymbol{\theta}) = \begin{bmatrix} \mathrm{E}\left[\frac{\partial L(\mathbf{r}|\boldsymbol{\theta})}{\partial \beta_1}s_{n-1}\right] & \cdots & \mathrm{E}\left[\frac{\partial L(\mathbf{r}|\boldsymbol{\theta})}{\partial \beta_1}s_{n-Q}\right] \\ \vdots & \ddots & \vdots \\ \mathrm{E}\left[\frac{\partial L(\mathbf{r}|\boldsymbol{\theta})}{\partial \beta_P}s_{n-1}\right] & \cdots & \mathrm{E}\left[\frac{\partial L(\mathbf{r}|\boldsymbol{\theta})}{\partial \beta_P}s_{n-Q}\right] \end{bmatrix} \tag{6.9}$$

The complete Information Matrix may then be found from the components as

$$\mathbf{J}(\boldsymbol{\theta}) = \begin{bmatrix} \mathbf{J}_{(\boldsymbol{\beta},\boldsymbol{\beta})}(\boldsymbol{\theta}) & \mathbf{J}_{(\boldsymbol{\beta},\mathbf{c})}(\boldsymbol{\theta})^T & \mathbf{0} \\ \mathbf{J}_{(\boldsymbol{\beta},\mathbf{c})}(\boldsymbol{\theta}) & \mathbf{J}_{(\mathbf{c},\mathbf{c})}(\boldsymbol{\theta}) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \frac{N}{2(\sigma^2)^2} \end{bmatrix} \tag{6.10}$$

The sub-matrix $\mathbf{J}_{(\boldsymbol{\beta},\boldsymbol{\beta})}(\boldsymbol{\theta})$ is the asymptotic Fisher Information Matrix of the pure autoregression characterised by root-space parameters $\boldsymbol{\beta}$, i.e. the auxilliary process in (6.5), and may be found by scaling the conditional Information Matrix presented in Section 4.5 by a factor of $N/(N-P)$. The sub-matrix $\mathbf{J}_{(\mathbf{c},\mathbf{c})}(\boldsymbol{\theta})$ is the asymptotic Information Matrix found by ignoring the autoregressive components of the ARMA process. The entries in sub-matrix $\mathbf{J}_{(\boldsymbol{\beta},\mathbf{c})}(\boldsymbol{\theta})$ are given by the cross-covariances between the first derivative of the negative log-likelihood of the auxilliary autoregression $\mathbf{r}$ and the other auxilliary autoregression $\mathbf{s}$.

This formulation of the Information Matrix removes the singularities as the roots of $\mathbf{a}$ approach the unit circle boundary but unfortunately does not avoid the issues due to the root-space parameterisation of the autoregressive process. However, unlike the case of the pure autoregressive model, a coefficient space representation of the autoregressive component is not without its own problems. It is a well known fact [149] that the asymptotic Information Matrix for an ARMA process in coefficient space is singular when the autoregressive and moving average polynomials contain shared roots, and is near singular in areas where these roots almost coincide. Thus, in the case of the mixed ARMA model, there appears to be no natural representation in which the Information Matrix is well behaved over the entire parameter space.

6.5 Parameter Estimation

Parameter estimation proceeds in a similar fashion to pure autoregressive or moving average processes. When using the scale-invariant prior, the MML estimates for the innovation variance may be found in closed form. The $\boldsymbol{\beta}$ and $\mathbf{c}$ estimates must be found numerically, with σ^2 estimated at each step of the process.

6.5.1 Maximum Likelihood Estimates of σ^2

To find the Maximum Likelihood estimator for σ^2 factorise the innovation variance out of the likelihood expression thus

$$L(\mathbf{y}|\boldsymbol{\theta}) = \frac{N}{2}\log(2\pi) + \frac{1}{2}\log|\mathbf{R}^*(\boldsymbol{\theta})| + \frac{N}{2}\log\sigma^2 + \frac{1}{2\sigma^2}\mathbf{y}\mathbf{R}^{*-1}(\boldsymbol{\theta})\mathbf{y}^T \tag{6.11}$$

where

$$\mathbf{R}^*(\boldsymbol{\theta}) = \frac{\mathbf{R}(\boldsymbol{\theta})}{\sigma^2} \tag{6.12}$$

is the process autocovariance matrix divided by the process variance. Differentiating (6.11) w.r.t. to σ^2 and solving yields the Maximum Likelihood estimator

$$\hat{\sigma}^2_{\mathrm{ML}}(\mathbf{y}) = \frac{\mathbf{y}\mathbf{R}^*(\boldsymbol{\theta})\mathbf{y}^T}{N} \tag{6.13}$$

This expression clearly involves manipulation of large matrices when N is large. If the Kalman Filter is used to evaluate the likelihood, the Maximum Likelihood estimator is given by

$$\hat{\sigma}^2_{\mathrm{ML}}(\mathbf{y}) = \frac{1}{N}\sum_{i=1}^{N}\frac{\hat{v}_i^2}{\hat{\sigma}_i^2} \tag{6.14}$$

where $\hat{\mathbf{v}}$ is the vector of estimated innovations, and $\hat{\boldsymbol{\sigma}}$ is the vector of estimated innovation standard deviations, as found by the Kalman Filter (with unity process innovation variance).

6.5.2 Minimum Message Length Estimates of σ^2

The MML estimator for σ^2 may be derived in a similar fashion to that for the pure AR and MA processes. Begin by noting the entries in sub-matrices $\mathbf{J}_{(\boldsymbol{\beta},\boldsymbol{\beta})}(\cdot)$ and $\mathbf{J}_{(\mathbf{c},\mathbf{c})}(\cdot)$ are linear functions of the theoretical autocovariances of the auxilliary sequences r_n and s_n, which are in turn linear functions of the innovation variance. Additionally, noting that the entries are also divided by the process autocoviance, the variance then cancels and disappears from both of these matrices. The log-determinant of the FIM may be written as

$$|\mathbf{J}(\boldsymbol{\theta})| = \left(|\mathbf{J}_{(\boldsymbol{\beta},\boldsymbol{\beta})}(\boldsymbol{\theta})| \cdot |\mathbf{J}_{(\mathbf{c},\mathbf{c})}(\boldsymbol{\theta})| - |\mathbf{J}_{(\boldsymbol{\beta},\mathbf{c})}(\boldsymbol{\theta})|^2\right) \cdot \frac{N}{2(\sigma^2)^2} \tag{6.15}$$

The message length may then be written as

$$I_{87}(\mathbf{y}, \boldsymbol{\theta}) = L(\mathbf{y}|\boldsymbol{\theta}) + \log \sigma^2 - \log \sigma^2 + K(\boldsymbol{\beta}, \mathbf{c}) \tag{6.16}$$

where $K(\boldsymbol{\beta}, \mathbf{c})$ are all the terms not dependent on σ^2. It can be seen that the prior term $(\log \sigma^2)$ and the Fisher term $(-\log \sigma^2)$ will cancel. Differentiating (6.16) with respect to σ^2 and solving yields

$$\hat{\sigma}^2_{87}(\mathbf{y}) = \frac{\mathbf{y}\mathbf{R}^*(\boldsymbol{\theta})\mathbf{y}^T}{N} = \hat{\sigma}^2_{\mathrm{ML}}(\mathbf{y}) \tag{6.17}$$

which is clearly coincides with the ML estimate. If the Kalman Filter is used to evaluate the negative log-likelihood, the estimator given by (6.14) may be used.

6.5.3 Minimum Message Length Estimates of $\boldsymbol{\beta}$ and c

The MML87 estimates for $\boldsymbol{\beta}$ and $\mathbf{c}$ cannot be found analytically and thus must be found by a numerical search. The Fisher Information Matrix of the $\boldsymbol{\beta}$ parameters is the same as conditional Information Matrix for the pure autoregressive model, and thus suffers from the same problems of ill-conditioning. The same diagonalisation-stabilisation technique is applied to this block of the complete FIM during the search, and the entries of the $\mathbf{J}_{(\boldsymbol{\beta},\mathbf{c})}(\boldsymbol{\theta})$ and $\mathbf{J}_{(\mathbf{c},\boldsymbol{\beta})}(\boldsymbol{\theta})$ blocks are all set to zero. The complete procedure for finding the MML87 estimates is then

1. Find initial coefficient estimates, $\hat{\mathbf{a}}_0$ and $\hat{\mathbf{c}}_0$, for an ARMA(P,Q) model using a suitable estimation scheme; in the case of this thesis, the ARMASel [73] or PEM estimates were used as they are very fast to compute.

2. Extract the root structure, $\hat{\boldsymbol{\beta}}_0$, of the initial estimates $\hat{\mathbf{a}}_0$.

3. Numerically search for the MML estimates, $\hat{\boldsymbol{\beta}}_{87}$ and $\hat{\mathbf{c}}_{87}$, within this given root structure, using the diagonal search modification of the conditional FIM matrix (see Section 4.6.1 for details), using the estimates $\hat{\boldsymbol{\beta}}_0$, $\hat{\mathbf{c}}_0$ as a starting point. The diagonal search modification is a simple robust approximation of the coding quantum, and in this case is given by

$$\frac{1}{2}\log|\mathbf{J}(\boldsymbol{\theta})| \approx \frac{1}{2}\sum_{i=1}^{P}\log\left(J_{(\beta_i,\beta_i)}(\boldsymbol{\theta}) + 1\right) + \frac{1}{2}\log|\mathbf{J}_{(\mathbf{c},\mathbf{c})}(\boldsymbol{\theta})| - \log(2\sigma^2) \tag{6.18}$$

If the scale-invariant prior is used the last term may be dropped as it will cancel with $-\log h(\sigma^2)$.

6.5.4 Comments on the MML87 ARMA Parameter Estimates

The ARMA model is formed as a combination of a moving average and an autoregressive process, and to this end the behaviour of the estimator is approximately the combined behaviour of the two estimators for the individual processes. Once again, the innovation variance has no effect on the accuracy to which the autoregressive and moving average parameters are stated, and the reason for this follows from a similar arguments to those presented in sections 4.8.1 and 5.6. A brief summary of the behaviour of the MML87 estimator, as compared to the ML estimator, follows:

- The poles and zeros of the MML87 estimates will be expected to be closer to the origin of the unit circle than the corresponding ML estimates due to the effect of the FIM term, and thus the resulting models will be expected to better capture longer term behaviour of the process.
- The moving average component of the FIM will eliminate the 'piling-up' effect from the MML87 estimates (see Section 5.8.3).
- The variance of the MML87 model will be slightly inflated as compared to the estimated variance of the ML model, due to the greater negative log-likelihood of the MML87 model.
- In a similar fashion to the AR and MA estimators, the MML87 estimator will converge to the ML estimator as $N \to \infty$, and is thus asymptotically efficient and consistent.

6.6 Results

The MML87 ARMA parameter estimation scheme was tested against the Maximum Likelihood estimates, the Matlab System Identification toolbox implementation of the Prediction Error Method and the ARMASel method [73]. Models were randomly generated by sampling from the priors (uniform over all root-space paramaters for the autoregressive component, uniform across the coefficients for the moving average component [147]).

For each (P, Q) order tested, three different sized sample sizes $N = \{k(3(P+Q)+1)\}$, $k = \{1, 2, 4\}$ were generated by simulating the one thousand randomly generated ARMA models. These sample sizes were chosen to provide a suitable range of data-to-parameter ratios, and were large enough for the Matlab implementation of the PEM method to be used. Experiments were performed for $(P, Q) = \{(2,1), (3,2), (4,3), (5,4), (6,5)\}$, which covered a wide range of orders. The orders were chosen so that the ARMASel estimator could be used, as that estimator requires that $(Q \leq P - 1)$. The estimated models were tested in terms of the normalised unconditional SPE_∞, SPE_1, KL_{100}, and J_{100} losses.

6.6.1 Parameter Estimation Experiments

The squared prediction errors and KL/J divergences of the estimated models are presented in Tables 6.2 and 6.3, with standard deviations shown in parenthesis. The main findings are now summarised.

- The MML87 estimates are almost uniformly the best in terms of both long-term and one-step ahead squared prediction errors. The ARMASel estimates attain overall higher squared prediction error scores (in some cases almost twice as large) but are a very significant improvement on the Maximum Likelihood and PEM estimates.

- The PEM estimates perform poorly in terms of all losses, particularly the KL divergence and J divergence scores. These arise from the very poor performance of a handful of estimated models, particularly those with zeros or poles extremely close to the unit circle. These models will yield large long-term squared prediction errors as they will generally grossly overestimate the model time-constants.

- The MML87 estimates largely outperform the ARMASel estimates in terms of KL/J divergences, but lose by a small margin in several cases. In all cases, for the smallest sample size, the MML87 estimates yield the lowest average KL/J divergences losses.

In summary, it can be seen that the MML87 estimates perform significantly better than the ML and PEM estimates, which suffer from problems with zeros and poles near the unit circle boundary. In no experiment undertaken was there any evidence of the 'piling up' effect being present in the MML87S estimates, and this is consistent with the findings in Section 5.8.3. Of the non-Bayesian estimation schemes, the ARMASel estimator appears to perform very well and the results suggest it is an excellent choice if a non-Bayesian

method is desired. Once again, given the chosen priors, the conventional MAP estimator coincides with the Maximum Likelihood estimator and is again seen to be outperformed by the MML87 estimator in terms of all losses.

6.6.2 Order Selection Experiments

A brief order selection experiment was then undertaken to determine if the problems that appeared when the MML87S criterion was used for order selection of pure AR and MA models would also be apparent in mixed ARMA model selection. One hundred $ARMA(P,Q)$ models, each with combined total number of parameters ($P+Q=4$) were sampled from the chosen parameter priors and each generated a dataset of size $N=20$. For each dataset, a set of candidate ARMA models $\{ARMA(P,Q) : P+Q \leq 7\}$ were fitted to the data. Each model selection criterion was then asked to nominate a candidate model from the set; the non-MML criteria using the models estimated by Maximum Likelihood, and the MML87S criterion using models estimated by MML87S, i.e. using the Fisher approximation given by (6.18). The results are presented in Table 6.1. The table

Table 6.1: Autoregressive Moving Average Order Identification Results

	Total Order (P+Q)										
Criterion	0	1	2	3	4	5	6	7	Correct	av(P)	av(P)
AIC	8	23	27	18	**6**	9	5	4	**6**	1.3700	1.2100
AICc	11	34	37	15	2	1	0	0	5	0.8300	0.8300
BIC	12	33	35	14	1	3	2	0	4	0.9300	0.8300
KICc	17	36	38	9	0	0	0	0	4	0.7300	0.6600
$MML87_S$	0	0	4	0	7	9	31	49	0	5.6500	0.4500

presents the number of times each criteria selected the correct total order, as well as the number of times each total order was chosen, and the number of times the exact correct order was chosen (i.e. when the nominated model was of the same P and Q order as the generating model). The total order is included as it is common for the 'complexity' of a model to be measured in terms of the total number of parameters, regardless of the type of parameters; this is the case with the AIC/BIC/KIC familes of model selection criteria, as an example. Furthermore, at least asymptotically, ARMA models of different individual P and Q orders but the same overall sum $(P+Q)$ yield similar predictive performance [141]. The average P and Q selected by each criteria are also displayed. The results show that the AIC criterion scores the highest number of correct total orders, but overfits more often

than the other non-MML methods, which all perform essentially the same. The MML87S criterion unfortunately overfits significantly, with almost all of the chosen models having more parameters than the generating models. The average P and Q of the models chosen by MML87S are also very biased in the favour of autoregressive components. This is explained by recalling that the MML87S estimates often possess small radial strength poles which violate the conditions under which the MML87S approximation is derived. This, coupled with the fact that the MA component is using the unconditional Fisher Information matrix which tends to overestimate the cost of the MA components means that the criterion tends to favour high order pure AR models over mixed ARMA structures of similar number of parameters. These results strongly suggest that a more robust message length approximation is required.

6.7 Conclusion

This chapter has examined the problem of parameter estimation and order selection of mixed autoregressive moving average models within the MML87 framework. The main results of this chapter are now summarised:

- A message length formulation for ARMA models based on the asymptotic Fisher Information Matrix was presented. The 'search' diagonalisation modification was introduced to avoid problems due to singularity of the Information Matrix due to coincident poles.

- The MML87 estimates using the 'Search' modification were compared against several conventional estimators and were shown to yield excellent results in terms of squared prediction errors and Kullback-Leibler losses.

- The MML87S message length approximation tended to overfit quite significantly when used for order selection. This was determined to be due to the MML87S estimates often yielding models which violate the assumptions under which the MML87S approximation was derived.

The next chapter introduces a new message length approximation that is designed to handle the conditions that violate the MML87 approximation for ARMA models.

Table 6.2: Autoregressive Moving Average Parameter Estimation Results, SPE Scores

	MML87		ML		PEM		ARMASel	
N	SPE_∞	SPE_1	SPE_∞	SPE_1	SPE_∞	SPE_1	SPE_∞	SPE_1
	ARMA(2,1)							
10	**0.3478**	**0.1587**	1.0295	0.2498	2.8582	0.3471	0.6119	0.1873
	(0.51)	*(0.20)*	*(1.69)*	*(0.31)*	*(8.75)*	*(0.45)*	*(0.98)*	*(0.24)*
20	**0.1992**	**0.0828**	0.3742	0.1065	0.6281	0.1279	0.2879	0.0893
	(0.24)	*(0.12)*	*(0.54)*	*(0.14)*	*(2.54)*	*(0.16)*	*(0.42)*	*(0.11)*
40	**0.1230**	**0.0402**	0.1839	0.0480	0.2946	0.0563	0.1474	0.0422
	(0.16)	*(0.05)*	*(0.27)*	*(0.07)*	*(1.49)*	*(0.07)*	*(0.20)*	*(0.05)*
	ARMA(3,2)							
16	**0.3083**	**0.1273**	0.9913	0.2669	2.3561	0.3196	0.5224	0.1561
	(0.39)	*(0.19)*	*(1.28)*	*(0.46)*	*(5.55)*	*(0.42)*	*(0.55)*	*(0.21)*
32	**0.1828**	**0.0698**	0.3683	0.1144	0.5847	0.1341	0.2566	0.0710
	(0.21)	*(0.10)*	*(0.43)*	*(0.17)*	*(1.32)*	*(0.24)*	*(0.29)*	*(0.09)*
64	**0.1058**	**0.0304**	0.1784	0.0445	0.2534	0.0532	0.1362	0.0340
	(0.14)	*(0.05)*	*(0.27)*	*(0.07)*	*(0.50)*	*(0.08)*	*(0.17)*	*(0.04)*
	ARMA(4,3)							
22	**0.3253**	**0.1108**	0.9942	0.2402	2.0175	0.3500	0.5325	0.1616
	(0.55)	*(0.16)*	*(1.01)*	*(0.35)*	*(4.31)*	*(0.56)*	*(0.49)*	*(0.27)*
44	**0.1869**	**0.0569**	0.4214	0.0893	0.6757	0.1251	0.2691	0.0647
	(0.21)	*(0.08)*	*(0.44)*	*(0.12)*	*(1.45)*	*(0.25)*	*(0.29)*	*(0.09)*
88	**0.1156**	**0.0270**	0.2040	0.0404	0.2436	0.0531	0.1412	0.0293
	(0.24)	*(0.04)*	*(0.30)*	*(0.06)*	*(0.41)*	*(0.09)*	*(0.18)*	*(0.04)*
	ARMA(5,4)							
28	**0.3887**	**0.0889**	0.9127	0.1833	1.7440	0.2835	0.5013	0.1190
	(1.63)	*(0.15)*	*(0.97)*	*(0.35)*	*(3.47)*	*(0.56)*	*(0.46)*	*(0.21)*
56	**0.1870**	0.0568	0.4122	0.0873	0.5922	0.1161	0.2688	**0.0543**
	(0.32)	*(0.16)*	*(0.47)*	*(0.24)*	*(1.76)*	*(0.19)*	*(0.29)*	*(0.10)*
112	**0.1070**	**0.0218**	0.1949	0.0338	0.2624	0.0473	0.1443	0.0219
	(0.13)	*(0.03)*	*(0.25)*	*(0.07)*	*(0.36)*	*(0.14)*	*(0.15)*	*(0.03)*
	ARMA(6,5)							
34	**0.3964**	**0.0862**	0.9399	0.1733	1.5634	0.2676	0.5125	0.1151
	(2.31)	*(0.14)*	*(0.89)*	*(0.30)*	*(2.72)*	*(0.45)*	*(0.47)*	*(0.21)*
68	**0.1821**	**0.0417**	0.4044	0.0753	0.5830	0.1092	0.2691	0.0500
	(0.20)	*(0.09)*	*(0.42)*	*(0.23)*	*(1.60)*	*(0.27)*	*(0.27)*	*(0.13)*
136	**0.1018**	**0.0194**	0.1901	0.0291	0.2249	0.0611	0.1335	0.0203
	(0.16)	*(0.03)*	*(0.27)*	*(0.05)*	*(0.29)*	*(0.64)*	*(0.16)*	*(0.03)*

Table 6.3: Autoregressive Moving Average Parameter Estimation Results, KL/J Divergences

	MML87		ML		PEM		ARMASel	
N	KL_{100}	J_{100}	KL_{100}	J_{100}	KL_{100}	J_{100}	KL_{100}	J_{100}
	ARMA(2,1)							
10	**0.5419**	**0.8381**	9.5568	9.8623	8.1035	12.544	0.8078	1.2365
	(1.74)	*(2.51)*	*(52.09)*	*(52.14)*	*(51.48)*	*(69.08)*	*(2.88)*	*(3.35)*
20	**0.1966**	**0.3168**	0.8700	1.0024	0.6449	0.9107	0.2066	0.3736
	(0.37)	*(0.46)*	*(3.03)*	*(3.06)*	*(2.32)*	*(2.60)*	*(0.36)*	*(0.51)*
40	0.0825	**0.1349**	0.1860	0.2474	0.1609	0.2807	**0.0773**	0.1525
	(0.20)	*(0.23)*	*(0.79)*	*(0.81)*	*(0.52)*	*(1.24)*	*(0.15)*	*(0.21)*
	ARMA(3,2)							
16	**0.4974**	**0.9087**	5.8858	6.3072	15.250	26.787	0.6976	1.2305
	(1.04)	*(3.65)*	*(21.3)*	*(21.4)*	*(172)*	*(208)*	*(1.54)*	*(2.87)*
32	0.2020	**0.3323**	0.9641	1.1338	0.7103	1.5380	**0.1822**	0.3997
	(0.37)	*(0.47)*	*(3.06)*	*(3.09)*	*(1.82)*	*(11.9)*	*(0.21)*	*(0.54)*
64	**0.0773**	**0.1330**	0.1878	0.2550	0.1556	0.2616	0.0791	0.1911
	(0.18)	*(0.21)*	*(0.57)*	*(0.59)*	*(0.29)*	*(0.38)*	*(0.12)*	*(0.24)*
	ARMA(4,3)							
22	**0.6388**	**0.9093**	8.8482	9.2200	10.934	19.195	0.8565	1.3014
	(5.74)	*(5.77)*	*(102)*	*(102)*	*(112)*	*(127)*	*(5.18)*	*(5.26)*
44	0.1697	**0.2865**	0.6154	0.7736	0.5469	0.9093	**0.1652**	0.3509
	(0.23)	*(0.30)*	*(1.14)*	*(1.19)*	*(1.13)*	*(1.64)*	*(0.17)*	*(0.34)*
88	**0.0628**	**0.1122**	0.1310	0.1984	0.1313	0.2295	0.0689	0.1608
	(0.07)	*(0.10)*	*(0.16)*	*(0.20)*	*(0.14)*	*(0.22)*	*(0.05)*	*(0.14)*
	ARMA(5,4)							
28	**0.3909**	**0.6466**	3.6759	4.0248	10.370	22.086	0.6324	1.0772
	(0.89)	*(0.97)*	*(15.20)*	*(15.25)*	*(143.17)*	*(168.65)*	*(1.86)*	*(2.06)*
56	**0.1539**	**0.2674**	0.4607	0.6133	0.4862	1.0208	0.1575	0.3280
	(0.17)	*(0.24)*	*(0.57)*	*(0.62)*	*(0.67)*	*(3.77)*	*(0.14)*	*(0.25)*
112	**0.0586**	**0.1052**	0.1113	0.1722	0.1288	0.2371	0.0656	0.1518
	(0.05)	*(0.08)*	*(0.11)*	*(0.13)*	*(0.12)*	*(0.25)*	*(0.04)*	*(0.11)*
	ARMA(6,5)							
34	**0.3702**	**0.6409**	2.5789	2.9244	3.8285	16.337	0.5227	1.0023
	(0.42)	*(0.87)*	*(8.23)*	*(8.26)*	*(26.4)*	*(188)*	*(1.00)*	*(2.37)*
68	**0.1563**	**0.2605**	0.3888	0.5317	0.4545	0.7501	0.1592	0.3208
	(0.22)	*(0.28)*	*(0.58)*	*(0.63)*	*(0.60)*	*(1.00)*	*(0.17)*	*(0.25)*
136	**0.0566**	**0.1011**	0.1016	0.1639	0.1182	0.2134	0.0649	0.1458
	(0.05)	*(0.08)*	*(0.08)*	*(0.11)*	*(0.11)*	*(0.24)*	*(0.05)*	*(0.11)*

Chapter 7

MMLD-S: A New Message Length Approximation

7.1 Chapter Overview

The previous three chapters presented an MML87 formulation of the ARMA model class and examined the performance of parameter estimation and model selection within this framework. Empirical results demonstrated that the parameter estimation capabilities of the MML87 criteria were good, generally yielding lower squared prediction errors and Kullback-Leibler divergences than the competing Maximum Likelihood and Burg estimators. Unfortunately, the order selection results were not quite as favourable when the MML87 estimates were used; this was primarily due to violations by the ARMA class of models of the assumptions upon which the MML87 criteria is based. This chapter presents a new MML approximation, dubbed MMLD-S (for *systematic sampling*), that is aimed at improving the Message Length estimates even in the face of the problems that the ARMA class of models presents while remaining computationally efficient.

7.1.1 Motivation for the I1F Approximation for ARMA Models

The main requirements for the MML87 message length to closely approximate the I1F message length (see Section 2.4) are:

- The priors over the uncertainty region $\boldsymbol{\Omega}$ are approximately flat, i.e.

$$\int_{\boldsymbol{\theta} \in \boldsymbol{\Omega}} h(\boldsymbol{\theta}) d\theta \approx h(\boldsymbol{\theta}) \cdot \mathrm{vol}(\Omega)$$

- The expected negative log-likelihood surface is locally quadratic around the point estimate $\boldsymbol{\theta}^*$ and within the region $\boldsymbol{\Omega}$, that is

$$\Delta(\boldsymbol{\theta}^* + \boldsymbol{\delta} || \boldsymbol{\theta}^*) \approx \frac{1}{2} \boldsymbol{\delta} \cdot \mathbf{J}(\boldsymbol{\theta}^*) \cdot \boldsymbol{\delta}^T, \text{ for } (\boldsymbol{\theta}^* + \boldsymbol{\delta}) \in \boldsymbol{\Omega}$$

  where $\mathbf{J}(\boldsymbol{\theta}^*)$ is the Fisher Information Matrix.

While the first assumption holds true, or at least may be adequately corrected for, the second assumption breaks down quite badly in the case of ARMA models for several reasons:

1. The quadratic assumption assumes that the KL surface is symmetric around the point estimate; ARMA models have hard boundaries at the edge of the parameter space where the KL approaches infinity. Thus, the KL curve is steeper in the direction of the stationarity boundary and shallower in the direction away from the stationarity boundary.

2. The quadratic assumption also assumes the curvature is constant over the entire parameter space; there are many combinations of parameters in an AR model that yield very flat KL curves over a small portion that then rise steeply as the parameters approach the stationarity boundary.

3. Minimisation of the trade-off between prior mass and excess coding cost in the MML87 formulation yields a solution that involves the log-determinant of the Fisher Information Matrix (FIM). The FIM for root-space autoregressive models with non-distinct roots is singular, which causes the MML87 approximation to break down.

Failure to take into consideration issue 1 may result in an underestimation of the coding volume and lead to a more conservative criterion. Failure to take into account considerations 2 and 3 may lead to gross overestimations of the coding volume and cause the approximation to catastrophically break down. The previous chapters applied several 'fixes' to the MML87 approximation that allow it to work adequately even when faced with these concerns. This chapter derives an approximation to the I1F equation that is explicitly designed to take into account all three issues.

7.1.2 Existing MMLD Algorithms

The I1F approximation has been newly proposed in this thesis and as such there have been no attempts to find efficient algorithms to evaluate the model cost integrals. However, the I1F approximation is closely related to the MMLD approximation proposed by Dowe [1], and there have been several methods proposed to find this message length. Under the MMLD approximation the message length required to briefly transmit data $\mathbf{y}$ relative to some model class is

$$I_{1\mathrm{D}}(\mathbf{y}) = -\log q(\boldsymbol{\Omega}) + \frac{1}{q(\boldsymbol{\Omega})} \int_{\boldsymbol{\theta}\in\boldsymbol{\Omega}} h(\boldsymbol{\theta}) L(\mathbf{y}|\boldsymbol{\theta}) d\boldsymbol{\theta} \tag{7.1}$$

where $L(\cdot)$ is the negative-log likelihood function and

$$q(\boldsymbol{\Omega}) = \int_{\boldsymbol{\theta}\in\boldsymbol{\Omega}} h(\boldsymbol{\theta}) d\boldsymbol{\theta} \tag{7.2}$$

is the prior-mass in the region $\boldsymbol{\Omega}$. There are two existing algorithms for implementing the MMLD approximation:

- The MMC (Message from Monte-Carlo) algorithm by Fitzgibbon [13] (described in greater detail in Section 2.3.3).
- The MMC$_{em}$ algorithm by Makalic [35].

Both of these methods are based on Monte-Carlo sampling to evaluate the integrals in (7.1). The MMC algorithm does this by importance sampling from the posterior distribution, and subsequently growing the region $\boldsymbol{\Omega}$ using the boundary rule to determine when the region growth should end. This method has the advantage of being able to handle regions of arbritary shape and group models of different model class, but suffers from the considerable time required to sample a suitable number of models from the posterior, particularly as the number of model parameters grows. The MMC$_{em}$ algorithm works slightly differently; it assumes that the geometry of the region is fixed as a hyperellipsoid, whose relative dimensions in each axis are determined by the eigenvalue decomposition of the FIM. The overall volume of the region increases or decreases based on a single scale factor K, and the integration is performed by fast Monte-Carlo sampling from within the hyperellipsoid. This method is significantly faster than Fitzgibbon's MMC algorithm, and

shows promising results on model classes with an unconstrained parameter space (in particular, MLP artificial neural networks) but like the MML87 approximation, it assumes the KL curves are symmetric, and furthermore it requires a large number of evaluations of the negative log-likelihood.

7.1.3 A Non-Sampling Based Method for One Parameter

There has been a small amount of work done on implementing the MMLD approximation without using sampling technqiues. Lam [150] proposed an algorithm for determining the MMLD region for a one parameter model using quadrature integration and a bisection search over the parameter space; this also yielded promising results for the model class it was applied to (the binomial model), though the problem of choosing higher dimensional region shapes was acknowledged.

7.1.4 A New Message Length Approximation

The new Message Length approximation is derived specifically to address the problems that cause MML87 to perform badly. The basic idea behind the approximation is to circumvent the quadratic assumption by examining the Kullback-Leibler divergence curves along a set of basis 'rays' through the parameter space. This curves are discretised into a set of discrete segments that provide a mechanism to allow the region to grow in an asymmetric manner while simultaneously computing the excess round-off cost integral. The MML87 approximation is popular because its time complexity is essentially constant with respect to the number of dimensions; however, this computational efficiency is only achieved at the expense of some restrictive assumptions. The MMLD-S algorithm relaxes the assumptions while still providing a solution that has a time complexity that varies only linearly with the number of model parameters; this modest increase in time complexity is considered a suitable price to pay for a great increase in robustness. This chapter begins with a discussion of the algorithm for models with a single free parameter, and then details the extensions required to handle multi-parameter models.

7.2 The MMLD-S Algorithm for Single Parameter Models

7.2.1 Genesis

This thesis has introduced a modification to the MMLD approximation to allow it to approximate the assertion cost and excess negative log-likelihood of a model θ^* without recourse to the data; this is termed the I1F (see Section 2.4) approximation. The 'model cost' component of the I1F message length expression is given by

$$\mathcal{D}_{1\mathrm{F}}(\theta^*) = -\log q(\theta^*) + \frac{1}{q(\theta^*)} \int_{\theta \in \Omega} h(\theta) \Delta(\theta || \theta^*) d\theta \tag{7.3}$$

where $\Delta(\theta||\theta^*)$ is the Kullback-Leibler divergence from 'true' model θ^* to approximating model θ, and $\Omega \subset \Theta$ is chosen to minimise (7.3). This function gives the assertion cost plus the expected excess negative log-likelihood of coding the data $\mathbf{y}$ using a suboptimal (in a negative log-likelihood sense) model. The complete message length is then given by

$$I_{1\mathrm{F}}(\mathbf{y}, \theta^*) = L(\mathbf{y}|\theta^*) + \mathcal{D}_{1\mathrm{F}}(\theta^*) \tag{7.4}$$

Under the single weak assumption that the KL divergence curve formed by varying the approximating model is uni-modal for all models (which is common) in the model space, the exact shape of region that minimises (7.3) is known to be (and indeed, may be nothing else than) a simple contiguous segment of the parameter line, i.e. $\Omega = \{\theta : \Omega^- \leq \theta \leq \Omega^+\}$.

7.2.2 Approximating the Model Cost

Assume there is parametric density function, $f(\mathbf{y}|\theta)$. The task is to find the model cost of a model θ^* from this family. The MMLD-S algorithm works by subdividing the KL divergence curve into discrete segments, and subsequently forming the region as a contiguous group of segments (each representing an interval of models in the model space); the integrals in (7.3) may then be approximated via summations of the set of segments. Assume there is an ordered set of $(N_P + 1)$ 'cuts' in the parameter space, $\mathbf{p} = \left\{p_1, ..., p_{(N_P+1)}\right\} \subset \mathbb{R}^{N_P}$. Also assume there is a set of associated KL divergences, $\mathbf{K}$, between the models at the 'cuts' and the model θ^*

$$\mathbf{K} = \left\{ \frac{1}{2} \left(\Delta(p_1||\theta^*) + \Delta(p_2||\theta^*)\right), \ldots, \frac{1}{2} \left(\Delta(p_{N_P}||\theta^*) + \Delta(p_{(N_P+1)}||\theta^*)\right) \right\} \tag{7.5}$$

and prior masses, $\mathbf{h}$

$$\mathbf{h} = \left\{ \int_{p_1}^{p_2} h(\theta)d\theta, \ldots, \int_{p_{N_P}}^{p_{(N_P+1)}} h(\theta)d\theta \right\} \tag{7.6}$$

There are therefore N_P segments, with the first and last 'cuts' forming the models bounding the area of the parameter space under consideration. The region, $\mathcal{R}$, can now be described as the set of the intervals that constitute the region, i.e.

$$\mathcal{R} \subset \{1, 2, \ldots, N_P\} \tag{7.7}$$

$$\forall i \in \mathcal{R}, \exists j \in \mathcal{R} : |i - j| \quad = \quad 1 \tag{7.8}$$

The restriction given by (7.8) ensures that the region spans a contiguous patch of the parameter space, Θ. The model cost of the region can then be approximated as

$$\mathcal{D}_{1\mathrm{F}}(\theta^*) \approx \mathcal{D}_{\mathrm{DS}}(\theta^*) = -\log \sum_{i \in \mathcal{R}} h_i + \frac{1}{\sum_{i \in \mathcal{R}} h_i} \sum_{i \in \mathcal{R}} h_i K_i \tag{7.9}$$

where $\mathcal{D}_{\mathrm{DS}}(\cdot)$ denotes this discrete approximation. The region $\mathcal{R}$ may now be determined by the discrete stepwise growth of the segmented KL/prior curve, i.e. by subsequent addition of the segments to the region until (7.9) is minimised. The discretisation of the KL and prior curves and the growth algorithm are detailed in the next subsections.

7.2.3 Determining the Uncertainty Region

The process of determining the region consists of the following steps

1. The *discretisation step*: successively 'cut' the KL curve N_K times to refine the piecewise linear approximation
2. The *interpolation step*: interpolate the discretised curve into N_P new segments
3. The *growth step*: grow the region from the N_P segments

These steps are now discussed.

The Discretisation Step

The discretisation step cuts the parameter line into intervals that are subsequently used to grow the uncertainty region and approximate the integral. The algorithm requires the specification of an initial span of the parameter line which will be considered for

discretisation such that $(\theta_L < \theta^* < \theta_U)$. As only a limited number of discretisations will occur, it is important to ensure they cover the area of the parameter axis that is most likely to contain the uncertainty region. For regular problems, the round-off cost of the I1F message length is expected to asymptotically converge to the MML87 round-off cost of $(P/2)$ nits; for finite N this may be larger, though empirical evidence suggested it rarely exceeds P nits. Thus, if attention is restricted to the area of the parameter line for which the KL divergences are less than some multiple of $(P/2)$ the region of the parameter space that is discretised will generally be expected to contain a large number of the segments that will be included in the final uncertainty region. A new segment of the parameter line $[\theta^-, \theta^+]$ is found over which the discretisation will occur such that the new endpoints satisfy

$$P \leq \Delta(\theta^- || \theta^*) \leq 2P, \theta^- \in [\theta_L, \theta^*) \tag{7.10}$$

and

$$P \leq \Delta(\theta^+ || \theta^*) \leq 2P, \theta^+ \in [\theta^*, \theta_U) \tag{7.11}$$

If no point in the parameter line satisfies these conditions, then the boundaries θ_L and θ_U may be used instead. A straightforward way to find the points θ^- and θ^+ is by a simple bisection search. Once these two points have been found the resulting discretised parameter vector $\mathbf{p}$ is simply found by uniformly dividing either side of θ^* into $NK/2$ points, i.e.

$$\mathbf{p} = \left[\mathbf{p}^{\{1\}}, \mathbf{p}^{\{2\}}\right] \tag{7.12}$$

where

$$\mathbf{p}^{\{1\}} = \theta_L + \left[0, \frac{(\theta^* - \theta_L)}{2NK}, \frac{2(\theta^* - \theta_L)}{2NK}, \ldots, \frac{(\frac{NK}{2} - 1)(\theta^* - \theta_L)}{2NK}, (\theta^* - \theta_L)\right] \tag{7.13}$$

and

$$\mathbf{p}^{\{2\}} = \theta^* + \left[\frac{(\theta_U - \theta^*)}{2NK}, \frac{2(\theta_U - \theta^*)}{2NK}, \ldots, \frac{(\frac{NK}{2} - 1)(\theta_L - \theta^*)}{2NK}, (\theta_U - \theta^*)\right] \tag{7.14}$$

The KL divergence vector $\mathbf{K}$ and prior mass vector $\mathbf{h}$ may then be found using (7.5) and (7.6).

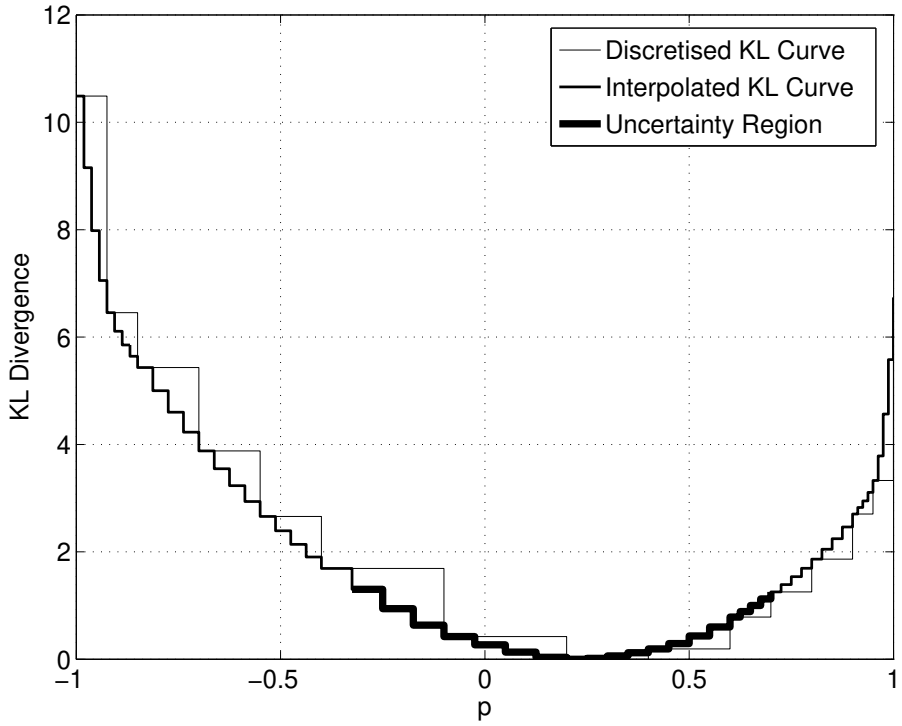


Figure 7.1: Discretised/Interpolated KL Curves and Estimated Uncertainty Region for an AR(1) Model with $N_K = 10$

The Interpolation Step

Once the refinement step has produced a suitable systematic sampling of KL divergence curve a further interpolation step is taken. This step increases the resolution of the segments, and improves the accuracy of the growth step dramatically. Additionally, the refinement step takes into consideration only the KL divergence curve; if the prior density is heavily curved the discretised segments will not necessary offer a good approximation of the prior curve. There are various choices for interpolation functions; in this thesis, the cubic spline interpolation method was used as it is quick to compute and offers a suitably smooth interpolation [151]. The selection of points for interpolation is simply chosen by uniformly subdividing the segments.

The Growth Step

At this stage, the vectors of cutpoints $\mathbf{p}$, KL divergences $\mathbf{K}$ and prior masses $\mathbf{h}$ are now available. The growth step begins by seeding the region

$$\mathcal{R} = \left\{ j : p_j \leq \theta^* \leq p_{(j+1)} \right\} \tag{7.15}$$

that is, the region initially contains only the segment nearest the point estimate. Given that the region $\mathcal{R}$ initially contains a single segment, the initial model cost is simply given by

$$\mathcal{D}_{\mathrm{DS}}(\mathcal{R}) = -\log h_j + K_j \tag{7.16}$$

The algorithm then proceeds as follows

1. Set $D \leftarrow \mathcal{D}_{\mathrm{DS}}(\mathcal{R})$ (save the cost)

2. If $(\sup\{\mathcal{R}\} < N_P)$, grow the region as $\mathcal{R}' \leftarrow \{\mathcal{R}, (\sup\{\mathcal{R}\} + 1)\}$; otherwise proceed to step 4.

3. If the new model cost $\mathcal{D}_{\mathrm{DS}}(\mathcal{R}') < \mathcal{D}_{\mathrm{DS}}(\mathcal{R})$, set $\mathcal{R} \leftarrow \mathcal{R}'$.

4. If $(\inf\{\mathcal{R}\} > 1)$, grow the region as $\mathcal{R}' \leftarrow \{(\inf\{\mathcal{R}\} - 1), \mathcal{R}\}$; otherwise proceed to step 6.

5. If the new model cost $\mathcal{D}_{\mathrm{DS}}(\mathcal{R}') < \mathcal{D}_{\mathrm{DS}}(\mathcal{R})$, set $\mathcal{R} \leftarrow \mathcal{R}'$.

6. If the model cost $\mathcal{D}_{\mathrm{DS}}(\mathcal{R}) < D$, return to step 1. Otherwise, terminate the algorithm.

The algorithm terminates when the growth of the region by the addition of a single new segment fails to reduce the cost. After the algorithm has ended, the complete message length may be found as

$$I_{1\mathrm{F}}(\mathbf{y}, \theta^*) \approx D + L(\mathbf{y}|\theta^*) \tag{7.17}$$

where $L(\mathbf{y}|\boldsymbol{\theta}^*)$ is the negative log-likelihood of the data. The resulting uncertainty region is given by

$$\hat{\Omega} = \left\{\theta : p_{(\inf\{\mathcal{R}\})} \leq \theta \leq p_{(\sup\{\mathcal{R}\})}\right\} \tag{7.18}$$

Figure 7.1 shows an example of the discretised and interpolated KL curve of an AR(1) model, with $N_K = 10$. The light line shows the initial discretisation into segments performed in the discretisation step and the medium weight line shows the discrete segments after the interpolation step. The heavy line shows the approximated uncertainty region found after the growth step.

7.2.4 Time Complexity of the Algorithm

The overall time complexity of the may be roughly determined by examining each step invidually. Assume that the time complexity of the KL divergence function is given by $O(\Delta)$. The time complexites of each step are given by

- **Refinement Step** $\approx N_K O(\Delta) + N_K^2$
- **Interpolation Step** $\approx N_P$
- **Growth Step** $\approx 2N_P$

In most cases the evaluation of the KL divergence represents the single most expensive component of the process, and dominates the other terms. In this case, the time complexity is

$$N_K \cdot \mathcal{O}(\Delta) \tag{7.19}$$

The larger the number of segments, N_K, in the refinement step, the better the accuracy of the interpolation curve, and thus the of cost, but the more expensive (in terms of computation time) the algorithm becomes.

7.2.5 Properties of the MMLD-S One Parameter Estimator

As the number of cuts $N_K \to \infty$, $\mathcal{D}_{\mathrm{DS}}(\cdot) \to \mathcal{D}(\cdot)$, i.e. the MMLD-S message length estimate converges to the exact MMLD message length. This is straightforward to demonstrate; as N_K increases, the distance between each cut shrinks at roughly a rate of $1/N_K$. As the message length approximation equation (7.9) represents a Riemannian integral the average bin width approaches zero as $N_K \to \infty$, and thus for continuous curves the Riemman sum converges to the integral. As the MMLD-S message length converges to the I1F message length, the MMLD-S criterion inherits the properties of the I1F estimator, such as invariance under one-to-one transformations and data invariance, at least asymptotically in N_K.

7.3 The MMLD-S Algorithm for Multi-Parameter Models

This section now describes the extension of the MMLD-S algorithm to multiple parameter models. This proves to be a reasonably straightforward extension of the process for the

one dimensional model class, with several important differences. Recall the definition of the I1F model cost for multiple parameters:

$$\mathcal{D}_{1\mathrm{F}}(\boldsymbol{\theta}) = -\log q(\boldsymbol{\theta}^*) + \frac{1}{q(\boldsymbol{\theta}^*)} \int_{\boldsymbol{\theta}\in\boldsymbol{\Omega}} h(\boldsymbol{\theta})\Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*)d\boldsymbol{\theta} \tag{7.20}$$

where the uncertainty region $\boldsymbol{\Omega} \subset \boldsymbol{\Theta}$ is chosen to minimise (7.20). The main difference between the multi-parameter and single parameter model cost is that the region $\boldsymbol{\Omega}$ is no longer a simple patch of the parameter line, but some higher dimensional shape. The approximation of the model cost for multiple parameter model classes under the MMLD-S algorithm is found by exploiting the MML boundary rule, and by assuming the region $\boldsymbol{\Omega}$ is adequately described by a regular geometry in parameter space; this method is more restrictive than for the one-dimensional case, due to the increased complexity of finding closed regions in higher dimensional space. The assumptions required for the approximation to be valid are

- The priors are roughly flat around the point estimate, so that $\int_{\boldsymbol{\theta}\in\boldsymbol{\Omega}} h(\boldsymbol{\theta})d\boldsymbol{\theta} \approx \mathrm{vol}(\boldsymbol{\Omega})h(\boldsymbol{\theta}^*)$. This assumption may be relaxed with appropriate modifications presented later, though computational requirements of the approximation will increase.

- The KL divergence surface around $\boldsymbol{\theta}^*$ is locally bowled; that is, in all directions away from $\boldsymbol{\theta}^*$ the KL divergence increases.

- That the volume of the region $\boldsymbol{\Omega}$ is roughly proportional to a hyper-rectangle, the constant of proportionality depending only on the dimension of the model class (and possibly, the number of data points).

These are less restrictive assumptions than the MML87 approximation; in particular, they relax the requirement of the locally quadratic KL divergence surface and its associated problems of non-symmetric coding regions, and break downs in the approximation due to flat spots and singularities.

7.3.1 Outline of the Algorithm

The process is quite similar to the one parameter case. A set of KL curves are extracted from the KL surface, and a discrete growing algorithm is applied to construct the region. The main differences, however, are that the KL curves do not necessarily lie on the natural

parameter axis, and that the region must now be a higher dimensional geometry rather than a simple segment of the parameter line, which complicates the evaluation of the expected round-off cost. The method exploits the boundary rule that states the boundary of an I1F region must lie on a level set of the KL surface. Thus, assuming that the boundary of the region shape selected roughly corresponds to the boundary of the uncertainty region, and that the region is grown in such a manner that the boundary always lies on (or close to) a level set, the integration may be performed by successive summation of the boundaries as the region is grown, i.e. a crude form of Lebesgue integration.

Let P denote the number of model parameters, i.e. $P = \dim(\boldsymbol{\theta}^*)$. Under the assumption that the volume of the region is proportional to a hyper-rectangle, there is only the need to determine the P orthogonal spans that describe the hyper-rectangle. Two important advantages arise from this approximation. Given that only P spans are required to describe the region, the time complexity of the algorithm is linearly proportional to the number of dimensions. The second advantage is that the discretisation, as in the single parameter algorithm, provides a natural and straightforward method in which to grow the region in a non-symmetric fashion, while simultaneously evaluating the integral. The main steps involved in the MMLD-S algorithm are:

1. Determination of the parameter rays
2. Discretisation of the KL curves along the parameter rays
3. Determination of the region and integration via the Growth Step

The first step is required as simply selecting the natural parameter axis as the rays along which to discretise the curves in parameter space will in most cases be a sub-optimal choice. The second step produces the discrete KL curves along the parameter rays, and these are used in the third step along with a suitable region geometry to estimate the model cost. All these steps are discussed in detail in the next subsections.

7.3.2 Describing the Region

Any method attempting to solve the MMLD or I1F message length equations, even approximately, must somehow choose a suitable region shape, either explicitly or implicitly. The MMLD-S approximation is no exception. A summary of the various approximations to the MMLD/I1F message length equation follows:

- MML87: The MML87 (or Wallace-Freeman) approximation [15] selects the region to be congruent to a scaled, translated and rotated cell of an optimal quantising lattice in P-dimensions. This choice, coupled with the assumptions of quadratricity of the KL divergence surface leads to closed form solutions of the I1F message length equation (see Section 2.5 for more details).

- MMC: The MMC algorithm [41] does not explicitly choose any shape for the region; instead, the shape is implicitly defined by the models that are included in the region. This leads to a method that is flexible, but at the same time, it is difficult to quantify the region, especially in higher dimensions where simply plotting the samples becomes impractical.

- MMC$_{em}$: The MMC$_{em}$ approximation [35] makes a specific choice that the region will be a hyper-ellipse, the relative dimensions of which are determined by the eigenvalues of the Information Matrix. The only free parameter then is the scale of the ellipse; clearly, within this framework, the regions will always be symmetric.

None of these choices are appropriate for the MMLD-S algorithm. The choice made by MML87 is driven by the simplifying assumptions made about the KL divergence surface; in contrast, the MMLD-S algorithm is based on weaker assumptions. Although the MMC$_{em}$ approximation makes no assumptions about the likelihood surface, the evaluation of the integrals is performed by Monte-Carlo methods and thus will yield sensible values regardless of the shape of the region. The MMLD-S algorithm ties the evaluation of the integral to the growth of the region, and in particular has been designed to allow for asymmetric region shapes. Thus a more flexible specification of the region shape must be adopted.

7.3.3 The Least Bounding Hyperbox

The MMLD-S algorithm defines the approximate uncertainty region $\hat{\boldsymbol{\Omega}}$ and computes the model cost by examining only those models that are constrained to lie along P orthogonal rays through the parameter space $\boldsymbol{\Theta} \subset \mathbb{R}^P$. For convenience, it is easier to construct the uncertainty region in the *uncertainty region* space $\boldsymbol{\Theta}' \subset \mathbb{R}^P$, in which the rays are chosen to be aligned with the dimension axis, and to transform back to the parameter space (where the rays are no longer necessarily aligned with the dimension axis) when required

by the relation

$$\boldsymbol{\theta} = \boldsymbol{\theta}'\mathbf{B}^T + \boldsymbol{\theta}^* \tag{7.21}$$

where $\mathbf{B} = \left[\mathbf{b}_1^T, \ldots, \mathbf{b}_P^T\right]^T \in \mathbb{R}^{(P\times P)}$ are a set of orthonormal basis vectors used to define the rays in parameter space, and $\boldsymbol{\theta}^*$ is the point estimate for which the model cost is sought. An appropriate choice of $\mathbf{B}$ is considered later in this chapter; for the time being, assume that a suitable $\mathbf{B}$ exists. Let $(\cdot)'$ denote a quantity in the uncertainty region space, with the unprimed variant denoting the same quantity in the parameter space. The MMLD-S approximation makes the assumption that the uncertainty region will be widest in each dimension along the dimension axis; the uncertainty region may then be defined by P contiguous sets of the reals that will determine the maximum width of the region in each of the P dimensions. Let $\boldsymbol{\Xi} = \{\boldsymbol{\xi}_1 \subset \mathbb{R}, \ldots, \boldsymbol{\xi}_P \subset \mathbb{R}\}$ be the set of contiguous subsets along each of the uncertainty region axis' such that

$$\bigcap_{i=1}^{P} \boldsymbol{\xi}_i = 0 \tag{7.22}$$

and

$$\boldsymbol{\xi}_i = \left\{\xi : -r_{(i,1)} \leq \xi \leq r_{(i,2)}\right\} \tag{7.23}$$

The vectors $\mathbf{r}_i = \left[r_{(i,1)}, r_{(i,2)}\right] \in \mathbb{R}_+^2$ contain the absolute values of the lower and upper bounds of each set $\boldsymbol{\xi}_i$ in dimension i; these may be formed into the matrix $\mathbf{R} = \left[\mathbf{r}_1^T, \ldots, \mathbf{r}_P^T\right]^T \in \mathbb{R}^{(P\times 2)}$. Let $\boldsymbol{\mathcal{H}}'$ denote the minimum volume hyper-box that contains all the points referenced by all the segments, i.e.

$$\boldsymbol{\mathcal{H}}' = \left\{\boldsymbol{\theta}' : -r_{(1,1)} \leq \theta_1' \leq r_{(1,2)},\ -r_{(2,1)} \leq \theta_2' \leq r_{(2,2)},\ \ldots,\ -r_{(P,1)} \leq \theta_P' \leq r_{(P,2)}\right\} \tag{7.24}$$

The volume of this hyper-box is given by

$$\text{vol}(\boldsymbol{\mathcal{H}}') = \prod_{i=1}^{P} \left(r_{(i,1)} + r_{(i,2)}\right) \tag{7.25}$$

For any region $\boldsymbol{\Omega}' \subset \boldsymbol{\Theta}'$ in uncertainty region space, the resulting region $\boldsymbol{\Omega} \subset \boldsymbol{\Theta}$ in the parameter space is given by

$$\boldsymbol{\Omega} = \left\{\boldsymbol{\theta} : (\boldsymbol{\theta} - \boldsymbol{\theta}^*)\mathbf{B}^T \in \boldsymbol{\Omega}'\right\} \tag{7.26}$$

Due to the orthonormality of $\mathbf{B}$, it follows that $|\mathbf{B}| = 1$ and therefore that

$$\text{vol}(\boldsymbol{\Omega}) = \text{vol}(\boldsymbol{\Omega}') \tag{7.27}$$

i.e. that the transformation leaves the scale (and thus, the volume) of the region unchanged. Additionally, as the scale is unchanged, the width of the sets $\boldsymbol{\xi}_i$ also corresponds to the width of the sets in parameter space, and thus to the width of the region along each parameter ray i. The set of models $\boldsymbol{\Theta}^{\boldsymbol{\xi}_i}$ along parameter ray i in parameter space that the set $\boldsymbol{\xi}_i$ maps to is given by

$$\boldsymbol{\Theta}^{\boldsymbol{\xi}_i} = \{\boldsymbol{\theta} \in \boldsymbol{\Theta} : \boldsymbol{\theta} = \xi \mathbf{b}_i + \boldsymbol{\theta}^*, \xi \in \boldsymbol{\xi}_i\} \tag{7.28}$$

where $\mathbf{b}_i$ is the direction of the ray associated with set $\boldsymbol{\xi}_i$. It is sufficient (and convenient) to work in the uncertainty region space if only the volume of the region is required. The hyperbox $\boldsymbol{\mathcal{H}}$ is one possible choice for the P-dimensional region in the MMLD-S algorithm; it is completely defined by the segments along the region axis and allows for segments to grow more on one side of $\boldsymbol{\theta}^*$ than the other. However, investigations of the regions produced by Random Coding indicate that a hyperbox will rarely provide a close approximation to the true region shape, particularly as N and P grow.

7.3.4 Non-rectangular Regions

If the P axis' of the uncertainty region space are used to form binary decision planes, the hyperbox $\boldsymbol{\mathcal{H}}'$ may be subdivided into exactly 2^P disjoint hyperboxes, $\boldsymbol{\mathcal{H}}'_1, \ldots, \boldsymbol{\mathcal{H}}'_{(2^P)}$. The index space of the leaves of the resulting binary tree is $\boldsymbol{\Psi} = \{1, 2\}^P$, with 1 denoting a cut to the left (the negative half), and 2 denoting a cut to the right (the positive half) of the appropriate axis. Clearly the index space is finitely countable and contains only 2^P unique elements; define the one-to-one mapping $\boldsymbol{\psi} : \{1, \ldots, 2^P\} \rightarrow \boldsymbol{\Psi}$ by

$$\boldsymbol{\psi}(k) = [\psi_1(k), \ldots, \psi_P(k)] \tag{7.29}$$

$$\psi_i(k) = \text{mod}\left(\lfloor k/2^{(i-1)} \rfloor, 2\right) + 1 \tag{7.30}$$

The set of all 2^P sub hyper-boxes formed by the binary subdivision of $\boldsymbol{\mathcal{H}}'$ is given by

$$\boldsymbol{\mathcal{H}}'_k = \left\{ \boldsymbol{\theta}' : \theta'_1 \in \boldsymbol{\xi}_1^{\{\psi_1(k))\}},\ \ldots,\ \theta'_P \in \boldsymbol{\xi}_P^{\{\psi_1(k))\}} \right\},\ \left(k = 1, \ldots, 2^P\right) \tag{7.31}$$

where

$$\boldsymbol{\xi}_i^{\{k\}} = \begin{cases} \left\{\xi : -r_{(i,1)} \leq \xi \leq 0\right\} & \text{for } k = 1 \\ \left\{\xi : 0 \leq \xi \leq r_{(i,2)}\right\} & \text{for } k = 2 \end{cases} \tag{7.32}$$

are the segments $\boldsymbol{\xi}_i$ split into two subsets, one on either side of the origin, for each dimension i. Let the matrix $\mathbf{T} \in \mathbb{R}^{\left(2^P \times P\right)}$ contain the width of each of the sub-hyperboxes in each dimension. The entries of $\mathbf{T}$ are given by

$$t_{(k,i)} = r_{(i,\psi_i(k))},\ \left(k = 1, \ldots, 2^P, i = 1, \ldots, P\right) \tag{7.33}$$

$t_{(k,i)}$ then denotes the width of the k-th hyperbox along dimension i. It is now possible to define the region $\hat{\boldsymbol{\Omega}}$ as a union of subsectors of shapes other than hyper-boxes; this may be done by inscribing a subsector of an alternative shape, denoted $\boldsymbol{\mathcal{S}}'_k$, into each of the 2^P hyper-boxes, each subsector having $\left(1/2^P\right)$-th of the volume of the complete shape from which it is taken. The volume of the k-th subsector $\boldsymbol{\mathcal{S}}'_k$ inscribed into the k-th hyperbox $\boldsymbol{\mathcal{H}}'_k$ is

$$\begin{aligned} \text{vol}(\boldsymbol{\mathcal{S}}'_k) &= \kappa(P) \prod_{i=1}^{P} t_{(k,i)} && (7.34) \\ &= \kappa(P) \text{vol}(\boldsymbol{\mathcal{H}}'_k) && (7.35) \end{aligned}$$

where $\kappa(P)$ is a geometric scaling constant, dependent only on the number of dimensions and the shape which is being inscribed. The approximated uncertainty region is now defined by a union of all 2^P subsector regions $\boldsymbol{\mathcal{S}}'_k$, i.e. $\hat{\boldsymbol{\Omega}}' = \bigcup_{k=1}^{2^P} \boldsymbol{\mathcal{S}}'_k$, so that the complete volume of the region is now

$$\text{vol}(\hat{\boldsymbol{\Omega}}') = \sum_{k=1}^{2^P} \text{vol}(\boldsymbol{\mathcal{S}}'_k) \tag{7.36}$$

Computation of the region volume would then seem to be problematic, as the number of subsectors grows rapidly as P increases; however, it can be shown with some manipulation

that the volume reduces to

$$\begin{aligned} \mathrm{vol}(\hat{\boldsymbol{\Omega}}') &= \kappa(P)\mathrm{vol}(\boldsymbol{\mathcal{H}}') && (7.37) \\ &= \kappa(P)\prod_{i=1}^{P}\left(r_{(i,1)} + r_{(i,2)}\right) && (7.38) \end{aligned}$$

that is, the volume is simply proportional to the volume of the minimum bounding hyperbox that contains all models in the segments, by a scaling constant dependent on the type of inscribed shape that is chosen. This has two advantages:

1. There is no requirement to construct the $\mathbf{T}$ matrix (and thus, each of the sub-hyperboxes $\boldsymbol{\mathcal{H}}'_k$), resulting in a linear time complexity.

2. There is no need to explicitly find the members of the set $\hat{\boldsymbol{\Omega}}' \subset \boldsymbol{\mathcal{H}}' \subset \boldsymbol{\Theta}'$ to compute the volume, as $\mathrm{vol}(\hat{\boldsymbol{\Omega}}')$, and therefore $\mathrm{vol}(\hat{\boldsymbol{\Omega}})$, are related to the volume of the circumscribed hyperbox via a simple scaling.

These two points are very important as the MMLD-S algorithm requires the rapid computation of the volume of regions. The volumes of several shapes may be found from the scaled volume of a circumscribed hyperbox, for example, hyperellipsoids and hypersimplices; these are discussed in the sequel. The approximated uncertainty region, $\hat{\boldsymbol{\Omega}}$, is completely defined by the sets $\boldsymbol{\Xi}$, the matrix $\mathbf{B}$ and the point estimate $\boldsymbol{\theta}^*$. From $\boldsymbol{\Xi}$ the matrix $\mathbf{R}$ may be constructed from (7.23), and $\mathrm{vol}(\hat{\boldsymbol{\Omega}})$ may be computed by (7.38). If an exact specification of the uncertainty region is required, the matrix $\mathbf{T}$ may be found from $\mathbf{R}$ by (7.30) and (7.33), and depending on the shape inscribed inside the hyper-boxes the models in the uncertainty region may be determined via the transformation in (7.26).

Figure 7.2 shows two segments along two orthogonal rays in parameter space, along with the minimum bounding hyperbox $\boldsymbol{\mathcal{H}}$ (heavy line) and the entries of the $\mathbf{R}$ matrix. The dotted lines are the two subsets of models $\boldsymbol{\Theta}^{\boldsymbol{\xi}_1}$ and $\boldsymbol{\Theta}^{\boldsymbol{\xi}_2}$ along the orthogonal rays, whose width clearly determines the minimum bounding hyperbox. It is clear that these sets bisect $\boldsymbol{\mathcal{H}}$ into $2^P = 4$ sub boxes. The medium weight solid lines clearly show the inscribed elliptical and triangular subsectors in each of these boxes. This plot also clearly demonstrates the ability of this representation to allow for the resulting region $\hat{\boldsymbol{\Omega}}$ to be asymmetric.

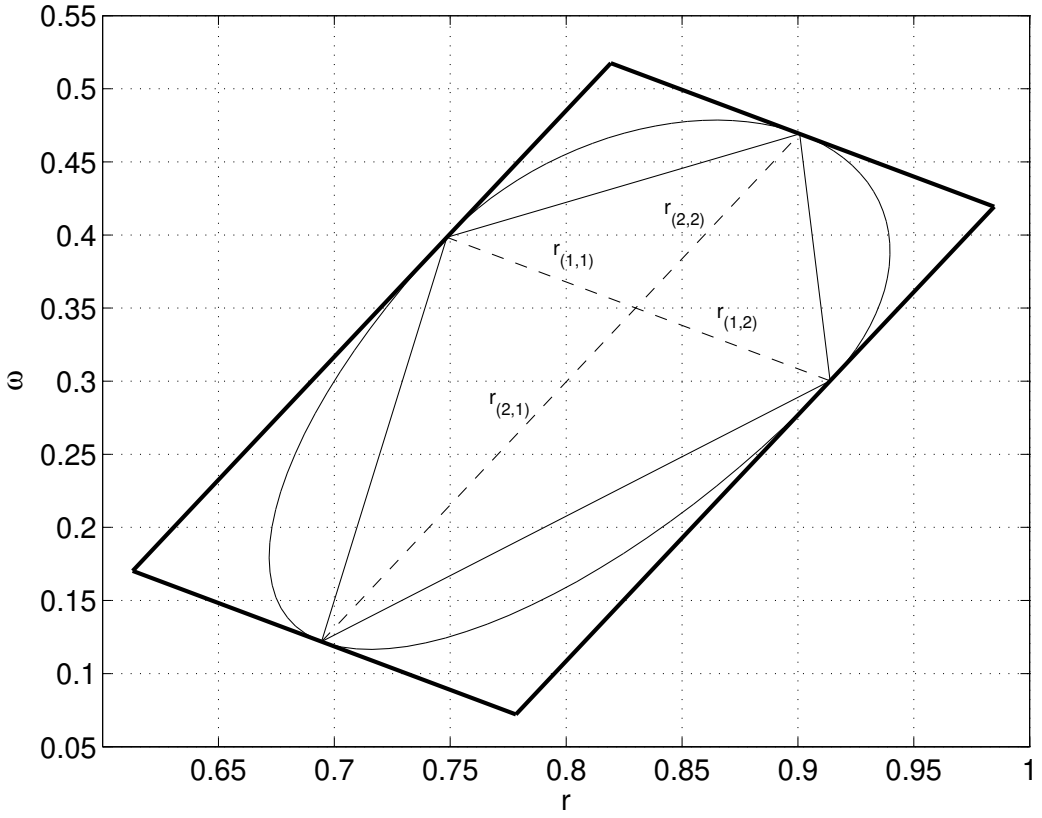


Figure 7.2: Circumbscribed Rectangle and Inscribed Regions; Patch Regions are Dotted Lines

7.3.5 Determination of the Parameter Rays

The previous subsections detailed a representation for asymmetric uncertainty regions; however, it relied on the specification of a suitable orthonormal basis' $\mathbf{B}$ to which the uncertainty region is aligned. In the one-parameter case, the choice is clearly straightforward: the single natural parameter axis is the only (and therefore, optimal) choice. When there are more than two parameters, there are an infinite number of choices. This algorithm makes use of the method used in the MMC$_{em}$ approximation to determine the principal axis of the hyper-ellipsoid. In that algorithm, the eigenvectors of the Information Matrix are chosen. This involves finding a matrix $\mathbf{B} = \left[\mathbf{b}_1^T, \ldots, \mathbf{b}_P^T\right]^T$ such that

$$\mathbf{J}(\boldsymbol{\theta}^*) = \mathbf{B}^T\mathbf{S}\mathbf{B} \tag{7.39}$$

where $\mathbf{S}$ is a diagonal matrix of singular values, i.e. a Singular Value Decomposition [46] of the Information Matrix $\mathbf{J}(\cdot)$. The rows $(\mathbf{b}_1, \ldots \mathbf{b}_P)$ of matrix $\mathbf{B}$ are then chosen as the unit vectors defining the principle axis of the region, as per the MMC$_{em}$ algorithm.

In effect, this selects the optimum set of parameter basis under the assumption that the KL divergence varies in a quadratic fashion, at least locally around $\boldsymbol{\theta}^*$. While this choice may not be completely optimal for the MMLD-S approximation, it is straightforward to compute and is a better choice than simply aligning $\mathbf{B}$ with the natural parameter space axis (i.e. $\mathbf{B} = \mathbf{I}_P$).

7.3.6 Discretisation of the KL Curves

This step cuts the parameter space into segments along each of the parameter rays. This amounts to discretising the space over which the $\boldsymbol{\xi}_i$ sets are defined, with the discretisation different for each dimension i. Again, it is most convenient to work in the uncertainty region space $\boldsymbol{\Theta}'$ and transform to the parameter space when required. In uncertainty region space, discretising along parameter space ray i involves varying θ'_i while holding the rest of $\boldsymbol{\theta}'$ to zero. For any $\theta'_i \in \mathbb{R}$, assuming the rest of the vector $\boldsymbol{\theta}'$ is zero, the resulting model in parameter space is given by

$$\theta'_i \mathbf{b}_i + \boldsymbol{\theta}^* \tag{7.40}$$

so that discretising along θ'_i amounts to discretising along the parameter ray aligned with $\mathbf{b}_i$. The one-dimensional discretisation and interpolation are undertaken as per Section 7.2.3 for each component θ'_i, $i = 1, \ldots, P$ of $\boldsymbol{\theta}'$. This process yields a matrix $\mathbf{U} = \left[\mathbf{u}_1^T, \ldots, \mathbf{u}_P^T\right]^T \in \mathbb{R}^{(P \times N_P)}$ of discrete points along each uncertainty region space axis, and a matrix $\mathbf{D} = \left[\mathbf{d}_1^T, \ldots, \mathbf{d}_P^T\right]^T \in \mathbb{R}_+^{(P \times N_P)}$ of KL divergences evaluated at each point in $\mathbf{U}$, so that

$$D_{(i,j)} = \Delta\left(u_{(i,j)}\mathbf{b}_i + \boldsymbol{\theta}^* || \boldsymbol{\theta}^*\right) \tag{7.41}$$

is the KL divergence of the model referenced by the j-th discrete point along ray i from the point estimate $\boldsymbol{\theta}^*$.

7.3.7 Computing the Model Cost

After discretisation has occurred, there two matrices $\mathbf{U}$ and $\mathbf{D}$ containing vectors of discretised points along the parameter rays and the associated KL divergences at these points. Now, assume there are discrete approximations of the $\boldsymbol{\Xi}$ sets that determine the width of the minimum bounding hyperbox in each dimension; these sets will now contain a list

of indices to intervals of the parameter rays in the $\mathbf{U}$ matrix, and therefore also to KL divergences. Let $\boldsymbol{\mathcal{R}} = \{\mathcal{R}_1, \ldots, \mathcal{R}_P\}$ be the set of these sets, which each satisfy

$$\mathcal{R}_i \subset \{1, \ldots, N_P\} \tag{7.42}$$

$$\forall j \in \mathcal{R}_i, \exists k \in \mathcal{R}_i : |j - k| = 1 \tag{7.43}$$

that is, they span a contiguous sequence of the natural numbers (which in turn, are indices into $\mathbf{U}$ and $\mathbf{D}$). As detailed in Section 7.3.4 the volume enclosed by the region may be determined by circumscribing a regular shape into the least hyper-box that encloses all points in the segments along each parameter ray (in this case, determined by the discrete sets $\boldsymbol{\mathcal{R}}$); this is given by

$$\text{vol}(\boldsymbol{\mathcal{R}}) = \kappa(P) \cdot \prod_{i=1}^{P} \left(u_{(i,\sup\{\mathcal{R}_i\})} - u_{(i,\inf\{\mathcal{R}_i\})} \right) \tag{7.44}$$

that is, the product of the widths of the segments spanning each parameter ray scaled by a geometric constant $\kappa(P)$ which depends only on P. Let $(\cdot)^{\{k\}}$ denote a quantity at step k of the growth algorithm. The algorithm begins by seeding each segment region $\mathcal{R}_i$ to contain just the single interval along each parameter ray that is closest to the point estimate $\boldsymbol{\theta}^*$ (and therefore to the origin in uncertainty region space), i.e.

$$\mathcal{R}_i^{\{1\}} = \left\{ j : u_{(i,j)} \leq 0 \leq u_{(i,j+1)} \right\} \tag{7.45}$$

for $i = 1, \ldots, P$. At each iteration the region is grown by adding a new interval to either end of one of the segments in $\boldsymbol{\mathcal{R}}$. The boundary rule states that the boundary of the optimal region must lie on a level set of the KL surface; thus the region may be restricted to grow in such a fashion that its boundary is always approximately on a level set of the KL surface. This is done by requiring that the interval that is added at each iteration is always the one that brings with it the lowest KL divergence. This means that at any point in time all of the intervals at the edges of each patch region have roughly similar KL divergences, and are thus roughly lying on a level set. The vector of total region volumes at each of the steps $(1 \ldots k)$ is

$$\mathbf{V} = \left[\text{vol}\left(\boldsymbol{\mathcal{R}}^{\{1\}}\right), \ldots, \text{vol}\left(\boldsymbol{\mathcal{R}}^{\{k\}}\right) \right] \tag{7.46}$$

and the *increase* in region volume that occurred at each step due to the addition of a new interval to one of the patch regions is

$$\boldsymbol{\delta}\mathbf{V} = \left[V_1, (V_2 - V_1), (V_3 - V_1 - V_2), \ldots, \left(V_k - \sum_{i=1}^{k-1} V_i \right) \right] \tag{7.47}$$

Finally, let $\mathbf{E}$ be the vector of KL divergences associated with the interval that was added to one of the segment regions $\mathcal{R}_i$ at each step, i.e. E_k is the KL divergence of the interval added in the k-th step of the algorithm. Now consider the I1F model cost approximation

$$\mathcal{D}_{\mathrm{1F}}(\boldsymbol{\theta}^*) = -\log q(\boldsymbol{\theta}^*) + \frac{1}{q(\boldsymbol{\theta}^*)} \int_{\boldsymbol{\theta} \in \boldsymbol{\Omega}} h(\boldsymbol{\theta}) \Delta(\boldsymbol{\theta} || \boldsymbol{\theta}^*) d\boldsymbol{\theta} \tag{7.48}$$

where

$$q(\boldsymbol{\theta}^*) = \int_{\boldsymbol{\theta} \in \boldsymbol{\Omega}} h(\boldsymbol{\theta}) d\boldsymbol{\theta} \tag{7.49}$$

Under the assumption of roughly flat priors, at step k the integral in (7.49) may be approximated as

$$q(\boldsymbol{\theta}^*) \approx \mathrm{vol}(\boldsymbol{\Omega}) h(\boldsymbol{\theta}^*) \approx h(\boldsymbol{\theta}^*) V_k = h(\boldsymbol{\theta}^*) \sum_{i=1}^{k} \delta V_i \tag{7.50}$$

The second integral in (7.48) may be first simplified by taking into account the assumption of local uniformity of the prior

$$\int_{\boldsymbol{\theta} \in \mathcal{R}} h(\boldsymbol{\theta}) \Delta(\boldsymbol{\theta} || \boldsymbol{\theta}^*) d\boldsymbol{\theta} \approx \frac{h(\boldsymbol{\theta})}{q(\boldsymbol{\theta}^*)} \int_{\boldsymbol{\theta}^* \in \boldsymbol{\Omega}} \Delta(\boldsymbol{\theta} || \boldsymbol{\theta}^*) d\boldsymbol{\theta} \tag{7.51}$$

The integral may now be approximated discretely by summing the height of each of the level sets (the entries of the $\mathbf{E}$ vector) multiplied by the volume they occupy in parameter space (the entries of the $\boldsymbol{\delta}\mathbf{V}$ vector), and the resulting model cost is given by

$$\mathcal{D}_{\mathrm{1F}}(\boldsymbol{\theta}^*) \approx \mathcal{D}_{\mathrm{DS}}(\boldsymbol{\theta}^*) = \mathcal{D}(\boldsymbol{\mathcal{R}}) = -\log \sum_{i=1}^{k} \delta V_i - \log h(\boldsymbol{\theta}^*) + \frac{1}{\left(\sum_{i=1}^{k} \delta V_i \right)} \sum_{i=1}^{k} \delta V_i E_i \tag{7.52}$$

where the prior in the excess coding length term cancels due to the assumption of local uniformity. The equation (7.52) is now denoted the MMLD-S message length. The segment regions $\mathcal{R}_i$ are grown by addition of single intervals until the model cost fails to decrease; the next section details the growth algorithm more explicitly.

7.3.8 Multi-parameter Region Growth

This section describes the growth algorithm to determine the region that minimises (7.52). The growth algorithm begins by seeding each of the segment regions $i = 1, \ldots, P$ thus

$$\mathcal{R}_i = \left\{ j : u_{(i,j)} \leq 0 \leq u_{(i,j+1)} \right\} \tag{7.53}$$

that is, each region includes the interval that is closest to the true parameter vector $\boldsymbol{\theta}^*$. Additionally, create a matrix $\mathbf{C} \in \mathbb{R}^{(P \times 2)}$ of KL divergences of all intervals neighbouring the segment regions $\mathcal{R}_i$, with entries

$$\mathbf{c}_j = \left[d_{(j,\inf\{\mathcal{R}_j\}-1)},\ d_{(j,\sup\{\mathcal{R}_j\}+1)} \right],\ (j = 1, \ldots, P) \tag{7.54}$$

The algorithm then proceeds as follows:

1. Set $\mathcal{D}' \leftarrow \mathcal{D}$, $\boldsymbol{\mathcal{R}}' \leftarrow \boldsymbol{\mathcal{R}}$ (save model cost and region)

2. Determine the neighbouring interval with the minimum KL divergence by solving

$$\{i, j\} \leftarrow \arg \min_{(j \in P, i \in 2)} \left\{ c_{(j,i)} \right\}$$

3. If $i = 1$ (expanding left), set

$$\begin{aligned} k &\leftarrow \inf\{\mathcal{R}_j\} - 1 \\ \mathcal{R}'_j &\leftarrow \{k, \mathcal{R}_j\} \\ c_{(j,i)} &= d_{(j,k-1)} \end{aligned}$$

else if $i = 2$ (expanding right), set

$$\begin{aligned} k &\leftarrow \sup\{\mathcal{R}_j\} + 1 \\ \mathcal{R}'_j &\leftarrow \{\mathcal{R}_j, k\} \\ c_{(j,i)} &= d_{(j,k+1)} \end{aligned}$$

4. Grow the region and update the model cost

$$\begin{aligned}
\mathbf{E}' &\leftarrow \left[\mathbf{E}, d_{(j,k)}\right] \\
\boldsymbol{\delta}\mathbf{V}' &\leftarrow \left[\boldsymbol{\delta}\mathbf{V}, \left(\mathrm{vol}(\boldsymbol{\mathcal{R}}') - \sum_{i=1}^{\dim(\boldsymbol{\delta}\mathbf{V})} \delta V_i\right)\right] \\
\mathcal{D}' &\leftarrow -\log\left(\sum_{i=1}^{\dim(\boldsymbol{\delta}\mathbf{V}')} \delta V_i'\right) + \boldsymbol{\delta}\mathbf{V}' \cdot \mathbf{E}'^{T} / \left(\sum_{i=1}^{\dim(\boldsymbol{\delta}\mathbf{V}')} \delta V_i'\right)
\end{aligned}$$

5. If $\mathcal{D}' < \mathcal{D}$, set $\mathbf{E} \leftarrow \mathbf{E}'$, $\boldsymbol{\delta}\mathbf{V} \leftarrow \boldsymbol{\delta}\mathbf{V}'$, $\boldsymbol{\mathcal{R}} \leftarrow \boldsymbol{\mathcal{R}}'$ and return to Step 1; otherwise, algorithm ends.

The algorithm terminates when the addition of a new interval to one of the segment regions fails to decrease the model cost, and upon termination $\mathcal{D}$ contains the approximate model cost for parameter vector $\boldsymbol{\theta}^*$, less the prior term. The complete message length for the data $\mathbf{y}$ sent using model $\boldsymbol{\theta}^*$ may be approximated as

$$I_{1\mathrm{F}}(\mathbf{y}, \boldsymbol{\theta}^*) \approx L(\mathbf{y}|\boldsymbol{\theta}^*) + \mathcal{D} - \log h(\boldsymbol{\theta}^*) \tag{7.55}$$

Several refinements to the scheme are covered in the next subsections, including a discussion of selection of suitable geometries for the regions and a modification for curved priors.

7.3.9 An Example

An example of the algorithm in action is now detailed. The model under consideration is a second order autoregression with parameters $\boldsymbol{\theta}^* = [r, \omega] = [0.83, 0.35]$ for $N = 50$ datapoints. Figure 7.3 shows the KL surface and estimated uncertainty region in the main figure. The new parameter axis p_1 and p_2 are shown with the patch regions emphasised. Subfigures 7.3(a) and 7.3(b) show the discretised KL curves, and 7.3(c) and 7.3(d) show the discrete curves post-interpolation, with the patch regions again in bold. The asymmetric nature of the uncertainty region is clear from examination of 7.3(d).

Figure 7.4 shows ten iterations of the growth algorithm. The main figure plots the region as it increases in volume at each step, the numerals denoting which step the region belongs to. Figures 7.4(a) and 7.4(b) show the contents of the $\mathbf{E}$ and $\boldsymbol{\delta}\mathbf{V}$ vectors, respectively, at each of the iterations. It is clear to see that the KL divergence vector $\mathbf{E}$ is

non-decreasing, indicating that the region boundary is as close to a level set at each step as is possible within the constraints of the discrete KL curves. Figure 7.4(c) shows the total volume of the region at each step, and 7.4(d) the total estimate model cost. It is clear to see, as would be expected, that the region is growing at each step and the model cost shrinking. Although the plots show only the first ten steps, the algorithm proceeded to grow the region 450 times to produce the uncertainty region in the main panel of Figure 7.3, with an estimated model cost of 5.23 nits.

7.3.10 Region Geometries

An important issue in determining the volume of the region is to select a suitable shape to inscribe into the sub hyperboxes (see Section 7.3.4). Before that is considered, an important property of the MMLD-S equation is examined.

Proposition 7.1: Assuming that $\text{vol}(\boldsymbol{\mathcal{R}})$ is given by (7.44), the sets $\boldsymbol{\mathcal{R}}$ that minimise (7.52) are invariant to the choice of $\kappa(P)$.

Proof: Define

$$\boldsymbol{\delta \mathbf{W}} = \frac{\boldsymbol{\delta \mathbf{V}}}{\kappa(P)}$$

where $\boldsymbol{\delta \mathbf{W}}$ are the incremental volumes assuming a rectangular geometry is chosen (i.e. a $\kappa(P) = 1$). Letting $N_V = \dim(\boldsymbol{\delta \mathbf{W}})$, equation (7.52) may be re-written as

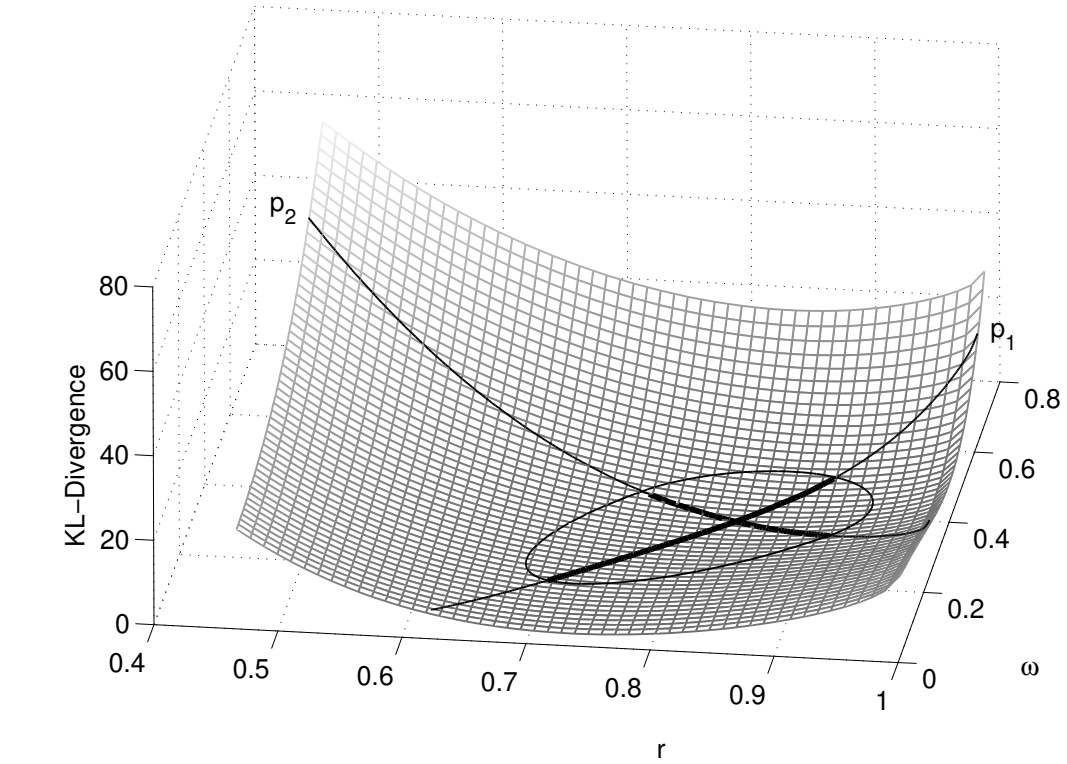
$$\mathcal{D}(\boldsymbol{\mathcal{R}}) = -\log \sum_{i=1}^{N_V} \delta W_i - \log h(\boldsymbol{\theta}^*) - \log \kappa(P) + \frac{\kappa(P)}{\kappa(P)\left(\sum_{i=1}^{N_V} \delta W_i\right)} \sum_{i=1}^{N_V} \delta W_i E_i \qquad (7.56)$$

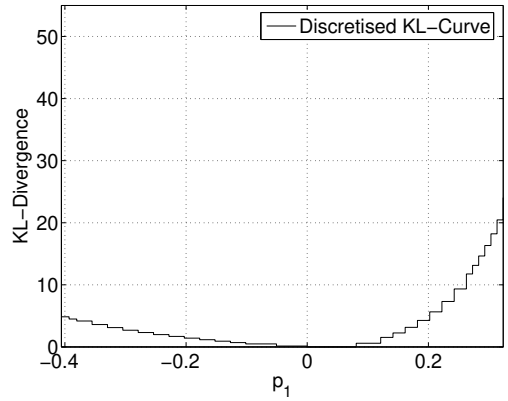
Clearly, the $\kappa(P)$ in the last term of (7.56) cancel, and the only dependence of $\mathcal{D}(\cdot)$ on $\kappa(P)$ is through the log term, which is constant for a given P. **Q.E.D.**

Corollary 7.2: Given a model class of dimension P, the parameter estimates $\hat{\boldsymbol{\theta}}$ that solve

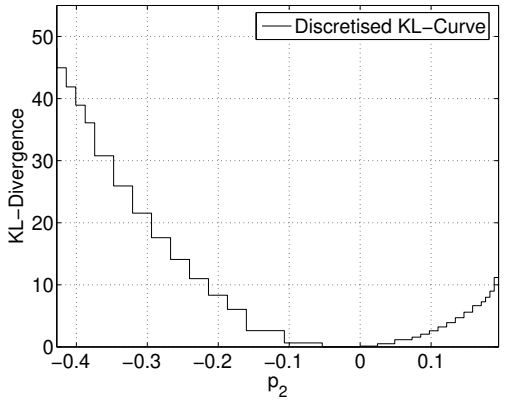
$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}^* \in \boldsymbol{\Theta}} \{L(\mathbf{y}|\boldsymbol{\theta}^*) + \mathcal{D}_{\text{DS}}(\boldsymbol{\theta}^*)\} \qquad (7.57)$$

are invariant to the choice of $\kappa(P)$.

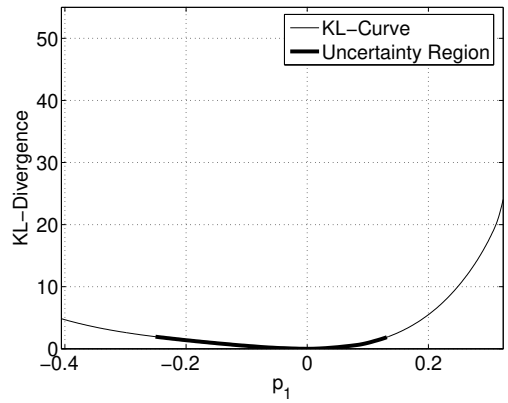




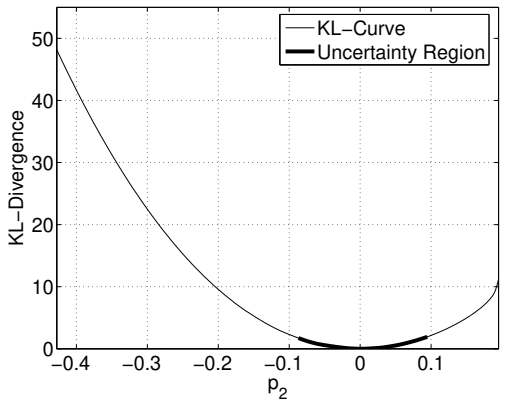
(a) Discretised KL Curve for p_1



(b) Discretised KL Curve for p_2

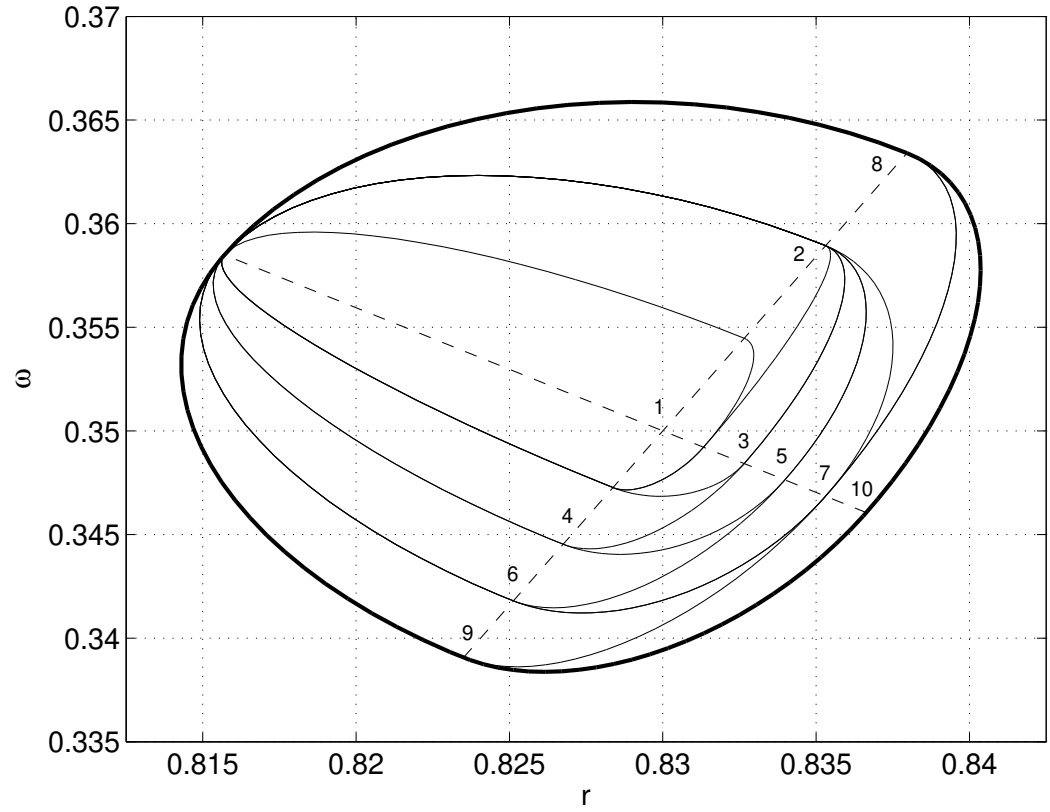


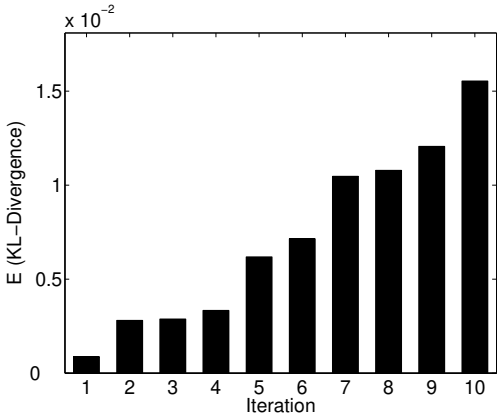
(c) Uncertainty Region for p_1



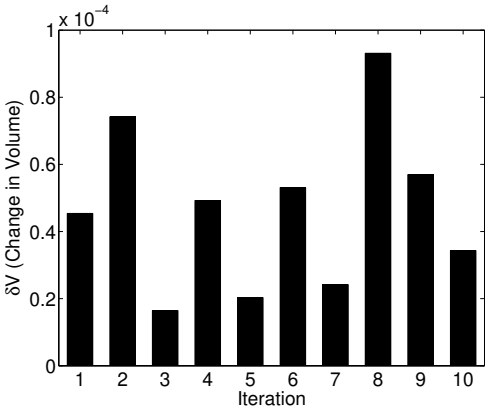
(d) Uncertainty Region for p_2

Figure 7.3: KL Surface and Estimated Uncertainty Regions for an AR(2) Model

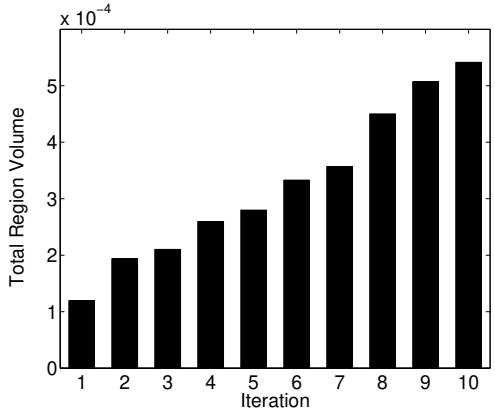




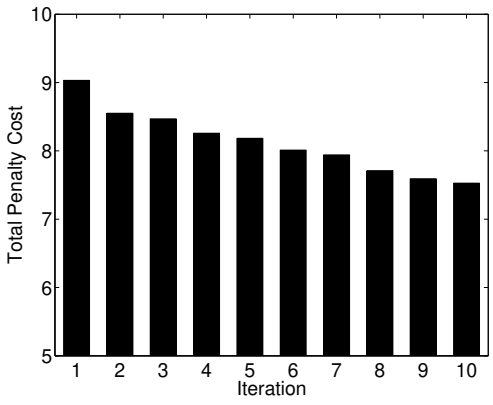
(a) KL Divergences at each Iteration, $\mathbf{E}$



(b) Increase in Volume per Iteration, $\boldsymbol{\delta}\mathbf{V}$



(c) Total Region Volume each Iteration



(d) Total Model Cost each Iteration

Figure 7.4: Ten Iterations of Region Growth for an AR(2) Model

Proof: This follows simply from (7.56). As $\kappa(P)$ only appears only as a constant log term it has only a constant effect on the model cost, irrespective of the value of $\boldsymbol{\theta}^*$. **Q.E.D.**

This demonstrates that given a fixed dimension, the entries of $\boldsymbol{\mathcal{R}}$ that minimise the MMLD-S message length equation do not depend on the shape inscribed into the sub hyperboxes. However, if the candidate models differ in dimensionality, the choice of shape has an effect on which model will have the minimum message length, as it appears as a dimensionality constant in (7.56). There are three simple choices: hyper-rectangular, hyper elliptical or hyper-tetrahedronal; their respective scaling functions are given below

$$\kappa(P) = \begin{cases} 1 & \text{(Rectangular)} \\ (P!)^{-1} & \text{(Hyper-tetrahedronal)} \\ \pi^{\frac{P}{2}} \cdot \left(\Gamma\left(\frac{P}{2}+1\right)\right)^{-1} \cdot \frac{1}{2^P} & \text{(Hyper-elliptical)} \end{cases} \tag{7.58}$$

As is evident from Figures 7.2 and 7.4, the choice of hyper-elliptical scaling function does not mean the region is a hyper-ellipse; instead, it essentially means that the 2^P sectors are each hyper-sectors of hyper-ellipses, with radii determined by the size of the segment regions. If all the segment regions are symmetric around the origin, the resulting region will be a true ellipse. The most natural choice is the hyper-elliptical; it would be expected that the KL curves are in general smooth and continuous, and the hyper-elliptical region would be expected to give a good approximation to the bowl around the parameter estimate, especially as N and P increase and the KL curves become steeper and the Uncertainty Region smaller. Indeed, for some model classes (such as linear regressions) the hyper-ellipsoids are the true optimal regions. Furthermore, asymptotically as $N \to \infty$ the true uncertainty region appears to be converge to a hypersphere for almost all common model classes. Any knowledge of the shape of the KL surface of the model class under examination should be taken into account when selecting a suitable region geometry.

7.3.11 Parameter Restrictions

Some classes of models have restrictions on the parameter space, i.e. $\boldsymbol{\Theta} \subset \mathbb{R}^P$. The boundaries of the parameter space defined by restrictions present in these models are

divided into two different types: *hard* boundaries and *soft* boundaries. These are defined as:

- Hard parameter boundaries are defined as those that separate the model space between parameter values that are permissible (in the sense that they yield a likelihood function) and those that produce a model that is nonsensical or does not admit a suitable likelihood function. An example of such boundaries would be the those induced by the restriction that $\theta \in [0, 1]$ in the binomial model class; in this case, parameter values outside this range are completely nonsensical, as negative probabilities or probabilities greater than unity are undefined.

- Soft parameter boundaries are usually induced by extra restrictions that are not inherent in the model class; in this case, it is common that parameters lying outside the 'valid' parameter space still lead to models that are valid in terms of producing a sensible likelihood function. An example of this occurs when a uniform prior is placed on the μ parameter of a Normal distribution; values of μ lying outside the support of the prior density still lead to completely valid likelihoods even though they are not valid in the sense that they violate the artificially enforced parameter range restriction.

Of the two, the hard boundaries are more straightforward to handle. The simplest procedure, and the one adopted in this thesis, is to simply restrict the parameter rays from passing beyond the hard parameter boundaries, and this is made much easier if the parameter boundaries form a hyper-box. The handling of soft boundaries presents a more difficult challenge and this is discussed in the next section.

Handling Soft Boundaries

Soft boundaries present more of an issue because of the nature of the KL divergence function near the boundaries. When the boundaries are hard, it is usual that the KL divergence rises rapidly to infinity as the model approaches the bounds (such as in the binomial model); this usually means that the elliptical estimated region makes for a reasonable approximation because the region is squashed near the boundary. In contrast, the soft boundaries are usually artificial in the sense that models lying outside the boundaries are still valid in terms of providing a suitable likelihood function, and this means that in

most cases the KL divergence will not rise steeply near the soft boundaries. If the parameter rays are simply prevented from passing through the soft boundary, the resulting regions estimated by hyperellipsoids may seriously underestimate the true region size. The solution adopted in this thesis is to ignore the soft boundaries during the discretisation and region growth stage, and subsequently determine how much of the region should be amputated once it has been grown, that is the final region is

$$\hat{\boldsymbol{\Omega}} = \hat{\boldsymbol{\Omega}}' \cap \boldsymbol{\Theta} \tag{7.59}$$

where $\hat{\boldsymbol{\Omega}}'$ is the optimal region found by disregarding soft parameter boundaries, and $\boldsymbol{\Theta}$ is the complete parameter space (enforcing all boundaries). The primary difficulty then becomes one of determining the intersection in (7.59), especially if constraints lead to complex boundaries. One approach that is relatively straightforward to implement is to determine the amount of the region that needs to be amputated based on sampling. Begin by uniformly sampling from the uncertainty region N_S times; store these models as $\boldsymbol{\Theta}^S = \left\{\boldsymbol{\theta}_1^S, \ldots, \boldsymbol{\theta}_{N_S}^S\right\}$. Then build the new set

$$\boldsymbol{\Theta}^R = \left\{\boldsymbol{\theta} \in \boldsymbol{\Theta}^S : \boldsymbol{\theta} \in \boldsymbol{\Theta}\right\} \tag{7.60}$$

of the sampled models that satisfy all parameter constraints. The volume of the uncertainty region that lies completely inside the parameter space may then be estimated as

$$\mathrm{vol}(\hat{\boldsymbol{\Omega}}) = \frac{\dim\left(\boldsymbol{\Theta}^R\right)}{N_S} \cdot \mathrm{vol}(\hat{\boldsymbol{\Omega}}') \tag{7.61}$$

Under this modification, and given that the region's geometric constant has no effect on the excess round-off cost, the combined MMLD-S message length equation may be approximated as

$$L(\mathbf{y}|\boldsymbol{\theta}^*) + \mathcal{D}_{\mathrm{DS}}(\boldsymbol{\theta}^*) - \log\dim\left(\boldsymbol{\Theta}^R\right) + \log N_S \tag{7.62}$$

This approximation may be derived by simply scaling the region's geometric constant by $\kappa(P)$ by a factor of $\left(\frac{\dim\left(\boldsymbol{\Theta}^R\right)}{N_S}\right)$. This approximation is only valid under the assumption that

$$\int_{\boldsymbol{\theta}\in\hat{\boldsymbol{\Omega}}} h(\boldsymbol{\theta})\Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*)d\boldsymbol{\theta} \approx \int_{\boldsymbol{\theta}\in\hat{\boldsymbol{\Omega}}'} h(\boldsymbol{\theta})\Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*)d\boldsymbol{\theta} \tag{7.63}$$

that is, if the round-off cost of the amputated region is roughly the same as the round-off cost of the complete region. While this may seem quite restrictive, it should be first noted the round-off cost is an *average* round-off cost, and that many regions of different volume may yield the same average round-off cost. Furthermore, is often the case that soft-parameter boundaries are formed by removing redundancies from the model class, and that the KL surface is often symmetric around these boundaries. If the priors are chosen coherently, they will assign the same prior mass to models that yield the same likelihood function regardless of the values of the parameters. In these cases, the volume of the complete region will often be an integer multiple of the volume of amputated region, and the assumption in (7.63) will hold. The next chapter will demonstrate that even if this is not the case, a suitable reparameterisation may be often be sufficient to allow (7.63) to hold.

It should be stressed that as the sampling is done only once, after the region has been grown, and that all that is done is to prune out those models that do not satisfy the constraints, the process is extremely fast, as long as a fast sampler is available. An algorithm for generating samples rapidly from the uncertainty region is detailed next.

Sampling from the Uncertainty Region

Until now the only way in which the geometry of the uncertainty region manifested itself was through the geometric constant $\kappa(P)$. However, if samples are required to be drawn from the uncertainty region then the geometry must be explicitly taken into account. If the region is chosen to be a hyper-rectangle (i.e. $\kappa(P) = 1$) the problem is trivial; if the more natural elliptical geometry is chosen the problem is more difficult. Begin by recalling the definition of the matrix $\mathbf{R} \in \mathbb{R}^{(P\times 2)}$ in Section 7.3.3 that holds the lengths of each of the segment regions $\boldsymbol{\Xi}$ on either side of the origin. Further, recall that the MMLD-S approximation uses a discrete approximation of these sets, denoted $\boldsymbol{\mathcal{R}}$ (see Section 7.3.7). The $\mathbf{R} \in \mathbb{R}^{(P\times 2)}$ matrix is then given by

$$\mathbf{r}_i = \left[-u_{(i,\inf\{\mathcal{R}_i\})},\ u_{(i,\sup\{\mathcal{R}_i\})}\right] \tag{7.64}$$

Table 7.1: Matrix $\mathbf{S}$ of Radii of Sectors for $P = 3$

Sector	t_1	t_2	t_3
1	$-r_{(3,1)}$	$-r_{(2,1)}$	$-r_{(1,1)}$
2	$-r_{(3,1)}$	$-r_{(2,1)}$	$+r_{(1,2)}$
3	$-r_{(3,1)}$	$+r_{(2,2)}$	$-r_{(1,1)}$
4	$-r_{(3,1)}$	$+r_{(2,2)}$	$+r_{(1,2)}$
5	$+r_{(3,2)}$	$-r_{(2,1)}$	$-r_{(1,1)}$
6	$+r_{(3,2)}$	$-r_{(2,1)}$	$+r_{(1,2)}$
7	$+r_{(3,2)}$	$+r_{(2,2)}$	$-r_{(1,1)}$
8	$+r_{(3,2)}$	$+r_{(2,2)}$	$+r_{(1,2)}$

where $\mathbf{U}$ is described in Section 7.3.6. The volume of the region may then be found as

$$\text{vol}(\mathbf{\Omega}) = \kappa(P) \prod_{i=1}^{P} \left(r_{(i,1)} + r_{(i,2)} \right) \tag{7.65}$$

which is simply the product of the widths of each patch region, scaled by the geometric constant. If an elliptical constant is chosen, the region may be described as a union of 2^P sectors of hyper-ellipses, the radii of each of the sectors determined by the patch regions. Assume that the matrix $\mathbf{S} \in \mathbb{R}^{(2^P \times P)}$ contains the radii of each of the sectors, with the signs of the elements of $\mathbf{S}$ determining which side of the origin they lie in each dimension. The entries of this matrix are given by

$$s_{(k,i)} = r_{(i,\psi_i(k))} \cdot \text{sign}\left(\psi_i(k) - \frac{3}{2} \right), \ \left(k = 1, \ldots, 2^P, i = 1, \ldots, P \right) \tag{7.66}$$

where

$$\psi_i(k) = \text{mod}\left(\lfloor k/2^{(i-1)} \rfloor, 2 \right) + 1 \tag{7.67}$$

The matrix $\mathbf{S}$ is simply a signed version of the matrix $\mathbf{T}$ (see Section 7.3.4), the signs indicating which side of the origin the length projects into. An example for $P = 3$ is shown in Table 7.1. From this matrix the volume of the region may be expressed as a sum of the volumes of each of the 2^P sectors thus

$$\text{vol}(\boldsymbol{\mathcal{R}}) = \kappa(D) \sum_{i=1}^{2^P} \left(\prod_{j=1}^{P} |s_{(i,j)}| \right) \tag{7.68}$$

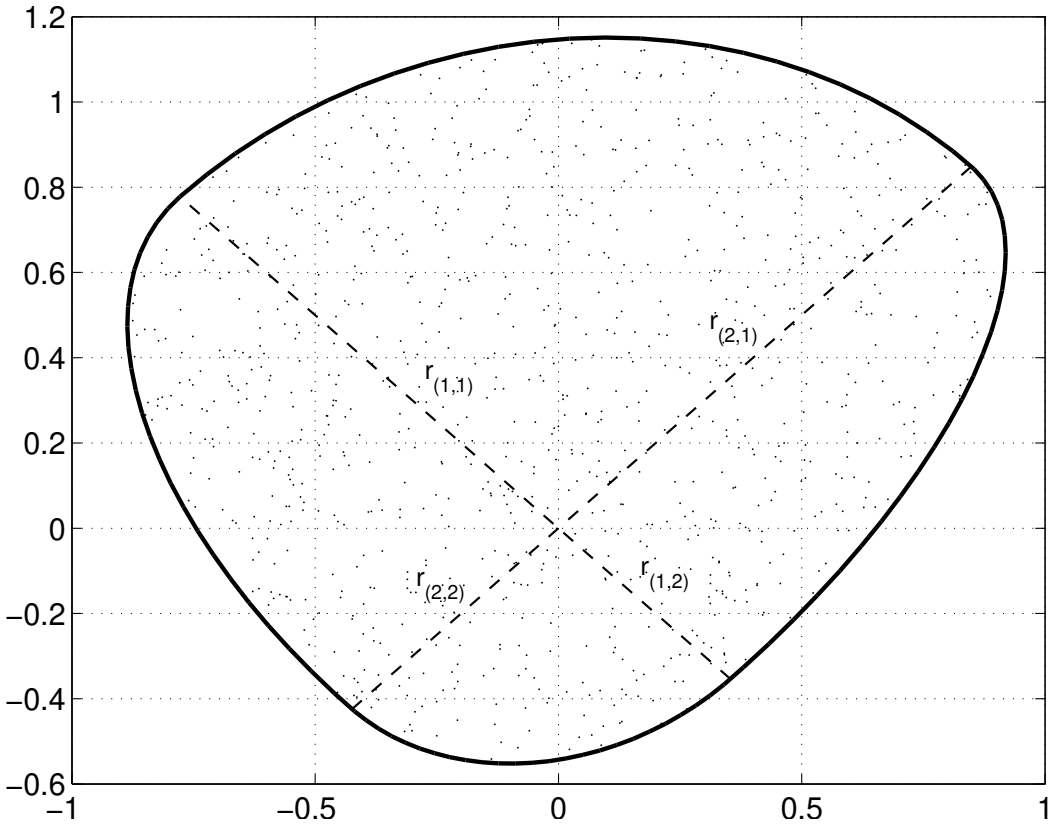


Figure 7.5: Example Uncertainty Region and Uniformly Distributed Random Samples

which may be easily shown to be equivalent to (7.65). The sampling process begins by sampling a set of points $\mathbf{\Lambda} \in \mathbb{R}^{(N_S \times P)}$ from a unit P-dimensional hypersphere. If the region was symmetrical, then $r_{(i,1)} = r_{(i,2)}$ for all dimensions, and each component could be simply scaled by the radius of the hyper-ellipse in each dimension, as per [35]. However, the uncertainty region may well be asymmetric in the MMLD-S approximation, and so this simple approach will not work. Instead, the set of points sampled from the unit hypersphere are 'allocated' to each of the sectors and scaled accordingly based on the radii of each sector (stored in $\mathbf{S}$). For the sampling to be uniform, the allocation of points to sectors must be in proportion to their relative volumes, i.e. a sector that has twice the volume of another sector should recieve twice as many points as it occupies twice the space. This may be done by randomly choosing the sector that a sampled point should be allocated to from a 2^P valued multinomial with probabilities

$$p(k) = \frac{\kappa(P)}{\text{vol}(\boldsymbol{\mathcal{R}})} \left(\prod_{j=1}^{P} |t_{(k,j)}| \right), \ \left(k = 1 \ldots 2^P\right) \tag{7.69}$$

Once a vector of allocations $\mathbf{k} \in \mathbb{N}^{N_S}$ has been randomly chosen from the multinomial distribution, the points sampled from the unit hypersphere may be transformed as

$$\boldsymbol{\lambda}'_i = \left[s_{(k_i,1)} \cdot |\lambda_{(i,1)}|,\ \ldots,\ s_{(k_i,P)} \cdot |\lambda_{(i,P)}| \right],\ i = 1 \ldots N_S \tag{7.70}$$

The matrix $\boldsymbol{\Lambda}' = \left[\boldsymbol{\lambda}'^T_1, \ldots, \boldsymbol{\lambda}'^T_{N_S} \right]^T$ now contains a set of points uniformly sampled from uncertainty region composed of the union of a elliptical sectors, assuming the parameter rays are aligned with the natural axis and the model under consideration $\boldsymbol{\theta}^*$ coincides with the origin. Generally this will not be the case, and the final transformation

$$\boldsymbol{\lambda}''_i = \left(\mathbf{B}^T \boldsymbol{\lambda}'^T_i \right)^T + \boldsymbol{\theta}^*,\ i = 1 \ldots N_S \tag{7.71}$$

renders the points in the matrix $\boldsymbol{\Lambda}'' = \left[\boldsymbol{\lambda}''^T_1, \ldots, \boldsymbol{\lambda}''^T_{N_S} \right]^T$ to be uniformly distributed within the uncertainty region. Figure 7.5 shows an example two-dimensional uncertainty region, with the radii of each elliptical sector indicated, and 1000 randomly sampled points plotted.

7.3.12 Handling Non-uniform Priors by Sampling

The MMLD-S approximation has been developed under the assumption that the priors are roughly flat within the confines of the uncertainty region. This section presents a method to relax this assumpion that is based on sampling; in this sense it is similar to the part of the MMC$_{em}$ algorithm that computes the prior mass, though of course within the MMLD-S framework the regions may be asymmetrical, and the round-off cost is not estimated via a sampling approach. Assuming that the evaluation of the KL divergence (or negative log-likelihood) is computationally considerably more expensive than evaluation of the priors (which is often the case), even when sampling is used to compute the prior mass in the MMLD-S algorithm the approximation remains significantly faster that those methods which also use sampling to compute the round-off cost. The basic approach is to compute the average prior value over the uncertainty region with the addition of each new interval to one of the segment regions. At step k of the algorithm, given a set $\boldsymbol{\Lambda} \in \mathbb{R}^{(N_S \times P)}$ of N_S uniform samples from the uncertainty region, compute

$$\bar{H}_k = \frac{1}{N_S} \sum_{i=1}^{N_S} h(\boldsymbol{\lambda}_i) \tag{7.72}$$

where the vector $\bar{\mathbf{H}}$ contains an estimate of the average prior over the uncertainty region. From this the vector $\mathbf{H}$ of *prior masses* at steps $(1 \dots k)$ may be estimated as

$$H_i = \bar{H}_i V_i, \ (i = 1 \dots k) \tag{7.73}$$

where V_i is the volume of the uncertainty region at step i of the algorithm. From this, a new vector of incremental prior-masses $\boldsymbol{\delta}\mathbf{H}$ may be formed as

$$\boldsymbol{\delta}\mathbf{H} = \left[H_1, (H_2 - H_1), (H_3 - H_1 - H_2), \dots, \left(H_k - \sum_{i=1}^{k-1} H_i \right) \right] \tag{7.74}$$

in a similar fashion to the $\boldsymbol{\delta}\mathbf{V}$ vectors. From this the MMLD-S message length equation may be rewritten as

$$\mathcal{D}_{\mathrm{DS}}(\boldsymbol{\mathcal{R}}) = -\log \sum_{i=1}^{N_V} \delta H_i + \frac{1}{\left(\sum_{i=1}^{N_V} \delta H_i \right)} \sum_{i=1}^{N_V} \delta H_i E_i \tag{7.75}$$

If a small number of samples is used at each step to estimate $\bar{H}_i$ to speed the process up, it is quite possible that $H_{(i+1)} < H_i$, which given the positive semi-definite nature of probability density functions, is clearly not valid. A simple solution is to choose $\bar{H}_i$ as

$$\bar{H}_i = \begin{cases} (N_S)^{-1} \sum_{i=1}^{N_S} h(\boldsymbol{\lambda}_i) & \text{for } V_i \left((N_S)^{-1} \sum_{i=1}^{N_S} h(\boldsymbol{\lambda}_i) \right) \geq H_{(i-1)} \\ \bar{H}_{(i-1)} & \text{for } V_i \left((N_S)^{-1} \sum_{i=1}^{N_S} h(\boldsymbol{\lambda}_i) \right) < \bar{H}_{(i-1)} \end{cases} \tag{7.76}$$

which at least guarantees that $\delta H_i \geq 0$, for all $i = 1 \dots k$.

7.3.13 Parameter Estimation by MMLD-S

Explicit estimation of parameters may be undertaken via MMLD-S by as per I1F and MML87, i.e.

$$\hat{\boldsymbol{\theta}}_{\mathrm{DS}} = \arg \min_{\boldsymbol{\theta} \in \boldsymbol{\Theta}} \{ L(\mathbf{y}|\boldsymbol{\theta}) + \mathcal{D}_{\mathrm{DS}}(\boldsymbol{\theta})(\mathbf{y}) \} \tag{7.77}$$

that is, it seeks the parameters that minimise the sum of the negative log-likelihood and model cost. Closed form solutions will typically not exist, or will at least be very difficult to derive, and thus the search must be done numerically. As gradient information is difficult to obtain, given the nature of the approximation, a weighted optimisation [55, 152, 153, 154]

in a neighbourhood of the MML87 or Maximum Likelihood estimates might be a reasonable approach to use.

7.3.14 MMLD-S Estimator Based on the Likelihood

For some problems it may be impossible or impractical to compute the Kullback-Leibler divergences that are used in the model costs. In this case, one solution is to refer to the MMLD message length formula

$$I_{1\mathrm{D}}(\mathbf{y}) = -\log q(\boldsymbol{\Omega}) + \frac{1}{q(\boldsymbol{\Omega})} \int_{\boldsymbol{\theta}\in\boldsymbol{\Omega}} h(\boldsymbol{\theta}) L(\mathbf{y}|\boldsymbol{\theta}) d\boldsymbol{\theta} \tag{7.78}$$

To implement this equation in the MMLD-S framework is reasonably straightforward; first the vectors of Kullback-Leibler divergences $\mathbf{D}$ must be redefined. They now become

$$\mathbf{d}_i = \left\{ L(\mathbf{y}|\hat{\boldsymbol{\theta}}_{\mathrm{ML}}(\mathbf{y}) + p_{(i,1)}\mathbf{b}_i), \ldots, L(\mathbf{y}|\hat{\boldsymbol{\theta}}_{\mathrm{ML}}(\mathbf{y}) + p_{(i,N_P)}\mathbf{b}_i) \right\} \tag{7.79}$$

They now contain the negative log-likelihoods (which may be viewed as empirical KL divergences [76]) of the dataset $\mathbf{y}$ evaluated at various models determined by the matrix $\mathbf{U}$ (which is found via the discretisation). The other issue to consider is that the basis are now best found from the observed Information Matrix $\mathbf{J}(\hat{\boldsymbol{\theta}}_{\mathrm{ML}}(\mathbf{y}), \mathbf{y})$, or Hessian of the negative log-likelihood, as

$$\mathbf{J}(\hat{\boldsymbol{\theta}}_{\mathrm{ML}}(\mathbf{y}), \mathbf{y}) = \mathbf{B}^T \mathbf{S} \mathbf{B} \tag{7.80}$$

The final message length is no longer approximated as per (7.55); instead it is now given directly by (7.52) as the 'model cost' term is no longer computed from expected excess coding lengths but is found from the negative log-likelihood. Explicit parameter estimation via (7.77) is now no longer possible.

7.3.15 Discussion of the Algorithm

Due to the assumption that the region volume is proportional to a hyper-rectangle, the MMLD-S algorithm needs only to examine P parameter rays. Assuming that the evaluation of the KL divergence is the most expensive (in computational terms) factor in the approximation, this yields a time complexity roughly

$$\mathcal{O}(P, N_K) \propto P \cdot N_K \tag{7.81}$$

that is, the time complexity increases linearly with as the number of model parameters and number of discretisation cuts (and therefore, evaluation of the KL divergence) that are required. Thus, finding the Uncertainty Region and model cost of a 10 parameter model is expected to take only roughly twice as long as that of a 5 parameter model. This is clearly advantageous as the number of parameters grows.

If the assumptions under which the algorithm is derived are met, the approximation will give converge to $\mathcal{D}_{1\mathrm{F}}(\cdot)$ as $N_K \to \infty$, in a similar fashion to the single parameter approximation. For moderate amounts of data, and elliptical regions, the assumptions are often satisfied to a reasonable level by many model classes, and the approximation yields message length estimates that are quite good. Even in the case that region geometry assumptions are seriously violated, the estimated model cost is bounded below by zero, due to the explicit handling of parameter boundaries (and subsequent restriction of the region to at most encompass the entire parameter space). The algorithm may never completely break down and estimate negative (i.e. nonsensical) model costs, as the MML87 approximation may do. However, this advantage does come with a price.

The MMLD-S estimator is not invariant in those situations where a one-to-one transformation removes or introduces hard boundaries into the parameter space. Unlike the MML87 estimator, the MMLD-S approximation does not allow the region to grow beyond the hard boundaries of the parameter space and thus given a fixed region geometry, the region volume and round off cost may differ slightly between the two parameterisations, particularly if the model under consideration lies near the parameter boundaries. The question then becomes: is it better to use an approximation that is invariant but may lead to poor message length estimations, or accept an approximation that suffers a small loss of invariance in exchange for a robust message length estimate that never breaks down in a disastrous fashion? Obviously the answer depends on the model class in question, and how badly the MML87 estimates may fail.

7.4 Examples: MMLD-S Applied to Two Problems

This section demonstrates the application of the MMLD-S approximation to two basic model classes. This is done to show the use of the MMLD-S approximation, highlight a few of the improvements it brings over to MML87 and simply to demonstrate that it

yields useful message lengths on simple, well defined problems. The two model classes chosen are the Binomial (a single parameter distribution), and the univariate Normal (a two parameter distribution). The results show that the approximation provides at least as good as and in some cases better message length and uncertainty region estimates than the corresponding MML87 approximation.

7.4.1 A Single Parameter Example: The Binomial Distribution

First the well known binomial distribution is examined. There is only a single free parameter: θ, the probability of a 'head' occuring in a Bernoulli trial. The negative log-likelihood of a sequence of N symbols, H of which are a 'head', is then given by

$$L(\mathbf{y}|\theta) = -H \log \theta - (N - H) \log(1 - \theta) - \log \binom{N}{2} \tag{7.82}$$

A suitably uniformative prior for θ is the uniform prior, i.e.

$$h(\theta) = 1, \ \theta \in [0, 1] \tag{7.83}$$

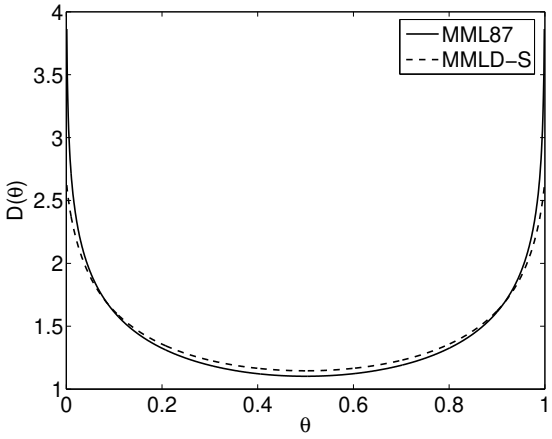
Finally, the KL divergence between true model θ^* and approximating model $\hat{\theta}$ is given by

$$\Delta(\hat{\theta}||\theta^*) = -N \left(\theta^* \left(\log \hat{\theta} + \log \theta^* \right) + (1 - \theta^*) \left(\log(1 - \hat{\theta}) + \log(1 - \theta^*) \right) \right) \tag{7.84}$$

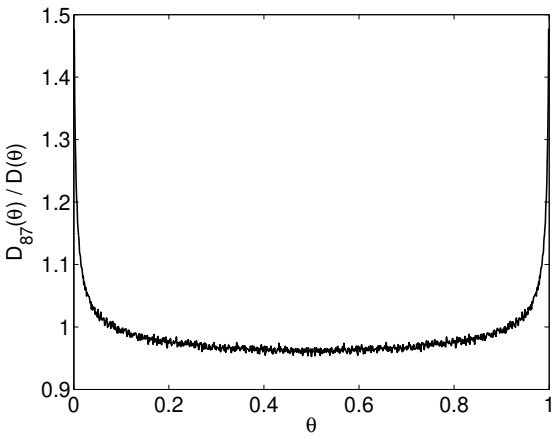
The MML87 formulation of the Binomial is well known [1], and the model cost term is given by

$$\mathcal{D}_{87}(\theta) = \frac{1}{2} (\log N - \log \theta - \log(1 - \theta) - \log 12 + 1) \tag{7.85}$$

Several tests shall be undertaken to compare the model costs estimated by MML87 and MMLD-S versus the exact I1F model cost. To find the exact model costs the simple method of Random Coding shall be employed, with the Random Coding model cost estimates denoted $\mathcal{D}_{\mathrm{R}}(\cdot)$; a description of this algorithm may be found in Section 3.3.2. The first tests will compare the model costs estimated by MML87 and MMLD-S over the parameter range for a moderate amount of data, $N = 10$. This is small enough that the parameter-dependent aspect of the Fisher Information Matrix is not completely swamped by the $\left(\frac{1}{2} \log N\right)$ term, but large enough that the MML87 approximation will not break down badly. The penalties are estimated with an $N_K = 100$, and the Random Coding



(a) MML87/MMLD-S Model Costs, $N = 10$



(b) $\mathcal{D}_{87}(\boldsymbol{\theta})/\mathcal{D}_{\mathrm{R}}(\boldsymbol{\theta})$, for $N = 10$

Figure 7.6: Model Cost Estimates and Errors for the Binomial Model

scheme uses $N_M = 7.5 \times 10^5$ samples. The model curves estimated by MMLD-S and MML87 are plotted in Figure 7.6(a), and Figure 7.6(b) presents the percentage errors between the MML87 curve as compared to the Random Coding model cost curve. The MMLD-S error ratio is not presented as it was within approximately 1% for all values of θ. The MMLD-S and exact curves are thus expected to coincide in the limiting case of $N_K \to \infty, N_M \to \infty$ as would be expected given the nature of the MMLD-S approximation for one parameter models. Examination of Figure 7.6 quickly reveals where the MMLD-S and MML87 penalties differ: for approximately $|\theta - \frac{1}{2}| > 0.4$ the MML87 model cost overestimates the I1F model cost, and for $|\theta - \frac{1}{2}| < 0.4$ it underestimates the I1F model cost. These two regimes are easily explained: when $|\theta - \frac{1}{2}| < 0.4$ the quadratic approximation of the KL divergence is too shallow and the rate of increase due to round-off is underestimated. For $|\theta - \frac{1}{2}| > 0.4$ the quadratic expansion becomes too steep, particularly as it enforces a symmetric region. The MMLD-S algorithm admits a non-symmetric region which allows it to save some fractions of nits for that part of the region that lies to the interior of the parameter space. It can be seen that as $|\theta - \frac{1}{2}| \to \frac{1}{2}$ the relative error in the MML87 estimated model cost grows rapidly.

The second set of tests involves examining the behaviour of the Binomial model for very small amounts of data, i.e. $(N \leq 1)$. While this situation may seem impractical, it may actually arise in mixture modelling applications where partial assignment to classes is used [23]. In addition to estimating the model costs via MML87 and MMLD-S, the small sample size modification to MML87 [1] is also employed, denoted $\mathcal{D}_{\mathrm{SM}}(\theta)$. The

Table 7.2: Binomial Message Lengths for Small Data Sets

N	θ	$\mathcal{D}_{\mathrm{R}}(\theta)$	$\mathcal{D}_{87}(\theta)$	$\mathcal{D}_{\mathrm{SM}}(\theta)$	$\mathcal{D}_{\mathrm{DS}}(\theta)$
2	$\theta = \frac{1}{2}$	0.4892	0.2973	1.5183	0.4887
1	$\theta = \frac{1}{2}$	0.2876	-0.0493	1.2330	0.2868
$\frac{1}{2}$	$\theta = \frac{1}{2}$	0.1534	-0.3959	0.9903	0.1522
$\frac{1}{2}$	$\theta = \frac{1}{10}$	0.3198	0.1149	1.3639	0.3201

results presented in Table 7.2 demonstrate the MMLD-S algorithm yields useful message length approximations even in the face of very small data, being extremely close to those estimated by Random Coding. When $N = 2$ the MML87 approximation underestimates the model cost by around a factor of two, and for $N \leq 1$ it completely breaks down and estimates negative message lengths (which are nonsensical) for $\theta = \frac{1}{2}$, i.e. the parameter which is the least informative in a Fisher sense. This breakdown is due primarily to the quadratic expansion yielding a slope that is much too shallow, resulting in an Uncertainty Region estimate that exceeds the parameter space. Figure 7.7 demonstrates the exact KL surface, the Uncertainty Region as estimated by MMLD-S and the quadratic approximation to the KL curve that is employed by MML87. In this case $J(\theta) = 4$, and the resulting MML87 uncertainty region estimate is $\Omega_{87} = \left\{x : |x - \frac{1}{2}| < 0.866\right\}$, yielding a complete region width of approximately 1.7, i.e. clearly exceeding the parameter space. For $N = \frac{1}{2}$ and $\theta = \frac{1}{10}$, the parameter yields a sufficiently steep Fisher Information that the MML87 region does not exceed the parameter space, though it can be observed that the MML87 approximation still underestimates the model cost by more than a factor of two. In all cases the small-sample size correction prevents the model cost from being negative (as is expected), but results in grossly overestimated model costs. However, it should be pointed out that this overestimation of model costs in a practical situation is not a concern as all that is required is a suitable positive model cost estimimation in the face of tiny data to ensure tiny classes are rejected by the message length criteria. The purpose of the MMLD-S comparison is not to claim the extra accuracy in model cost estimation for tiny sample sizes will have any particularly great impact on mixture modelling results; rather it is to demonstrate that the method more gracefully handles these difficult situations than MML87 and its modifications.

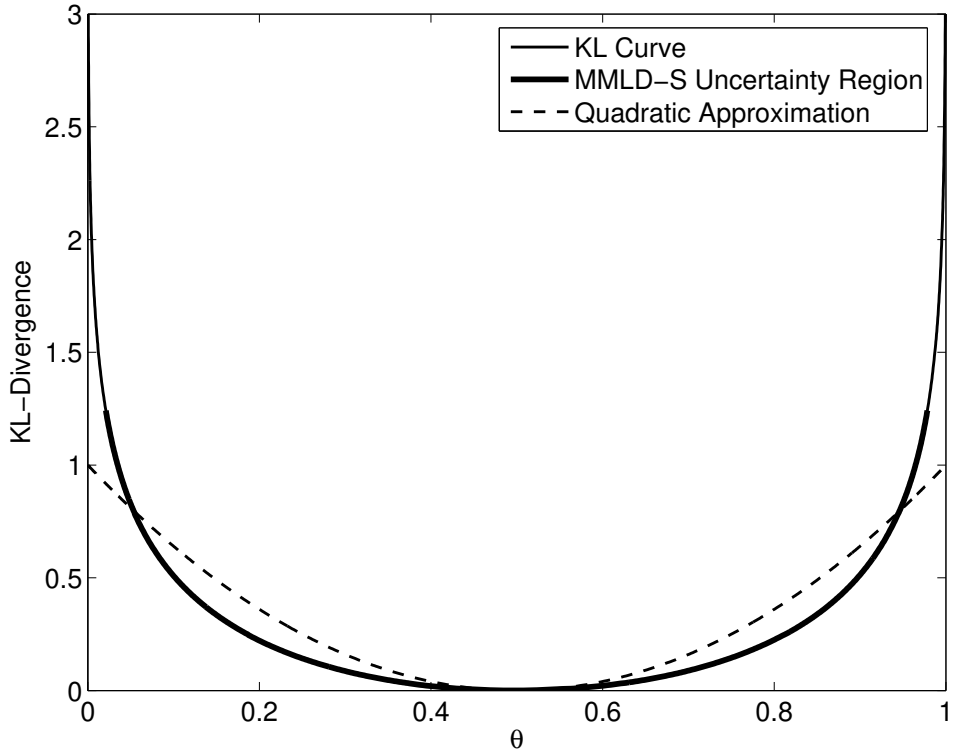


Figure 7.7: KL Curve and approximation for the Binomial Model with $\theta = \frac{1}{2}, N = 1$

7.4.2 A Multiple Parameter Example: The Normal Distribution

The second model considered is the univariate Normal distribution. There are two free parameters: the mean μ and variance $\tau = \sigma^2$. The negative log-likelihood for N observations $\mathbf{y} \in \mathbb{R}^N$ is given by

$$L(\mathbf{y}|\mu, \tau) = \frac{N}{2}\log(2\pi) + \frac{N}{2}\log\tau + \frac{1}{2\tau}\sum_{i=1}^{N}(y_i - \mu)^2 \tag{7.86}$$

To ensure that the priors have no effect on the study of the uncertainty regions, uniform distributions are placed over both μ and τ thus

$$h(\mu, \tau) = \frac{1}{800},\ \mu \in [0.01, 40.01], \tau \in [-10, 10] \tag{7.87}$$

The KL divergence between a generating model $\boldsymbol{\theta}^* = \{\mu^*, \tau^*\}$ and approximating model $\boldsymbol{\theta} = \{\mu, \tau\}$ is given by

$$\Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*) = \frac{N}{2}\log\frac{\tau}{\tau^*} + \frac{N}{2\tau}\left(\tau^* + (\mu - \mu^*)^2\right) - \frac{N}{2} \tag{7.88}$$

The (μ, τ) entries of Fisher Information matrix are zero, so the basis vectors for the MMLD-S algorithm are $\mathbf{B} = \mathbf{I}_2$, i.e. the parameter rays are aligned with the natural component axis. The MML87 formulation of the Normal is well known [1] and the model cost is given by

$$\mathcal{D}_{87}(\boldsymbol{\theta}) = \log N - \frac{1}{2}\log 2\tau^3 + \log \kappa_2 + 1 \tag{7.89}$$

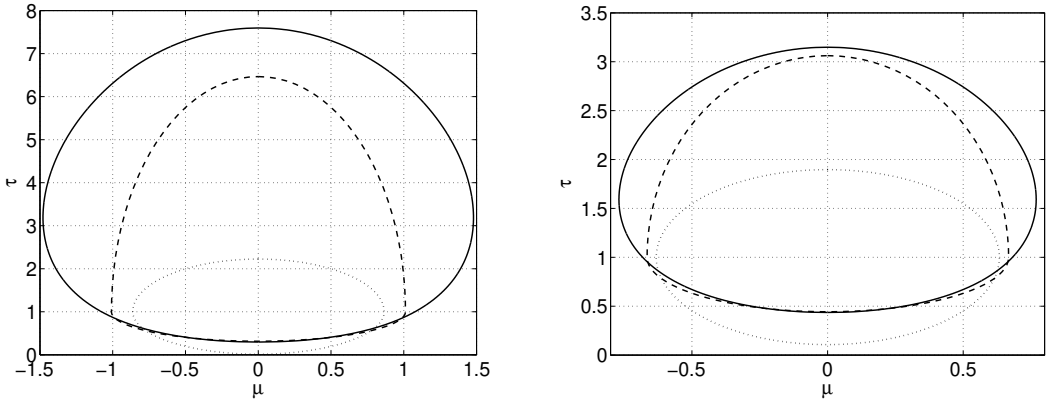
However, for a fairer comparison a slightly modified version of MML87 based on elliptical regions (rather than cells of quantising lattices) was employed, which in all cases lead to reduced approximation errors; the key aspect of MML87 under examination being the quadratic approximation of the KL divergence rather than the exact choice of region shape. For the univariate Normal distribution the problem of finding the I1F uncertainty region and determining the values of the assertion and round-off integrals may be greatly simplified to an expression involving univariate numerical integration. This is new work and a complete derivation is given in Appendix B of this thesis. The univariate Normal has two special properties when uniform priors are used for μ and τ; the first is that the model cost is invariant to the value of μ^*, i.e.; this may be obviously seen by examining (7.88); the μ^* parameter is seen only as a difference from μ; thus translating μ^* and μ by any amount has no effect. The second property is revealed when model costs for varying τ^* are examined versus model costs estimated via MMLD-S and MML87, and is conjectured to arise from the behaviour of the KL divergence function when τ^* is varied. If a fixed geometry is used to approximate the uncertainty region, the difference in nits between the exact model cost and the approximate model cost will be constant depending only on the datasize N, and not on particular values of τ^*. The immediate result of this is that a comparison between the exact model costs and those estimated by methods with a fixed geometry need only examine a single (μ^*, τ^*) (the particular values being inconsequential) for various N to gauge the performance of the methods. The MMLD-S and MML87 approximations both assume fixed geometries (glued hyper-ellipses for MMLD-S, and an ellipse for this modified form of MML87). Table 7.3 lists model cost estimates for MMLD-S and MML87 versus the exact values. In the table $\mathcal{D}_{\mathrm{1F}}$ denotes the exact I1F model cost, $\mathcal{D}_{\mathrm{DS}}$ the model cost estimated by MMLD-S, $\mathcal{D}_{87}$ the model cost estimated by MML87, the $(-\log q)$ columns give the assertion cost, and the $\Delta(\cdot)$ columns list the expected round-off

costs. The exact and estimated uncertainty regions for a range of N are shown in Figures 7.8(a)-7.8(f).

Table 7.3: Normal Model Costs for ($\mu^* = 0$, $\tau^* = 1$)

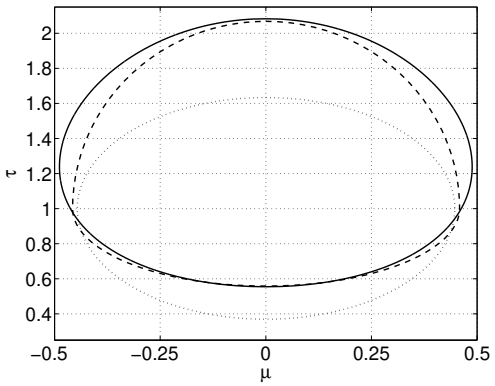
	Exact			MMLD-S			MML87	Error	
N	$\mathcal{D}_{1F}$	$-\log q$	$\Delta(\cdot)$	$\mathcal{D}_{DS}$	$-\log q$	$\Delta(\cdot)$	$\mathcal{D}_{87}$	$\mathcal{D}_S - \mathcal{D}_{1F}$	$\mathcal{D}_{87} - \mathcal{D}_{1F}$
5	5.746	3.847	1.898	5.957	4.403	1.553	6.424	0.211	0.678
10	6.822	5.496	1.325	6.899	5.674	1.225	7.117	0.077	0.295
20	7.674	6.530	1.144	7.701	6.600	1.101	7.810	0.027	0.136
50	8.665	7.610	1.054	8.679	7.641	1.038	8.726	0.014	0.061
100	9.385	8.358	1.026	9.391	8.372	1.019	9.419	0.006	0.034
1000	11.71	10.70	1.002	11.71	10.70	1.002	11.72	0.000	0.001

Examination of Table 7.3 reveals that the MMLD-S approximation yields more accurate model cost estimates than MML87. Further examination reveals that the MMLD-S approximation underestimates the prior mass allocated to the assertion (in this case, simply the area of the region) and slightly underestimates the round-off cost. The MML87 approximation estimates the round-off cost as ($D/2 = 1$) for all values of N and thus underestimates the round-off cost in all cases. Asymptotically both the exact and MMLD-S round-off costs approach 1, and this is not unexpected as for very large N the uncertainty region will shrink and is expected to converge to an ellipse, with the KL divergence within the region being quadratic (i.e. the assumptions under which MML87 is derived). Examination of Figures 7.8(a)-7.8(f) is quite revealing. Both the MMLD-S regions and the MML87 regions are smaller than the exact uncertainty regions in all cases (though only marginally so for $N = 1000$), which corroborates the increased assertion costs in Table 7.3. The differences between MML87 and MMLD-S are also evident in the diagrams; in particular, it is clear that the quadratic approximation leads to the underestimation of the uncertainty region by treating both sides of τ^* as having the same curvature. The MMLD-S approximation is able to take advantage of the fact that the curvature of the KL divergence for ($\tau > \tau^*$) is shallower than for ($\tau < \tau^*$) and grows the two ellipses on the upper side of τ^* to a larger size than those on the lower side. An interesting point arises when comparing MMLD-S to the exact region: the geometry of the exact region would suggest that a glueing of four elliptical sectors could produce a very close approximation, closer than achieved by the MMLD-S approximation. There is, however, a reason why MMLD-S fails to approximate the uncertainty region even closer than it does. The

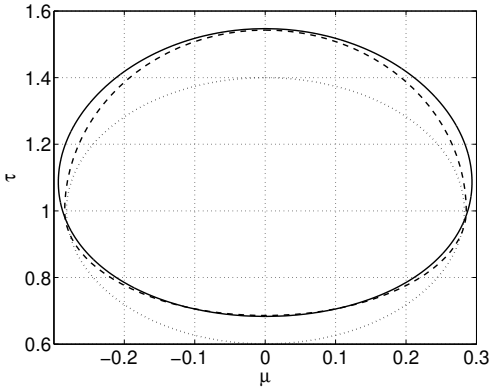


(a) $N = 5$

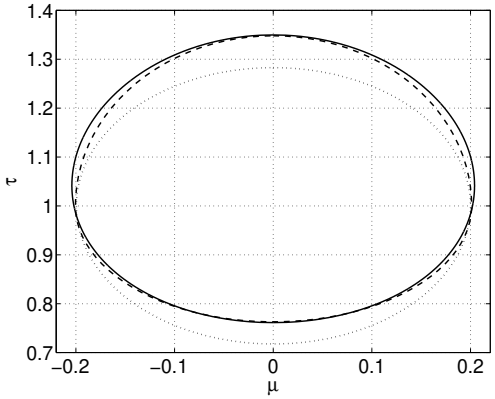
(b) $N = 10$



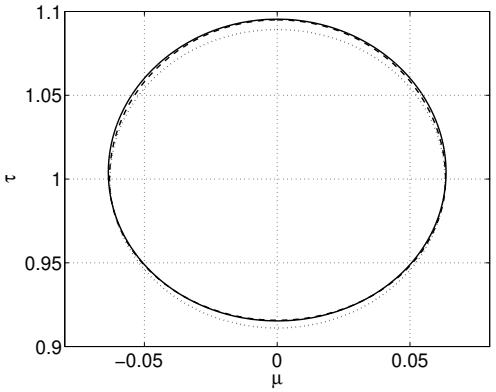
(c) $N = 20$



(d) $N = 50$



(e) $N = 100$



(f) $N = 1000$

Figure 7.8: I1F, MMLD-S and MML87 Uncertainty Regions for $\mathcal{N}(0,1)$ (I1F=Solid, MMLD-S=Dashed, MML87=Dotted)

MMLD-S approximation defines the region by growing patch regions along each of the component axis; if these component axis are not aligned with the axis that define the minimum circumscribed rectangle around the exact uncertainty region, the MMLD-S region may well underestimate the region area. In the case of the normal distribution, the maximum τ-span lies along the $\mu = \mu^*$ axis. However, the maximum μ-span does not lie along the $(\tau = \tau^*)$ axis; rather it can be shown to lie along

$$\tau_{MAX} = \exp\left(\frac{2\xi}{N}\right)\tau^* \tag{7.90}$$

where ξ is the KL divergence of any model on the uncertainty region boundary. Clearly, for small N the maximum-span axis lies further away from τ^* than for large N, and as $N \to \infty$, $\tau_{MAX} \to \tau^*$. The plots demonstrate this is the case; as N grows larger, the approximation improves as $\tau_{MAX} \to \tau^*$.

7.5 Conclusion

This chapter has introduced the new MMLD-S message length approximation that is more robust in the face of non-quadratic KL divergence surfaces and asymmetrical Uncertainty Regions. The salient points of this chapter were:

- The MMLD-S approximation defines the uncertainty region as a union of hyper-elliptical sectors. In contrast to the MML87 approximation, this allows for asymmetrical uncertainty regions. The time complexity of the estimator is linear in the number of dimensions and works by exploiting a property of the uncertainty region boundary rule.

- Modifications for handling 'soft' parameter boundaries, and non-uniform prior densities based on sampling were presented.

- The approximation was applied to two test cases: the Binomial distribution and the Normal distribution. In the case of the former, it provides excellent estimates of the message length even when the amount of data is tiny, in the latter, the ability to define asymmetric uncertainty regions lead to closer estimates of the exact regions than MML87 .

The next chapter details the application of MMLD-S to the Autoregressive-Moving Average (ARMA) class of models.

Chapter 8

MMLD-S Inference of ARMA Models

8.1 Introduction

The previous chapter introduced the new MMLD-S message length approximation that addressed some of the problems of MML87 that, in particular, arose for the ARMA class of models. This chapter examines the performance of this approximation when applied to ARMA models. As a brief recap, it can be recalled that the MMLD-S approximation requires the following to be available:

- A negative log-likelihood function, $L(\mathbf{y}|\boldsymbol{\theta})$
- Priors over the parameters, $h(\boldsymbol{\theta})$
- The KL divergence function, $\Delta(\boldsymbol{\theta}_1||\boldsymbol{\theta}_2)$
- The exact Fisher Information Matrix, $\mathbf{J}(\boldsymbol{\theta})$

This chapter is organised as follows: Section 8.2 works the Autoregressive model into the MMLD-S framework, Section 8.3 examines Moving Average models, Section 8.4 presents the combined ARMA model, Section 8.5 examines the performance of the approximation, and Section 8.6 covers concluding remarks.

8.2 Autoregressive Models

This section examines the MMLD-S estimator for the Autoregressive model class. As per Chapter 3, the model is parameterised in root-space rather than coefficient space, and subsequently the full parameter vector is $\boldsymbol{\theta} = [\boldsymbol{\beta}, \sigma^2, S]$ where $\boldsymbol{\beta}$ is the vector of root-space parameters, σ^2 is the innovation variance and S is the autoregressive structure (see Section 4.3.4). The negative log-likelihood is given by

$$L(\mathbf{y}|\boldsymbol{\theta}) = \frac{N}{2}\log(2\pi) + \frac{1}{2}\log|\boldsymbol{\Gamma}(\boldsymbol{\theta})| + \frac{1}{2}\mathbf{y}\boldsymbol{\Gamma}^{-1}(\boldsymbol{\theta})\mathbf{y}^T \tag{8.1}$$

where $\mathbf{y} \in \mathbb{R}^N$ is the vector of data, and $\boldsymbol{\Gamma}(\cdot)$ is the autoregressive process autocovariance matrix. This may be efficiently evaluated via the Kalman Filter, or by reformulating it into the conditional form. The parameter priors are chosen in the same way as in Chapter 3; for completeness these are re-summarised here

$$\begin{aligned}
h(\boldsymbol{\theta}) &= h(S)\cdot h(\sigma^2)\cdot h(\boldsymbol{\beta}) && (8.2)\\
h(S) &= \left(\sum_{i=1}^{P_{MAX}} \lfloor i/2 \rfloor + 1\right)^{-1} && (8.3)\\
h(\boldsymbol{\beta}) &= \left(\frac{1}{\pi}\right)^{N_C}\cdot\left(\frac{1}{2}\right)^{N_R}\cdot N_C!\cdot N_R! && (8.4)\\
h(\sigma^2) &= \frac{1}{\sigma^2}\left(\log(\sigma_U^2) - \log(\sigma_L^2)\right)^{-1} && (8.5)
\end{aligned}$$

where P_{MAX} is the maximum order the AR process may assume, S is the root structure, N_C the number of complex poles and N_R the number of real poles . Finally, the KL divergence function may be found as the KL divergence between two multivariate Normal distributions. This may be efficiently expressed in the conditional form

$$\begin{aligned}
\Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*) &= \frac{1}{2}\left(\log\frac{|\boldsymbol{\Gamma}(\boldsymbol{\theta})|}{|\boldsymbol{\Gamma}(\boldsymbol{\theta}^*)|} + \mathrm{Tr}\left(\boldsymbol{\Gamma}(\boldsymbol{\theta}^*)\cdot\boldsymbol{\Gamma}^{-1}(\boldsymbol{\theta})\right)\right) + \frac{(N-P)}{2}\log\frac{\sigma^2}{\sigma^{2*}}\\
&\quad + \frac{(N-P)}{2\sigma^2}\left(\sigma^{2*} + (\mathbf{a}(\boldsymbol{\theta}^*) - \mathbf{a}(\boldsymbol{\theta}))\cdot\boldsymbol{\Gamma}(\boldsymbol{\theta})\cdot(\mathbf{a}(\boldsymbol{\theta}^*) - \mathbf{a}(\boldsymbol{\theta}))^T\right) - \frac{N}{2} \qquad (8.6)
\end{aligned}$$

where $\boldsymbol{\Gamma}(\boldsymbol{\theta})$ is a $(P \times P)$ autocovariance matrix, and $\mathbf{a}(\boldsymbol{\theta})$ are the parameters expressed in coefficient space. The way in which the innovation variance is handled may be simplified considerably by the following proposition.

Proposition 8.1: Treating σ^{2^*} as known, the uncertainty region $\boldsymbol{\Omega}(\boldsymbol{\beta}^*)$ for the parameters $\boldsymbol{\beta}^*$ is identical for all values of $\sigma^{2^*} > 0$.

Proof: Express the KL divergence in unconditional form as

$$\Delta(\boldsymbol{\theta}_2||\boldsymbol{\theta}_1) = \frac{N}{2}\log\frac{\sigma^2}{\sigma^{2^*}} + \frac{1}{2}\left(\log\frac{|\boldsymbol{\Gamma}'|}{|\boldsymbol{\Gamma}'^*|}\right) + \frac{\sigma^{2^*}}{\sigma^2}\mathrm{Tr}\left(\boldsymbol{\Gamma}'^* \cdot \boldsymbol{\Gamma}'^{-1}\right) - \frac{N}{2} \tag{8.7}$$

where $\boldsymbol{\Gamma}' = \sigma^{-2}\boldsymbol{\Gamma}$ and $\boldsymbol{\Gamma}'^* = \sigma^{-2^*}\boldsymbol{\Gamma}^*$. Clearly, if only the uncertainty region for the $\boldsymbol{\beta}^*$ parameters is being determined, then $\sigma^2 = \sigma^{2^*}$ for all models $\boldsymbol{\theta} \in \boldsymbol{\Omega}(\boldsymbol{\beta}^*)$, as σ^2 is never varied. As σ^2 and σ^{2^*} are present in (8.7) only as a ratio, the KL divergence for any model in the region is unchanged by varying σ^{2^*}. **Q.E.D.**

Of course, when the uncertainty region is expanded to include the innovation variance (and thus encompass different values of σ^2), the model cost will vary with σ^{2^*}; however, given that only one innovation variance parameter is required irrespective of the order of the ARMA models, the accuracy to which it should be stated may be adequately found via the MML87 approximation without majorly affecting the order selection process. This simplification to the message length estimates may be made by fixing the variance to a constant (for convenience, assume $\sigma^2 = 1$) and finding the model cost for $\boldsymbol{\beta}$ parameters only, denoted $\mathcal{D}(\boldsymbol{\beta})$. Once this has been computed, the final message length may be found as

$$I_{\mathrm{DS}}(\boldsymbol{\theta}, \mathbf{y}) = L(\mathbf{y}|\boldsymbol{\theta}) + \mathcal{D}_{\mathrm{DS}}(\boldsymbol{\beta}) - \log h(\sigma^2) + \frac{1}{2}\left(\log\left(\frac{N}{2\sigma^4}\right) - \log 12 + 1\right) \tag{8.8}$$

where the variance is encoded using the MML87 approximation for a single parameter. The Information Matrix used to determine the parameter basis' for the $\boldsymbol{\beta}$-parameters may be computed as per Section 3.3.4.

8.2.1 Parameter Boundaries

The MMLD-S algorithm explicitly takes parameter boundaries into account. The redundancy of the parameterisation in root-space poses the most interesting problem. Furthermore, the N_R real poles $[\alpha_1, \ldots, \alpha_{N_R}]$ must satisfy

$$\alpha_1 \leq \alpha_2 \leq \ldots \leq \alpha_{N_C} \tag{8.9}$$

and the N_C complex poles with radii $[r_1, \ldots, r_{N_C}]$

$$r_1 \leq r_2 \leq \ldots \leq r_{N_C} \tag{8.10}$$

to ensure that all the models spanned by the parameter space are unique. Simply treating all of these restrictions as 'hard' boundaries (as per Section 7.3.11) by preventing the region to grow past them leads to suboptimal message length estimates. These parameter restrictions are therefore treated as soft. The requirement for the poles to satisfy $|p_i| < 1$ is a strict requirement for stationarity, and KL divergences may not be computed for models lieing outside the stationarity region; thus the only way to handle these restrictions is to treat them as 'hard'.

8.3 Moving Average Models

The MMLD-S algorithm is easiest to implement when any restricted parameters are bounded by a simple polytope in parameter space; if invertibility is required, the moving average model expressed in coefficient space, $\mathbf{c}$, has parameters bounded by a complex, non-rectangular polytope. It is possible, if a uniform prior over the moving average parameters is assumed, to re-express the model in partial autocorrelation space, $\boldsymbol{\varrho}$. A one-to-one transformation between $\boldsymbol{\varrho}$ and the moving average parameters $\mathbf{c}$ exists [2], and in partial autocorrelation space the invertibility restriction reduces to a requirement that $\varrho_i \in (-1, 1)$. To transform from partial autocorrelations to coefficients, a recurrence relation may be employed. The full parameter vector is $\boldsymbol{\theta} = \left[\boldsymbol{\varrho}, \sigma^2, Q\right]$. The negative log-likelihood of an MA(Q) process may then be expressed in the usual unconditional form

$$L(\mathbf{y}|\boldsymbol{\theta}) = \frac{N}{2}\log(2\pi) + \frac{1}{2}\log|\boldsymbol{\Gamma}(\boldsymbol{\theta})| + \frac{1}{2}\mathbf{y}\boldsymbol{\Gamma}^{-1}(\boldsymbol{\theta})\mathbf{y} \tag{8.11}$$

The priors are no longer over $\mathbf{c}$ but instead over $\boldsymbol{\varrho}$. A uniform prior $h(\mathbf{c}) \propto 1$ over the invertibility region induces a set of Beta priors over $\boldsymbol{\varrho}$ [147], and thus the complete prior

density over an MA(Q) model is given by

$$\begin{aligned} h(\boldsymbol{\theta}) &= h(Q) \cdot h(\sigma^2) \cdot h(\boldsymbol{\varrho}) && (8.12) \\ h(Q) &= (Q_{MAX})^{-1} && (8.13) \\ h(\boldsymbol{\varrho}) &= \prod_{i=1}^{Q} \text{Beta}\left(\varrho_i, \lfloor (i+1)/2 \rfloor, \lfloor i/2 \rfloor + 1\right) && (8.14) \\ h(\sigma^2) &= \frac{1}{\sigma^2}\left(\log(\sigma_U^2) - \log(\sigma_L^2)\right)^{-1} && (8.15) \end{aligned}$$

where $\text{Beta}(\cdot,\cdot)$ is a Beta distribution and $\lfloor \cdot \rfloor$ denotes the integer component. The KL divergence between the N data points generated by a 'true' Moving Average model $\boldsymbol{\theta}^*$ and an approximating model $\boldsymbol{\theta}$ is given by

$$\Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*) = \frac{N}{2}\log\frac{\sigma^2}{\sigma^{2*}} + \frac{1}{2}\log\frac{|\boldsymbol{\Gamma}|}{|\boldsymbol{\Gamma}^*|} + \frac{\sigma^{2*}}{\sigma^2}\text{Tr}\left(\boldsymbol{\Gamma}^* \cdot \boldsymbol{\Gamma}^{-1}\right) - \frac{N}{2} \tag{8.16}$$

where $\boldsymbol{\Gamma}$ is the autocovariance matrix of the process defined by $\boldsymbol{\theta}$, and $\boldsymbol{\Gamma}^*$ is the autocovariance matrix of the process defined by $\boldsymbol{\theta}^*$. Proposition 8.1 is applicable to the Moving Average KL divergence (8.16) and thus the coding of the coefficients $\mathbf{c}$ and the innovation variance σ^2 may be decoupled as per Section 8.2. The Beta priors that are induced over $\boldsymbol{\varrho}$ through the transformation to partial autocorrelation space may be sufficiently 'curved' as to seriously violate the MMLD-S assumption of uniform priors. To this end, the prior mass is estimated via the sampling extension to MMLD-S described in Section 7.3.12.

8.3.1 Parameter Restrictions

The parameter restrictions for a moving average model characterised in partial autocorrelation space to be unique are

$$\varrho_i \in (-1, 1) \tag{8.17}$$

for all $\varrho \in \boldsymbol{\varrho}$. Simply treating these restrictions as 'hard' may lead to suboptimal results. Models that lie outside of the invertibility region are not invalid in the same way as autoregressions that lie outside of the stationarity region; they are simply indistinguishable from an equivalent model that lies within the invertibility region. The KL divergence of the Moving Average model, for finite amounts of data does not rise to infinity as $|\varrho_i| \to 1$, so the parameter may be treated as soft. The simplest approach would be to expand

the hard parameter boundaries during the growth phase to allow the region to cross the soft parameter bounds. The problem with this approach is that the models outside of the invertibility region are distributed more sparsely than those within the invertiblity region. This is intuitive as the volume of the space outside of the invertibility region is infinite, while the invertibility region has a finite volume. This causes problems for the grow-and-prune approach to handling soft boundaries as the region will grow much larger in the space ouside the invertibility region. While the pruning will mean that the final region volume is approximately correct, the average round-off KL divergence will be underestimated. The solution to this is to impose the simple transformation

$$\varrho_i = \begin{cases} -2 - \varrho_i & \text{for } \varrho_i < -1 \\ \varrho_i & \text{for } \varrho \in [-1, 1] \\ 2 - \varrho_i & \text{for } \varrho > 1 \end{cases} \tag{8.18}$$

This simple transformation ensures the scale of models on both sides of the invertibility region is the same, and the region growth will be approximately the same, leading to better estimates of average round-off cost.

8.4 Autoregressive Moving Average Models

The mixed Autoregressive Moving Average model is now considered. This is literally a combination of the two preceding model classes, and may be implemented in an MMLD-S framework in this manner. The full parameter vector is now $\boldsymbol{\theta} = \left[\boldsymbol{\beta}, \mathbf{c}, \sigma^2, S, Q\right]$. For the purposes of completeness, the priors used are summarised below

$$h(\boldsymbol{\theta}) = h(S) \cdot h(Q) \cdot h(\boldsymbol{\beta}) \cdot h(\boldsymbol{\varrho}) \cdot h(\sigma^2) \tag{8.19}$$

$$h(S) = \left(\sum_{i=1}^{P_{MAX}} \lfloor i/2 \rfloor + 1 \right)^{-1} \tag{8.20}$$

$$h(Q) = (Q_{MAX})^{-1} \tag{8.21}$$

$$h(\boldsymbol{\beta}) = \left(\frac{1}{\pi}\right)^{N_C} \cdot \left(\frac{1}{2}\right)^{N_R} \cdot N_C! \cdot N_R! \tag{8.22}$$

$$h(\boldsymbol{\varrho}) = \prod_{i=1}^{Q} \text{Beta}\left(\varrho_i, \lfloor (i+1)/2 \rfloor, \lfloor i/2 \rfloor + 1\right) \tag{8.23}$$

$$h(\sigma^2) = \frac{1}{\sigma^2} \left(\log(\sigma_U^2) - \log(\sigma_L^2) \right)^{-1} \tag{8.24}$$

The KL divergence between generating ARMA model $\boldsymbol{\theta}^*$ and approximating model $\boldsymbol{\theta}$ is given by:

$$\Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*) = \frac{N}{2}\log\frac{\sigma^2}{\sigma^{2*}} + \frac{1}{2}\log\frac{|\boldsymbol{\Gamma}|}{|\boldsymbol{\Gamma}^*|} + \frac{\sigma^{2*}}{\sigma^2}\mathrm{Tr}\left(\boldsymbol{\Gamma}^* \cdot \boldsymbol{\Gamma}^{-1}\right) - \frac{N}{2} \tag{8.25}$$

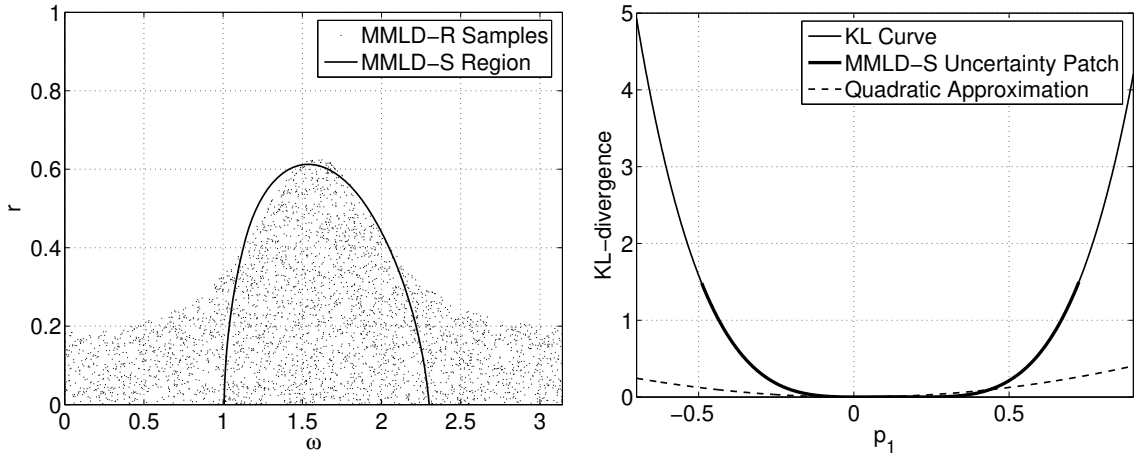
which is clearly of the same form as the Moving Average and unconditional Autoregressive KL divergences; thus Proposition 7.1 applies to the estimation of the uncertainty region for the ARMA model. The soft/hard parameter boundaries, transformation of coefficients and the application of the prior sampling modification for the Moving Average partial autocorrelation parameters was applied as per the pure AR and MA models previously detailed.

8.5 Performance of the MMLD-S Approximation

The effectiveness of the MMLD-S algorithm at estimating the message lengths of models is now compared against the MML87 message lengths, and the 'ground truth' message lengths as estimated via the Random Coding approach.

8.5.1 Autoregressive Models

The autoregressive model is examined first. This has been shown in Section 4.6 to pose some serious problems for the MML87 criteria when expressed in root-space, due to issues with the Information Matrix and the KL divergence surface, which have been addressed to some degree by the MML87 'Search Modification' approximation. To determine whether the MMLD-S approximation offers improvement over the 'Search Modification' approximation, as compared to the exact I1F message length, a series of experiments were undertaken. These consist of randomly sampling 250 models from the stationarity region for a range of model orders and data sizes, and comparing the MMLD-S and MML87S message lengths against the exact I1F message lengths, as estimated by the random coding process. Model orders ($P = 2, \ldots, 5$) are considered as the process of estimating the 'true' model cost via Random Coding became prohibitively slow for larger model sizes. The results are summarised in Tables 8.1 and 8.2; the results present the mean absolute relative errors, and the (signed) minimum and maximum relative errors for both approximations. The relative error between the approximate model cost $\mathcal{D}_{(\cdot)}(\boldsymbol{\theta})$ and the true model cost (as



(a) Uncertainty Regions (b) KL Divergence Curves

Figure 8.1: Uncertainty Regions for AR(2) Model $\boldsymbol{\beta}^* = [r, \omega] = [0.08, 1.65]$, $N = 20$

estimated by random coding) $\mathcal{D}_{\mathrm{R}}(\boldsymbol{\theta})$ is given by

$$\delta_S = \frac{\mathcal{D}_{(\cdot)}(\boldsymbol{\theta})}{\mathcal{D}_{\mathrm{R}}(\boldsymbol{\theta})} - 1 \tag{8.26}$$

where $\mathcal{D}_{(\cdot)}(\cdot)$ is the model cost of the particular approximation.

Table 8.1: Autoregressive Model Cost Relative Errors for MML87S

	$N = 20$			$N = 35$			$N = 50$		
P	E $[\lvert\delta_S\rvert]$	inf δ_S	sup δ_S	E $[\lvert\delta_S\rvert]$	inf δ_S	sup δ_S	E $[\lvert\delta_S\rvert]$	inf δ_S	sup δ_S
2	0.1143	-0.8758	0.4916	0.1093	-0.7252	0.4499	0.1068	-0.6281	0.4040
3	0.2006	-0.3414	0.8254	0.2000	-0.3071	0.7684	0.2004	-0.2834	0.7144
4	0.2350	-0.3929	0.8246	0.2588	-0.2050	0.8181	0.2729	-0.1473	0.8674
5	0.3136	-0.4090	0.9726	0.3295	-0.2277	1.0093	0.3471	-0.1356	1.0926

Table 8.2: Autoregressive Model Cost Relative Errors for MMLD-S

	$N = 20$			$N = 35$			$N = 50$		
P	E $[\lvert\delta_S\rvert]$	inf δ_S	sup δ_S	E $\left[\delta_S^2\right]$	inf δ_S	sup δ_S	E $\left[\delta_S^2\right]$	inf δ_S	sup δ_S
2	0.0844	-0.1322	0.4451	0.0684	-0.1021	0.4465	0.0607	-0.1206	0.3933
3	0.1477	-0.1871	0.5947	0.1217	-0.0666	0.5306	0.1067	-0.0455	0.5582
4	0.2212	-0.0740	0.7294	0.1980	-0.1012	0.6964	0.1871	-0.1501	0.6612
5	0.2884	-0.0206	0.8877	0.2465	-0.0115	0.7531	0.2331	0.0120	0.7826

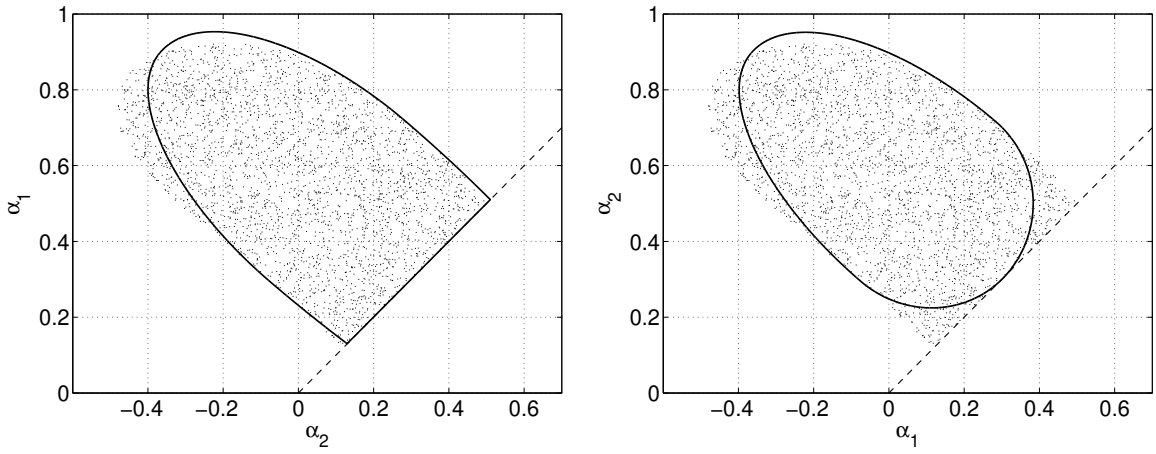
The results indicate that for all data sizes and all model orders, the MMLDS approach has consistently lower mean model cost errors, and a smaller range of model cost errors. As N increases, the MMLD-S errors decrease more rapidly than for the MML87S approximation, an indication that while the conservative nature of the MML87S approximation yields better message costs for it for small data sizes, it is too conservative once the data

sizes start to grow. Similar results were also observed for several further tests undertaken on a sample size of $N = 10$. It is interesting to note that the maximum underestimation scores for MML87S are particulary great, especially for small data sizes; this is due to the fact that even with the modifications made the approximation is performing poorly in the cases that cause the exact MML87 approximation to breakdown. The model for $(P = 2, N = 20)$ that yielded a -0.8758% underestimation of the exact I1F message length cost is

$$A(q) = 1 + 0.0141q^{-1} + 0.006996q^{-2} \tag{8.27}$$

which corresponds to a root-space model with $\boldsymbol{\beta} = [r, \omega] = [0.08, 1.65]$. The MMLD-S model cost estimate is 2.16 nits (with $N_K = 100$), the exact MML87 model cost is -1.0421 nits, and the 'Search' modification model cost is 0.19; compare this with the exact I1F model cost of 1.5 nits. Obviously the two MML87 approximations break down quite badly in this case, the exact MML87 approximation significantly more so than the MML87S approximation which at least yields a positive model cost. The MMLD-S algorithm overestimates the model cost by around 40%, and Figure 8.1(a) reveals why. For the small sample size, and the highly 'uninformative' (in a Fisher sense) parameters, the exact uncertainty region is not easily approximated by a rectangle (or four elliptical subsectors). The MMLD-S region has captured the central component of the true uncertainty region, but has failed to cover the sections on either side of the centre. The KL curve, MMLD-S patch region and MML87S quadratic approximation for parameter axis 1 are shown in Figure 8.1(b). It can be observed that near the parameter estimates the curvature of the KL surface is extremely shallow; the quadratic expansion fails to capture the rapid rise of the true KL divergence as the parameters approach the stationarity boundary, leading to an overestimation of the coding volume, and subsequent underestimation of the model cost. In general, the results indicated that the largest approximation errors for both MMLD-S and MML87S appeared to arise when the model contained complex poles, particulary for smaller radial components, as the uncertainty regions in these cases are not very well approximated by convex shapes (unions of hyperellipsoidal sectors in the case of MMLD-S, and lattice cells for MML87, as the extreme case in Figure 8.1(a) demonstrates).

Finally, the effectiveness of the 'amputation' approach to handling soft boundaries for the AR models is examined. Figures 8.2(a) and 8.2(b) show the uncertainty regions of the



(a) 'Soft' boundaries

(b) 'Hard' boundaries

Figure 8.2: Uncertainty Regions for the AR(2) Model $\boldsymbol{\beta}^* = [\alpha_1, \alpha_2] = [0.1, 0.5]$, $N = 20$

model AR model with parameters $\boldsymbol{\beta} = [\alpha_1, \alpha_2] = [0.1, 0.5]$ for $N = 20$. Figure 8.2(a) shows the uncertainty region using the amputation procedure (see Section 7.3.11 for details) and Figure 8.2(b) shows the uncertainty region treating the redundancy boundary as 'hard'. It is clear that merely treating the boundary as hard leads to a further underestimation of the uncertainty region volume, while the soft boundary approach produces a much closer fit. In this case, the 'true' model cost as estimated by random coding was approximately 2.17 nits; using 'soft' boundaries yielded a model cost of 2.27 nits (a 4.5% error), using 'hard' boundaries yielded a cost of 2.34 nits (a 7.8% error) and ignoring restrictions altogether yielded a cost of 1.7 nits (a 21% error). Clearly, ignoring the restrictions altogether leads to a very poor estimate of the uncertainty region (a large overestimation).

For the two parameter case, the difference between treating the boundaries as hard and soft is not too great. However, as the dimensionality of the parameter space grows the error can become quite large (as the ratio of volume of the hyper-ellipses to hyper-boxes begins to shrink). As an example, consider the AR(4) model with four real poles $[\alpha_1, \ldots, \alpha_4]$, and a sample size of $N = 20$. Table 8.3 summarises the model costs using soft, hard and no boundaries for several values of α_1, with $\alpha_{(2:4)} = [0.2, 0.3, 0.4]$. This table is interesting for several reasons. The most obvious is that the soft boundary scheme achieves much better approximation accuracy than either the hard boundary scheme, or simply ignoring boundaries. Failing to take the boundaries into account leads to a gross underestimation of the model cost for all four values of α_1, and simply treating the boundaries as 'hard' also leads to an overestimation of the model costs. It can also be observed that treating

Table 8.3: Autoregressive Model Cost Relative Errors with Different Boundary Schemes

		Soft		Hard		None	
α_1	True Cost	Cost	Error	Cost	Error	Cost	Error
0	3.5823	3.7331	0.0421	5.5375	0.5458	0.9117	-0.7455
0.05	3.6328	3.8288	0.0539	5.7004	0.5691	0.9225	-0.7461
0.1	3.6995	3.9448	0.0663	5.9688	0.6134	0.9362	-0.7469
0.15	3.7549	4.0789	0.0863	6.6058	0.7593	0.9657	-0.7428

the boundaries as hard leads to a estimate that appears to be quite sensitive to values of α_1. As α_1 is increased, the model costs estimated by the soft boundary scheme grow at a similar rate to the true model costs; in contrast, the hard boundary model costs grow much more rapidly. This appears to be due to the fact that α_1 is approaching α_2, i.e. two poles are close to coinciding. As this happens, the rate of change of rotation of the eigenvectors increases as the two poles come closer. The width of the region in the hard boundary scheme is determined by the intersection of the parameter rays (which are aligned with the eigenvectors of the FIM) with the redundancy boundary; therefore, as two poles move closer together the direction of the eigenvectors, and thus the point of intersection with the redundancy boundary, and hence the region volume, changes more rapidly. These examples suggests that the extra work required to implement the boundaries as 'soft' is justified by improved approximations of the model cost whenever this is possible.

8.5.2 Moving Average Models

The effectiveness of the MMLD-S algorithm for estimating the model costs of moving average models is now examined.

Comparisons with MML87

A similar examination of the MMLD-S message length estimations versus those found via the MML87 approximation are also now undertaken. The MML87 implementation was done using the *asymptotic* Information Matrix explicitly to avoid the problems of catastrophic breakdowns near the boundaries of invertibility. Thus, assuming uniform priors over the invertibility region are used, this formulation has no simply identifiable cases in which it will grossly underestimate the model cost. In fact, the minimum model cost will be found when all the coefficients are zero, i.e. when the Q-th order MA model coincides with a model that includes no time correlation. Even in this case, the information

matrix is given by $\mathbf{J}(\mathbf{c}) = N\mathbf{I}_Q$, and $\frac{1}{2}\log|\mathbf{J}(\mathbf{c})|$ is significantly greater than $\frac{1}{2}(\log\kappa_Q + 1)$ for most reasonable values of Q and N; of course, for tiny N and large Q there may be issues but such models are usually removed from the candidate set by requiring that $(Q \leq N/2)$, or a similarly basic condition. A Monte-Carlo experiment designed to give an estimate of the quality of the approximation of the 'true' model costs (estimated once again via Random Coding) by both MML87 and MMLD-S was undertaken. The results are summarised in Tables 8.4 (MML87) and 8.5 (MMLD-S).

Table 8.4: Moving Average Model Cost Relative Errors for MML87

	$N = 20$			$N = 35$			$N = 50$		
Q	E $[\lvert\delta_S\rvert]$	inf δ_S	sup δ_S	E $[\lvert\delta_S\rvert]$	inf δ_S	sup δ_S	E $[\lvert\delta_S\rvert]$	inf δ_S	sup δ_S
2	0.1552	-0.0123	1.3587	0.0974	-0.0226	1.0375	0.0707	-0.0318	0.8964
3	0.2149	0.0024	1.1323	0.1351	-0.0102	0.8805	0.1034	-0.0062	0.6809
4	0.2751	0.0565	0.9802	0.1753	0.0188	0.6563	0.1419	0.0145	0.5467
5	0.3444	0.1047	0.9325	0.2341	0.0211	0.7398	0.2117	0.0274	0.7733

Table 8.5: Moving Average Model Cost Relative Errors for MMLD-S

	$N = 20$			$N = 35$			$N = 50$		
Q	E $[\lvert\delta_S\rvert]$	inf δ_S	sup δ_S	E $[\lvert\delta_S\rvert]$	inf δ_S	sup δ_S	E $[\lvert\delta_S\rvert]$	inf δ_S	sup δ_S
2	0.0235	-0.1034	0.0501	0.0208	-0.1005	0.0600	0.0154	-0.0782	0.0490
3	0.0505	-0.1337	0.3347	0.0356	-0.0547	0.2606	0.0305	-0.0547	0.2124
4	0.1114	-0.1578	0.4064	0.0810	-0.0347	0.3276	0.0784	-0.0571	0.3166
5	0.1745	-0.0486	0.6926	0.1389	-0.0209	0.5050	0.1470	-0.0265	0.5038

In all the tests considered the MMLD-S approximation achieved significantly lower average model cost approximation errors than the MML87 approximation, and in all cases attained lower maximum overestimation approximation errors, particularly for $Q = 2$ and $Q = 3$. As is to be expected, the mean approximation errors increase for larger Q and smaller N, with several minor exceptions (such as $Q = 5$, $N = 50$). These trends were discussed in the previous Section, and similar arguments will hold for the Moving Average case. The MML87 criterion can be noted to never underestimate the model cost for $Q > 3$, and for $Q \leq 3$ the underestimations are minor. In contrast, the MMLD-S approximation shows larger underestimations in all cases; this is not completely unsurprising and may be explained by recalling that the MML87 criterion uses the asymptotic Information Matrix to estimate the coding volume. For an MA model, the *per datum* KL divergence increases as $N \to \infty$, asymptoting to the KL divergence of an identically parameterised autoregressive model. As the Information Matrix is present in the second order Taylor series expansion

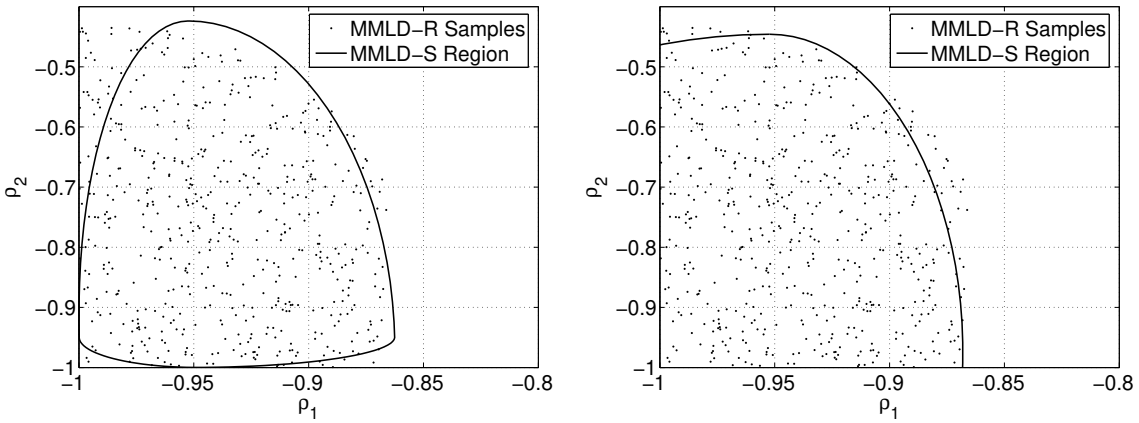
of the KL divergence, the larger the N the steeper the curvature of the KL divergence and the larger the terms in the *per datum* Information Matrix. The asymptotic Information Matrix is based on the asymptotic behaviour of the KL divergence, and as the asymptotic KL divergence is always steeper than the finite-sample KL divergence, the coding quantum volume is expected to be underestimated in most cases. The model cost approximation error results certainly support this hypothesis, as the MML87 criterion overestimates the model cost on average by a significantly larger amount than the MMLD-S criterion, which is based on the exact finite-sample KL divergence.

Comparisons to 'True' Model Costs

The MMLD-S approximation is now compared to the 'true' model costs estimated via the Random Coding procedure. As in the case of the AR models, many of the interesting cases arise at the boundaries of the parameter space, when the difference between 'hard' and 'soft' boundaries can be significant. Figures 8.3(a) and 8.3(b) show the approximated uncertainty regions for both hard and soft boundaries. It is clear to see that the uncertainty region based on the soft boundary scheme approximates the true uncertainty region much closer, and that the uncertainty region based on the hard boundaries clearly underestimates the region volume. The 'true' model cost, found by Random Coding, was 4.43 nits; the hard boundary scheme approximation resulted in a model cost of 4.62 nits, and the soft boundary scheme a model cost of 4.43 nits. In this case, the usage of hard boundaries results in a modest overestimation of the model cost (by approximately 4.5%). As the order of the model increases the absolute errors of both approximations increases in magnitude, though in general the excess error of hard boundaries over soft did not appear to be as significant as in the case of the autoregressive models.

8.5.3 Parameter Estimation of MA(1) Models

To conclude this section the parameter estimation capabilities of the MML87 and MMLD-S approximations for MA(1) models is examined. Parameter estimation of AR(1) models by MMLD-S was also considered, but the model costs of the AR(1) model class as estimated by MMLD-S and MML87 are very close over the entire parameter space, and the resulting parameter estimators are also very similar in behaviour. However, for the MA(1) model class, the MMLD-S and MML87 model costs differ significantly for finite sample sizes.



(a) 'Hard' Boundaries (b) 'Soft' Boundaries

Figure 8.3: Uncertainty Regions for MA(2) Model $\boldsymbol{\varrho}^* = [-0.95, -0.95]$, $N = 20$

For higher order moving average models the evaluation of the model cost by the MMLD-S algorithm becomes too slow to perform a gradient free search (in which the cost function must be evaluated many times) and therefore higher order parameter estimation tests are not undertaken. The MMLD-S and MML87 model cost curves for $N = \{8, 16, 32\}$ are presented in Figure 8.4(a). It is obvious that the MML87 scheme based on the asymptotic Information Matrix overestimates the model cost quite significantly, particularly near the boundaries. The results of parameter estimation over a range of c_1^* values, for $N = 8$ are presented in Tables 8.6 and 8.7. The results are quite interesting; both estimators show

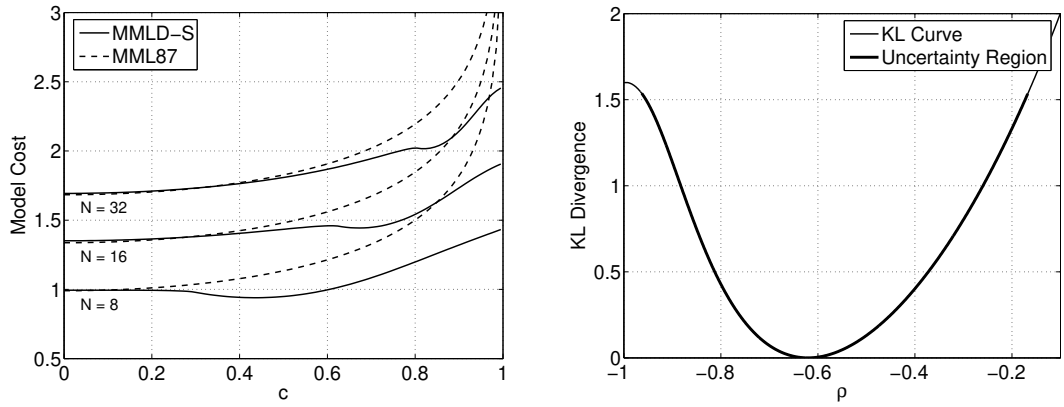
Table 8.6: Parameter Estimation Results for MMLD-S, $N = 8$

	c_1^*									
Score	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
Bias	0.0132	0.0299	0.0468	0.0643	0.0873	0.1210	0.1662	0.2224	0.2908	0.3802
Var	0.2025	0.1986	0.1866	0.1745	0.1564	0.1411	0.1285	0.1192	0.1105	0.1096
SE	0.2025	0.1992	0.1886	0.1784	0.1639	0.1556	0.1561	0.1685	0.1949	0.2541
KL	1.9170	1.9093	1.8928	1.8788	1.8561	1.8441	1.8396	1.8464	1.8654	1.9092

Table 8.7: Parameter Estimation Results for MML87, $N = 8$

	c_1^*									
Score	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
Bias	0.0255	0.0512	0.0741	0.1002	0.1288	0.1682	0.2158	0.2738	0.3497	0.4411
Var	0.1609	0.1576	0.1514	0.1429	0.1300	0.1182	0.1074	0.0984	0.0952	0.0945
SE	0.1614	0.1600	0.1568	0.1528	0.1465	0.1463	0.1539	0.1733	0.2173	0.2890
KL	1.8499	1.8444	1.8359	1.8268	1.8154	1.8112	1.8117	1.8240	1.8542	1.8996

the same basic trend. For smaller c_1^* values the bias of both is smaller, and grows as $c_1^* \to 1$. At the same time, the variance of the estimates is largest for $c_1^* = 0.1$ and decreases as



(a) MA(1) Model Costs

(b) KL Divergence Curve, $N = 8$

Figure 8.4: MA(1) Model Costs and KL Divergence Curves

c_1^* grows. This behaviour is not surprising, given that for small c_1^* the Fisher Information of the MA(1) model is small, indicating that an unbiased estimator would possess larger variance. As c_1^* grows, the effect of the model cost on the estimates becomes larger as there is more of a 'shrinkage' effect. The model cost of the MML87 estimator is much larger than that of the MMLD-S estimator, which explains why the MML87 estimator possesses uniformly lower variance and uniformly higher bias as compared to MMLD-S.

The squared-error risk of an estimator is a combination of its variance and bias, i.e. $\mathrm{SE}(\hat{c}) = \mathrm{bias}(\hat{c})^2 + \mathrm{var}(\hat{c})$ (see [155], page 19). It is therefore unsurprising that it is large at both ends of c_1^* with a minima at around $c_1^* = \frac{6}{10}$, with similar behaviour exhibited by the KL risk. For smaller values of c_1^*, the decreased variance of the MML87 estimator leads to lower squared-error risks than the MMLD-S estimator, but for larger value of c_1^*, the decreased variance is not enough to counter the larger bias and the MMLD-S estimator performs better. In terms of KL risk, the MML87 estimator dominates the MMLD-S estimator for all values of c_1^*. This is not particularly surprising either, as the stronger shrinkage effects of the larger MML87 model cost will lead to larger innovation noise variance estimates than the MMLD-S estimator, and the KL risk is known to favour overestimations of scale parameters (for example, the Minimum Expected KL divergence estimator for Neyman-Scott, see [1], pages 206-208).

The model costs of the MML87 and MMLD-S estimators for several data sizes are presented in Figure 8.4(a). While the MML87 curve grows monotonically as $c \to 1$, the MMLD-S model costs exhibit unusual dips. It appears that the nature of the KL divergence of the MA model class is responsible for these unusual model cost curves. Due

to the fact that every model inside the invertibility region maps to exactly one model outside the invertibility region, the KL divergence for any model $c_1^* \in \mathbb{R}$ exhibits a 'flat' regime around the borders of the invertibility region [64, 156]. Consider the case when $c_1^* > 0$. For a suitably large value of c_1^* the uncertainty region will cover part of this 'flattish' area of KL divergence; when this happens the uncertainty region is able to cover larger amount of the parameter space for a smaller increase in KL divergence (and hence, round-off cost). For a large enough c_1^* the uncertainty region grows suitably large and is clipped against the invertibility boundary. From this point on, the uncertainty region begins to shrink again as $c_1^* \to 1$. This pattern of uncertainty region volume shrink-growth-shrink is responsible, at least in the case of MA(1) models, for the dip in the model cost curve. It can be observed that as N increases, the location of the 'dip' in the model cost begins to approach the invertibility boundary (as c_1^* must be progressively closer to 1 for the uncertainty region to cover the 'flat' regime), and asymptotically vanishes as $N \to \infty$ (as it is clearly not present in the MML87 model cost).

Figure 8.4(b) shows the KL curve for the model $c_1^* = \frac{6}{10}$ (recalling that partial autocorrelation $\rho_1^* = -c_1^*$ for the MA(1) model). While the curve is roughly quadratic around $\rho_1^* = -\frac{6}{10}$, it is clear to see that the KL divergence curve begins to flatten out for $\rho < -0.95$; a small decrease in ρ_1^* would result in a larger decrease in the left boundary of the uncertainty region (and hence, overall region growth) as it has begun to enter the 'flat' regime, and may grow more rapidly as the slope of the KL divergence curve is no longer as steep. This behaviour is not observed in the autoregressive model due to the fact that the KL divergence (and its second derivative) approach infinity at the stationarity boundary, and there is no 'flattening' out of the KL divergence curve. Despite the reduced model cost, and hence reduced shrinkage effect of the MMLD-S estimator, none of the tests revealed any evidence of the presence of the 'piling-up' effect in the MMLD-S estimates (see Section 5.8.3 for further details on this issue).

8.5.4 Autoregressive Moving Average Models

Finally the performance of the MMLD-S approximation when applied to ARMA models was investigated. The comparison against the true model costs and the MML87 approximation was undertaken in a similar fashion to the previous two sections. The true

model costs are approximated by Random Coding, and the MMLD-S and MML87 approximations are assessed in terms of their mean absolute relative errors, and minimum and maximum relative errors. The results are presented in Tables 8.8 and 8.9. The results

Table 8.8: Autoregressive Moving Average Model Cost Relative Errors for MML87S

	$N = 20$			$N = 35$			$N = 50$		
(P,Q)	E $[\lvert\delta_S\rvert]$	inf δ_S	sup δ_S	E $[\lvert\delta_S\rvert]$	inf δ_S	sup δ_S	E $[\lvert\delta_S\rvert]$	inf δ_S	sup δ_S
(1,1)	0.4086	-0.1557	2.4106	0.3400	0.0006	2.0352	0.3050	0.0006	1.7860
(2,1)	0.3858	0.0004	1.7518	0.3405	0.0030	1.4696	0.3174	0.0005	1.2836
(3,1)	0.4943	0.0033	1.5789	0.4751	0.0097	1.3592	0.4656	0.0045	1.2752
(3,2)	0.5032	0.0206	1.3775	0.4571	0.0231	1.2510	0.4406	0.0186	1.2334

Table 8.9: Autoregressive Moving Average Model Cost Relative Errors for MMLD-S

	$N = 20$			$N = 35$			$N = 50$		
(P,Q)	E $[\lvert\delta_S\rvert]$	inf δ_S	sup δ_S	E $[\lvert\delta_S\rvert]$	inf δ_S	sup δ_S	E $[\lvert\delta_S\rvert]$	inf δ_S	sup δ_S
(1,1)	0.0847	-0.1709	1.1216	0.0695	-0.2001	0.9516	0.0574	-0.0865	0.7677
(2,1)	0.1730	-0.0676	0.9600	0.1456	-0.0930	0.7949	0.1266	-0.1212	0.6766
(3,1)	0.2843	-0.1002	1.1506	0.2403	-0.0015	1.1120	0.2210	0.0173	0.9447
(3,2)	0.3019	0.0256	1.1516	0.2644	0.0404	1.0032	0.2482	-0.0069	1.2299

indicate that the MMLD-S approximation in general attains lower mean relative errors, and a smaller range of relative errors for almost all model orders and sample sizes. As the ARMA model class is a combination of both AR and MA model classes, the particular models that yield large approximation errors generally appeared to include AR or MA models that caused problems for the pure model classes. The improvements that soft boundaries yielded over hard boundaries were also found to be on models that included AR or MA components that would cause problems in their pure form.

8.6 Conclusion

This chapter has examined the formulation of AR, MA and ARMA models in the MMLD-S framework. The main contributions were:

- MMLD-S Message Length expressions were developed, and the issue of parameter redundancy and clipping in both AR and MA models was addressed by the application of the 'soft' boundaries clipping scheme presented in the previous chapter.

- Experiments were undertaken to compare the performance of the MMLD-S approximation and the MML87S approximation against the 'true' model costs (as approximated by Random Coding). The MMLD-S approximation attained generally lower mean absolute relative errors and a smaller error range for the AR, MA and ARMA models as compared to the MML87 and MML87S approximations.

- Examples demonstrated the improvements to the construction of uncertainty regions made by the 'soft' boundary scheme over the more straightforward 'hard' scheme.

- Parameter estimation of AR(1) and MA(1) models by MMLD-S was examined. In the case of AR(1) models, it was found to be almost identical to the MML87 approximation. In the case of the MA(1) model, the MMLD-S approximation was shown to 'shrink' the parameter estimate significantly less than the MML87 approximation, resulting in higher parameter variance but lower bias. Additionally, experimental evidence indicated no presence of 'piling-up' in the MMLD-S estimates.

The next chapter presents a large number of experiments to evaluate the model selection performance of the message length criteria developed over the previous five chapters.

Chapter 9

Order Selection Results

The previous chapters of this thesis have been concerned with the development of several new criteria based on the Minimum Message Length principle for the inference of autoregressive moving average models. This chapter now examines where these criteria fit in with regards to the slew of existing criteria for ARMA model selection. Empirical experiments are used to examine the behaviour of the MML criteria for various synthetic and real data sets.

9.1 Alternative Criteria and Performance Metrics

For comparison purposes a range of alternate order selection criteria were also tested. These are summarised below.

$$AIC(P) = 2L(\mathbf{y}|\hat{\boldsymbol{\theta}}_{\mathrm{ML}}) + 2P \tag{9.1}$$

$$AICc(P) = 2L(\mathbf{y}|\hat{\boldsymbol{\theta}}_{\mathrm{ML}}) + 2\frac{(P+1)N}{N-P-2} \tag{9.2}$$

$$BIC(P) = 2L(\mathbf{y}|\hat{\boldsymbol{\theta}}_{\mathrm{ML}}) + P\log N \tag{9.3}$$

$$KIC(P) = 2L(\mathbf{y}|\boldsymbol{\theta}_{\mathrm{ML}}) + 3P \tag{9.4}$$

$$KICc(P) = 2L(\mathbf{y}|\hat{\boldsymbol{\theta}}_{\mathrm{ML}}) + 2\frac{(P+1)N}{N-P-2} - N\psi\left(\frac{N-P}{2}\right) + N\log\frac{N}{2} \tag{9.5}$$

$$SC(P) = L(\mathbf{y}|\hat{\boldsymbol{\theta}}_{\mathrm{ML}}) + \frac{P}{2}\log\frac{N}{2\pi} + \log\int_{\boldsymbol{\theta}\in\boldsymbol{\Theta}}\sqrt{|\mathbf{J}(\boldsymbol{\theta})|}d\boldsymbol{\theta} \tag{9.6}$$

$$CIC(P) = \log\hat{\sigma}^2 + \max\left\{\prod_{i=0}^{P}\frac{1+v(i,\cdot)}{1-v(i,\cdot)} - 1, 3\sum_{i=0}^{P}v(i,\cdot)\right\} \tag{9.7}$$

where P is the number of free parameters, $\boldsymbol{\theta}_{\mathrm{ML}}$ are the Maximum Likelihood estimates, N is the number of data points, $v(i,\cdot)$ the variance of the estimated coefficients, and $\mathbf{J}(\cdot)$ is the Fisher Information Matrix. The selection covers those criteria which assume classic status in the literature (AIC and BIC), criteria especially designed for small sample sizes (AICc and KICc) and modern criteria based on finite sample considerations (CIC and SC). The following loss functions were employed to assess the performance of the various criteria:

1. Multi-step squared prediction errors (SPE_∞)
2. One-step ahead unconditional squared prediction error (SPE_1)
3. One-hundred step ahead unconditional Kullback-Leibler divergence (KL_{100})
4. One-hundred step ahead unconditional symmetric Kullback-Leibler divergence (J_{100})
5. One-step ahead conditional squared prediction error (SPE_{1c})
6. One-step ahead conditional Kullback-Leibler divergence (KL_{1c})
7. One-step ahead conditional symmetric Kullback-Leibler divergence (J_{1c})
8. Correct order selection

The SPE_1 loss measures the models ability to make single estimates on the next time instant, and SPE_∞ assesses how well the long-term effect of shocks is captured by the inferred models. The Kullback-Leibler divergence is a general measure of a model's overall fit that also incorporates the variance parameter, while the symmetric variant punishes underestimation and overestimation of variance in a more equitable fashion than the directed version. These loss functions are examined in more detail in Section 3.8.

9.2 Minimum Message Length Based Criteria

Several criteria based on the Minimum Message Length principle are tested. The first two are based on the MML87 approximation, and the others are based on the MMLD-S approximation. The first MML87 criterion is based exactly on the Message Lengths as estimated by MML87, denoted 'MML87' in the results; the second uses the 'Search' modification (see Section 4.6) to estimate the message lengths, which circumvents some of

the problems with the full Fisher Information matrix. This criteria is denoted 'MML87S'. All the remaining MML criteria begin by creating an MMLD style grouping of model dimensions, and subsequently selecting a representative model from this collection. Assume there is a countable set of candidate models $\boldsymbol{\Theta}' = \{\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_N\}$, each of a different class, inferred from the data. The first step is to select a subset, $\boldsymbol{\Omega}$, of these models; this is the uncertainty region over the model class and contains models that are considered indistinguishable on the basis of the observed data and prior beliefs. Begin by decomposing each model into two components, $\boldsymbol{\theta}_i = \left\{\boldsymbol{\theta}_i^S, \boldsymbol{\theta}_i^T\right\},\ \boldsymbol{\theta}_i \in \boldsymbol{\Theta}'$ where $\boldsymbol{\theta}_i^S$ denotes the structural parameters that describe which model class the model belongs to, and $\boldsymbol{\theta}_i^T$ denotes the remaining parameters that describe the continuous parameters of the model. Typically each $\boldsymbol{\theta}_i \in \boldsymbol{\Theta}$ comes from a different model class (for example, different order AR models), and the models $\boldsymbol{\theta}_i^T$ are generally chosen as suitable point estimates from the model class specified by $\boldsymbol{\theta}_i^S$. The region $\boldsymbol{\Omega}$ is found by solving

$$\boldsymbol{\Omega} = \arg\min_{\boldsymbol{\Omega} \subset \boldsymbol{\Theta}'} \left\{ -\log \sum_{\boldsymbol{\theta} \in \boldsymbol{\Omega}} h(\boldsymbol{\theta}^S) + \left(\sum_{\boldsymbol{\theta} \in \boldsymbol{\Omega}} h(\boldsymbol{\theta}^S) \right)^{-1} \left(\sum_{\boldsymbol{\theta} \in \boldsymbol{\Omega}} h(\boldsymbol{\theta}^S) I_{\mathrm{DS}}(\mathbf{y}, \boldsymbol{\theta}^T) \right) \right\} \tag{9.8}$$

In the case of the ARMA model class, $\boldsymbol{\theta}^S = \{S, Q\}$ and $\boldsymbol{\theta}^T = \left\{\boldsymbol{\beta}, \mathbf{c}, \sigma^2\right\}$. The message length of each individual element $I_{\mathrm{DS}}(\mathbf{y}, \boldsymbol{\theta}^T)$, $\boldsymbol{\theta}_i \in \boldsymbol{\Theta}'$ is the message length of transmitting data $\mathbf{y}$ using the model $\boldsymbol{\theta}_i^T$. The group $\boldsymbol{\Omega}$ then minimises the average message length of transmitting data $\mathbf{y}$ by random coding, using the set of candidate model classes represented by point estimates given by $\boldsymbol{\Theta}$, and may be viewed as a way of expressing uncertainty over the structural parameters that define the candidate model classes. Standard order selection methods applied to ARMA models when the candidate set spans a large number of different structures have been shown to exhibit high variance [141]. It is of interest to test whether MMLD grouping of models with similar message lengths will help to reduce the high variance typical of subset selection methods in a similar fashion to ensemble techniques such as bagging [157] and model averaging [158]. The extension of the MML principle to stating the structural parameters to a suitable accuracy to introduce uncertainty over model order selection is an intriguing idea that has received very little attention in the MML literature; the most successful application being the discovery of Causal models (see [24] and [1], pages 326–336). The problem of selecting a final representative point estimate from the group (i.e. making a choice of model order) remains. Given a group of

candidate models $\mathbf{\Omega}$ spanning different model classes/dimensions, a posterior probability mass function $p(\cdot)$ may be found as

$$p\left(\boldsymbol{\theta}_i \in \mathbf{\Omega}|\mathbf{y}\right) = \exp\left(-I_{\mathrm{DS}}(\mathbf{y}, \boldsymbol{\theta}_i)\right) \cdot \left(\sum_{\theta_i' \in \mathbf{\Omega}} \exp\left(-I_{\mathrm{DS}}(\mathbf{y}, \boldsymbol{\theta}_i')\right)\right)^{-1} \tag{9.9}$$

From this basis, several selection criteria can be defined. The most straightforward is Maximum a Posterior estimation, i.e.

$$\hat{\boldsymbol{\theta}}_{\mathrm{MML}} = \arg\min_{\boldsymbol{\theta} \in \mathbf{\Omega}} \{p(\boldsymbol{\theta}|\mathbf{y})\} \tag{9.10}$$

$$= \arg\min_{\boldsymbol{\theta} \in \mathbf{\Theta}'} \{p(\boldsymbol{\theta}|\mathbf{y})\} \tag{9.11}$$

given it is trivial to show that the model with the minimum message length from $\mathbf{\Theta}'$ must always be in $\mathbf{\Omega}$. This is identical to simply selecting the model with the smallest message length, i.e. the Minimum Message Length principle extends the MAP principle to operate in an invariant fashion across varying model classes. This is denoted MMLD-S' in the following tables. Given that a posterior probability mass function is available, it is also possible to apply statistical decision theory [40] to select estimates that will minimise expected loss functions, the expectation taken with respect to the posterior distribution. This is suggested in [159], and is analogous to the proposed methods for point estimation from a model class within the MMLD framework [41]. The two Bayes decision criteria chosen in this thesis are

$$\hat{\boldsymbol{\theta}}_{\mathrm{MEKL}} = \arg\min_{\boldsymbol{\theta}^* \in \mathbf{\Omega}} \left\{ \sum_{\boldsymbol{\theta} \in \mathbf{\Omega}} p(\boldsymbol{\theta}|\mathbf{y}) \Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*) \right\} \tag{9.12}$$

$$\hat{\boldsymbol{\theta}}_{\mathrm{MEJ}} = \arg\min_{\boldsymbol{\theta}^* \in \mathbf{\Omega}} \left\{ \sum_{\boldsymbol{\theta} \in \mathbf{\Omega}} p(\boldsymbol{\theta}|\mathbf{y}) \left(\Delta(\boldsymbol{\theta}||\boldsymbol{\theta}^*) + \Delta(\boldsymbol{\theta}^*||\boldsymbol{\theta})\right) \right\} \tag{9.13}$$

where (9.12) defines the Minimum Expected Kullback-Leibler (MEKL) divergence criterion, and (9.13) the Minimum Expected J divergence (symmetric Kullback-Leibler divergence) criterion (MEJ) . These two minimum cost criteria possess the advantage that they, like the MML criteria (9.9) on which they are based, are invariant under one-to-one reparameterisations. Of course, these two criteria do not make an inductive guess at the generating source; rather they attempt to minimise the cost of encoding new data based on what has been observed, and thus they are selected with a specific purpose (prediction)

in mind. The posterior probabilities of each model are based on the message lengths of the models. The results in the previous chapter demonstrate that the MMLD-S approximation attains lower approximation errors (using the I1F message length as ground truth) as compared to the MML87 and MML87S 'Search' approximations, and in the sequel the MMLD-S message lengths will be used to calculate the posterior probabilities used to find the discrete MMLD group and the MEKL/MEJL estimates.

9.2.1 Tests on Synthetic Data

Tests on synthetic data are illuminating as the 'true' generating model is known and exact losses may be computed. The tests on synthetic data were undertaken in 'controlled' conditions so that the order selection criteria would not be influenced by violation of assumptions in their derivation. For all the tests, the data was generated by ARMA models with normally distributed innovations, with the ARMA parameters being randomly drawn from the prior distributions used in the MML criteria. It may be argued that this gives an unfair advantage to the MML-based criteria; to counter these arguments it may be noted that the non-MML criteria are explicitly *non-Bayesian* and attempt to work well without any requirement for prior knowledge. The rationale of their deriviation suggests that they should be expected to perform satisficatorily regardless of the method used to generate models, assuming it is not degenerate or too unrealistically biased. The uniform priors chosen in these experiments yielded a large variety of models with time responses that included a wide range of weak and strong periodic and non-periodic components, and so were deemed a suitable test set.

9.2.2 Sample Sizes

For almost all experiments the sample sizes considered were kept small; it is well known that the MML and SC criteria both differ from BIC by only an $o(1)$ term, and this will have the most effect for small sample sizes. For larger sample sizes MML will perform quite similarily to BIC, whose performance has been well documented in literature (see for example [83]). Small sample performance is also very important if ARMA models are used as building blocks in more complex model structures such as mixture models or decision trees in which the number of samples allocated to particular classes or leaves may be small.

9.2.3 Tests on Real Data

Tests on 'real' datasets were also undertaken. This made it possible to assess the behaviour of the MML criteria outside of the ideal conditions, i.e. in cases where the true underlying process is no longer known to be of an ARMA form, and the innovations are no longer necessarily normally distributed.

9.3 Autoregressive Results

The first set of results presented are from tests performed on inference of Autoregressive models. An initial set of results for MML87 have been examined in Chapter 4, and the experiments performed in this chapter are more thorough, focussing primarily on order selection. Experiments on both synthetic and real data were undertaken, and demonstrate the general competitiveness of the MMLD-S approximation, as compared to the MML87 approximation, more traditional Information Criteria and modern alternative model selection methods.

9.3.1 AR Order Estimation on Synthetic Data

The first set of autoregressive order selection results were focussed on model selection from synthetically generated data. These experiments were designed to push the model selection criteria reasonably hard, and to determine their ability to select models from varying complexities of 'true' model sources. In each experiment, 5000 source AR models were randomly generated by sampling from the model priors; each 'true' model order $P^* = \{1, 2, \ldots, 10\}$ had 500 representatives, giving an even spread of model complexity across the test set. It is quite common in the literature to test criteria on low order generating models with several strong components on moderate to large sample sizes (see for example [91], [6], [84]); the choice of AR(5) and higher order models tests the model selection criteria's ability to select good approximating models even when the generating model is of higher complexity than could be accurately estimated from the data. As the 'generating' machinery of real life processes is unknown and can not readily assumed to be low order, this seemed a reasonable testing regime.

In these experiments, two sets of results are reported. First, the results of the alternate criteria which require the Maximum Likelihood estimates are presented; additionally, the

results of the MML87, MM87-Search and MMLD-S criteria using the Maximum Likelihood estimates are also presented for comparison purposes. For the second set of test results, the MML-based criteria are given the MML87S estimates based on the search modification (which strictly they should always use, as in general they yield smaller message lengths than the Maximum Likelihood estimates even for the full MML87 and MMLD-S criteria).

9.3.2 Discussion Of ML Results

The results of the various order selection tests using Maximum Likelihood estimates for each candidate order are presented in Tables 9.1 - 9.4. In all tables, unless otherwise specified, the scores presented for prediction errors and KL/J divergences are the means of the results obtained over the complete run of replications. In general the distributions of these scores were positively skewed with heavy right tails. In almost all cases examined the variances of the scores were found to be increasing with increasing means, and standard error for the means is roughly $1/\sqrt{N_I}$ where N_I is the number of replications. The results are now elaborated on.

Table 9.1: AR Order Selection Results (ML Estimates), $N = 10$

$N = 10$, $P \in [0, 4]$, $P^* \in [1, 10]$

Criteria	SPE_∞	SPE_1	KL_{100}	J_{100}	av(P)	$< P^*$	$= P^*$	$> P^*$
AIC	1.1875	0.1908	2.2690	2.6565	1.9348	4359	**403**	238
AICc	0.7166	0.1661	1.2664	2.7507	1.1770	4697	277	26
BIC	1.1266	0.1813	2.1433	2.5426	1.7862	4458	371	171
KIC	1.0137	0.1707	1.8916	2.3543	1.5386	4579	330	91
KICc	0.6870	0.1887	1.2333	3.1010	1.0126	4765	225	10
SC	0.8612	0.1655	1.6108	2.2479	1.2906	4677	288	35
CIC	1.0576	0.1758	1.9553	2.3829	1.6366	4530	345	125
MML87	0.7244	**0.1554**	1.2166	**2.3080**	2.0722	4222	396	382
$MML87_S$	**0.6875**	0.1570	**1.0971**	2.3501	1.4968	4593	316	91
MMLD-S	0.6898	0.1604	1.1162	2.3338	1.4772	4592	318	90
MEKL	**0.6562**	0.1551	**1.0004**	2.0491	1.5798	4503	363	134
MEJ	0.6909	**0.1417**	1.0483	**1.6566**	1.7294	4435	402	163

Comparison of the MML Based Criteria

For $N = 10$ the MML87S criterion attains lower scores than the MMLD-S criterion for all losses, barr the J divergence. This is interesting as the MML87S approximation achieves worse approximation errors when compared to the ideal model costs (see Chapter 8) as compared to MMLD-S. However, because many of the maximum likelihood candidate

Table 9.2: AR Order Selection Results (ML Estimates), $N = 20$

$N = 20$, $P \in [0, 7]$, $P^* \in [1, 10]$

Criteria	SPE_∞	SPE_1	KL_{100}	J_{100}	av(P)	$< P^*$	$= P^*$	$> P^*$
AIC	0.6343	0.1069	0.5988	0.7765	2.7764	3963	**566**	471
AICc	0.4696	**0.0748**	0.3798	0.5404	1.9762	4406	501	93
BIC	0.5216	0.0855	0.4491	0.6152	2.1126	4353	499	148
KIC	0.5213	0.0855	0.4492	0.6153	2.1114	4353	499	148
KICc	0.4470	0.0814	0.3686	0.5545	1.7170	4527	435	38
SC	0.4518	0.0857	0.3705	0.5713	1.6532	4554	429	17
CIC	0.5515	0.0896	0.4805	0.6478	2.2168	4303	501	196
MML87	0.5443	0.1009	0.3932	0.6132	2.9746	3899	464	637
$MML87_S$	**0.4167**	0.0767	0.3352	0.5008	1.9664	4427	476	97
MMLDS	0.4181	0.0756	**0.3340**	**0.4961**	2.0642	4357	523	120
MEK	**0.4123**	0.0723	0.3208	0.4822	2.2036	4281	520	199
MEJ	0.4163	**0.0689**	**0.3195**	**0.4692**	2.2700	4262	533	205

Table 9.3: AR Order Selection Results (ML Estimates), $N = 50$

$N = 50$, $P \in [0, 10]$, $P^* \in [1, 10]$

Criteria	SPE_∞	SPE_1	KL_{100}	J_{100}	av(P)	$< P^*$	$= P^*$	$> P^*$
AIC	0.2612	0.0338	0.1157	0.1832	3.3626	3668	767	565
AICc	0.2385	0.0274	0.0985	0.1586	2.8994	3957	762	281
BIC	0.2269	0.0275	0.0934	0.1542	2.4390	4224	711	65
KIC	0.2335	0.0270	0.0963	0.1560	2.6912	4083	753	164
KICc	0.2271	**0.0265**	0.0926	0.1518	2.5210	4176	728	96
SC	0.2304	0.0324	0.0990	0.1700	2.1768	4358	618	24
CIC	0.2364	0.0275	0.0974	0.1576	2.7272	4058	753	189
MML87	0.2201	0.0273	0.0923	0.1506	3.6408	3400	**771**	829
$MML87_S$	0.2141	0.0274	0.0894	0.1486	2.5100	4183	735	82
MMLD-S	**0.2136**	0.0270	**0.0867**	**0.1442**	2.6396	4115	757	128
MEKL	**0.2136**	0.0257	0.0857	0.1421	2.7638	4023	755	222
MEJ	0.2138	**0.0251**	**0.0855**	**0.1411**	2.7750	4013	**777**	210

models include poles with large magnitudes, the cost of overfitting, even by a single order, can become quite high. The more conservative nature of the MML87S criterion near the boundaries means such models are more likely to be rejected by MML87S than by MMLD-S.

For the remaining sample sizes, the MMLD-S criterion generally outperforms the MML87S criterion in all losses as well as in number of correct order identifications, although it does attain slightly more overestimations of order. When compared to the exact MML87 criterion, MMLD-S attains (sometimes significantly) lower scores for all losses and all sample sizes, excluding SPE_1 for $N = 10$. The exact MML87 criterion achieves a higher number of correct identifications for $N = 10$ and $N = 50$; however, for all sample

Table 9.4: AR Order Selection Results (ML Estimates), $N = 100$

$N = 100$, $P \in [0, 10]$, $P^* \in [1, 10]$								
Criteria	SPE_∞	SPE_1	KL_{100}	J_{100}	av(P)	$< P^*$	$= P^*$	$> P^*$
AIC	0.1452	0.0159	0.0461	0.0800	3.6768	3466	939	595
AICc	0.1400	0.0145	0.0431	0.0752	3.4164	3630	940	430
BIC	0.1321	0.0135	0.0402	0.0714	2.6800	4119	838	43
KIC	0.1344	0.0127	0.0400	0.0702	3.0232	3900	926	174
KICc	0.1338	**0.0125**	0.0394	0.0694	2.9368	3947	915	138
SC	0.1358	0.0157	0.0436	0.0787	2.4806	4215	775	10
CIC	0.1353	0.0127	0.0401	0.0703	3.0362	3893	924	183
MML87	0.1265	0.0126	0.0379	0.0669	3.7646	3338	919	743
MML87_S	0.1276	0.0131	0.0384	0.0681	2.8006	4041	894	65
MMLD-S	**0.1255**	0.0127	**0.0373**	**0.0662**	2.9536	3945	**945**	110
MEKL	0.1261	0.0124	0.0371	0.0656	3.0576	3888	925	187
MEJ	0.1260	**0.0123**	**0.0370**	**0.0655**	3.0560	3891	932	177

sizes it grossly overestimates the true order as compared to all other criteria. This phenomenon was observed in Section 4.9.4 and attributed to the fact that models estimated with complex poles and small angular components, or real poles with similar magnitudes will artificially attain low message lengths due to violations of the conditions under which the MML87 approximation was derived. It is clear that the chance two poles will be almost coincident increases with the number of poles being estimated, and so the models that violate the approximation tend to be higher order, leading to a large number of overestimations.

The MMLD-grouping MEKL and MEJ criteria generally achieve prediction scores that are at best superior and at worst slightly inferior to the best prediction scores achieved by the other MML criteria. This would suggest that there is a benefit in considering uncertainty in the model structure in addition to the uncertainty in the model parameters themselves. The price paid for these improvements appears to be a larger number of over estimations of model orders (at least in comparison to MMLD-S), which is itself interesting as it demonstrates that a criterion's performance at correctly determining the underlying order is not a necessarily a good indication of its performance in terms of prediction. While it may be argued that induction and prediction are not identical goals (see for example, [1] pages 206-208 for an example), a model's predictive performance, especially when considered over a range of different loss functions, is a reasonable way to assess how 'close' the chosen model is to the underlying data generation process.

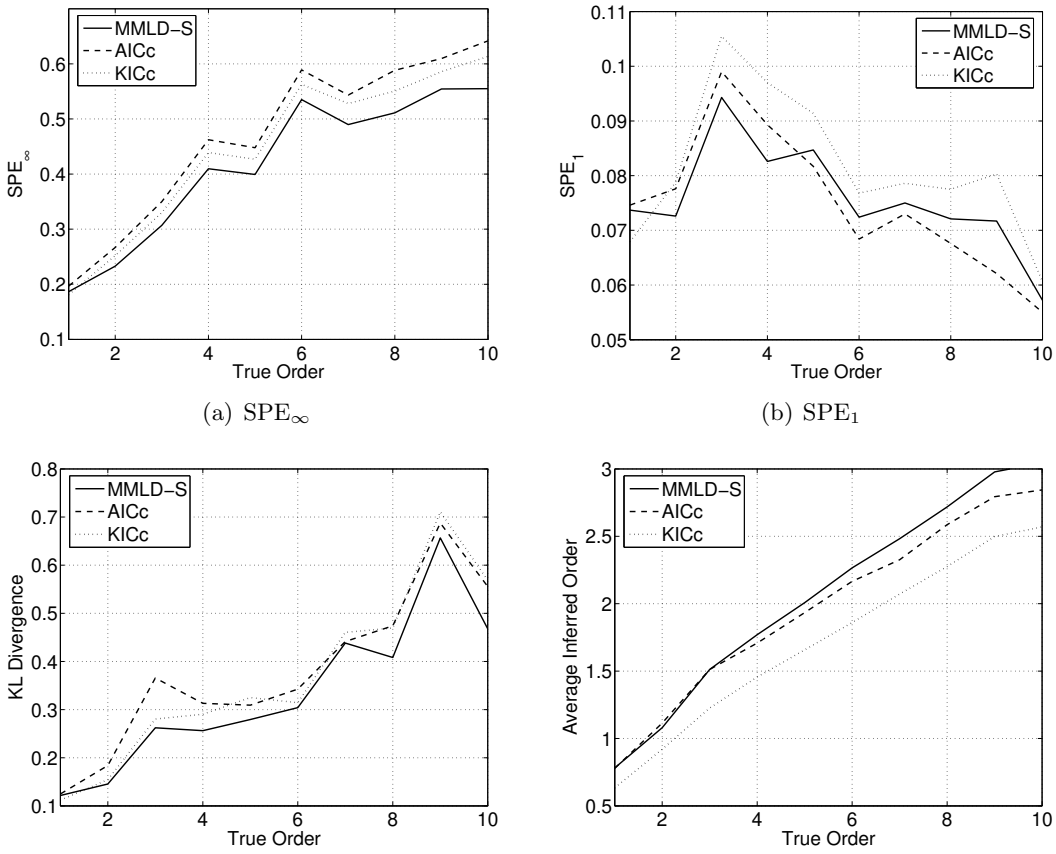
In general the MEKL criterion achieves lower SPE_∞ scores than MEJ, which in contrast achieves lower SPE_1 and J_{100} scores, while both attained similar KL_{100} scores. This is not surprising as the MEKL estimator would be expected to favour models with larger variance estimates (which are usually lower order) more than the MEJ estimator. The SPE_∞ score is more sensitive to parameter variance (because of the recurrent nature) and is thus more sensitive to overfitting than the SPE_1 and J_{100} scores. This largely explains the general trends of the two criteria. The performance of the message length based criterion against the conventional order selection schemes is considered next.

Comparison With Alternative Criteria

The salient points of the performance of the MMLD-S, MML87 and MML87S criterions as compared with the alternative criteria are now summarised. In general, across all sample sizes the MMLD-S criterion is very competitive as compared to the conventional order selection criteria, often achieving the lowest test scores in all losses. Even when a conventional criterion performs better in a particular test score, the MMLD-S criterion rarely loses by a great margin. As an example, in Table 9.2 the AICc criterion performs slightly better in terms of SPE_1, but the difference is marginal. In contrast, the MMLD-S criterion clearly outperforms the AICc criterion in terms of SPE_∞ and KL test scores.

In terms of correct order selection, the AIC, exact MML87 and MMLD-S criteria appear to perform the best on average. However, in the case of AIC and exact MML87, this performance comes with two prices: first, the number of overestimations of order achieved by these two criteria is in general the highest. Second, the performance in terms of test scores, particularly SPE_∞ and KL scores is generally significantly worse than the MMLD-S and MML87S criteria. It is interesting to note, however, that despite sometimes achieving higher number of overestimations of order, the exact MML87 criterion still outperforms the AIC criterion in general in terms of all losses. The MMLD-S criterion achieves a high number of correct order identifications while keeping the number of overestimations low.

Despite often selecting higher order models than AICc, KICc and SC, the MMLD-S criterion often outperforms them or achieves quite similar test scores, even for small sample sizes like $N = 10$ when it could be expected that a more conservative approach would yield better results. On the whole, the AICc and KICc criteria appear to perform amongst the best of the conventional criteria, often achieving the lowest prediction scores



(a) SPE_∞

(b) SPE_1

(c) KL_{100}

(d) Average P

Figure 9.1: AR results for several criteria, $N = 20$

of all the non-MML methods. This supports the experiments undertaken in [6] in which a similar finding was reported.

The fact that the MMLD-S criterion may often select more complex (in terms of number of parameters) models than the AICc, KICc and SC criteria and yet still achieves better prediction scores is primarily due to the fact that the MML based criteria treat each model as fully specified (and not just a model class) and that the values of a model's parameters have a large effect on the final computed message length. Models with poles close to the boundaries will generally have smaller scale estimates, and will often have poorer long-term prediction performance (because small overestimations near the boundary have a larger effect than small underestimations). They will also tend to have higher message lengths because the KL divergence surface is steeply bowled in these regimes, leading to small uncertainty regions. The MML based criterion will tend to shy away from models with large numbers of poles near the boundary while the conventional criterion make no distinction between a model with poles near the origin or one with poles close to the boundary.

Due to this key difference, the MMLD-S criterion may well select a higher order model with poles further from the boundary if the lower order options have poles very close to the boundaries. Such a higher order model will generally require only a small number of nits extra to state as the uncertainty region will be large, and may even have a lower model cost depending on how close the poles of the lower order models are to the boundary. This is very different behaviour from the alternative information criteria in which the complexity penalty terms are dependent soley on the number of parameter (for a given sample size N).

Figures 9.1(a)-9.1(d) show several plots of test scores against the true model order, taken from the $N = 20$ test set for the MMLD-S, AICc and KICc critera; these criteria were chosen as they performed quite well in all tests and could be considered amongst the best of their respective class of criteria (non-Bayesian for AICc and KICc, and MML-based for MMLD-S). This sample size was chosen as the criteria were given a large range of models (from $P = 0$ to $P = 7$) to choose between, and the results generalised quite well to larger sample sizes. The general trend in Figures 9.1(a) and 9.1(c) is easily visible: the MMLD-S criterion outperforms both the AICc and KICc criteria, with the difference growing larger as the true model order increases, with the KICc criterion coming overall

second best. The plot in 9.1(b) shows a less clearly defined trend; indeed, the only thing that is reasonably obvious from this plot is that for all but a true order of $P^* = 1$ the KICc criterion performs generally the worst; this is attributable to the overly conservative nature of the KICc criterion.

Figure 9.1(d) is also shows an interesting trend: the KICc criterion is clearly much more conservative than the MMLD-S and AICc criterion, and AICc is slightly more conservative than the MMLD-S, especially as the true order grows. It is interesting that despite MMLD-S selecting on average higher order models for all true model orders it remains competitive and even outperforms the more conservative criteria in terms of the various losses. This is attributed to fact that the MML principle takes into account the values of model parameters rather than just the number of parameters when selecting a model.

An Illustrative Example

An example is taken from the ($N = 20$) test set to demonstrate the previously mentioned points. The true model was the AR(3) model

$$y_n + 1.2y_{n-1} + 0.22y_{n-2} - 0.05y_{n-3} = v_n \tag{9.14}$$

In this case the AICc, SC and KICc criteria all selected the AR(1) model

$$y_n - 0.97y_{n-1} = v_n \tag{9.15}$$

while the MMLD-S criteria selected the AR(4) model

$$y_n + 1.2y_{n-1} - 0.09y_{n-2} - 0.4y_{n-3} - 0.06y_{n-4} = v_n \tag{9.16}$$

This AR(4) model had four real poles, $\boldsymbol{\alpha} = [-0.16, 0.56, -0.74, -0.84]$ and a model cost of 4.39 nits; compare this to the model cost for the AR(1) candidate of 3.44 nits. It

Table 9.5: Test Scores for Two AR Models Estimated by ML

Model	SPE_∞	SPE_1	KL_{100}	J_{100}
AR(1)	0.5686	0.0089	0.0514	0.1316
AR(4)	0.2317	0.0099	0.0489	0.1119

is clear to see that the AR(1) model costs 3.44 nits *per parameter* to state, and the AR(4) model costs approximately 1.1 nits per parameter to state. In contrast, the per parameter penalties for the AICc, KICc and SC criteria are always a nondecreasing function of the dimensionality of the AR order. Table 9.5 summarises the test scores of the two models selected by MMLD-S and by AICc, KICc and SC. It is clear that there is a small trade-off in SPE_1 prediction scores for a large decrease in multiple step prediction errors, and a moderate decrease in terms of KL and J divergences. This behaviour is typical of the MMLD-S criterion as the SPE_1 test scores are less affected by overestimation of pole magnitudes than the multi-step SPE or KL divergence scores.

This example is not intended to argue that in all cases it is better to select higher order models; rather it shows that considering a model on the merits of the value of its parameters rather than simply its order can be advantageous. Of course, the price paid for this behaviour is that a suitable subjective prior distribution must be chosen to apply the MML principle, which is avoided by the alternative information criteria.

9.3.3 Discussion of MML Results

Table 9.6 presents order selection results for the same set of datasets as previously discussed (excepting $N = 100$ for which the MML and ML candidate models were quite similar), this time using the MML87S estimator to estimate the candidate model parameters. The results are interesting and the key points are summarised below:

For all three sample sizes considered, the MMLD-S criterion using the MML87S estimates achieves lower test scores for all losses when compared to the best results attained when using the ML estimates (presented in Tables 9.1 - 9.3). The difference is particularly marked for the SPE_∞ and KL/J divergence losses and less so for the SPE_1 losses, indicating that the improvements can be attributed to less overestimations of pole strength and less underestimations of the variance due to the influence of the Fisher Information term. Of course, the models are generated according to the subjective priors and it would be hoped that the results would be at least be no worse than those obtained by estimating the candidate models by Maximum Likelihood; however, such a significant improvement does suggest that if one truly believes the model to have a pole structure similar to some proposed subjective prior the MML87S estimates should be strongly considered.

Table 9.6: AR Order Selection Results (MML87S Estimates)

$N = 10$, $P \in [0, 4]$, $P^* \in [1, 10]$								
Criteria	SPE_∞	SPE_1	KL_{100}	J_{100}	av(P)	$< P^*$	$= P^*$	$> P^*$
MML87	0.3880	0.1286	0.7591	**1.2691**	3.7882	3091	**519**	1390
MML87_S	0.3933	0.1416	0.7843	1.3113	3.1752	3470	484	1046
MMLDS	0.3906	0.1367	0.7960	1.3114	1.8480	4454	387	159
MEK	0.3807	0.1341	**0.7409**	1.6456	2.0742	4220	410	370
MEJ	**0.3802**	**0.1260**	0.7548	1.3409	2.2770	4142	455	403
$N = 20$, $P \in [0, 7]$, $P^* \in [1, 10]$								
Criteria	SPE_∞	SPE_1	KL_{100}	J_{100}	av(P+Q)	$< P^*$	$= P^*$	$> P^*$
MML87	0.2733	**0.0630**	0.2745	0.4104	6.0974	1859	539	2602
MML87_S	**0.2590**	0.0649	**0.2722**	**0.4034**	5.9380	2024	512	2464
MMLDS	0.2607	0.0667	0.2818	0.4134	2.5112	4183	**591**	226
MEK	0.2525	0.0649	**0.3427**	**0.4753**	2.7434	4029	557	414
MEJ	**0.2522**	**0.0620**	0.3722	0.4959	2.8264	3984	569	447
$N = 50$, $P \in [0, 10]$, $P^* \in [1, 10]$								
Criteria	SPE_∞	SPE_1	KL_{100}	J_{100}	av(P)	$< P^*$	$= P^*$	$> P^*$
MML87	0.1823	0.0249	0.0832	0.1386	8.8198	362	534	4104
MML87_S	0.1500	**0.0239**	0.0801	0.1309	8.7336	650	536	3814
MMLD-S	**0.1490**	0.0251	**0.0786**	**0.1291**	2.9744	3916	**833**	251
MEKL	**0.1470**	0.0237	0.0771	0.1261	3.2132	3761	797	442
MMLD_{EJ}	0.1470	**0.0233**	**0.0771**	**0.1256**	3.2354	3750	805	445

The MMLD-S criterion attains lower test scores than MML87S and the exact MML87 criterion on some sample sizes (particularly for $N = 50$) and is outperformed for others, and across all sample sizes the SPE_∞ attained by MMLD-S are very competitive. The differences between all the criteria in terms of all losses are not particularly great (with the exception of the exact MML87 SPE_∞ score for $N = 50$).

The MML87S and exact MML87 criteria tend to heavily over estimate the model order for all sample sizes tested. When compared to the results obtained using the ML estimates, the MMLD-S criterion achieves a higher number of correct order identifications for a small increase in over estimations. This is largely due to the fact that the MML estimated models will generally tend to have poles that are further away from the stationarity boundary than the corresponding ML estimates, leading to reduced message lengths, particularly for higher order candidate models. The message length can rise very rapidly as the poles approach the boundary, and a small decrease in magnitude can lead to a very large decrease in message length.

The MEKL and MEJ criteria almost always attain lower scores for all losses (the major exception being the MEKL J divergence score for $N = 10$), again indicating the benefits of

considering uncertainty across model structures when performing model selection. However, the price paid for the improvements to the prediction scores obtained by MEKL and MEJ is a large increase in overestimations of the model order (almost twice as often as MMLD-S for all sample sizes). Some of the findings present an interesting conundrum. The MML87S criterion heavily overestimates the model order (as compared to the true model order) as compared to MMLD-S, yet for $N = 20$ and $N = 50$ achieves better prediction scores on some of the losses. The overfitting is occuring because the candidate models as estimated by MML87S are usually models that yield low message lengths because they violate or go close to violating the assumptions under which the MML87S criterion produces sensible message lengths. This behaviour was already observed earlier in this thesis in Section 4.9.4. However, it is usually the case that large amounts of overfitting will lead to poor predictive performance [76, 110] because the variance in the excess parameters is too great. An examination of the candidate models estimated by MML87S indicates why this is not the case when the MML87S estimates are used.

Let a pole be deemed 'active' if it's magnitude exceeds 0.1; poles smaller in magnitude than this have little effect on the resulting frequency and time response of an AR model and may be for all intents and purposes considered non-existent. Table 9.7 shows the average number of 'active' poles for each order of candidate models for both the ML estimates and MML estimates for the $N = 20$ test set. It is clear that as the number of active poles for the ML estimates is roughly equal to the number of parameters in the model, while in the case of the MML estimates the number of active poles is significantly less than the number of parameters, especially for $P = 6$ and $P = 7$. This is due to the 'shrinkage' effect of the Fisher Information term; if the introduction of a new pole with magnitude greater than 0.1 yields only a marginal decrease in negative log-likelihood, it is usually cheaper under the MML87S approximation to essentially ignore the decrease in likelihood and simply make the pole very small so that the cost of the parameter is reduced. The higher order MML candidate models consist of several active poles and several very small poles; this shrinkage to the origin of the unit circle of pole magnitudes acts to reduce the variance of

Table 9.7: Number of Active Poles for ML and MML Estimates

P	1	2	3	4	5	6	7
ML	0.9454	1.9266	2.9098	3.9272	4.9312	5.9318	6.9254
MML87	0.9412	1.7630	2.4256	2.8746	3.2736	3.5570	3.8532

the parameter estimates significantly so that overestimation of order has a reduced effect. The low variances of the estimated coefficients in the parameter estimation experiment presented in Section 4.9.3 also support these findings.

That the KL divergences are actually lower is also a product of this 'shrinkage'. The MML87S approximation, while more robust than the exact MML87 criterion, still breaks down for models with large numbers of small poles. This leads to a situation where it is artificially cheaper to make a pole small even in the case that it actually results in an *increase* in the negative log-likelihood, as compared to a lower order model. The result is that many of the higher order models have noise variance estimates that are actually larger than the lower order candidates; this behaviour is not found in the ML estimated candidate models and explains why overestimating the order can actually reduce the KL divergence, which is so sensitive to the noise variance parameter. It is also not uncommon for higher order shrunken linear regression estimates to outperform lower order subset selections (see for example [140]) in terms of prediction scores, and the SPE_1 scores are essentially prediction errors between two linear regressions. Furthermore, the priors used to generate the test models (uniform over pole magnitudes and angular components) lead to models with generally small higher order coefficients. This means that overestimation of the order with shrunken estimates will be unlikely to result in very bad prediction scores, as shrinking the poles towards the origin of the unit circle generally leads to the coefficients being shrunk towards zero.

Conclusion

The overall findings of these experiments are summarised below:

- In terms of prediction scores, the MMLD-S criteria is very competitive with the MML87S and exact MML87 criteria, outperforming both in general as N increased when using the ML estimates, and either outperforming or remaining very competitive with both criteria when using the MML87S estimates.

- In contrast, MMLD-S generally outperformed the MML87S and exact MML87 criteria in terms of model order selection for all sample sizes and using both sets of estimates. When using Maximum Likelihood estimates, the MML87S and MMLD-S order selection results were close for small N but the MML87S criteria began to

degrade as N increased. When using the MML87S estimates, the MML87S and exact MML87 criteria perform very poorly in terms of order selection, on average overestimating the order by a large amount.

- The MMLD-S criterion performs very well when compared against the alternative information criteria, often attaining the lowest test scores. Even when another criterion outperforms the MMLD-S criterion, the difference is quite small. This indicates that while the MMLD-S criterion does not 'win' for all sample sizes (as no criterion can be expected to [65]) it never performs too badly.

- If a single, relatively parsimonious model that has good predictive performance is desired, the results would suggest that the MEKL and MEJ criterion offer a good choice.

- If a single model that has good predictive and inductive performance is desired, the MMLD-S criterion appears to offer an excellent trade-off of prediction and model selection capabilities (measured by number of correct and over estimations of model order) of all the considered criteria.

- When employing the MML87S estimates, all MML criteria outperform the best performing non-Bayesian criteria employing the Maximum Likelihood estimates. Although this may be somewhat attributed to the way in which the models are generated (i.e. from the priors), the differences are quite significant and indicate the MML87S estimator is quite effective, particularly for small sample sizes.

9.4 Moving Average Results

The pure moving average model was examined next. In a similar fashion to the autoregressive experiments, each experiment consisted of 5000 source MA models randomly generated by sampling from the model priors, with each 'true' model order $Q^* = \{1, 2, \ldots, 10\}$ having 500 representatives. For each dataset a set of candidate models was estimated from the data, and each criterion was required to select one of these candidates as the 'best' model. The large range of model orders was chosen so as not to heavily favour those criteria which are overly conservative, and to provide a challenging testing environment in which to assess the behaviour of each model selection scheme. In the following tests the

non-MML criteria all use the Maximum Likelihood estimates (as they are the estimates that minimise the criterion scores, and in many cases are the only estimates for which the criteria derivations are even valid). As the MML criteria are designed to compared fully specified models, they are tested using both the ML estimates (which are unlikely to minimise the message length) and the MML estimates.

9.4.1 Discussion of ML Results

Table 9.8: MA Order Selection Results (ML Estimates), $N = 10$

$N = 10$, $Q^* \in [0, 10]$									
Criteria	SPE_1	KL_{100}	J_{100}	SPE_{1c}	KL_{1c}	J_{1c}	$< Q^*$	$= Q^*$	$> Q^*$
AIC	0.5290	9.9425	12.4220	**0.2764**	0.5083	0.9707	4368	**453**	179
AICc	0.4590	3.1945	9.1889	0.3104	0.4200	1.0497	4732	256	12
BIC	0.5038	8.0766	10.8478	0.2807	0.4810	**0.9573**	4471	412	117
KIC	0.4749	5.2318	9.4838	0.2956	0.4452	0.9818	4627	315	58
KICc	0.4639	1.7942	8.8461	0.3443	**0.4191**	1.1807	4833	165	2
MML87	0.5178	**0.9953**	62.1363	0.4553	0.4936	2.3615	4855	120	25
MMLDS	**0.4521**	3.5144	**8.1824**	0.2865	0.4413	0.9614	4560	365	75
MEKL	**0.4450**	1.6822	7.8719	0.3146	**0.4070**	1.0461	4716	237	47
MEJ	0.4467	4.2532	**7.0854**	**0.2670**	0.4351	**0.9190**	4347	426	227

Table 9.9: MA Order Selection Results (ML Estimates), $N = 20$

$N = 20$, $Q^* \in [0, 10]$									
Criteria	SPE_1	KL_{100}	J_{100}	SPE_{1c}	KL_{1c}	J_{1c}	$< Q^*$	$= Q^*$	$> Q^*$
AIC	0.4127	3.5207	4.2671	**0.1802**	0.3508	0.6286	3613	**867**	520
AICc	0.3527	1.8872	3.1166	0.1963	0.2952	**0.6028**	4150	708	142
BIC	0.3639	2.0826	3.2005	0.1950	0.3083	0.6148	4108	720	172
KIC	0.3637	2.0806	3.2020	0.1954	0.3084	0.6155	4111	718	171
KICc	0.3529	1.3406	3.0060	0.2155	0.2941	0.6591	4379	572	49
MML87	0.4078	**1.0750**	7.6134	0.2807	0.3332	0.9659	4537	338	125
MMLDS	**0.3457**	1.3399	**2.7394**	0.1998	**0.2871**	0.6101	4209	622	169
MEKL	0.3479	**1.0237**	2.9762	0.2219	0.2848	0.6562	4396	455	149
MEJ	**0.3303**	1.3533	**2.6016**	0.1928	**0.2825**	**0.5914**	4017	592	391

The order selection results using the Maximum Likelihood estimates are presented in Tables 9.8 - 9.10. The MML87 criterion is clearly the most conservative, scoring the largest number of order under estimations and also the least amount of correct order identifications by a large margin. It does not perform particularly well in terms of SPE_1, J divergence or conditional scores but it uniformly achieves the lowest unconditional KL divergence scores. As with the autoregressive models, the AIC criterion attains the highest number of

Table 9.10: MA Order Selection Results (ML Estimates), $N = 50$

$N = 50$, $Q^* \in [0, 10]$									
Criteria	SPE_1	KL_{100}	J_{100}	SPE_{1c}	KL_{1c}	J_{1c}	$< Q^*$	$= Q^*$	$> Q^*$
AIC	0.2173	0.7350	0.8988	0.1057	0.2029	0.3563	1871	1789	1340
AICc	0.1700	0.5209	0.7115	**0.1010**	0.1717	0.3173	2565	**1820**	615
BIC	0.1706	0.3875	0.6773	0.1083	0.1607	0.3186	3287	1553	160
KIC	**0.1659**	0.4872	0.6961	0.1014	0.1673	0.3140	2810	1757	433
KICc	0.1676	0.3907	**0.6658**	0.1044	**0.1610**	**0.3156**	3220	1577	203
MML87	0.2326	**0.2974**	1.2231	0.1519	0.1932	0.4395	3696	969	335
MMLDS	0.1807	0.3436	0.6971	0.1136	0.1633	0.3331	3335	1365	300
MEKL	0.1915	0.3013	0.7535	0.1183	0.1643	0.3459	3549	1145	306
MEJ	0.1749	0.3365	0.6703	0.1091	**0.1594**	0.3240	3211	1270	519

correct order identifications but only at the expense of the most over estimations. However, unlike in the case of the autoregressive models, the effects of overfitting are reduced as the AIC prediction scores (with the exception of the unconditional KL divergence) are not dramatically greater than the other criteria. The AIC criterion also performs very well in terms of conditional squared prediction errors.

The MMLD-S criteria performs quite well in terms of all the losses, and particularly the SPE_1 and J divergence scores. However, an examination of the table indicates that even when the MMLD-S criterion does not 'win', the margin of loss is generally quite small. The MMLD-S criterion rarely loses very badly, the largest loss being in terms of unconditional KL divergence for $N = 10$. However, even in this case, it largely outperforms all the non-MML criteria barr KICc. Furthermore, it outperforms KICc on the SPE and J divergence losses and on number of correct order identifications.

The AICc and KICc criteria appear the overall best of all the non-MML order selection criteria, achieving similar SPE_1 scores for all sample sizes. The KICc also achieves lower unconditional KL divergences (and for $N = 10$, lower J divergence) than AICc, but tends to attain larger conditional SPE and J divergence scores. The KICc criteria is significantly more conservative than AICc and achieves a markedly smaller number of correct order identifications (though, a correspondingly smaller number of over estimations of order). The MMLD-S criteria appears to sit somewhere in between the AICc and KICc criteria, while achieving an overall larger number of outright 'wins' on the various losses than either of them.

The MEKL criterion performs better than MMLD-S in terms of unconditional KL divergences, but for $N = 20$ and $N = 50$ has generally higher conditional prediction scores

and unconditional J divergence scores. It also has less over estimations of order, though once again this is gained at the expense of number of correct order identifications. This is not unexpected as the KL divergence favours models with larger variance parameters, and these tend to be lower order models for which negative log-likelihood is lower.

The MEJ attains better test scores than MMLD-S on almost all loss functions excepting the unconditional KL divergence. In some cases the improvements are significant (such as the conditional SPE score for $N = 10$, or the unconditional SPE_1 score for $N = 20$) but in many other cases the improvements are marginal. These improvements also come at the cost of reduced correct order identifications and increased order over estimations. When compared to MML87 and the non-MML criteria, the MEJ criterion is very competetive and generally attains lower scores for all losses but directed KL divergences. As in the case of autoregressive models, the key difference between the MML-based criteria and the conventional criteria is that the MML based criteria consider not just the order of a model, but the values of its parameters. This explains why in many cases MMLD-S outperforms AICc and KICc in terms of prediction scores while either selecting the correct model order less frequently (in comparison to AICc) or overestimating the order more frequently (in comparison to KICc). For the MML-based criteria the order of model is less important than the values of the model parameters, while in the case of the conventional criteria the order (and sample size) completely determines the penalty.

The very conservative nature of the MML87 criterion for pure moving average models has been previously observed in Section 5.8.4. The use of the asymptotic Information Matrix leads to the overestimation of the model cost, particular for models with zeros near the invertibility border. It is well known that Maximum Likelihood estimation often produces models with zeros on the invertibility boundary (or close to, if the search space is restricted to exclude the invertibility boundary), the frequency increasing with decreasing N [63]. Many of the higher order models estimated by Maximum Likelihood in these test sets have zeros very close to the invertibility boundary and this leads to unrealistically large message lengths, due to the poor approximation offered by the asymptotic FIM to the KL divergence near the boundary. This results in an average higher model cost (than if the MML87 estimates were used) as the model order increases, and yields a more conservative criterion. The MMLD-S criterion is based on the exact KL divergence which is less extreme near the invertibility boundary and thus the model costs are often

significantly lower, leading to a less conservative criteria. If these arguments are correct it would be expected that using the MML87 estimates will result in higher order models being chosen with greater frequency by the MML87 criterion, while the behaviour (in terms of order of models chosen) of the MMLD-S criterion will not be expected to change as dramatically. This is examined next.

9.4.2 Discussion of MML Results

The order selection results using the MML87 parameter estimates are presented in Table 9.11. The first point to note is that the MML criteria using the MML estimates attain lower scores for all the losses when compared to the best results using the ML estimates. This is particularly noticable in the case of the unconditional KL and J divergence scores, and less so for the SPE and conditional scores.

It is immediately clear that when using the MML87 parameter estimates, the MML87 criterion is far less conservative than when using the ML parameter estimates. The MML87 criterion correctly estimates the model order 120 times for $N = 10$ using the ML estimates as compared to 383 times using the MML estimates (a threefold increase), and for $N = 20$ it scores 338 correct estimations using the ML estimates compared to 735 correct estimations using the MML estimates (roughly double). In contrast, the MMLD-S criterion scored 365 correct order estimations using the ML estimates for $N = 10$ and 443 correct order estimations using the MML estimates (roughly a third increase), with an even smaller increase of 622 to 769 for $N = 20$. This dramatic increase for the MML87 criterion supports the explanation presented in the Section 9.4.1 of the conservative behaviour of MML87 when the ML parameter estimates are employed.

Comparisons of the MML87 and MMLD-S criteria show that for $N = 10$ and $N = 20$ the MML87 criterion performs better than MMLD-S in terms of unconditional and predictive KL divergence scores, and performs worse in terms of all the other losses. For $N = 10$ the differences in scores are moderate, and for $N = 20$ they are small. Given that the MML87 criterion is more conservative (observe that it obtains less correct and overestimations of orders for $N = 10$ and $N = 20$) than the MMLD-S criterion even when using the MML87 estimates it is unsurprising that it achieves better KL divergence scores at the expense of degraded performance in the squared prediction error and J divergence losses. For $N = 50$ the MML87 criterion wins on all loss functions, but the differences

Table 9.11: MA Order Selection Results (MML Estimates)

$N = 10$, $Q^* \in [0, 10]$									
Criteria	SPE_1	KL_{100}	J_{100}	SPE_{1c}	KL_{1c}	J_{1c}	$< Q^*$	$= Q^*$	$> Q^*$
MML87	0.4061	**0.6597**	4.7875	0.2771	**0.3876**	0.8958	4534	383	83
MMLDS	**0.4018**	0.7140	**3.5608**	**0.2575**	0.3967	**0.8514**	4381	**443**	176
MEKL	0.4065	0.6411	4.2433	0.2655	**0.3771**	0.8776	4435	387	178
MEJ	**0.3865**	0.6910	**3.1513**	**0.2423**	0.3857	**0.8212**	4172	477	351
$N = 20$, $Q^* \in [0, 10]$									
Criteria	SPE_1	KL_{100}	J_{100}	SPE_{1c}	KL_{1c}	J_{1c}	$< Q^*$	$= Q^*$	$> Q^*$
MML87	0.2905	**0.3997**	1.4596	0.1803	**0.2516**	0.5303	4011	735	254
MMLDS	**0.2882**	0.4116	**1.3704**	**0.1754**	0.2526	0.**5234**	3934	**769**	297
MEKL	0.2928	0.3724	1.5294	0.1795	**0.2440**	0.5275	4021	592	387
MEJ	**0.2729**	0.3984	1.2520	0.1658	0.2460	**0.5009**	3614	692	694
$N = 50$, $Q^* \in [0, 10]$									
Criteria	SPE_1	KL_{100}	J_{100}	SPE_{1c}	KL_{1c}	J_{1c}	$< Q^*$	$= Q^*$	$> Q^*$
MML87	**0.1297**	**0.1688**	**0.3782**	**0.0883**	**0.1319**	**0.2593**	2775	**1753**	472
MMLDS	0.1320	0.1706	0.3867	0.0891	0.1328	0.2618	2814	1704	482
MEKL	0.1355	**0.1615**	0.4060	0.0898	**0.1281**	0.2577	2966	1469	565
MEJ	**0.1257**	0.1666	**0.3631**	**0.0852**	0.1291	**0.2518**	2554	1540	906

are very marginal. This is not unexpected given that as N grows the behaviour of the exact KL divergence and the asymptotic KL divergence begin to coincide more closely; coupled with the shrinking uncertainty region sizes, the MMLD-S and MML87 criteria will be expected to become very similar for larger N.

The behaviour of the MEKL and MEJ criteria using the MML87 estimates as compared to the MMLD-S criteria is similar to their behaviour using the ML estimates. The MEKL criteria is more conservative, and trades some loss in SPE and J divergence scores for an improvement in unconditional KL divergence. The MEJ criteria attains a smaller reduction in unconditional KL divergence, but additionally yields improvements in all other losses over the MMLD-S and MML87 criteria. This comes at the price of a large increase in overestimations of model order (almost doubling in the case of $N = 10$ and $N = 20$). This again appears to highlight the practical differences between induction (as embodied by the MMLD-S criteria) and prediction (as embodied by the MEKL and MEJ estimators).

9.4.3 Conclusion

The main findings of the Moving Average order selection tests are:

- The MML criteria equipped with MML estimates attain lower test scores than the best of the criteria using the ML estimates in almost all cases across all sample sizes, suggesting the MML87 estimator is a good choice if one believes that the underlying data generating mechanism is not strongly at odds with the chosen priors.

- When using the Maximum Likelihood estimates, and to a lesser estimate when using the MML87 parameter estimates, the MML87 criterion based on the asymptotic Fisher Information is conservative and selects the correct model order less frequently than the MMLD-S criterion. In general, it attains lower KL divergence scores but generally at the expense of worse performance in terms of all other losses.

- Of the non-MML based schemes, the AICc and KICc criteria appear to perform the best in terms of the various losses, with the KICc criteria being slightly more conservative (and scoring better in terms of KL divergences due to the larger variance estimates present in lower order models).

- The MMLD-S offered a good trade-off between correct order estimations and good performance in terms of all losses. Being less conservative than the MML87 criteria, it generally scored worse in terms of directed KL divergences (though generally superior to the non-MML criteria), but conversely often attained better SPE and J divergence scores. Additionally, when the MMLD-S criteria 'lost' to another criteria, it rarely performed very poorly and generally remained competitive.

- The MEJ offered general improvements in terms of prediction losses as compared to the MMLD-S/MML87 criteria while incurring a significant increase in overestimations of model order.

The main conclusion to be drawn here is that if prediction is the primary aim, then the MEJ would appear to be the an excellent candidate given its very competitive performance in all the loss functions. However, if induction was the aim, in which superfluous parameters were to be dispensed with while maintaining competitive prediction scores, then the MMLD-S criterion appears to be an excellent choice.

9.5 Autoregressive Moving Average Results

The final experiments on synthetic data involve fitting mixed autoregressive moving average models. These were conducted in similar fashion to the preceding experiments on pure autoregressive and moving average models with two exceptions. The first is that for each sample size only 1000 tests are undertaken; this was simply due to the large computational requirements of fitting many candidate ARMA models. The second difference is that the MMLD-S and MEJ/MEKL criteria were the only MML-based criteria considered for order selection; in light of the performances for the pure model structures, coupled with the tendency to underestimate the model cost for AR components and overestimate model costs for the MA components, the message lengths yielded by the MML87 and MML87S approximations were deemed to differ too greatly from the ideal code lengths that they are attempting to estimate. The simple order selection experiment undertaken in Section 6.6.2 also indicated that the tendency to underestimate AR costs and overestimate MA costs results in a model selection criterion that is heavily biased towards pure AR structures, which is obviously a problem when the generating models are mixed ARMA structures. The MMLD-S criterion much more closely approximates the ideal code lengths (see Section 8.5.4), and does so without the need for ad-hoc modifications, and so was chosen instead.

Other than these two differences, the tests are carried out in the same fashion as the previous tests: models are randomly generated from the chosen priors, for a range of *summed total* orders $(P + Q) = \{1, \ldots, 10\}$, the number of autoregressive and moving average components being randomly determined for each model such that $(Q < P)$, this restriction ensuring the models represent physically possible processes [141]. A set of candidates are then inferred and each criteria is asked to nominate a single model. Once again, one set of tests was conducted using the Maximum Likelihood estimates with the non-MML and MML-based criteria, and another test conducted using the MML87S estimates and the MML-based criteria. The MML87S point estimates were employed even though the MML87S criteria was not used for order selection because it was found that the MML87S estimates tended to possess lower MMLD-S message lengths than the ML estimates, and were thus closer to the ideal MMLD-S point estimates (which were far too computationally difficult to find).

In the tables of results the furthermost three columns now contain the number of times each criterion chose models whose *total number of parameters* was less than, equal to and greater than the total number of parameters of the true model, i.e. $(P^* + Q^*)$. The total number of parameters was chosen as all the non-MML criteria (including SC as it is based on the asymptotic Information Matrix) consider two different ARMA models to have the same complexity if they have the same number of parameters, regardless of the whether they are autoregressive or moving average parameters. The number of times the order selection criteria nominate a model with the correct total number of parameters will then serve as one measure of how often the methods choose the correct 'complexity' model, and how often models are chosen that possess more parameters than the true model and will give an indication of the parsimonity of the model various criteria.

9.5.1 Discussion of ML Results

Table 9.12: ARMA Order Selection Results (ML Estimates), $N = 11$

$N = 11$, $\{(Q, P) \in [0, 4] : P + Q \leq 4\}$, $\{(Q^*, P^*) \in [0, 10] : P^* + Q^* \leq 10\}$

Criteria	SPE_∞	SPE_1	KL_{100}	J_{100}	SPE_{1c}	KL_{1c}	J_{1c}	$<$	$=$	$>$
AIC	1.0753	0.3132	84.593	85.442	0.1807	0.8133	1.2048	858	**88**	54
AICc	0.6214	0.2637	1.9667	4.6634	0.1637	0.4528	0.8616	932	60	8
BIC	0.9645	0.2899	78.492	79.408	**0.1594**	0.6969	1.0705	884	83	33
KIC	0.8741	0.2739	77.780	78.928	0.1688	0.6436	1.0381	915	69	16
KICc	**0.5669**	0.2664	**1.4273**	**4.7031**	0.1839	**0.4027**	0.8912	955	43	2
SC	0.7234	**0.2662**	3.4347	5.1274	0.1732	0.5349	0.9760	936	58	6
MMLDS	0.6279	0.2793	4.9053	7.5030	0.1629	0.4631	**0.8375**	883	82	35
MEKL	**0.4945**	**0.2344**	1.6710	4.2363	0.1463	**0.3662**	**0.7177**	818	110	72
MEJ	0.5773	0.2494	2.9260	**3.9731**	**0.1388**	0.4035	0.7198	787	112	101

Table 9.13: ARMA Order Selection Results (ML Estimates), $N = 20$

$N = 20$, $\{(Q, P) \in [0, 7] : P + Q \leq 7\}$, $Q^*, P^* \in [0, 10] : P^* + Q^* \leq 10$

Criteria	SPE_∞	SPE_1	KL_{100}	J_{100}	SPE_{1c}	KL_{1c}	J_{1c}	$<$	$=$	$>$
AIC	0.7512	0.3971	25.6693	26.0844	0.1379	0.5190	0.8116	746	**127**	127
AICc	0.4770	0.3224	7.8927	8.3562	**0.1230**	0.3333	0.5895	851	118	31
BIC	0.5571	0.3333	11.1782	11.6456	0.1255	0.3837	0.6539	845	111	44
KIC	0.5552	0.3330	11.1457	11.6126	0.1252	0.3820	0.6512	845	111	44
KICc	0.4248	0.3125	1.8994	2.5001	0.1274	0.2903	0.5565	898	89	13
SC	0.4294	0.3173	2.8752	3.5761	0.1444	0.3048	0.5972	912	79	9
MMLDS	**0.3865**	**0.3055**	**1.5162**	**2.0423**	0.1244	**0.2766**	**0.5260**	855	115	30
MEKL	**0.3553**	**0.2778**	**0.8764**	**1.4457**	0.1155	**0.2518**	0.4883	757	**133**	110
MEJ	0.3697	0.2831	1.1749	1.6425	**0.1059**	0.2616	**0.4849**	729	130	141

Table 9.14: ARMA Order Selection Results (ML Estimates), $N = 50$

$N = 50$, $\{(Q, P) \in [0, 10] : P + Q \leq 10\}$, $Q^*, P^* \in [0, 10] : P^* + Q^* \leq 10$

Criteria	SPE_∞	SPE_1	KL_{100}	J_{100}	SPE_{1c}	KL_{1c}	J_{1c}	<	=	>
AIC	0.3195	0.1558	0.6740	0.8163	0.0827	0.1926	0.3326	552	199	249
AICc	0.2571	0.1210	0.4549	0.5928	0.0729	0.1576	0.2806	674	**201**	125
BIC	0.1990	0.0946	0.2824	0.4487	0.0636	0.1200	0.2297	809	157	34
KIC	0.2303	0.1079	0.3855	0.5277	0.0680	0.1421	0.2574	722	200	78
KICc	0.2020	0.0941	0.2949	0.4424	0.0631	0.1224	0.2304	784	172	44
SC	0.1959	0.0981	0.2184	0.4458	0.0736	0.1165	0.2352	852	140	8
MMLDS	**0.1710**	**0.0849**	**0.2051**	**0.3750**	**0.0585**	**0.1038**	**0.2020**	786	183	31
MEKL	0.1705	0.0791	**0.1729**	0.3653	0.0604	0.1009	0.1998	714	156	130
MEJ	**0.1698**	**0.0767**	0.1886	**0.3439**	**0.0543**	**0.1000**	**0.1928**	686	163	151

The results of ARMA order selection using ML estimates are presented in Tables 9.12-9.14. For $N = 11$, the KICc criterion outperforms MMLD-S on all losses but the conditional J divergence. The AICc and SC criteria perform quite similar to MMLD-S, with the exceptions of the unconditional KL and J divergences in which AICc and SC perform better, and the conditional scores in which SC performs worse. The AIC, KIC and BIC criteria all perform quite poorly due to large amounts of overfitting that characterise these methods for small sample sizes. As the sample size decreases the MMLD-S message lengths will obviously differ from the true message lengths by a greater amount as the behaviour of the KL divergence surface begins to differ more and more from the assumptions under which MMLD-S is derived. It would appear that for $N = 11$ the MMLD-S criterion is too optimistic and the resultant overfitting leads to degraded performance. It is interesting to note however, that although in terms of number of overfittings and number of underfittings MMLD-S and BIC are virtually indistinguishable, the MMLD-S criterion attains almost uniformly lower losses than BIC for $N = 11$. This is most likely due to the fact that MMLD-S is selecting fully specified models and avoiding those models with poles or zeros very close to the unit circle. It is also interesting to note that the MEKL criterion yields losses that are either lower or slightly worse than the KICc scores, while scoring a significantly higher number of true order selections (and a correspondingly higher number of overfittings). This again suggests that considering models in terms of more than just their total number of parameters can be beneficial.

For $N = 20$ the MMLD-S criterion outperforms or comes very close to the non-MML criteria in terms of almost all losses, and the only times it performs worse the differences are marginal (both SPE_1 losses). An interesting point to note is that for all criteria barr

AICc, the unconditional KL and J divergence losses are smaller than for $N = 11$, as would be expected. For $N = 20$ the AICc criterion however, yields worse unconditional KL and J divergence losses than for $N = 11$. This is attributed to the less conservative nature of AICc for $N = 20$, coupled with the very large growth in number of candidate models (and therefore, growth in selection risk) from $N = 11$ to $N = 20$. The KICc and SC criteria perform the overall best of the non-MML criteria. The MEKL and MEJ criteria yield improvements in terms of all losses, some minor (in the case of the unconditional SPE losses) and some greater (such as the KL and J divergence losses); however, this all arrives at a greatly increased number of overestimations of total number of parameters (from 30 to 141 in the case of MEJ).

For $N = 50$ MMLD-S attains uniformly lower scores for all losses as compared to the non-MML criteria. For a sample size as large as $N = 50$ the MMLD-S approximation yields message lengths that are much closer to the ideal message lengths than for the smaller sample sizes, which helps to explain its good performance as compared to the case of $N = 11$. The SC and KICc criteria once again perform the best of the non-MML methods, with the SC method yielding scores that are in some cases comparable to MMLD-S (particularly the SPE_∞ and unconditional KL losses). The MMLD-S method outperforms both in terms of correct order selections, while attaining less overestimations than KICc and more than SC. Despite the large increase in number of candidate models, the increase in sample size is sufficient to overcome the increased selection risk and all criteria attain lower losses than for $N = 20$. The MEKL and MEJ offer improvements over MMLD-S for all but the SPE_1 losses, but at the expense of reduced number of correct order estimations and a very significant increase in overestimations.

9.5.2 Discussion of MML Results

The results using the MML87S estimates are presented in Table 9.15. The first point to note is that all three MML-based criteria using the MML87S estimates attain lower losses than the best performing criteria using the ML estimates. This is comforting, as the models used for the tests were drawn from the priors used by the MML-based methods. However, the results show that in many cases, particularly for the SPE_∞ and KL/J divergence losses, the differences can be quite significant. This is most probably due to a combination of the shrinkage of the MML-estimates due to the effects of the Fisher

Table 9.15: ARMA Order Selection Results (MML Estimates)

$N = 11$, $\{(Q, P) \in [0, 4] : P + Q \leq 4\}$, $\{(Q^*, P^*) \in [0, 10] : P^* + Q^* \leq 10\}$

Criteria	SPE_∞	SPE_1	KL_{100}	J_{100}	SPE_{1c}	KL_{1c}	J_{1c}	$<$	$=$	$>$
MMLDS	0.3877	0.2283	0.7896	**1.9752**	0.1333	0.3479	0.6534	871	90	39
MEKL	**0.3578**	**0.2016**	**0.5842**	2.9183	0.1320	**0.3086**	0.6161	771	**124**	105
MEJ	0.3597	0.2039	0.6776	2.0303	**0.1245**	0.3249	**0.6122**	739	120	141

$N = 20$, $\{(Q, P) \in [0, 7] : P + Q \leq 7\}$, $Q^*, P^* \in [0, 10] : P^* + Q^* \leq 10$

Criteria	SPE_∞	SPE_1	KL_{100}	J_{100}	SPE_{1c}	KL_{1c}	J_{1c}	$<$	$=$	$>$
MMLDS	0.2749	0.1647	0.5367	1.0189	0.1124	0.2325	0.4523	597	**258**	145
MEKL	**0.2542**	**0.1338**	**0.4152**	0.9281	0.1045	0.2163	0.4305	410	178	412
MEJ	0.2566	0.1352	0.4381	**0.8457**	**0.0947**	**0.2152**	**0.4099**	440	195	365

$N = 50$, $\{(Q, P) \in [0, 10] : P + Q \leq 10\}$, $Q^*, P^* \in [0, 10] : P^* + Q^* \leq 10$

Criteria	SPE_∞	SPE_1	KL_{100}	J_{100}	SPE_{1c}	KL_{1c}	J_{1c}	$<$	$=$	$>$
MMLDS	0.1389	0.0773	0.1350	0.2787	0.0527	0.0930	0.1804	587	**288**	125
MEKL	0.1357	**0.0714**	**0.1197**	0.2778	0.0536	0.0898	0.1768	431	224	345
MEJ	**0.1346**	0.0700	0.1259	**0.2550**	**0.0495**	**0.0884**	**0.1707**	452	233	315

Information term, and the fact that the MML87S estimates appear not to suffer from the 'piling-up' problem (see Section 5.8.3 for more details). With the notable exception of the unconditional J divergence for $N = 11$, the MEKL and MEJ generally attain lower losses than the MMLD-S criterion, though as in the case of the ML estimates this is obtained only at a significant increase in the number of overestimations.

All three criteria score a higher number of correct total order selections as compared to the same criteria using the Maximum Likelihood estimates. As an example, it can be seen that for $N = 20$ the MMLD-S approximation scores 115 correct order selections using the ML estimates, and 258 correct order selections when using MML87S estimates. This is predominantly due to the fact that many of the ML estimated candidate models have poles and zeros close to the unit circle boundary, which in turn leads to very large message lengths. The MML87S estimates shy away from such pole/zero positions and the savings in message lengths, particularly for higher order models can be significant. However, by the same token the number of overestimations of total model order has increased quite dramatically; for $N = 20$ the number of overestimations of total model order has increased from 30 to 145, with even larger increases for the MEKL and MEJ criteria. However, once again it should be noted that despite the large number of overestimations of model order by the MEKL and MEJ methods, their overall losses are often lower than the MMLD-S criterion, suggesting that purely measuring a model selection techniques performance by how often it selects the correct model is not advisable.

9.5.3 Conclusion

The main conclusions to be drawn from the ARMA model selection tests are now summarised.

- The MMLD-S criterion appears to offer a good compromise between predictive performance (in terms of the various losses) and inductive performance (measured in terms of number of correct total model order selections).

- The MEKL and MEJ criteria appear to offer improvements over the MMLD-S criterion in terms of predictive performance, but this comes at the expense of a significantly increased number of overestimations of total model order. However, the models are still considerably more parsiminous than Bayesian average predictive densities over all possible model orders while achieving very good predictive performance.

- Of the non-MML criteria, the SC and KICc methods appeared to offer the overall best performance in terms of predictive losses, and generally yielded very parsiminous and compact models.

9.6 Experiments on Real Data

This section details experiments undertaken on real data. The MML criteria (MML87S and MMLD-S) have been shown to perform well in order selection and parameter estimation experiments when the models were generated from the chosen prior distributions. The real data experiments were undertaken to test how well the MML approximations perform when the data was not known to be explicitly generated from 'ideal' conditions (i.e. normally distributed noise, generated by a 'true' ARMA model, models generated from priors). The performance of the varying criteria was assessed by dividing the real data into contiguous blocks of (possibly overlapping) datasets that were used for fitting, and then using blocks of data for testing. For each dataset, a candidate set of models was fitted and each criteria was required to select a single 'best' model from the candidate set; the testing data was then used to produce various losses. Given a complete real data set $\mathbf{z} \in \mathbb{R}^{N_c}$, the process may be summarised as

1. The process begins by fitting a set of candidate AR models $\mathcal{M} = \{\text{AR}(P) : P \leq N_s\}$ to the complete dataset $\mathbf{z}$. A single point estimate is chosen by the BIC criterion, and the fitted residuals $\hat{\mathbf{v}}$ of the chosen model are used to estimate the innovation sequence for time steps $(N_s + 1 : N_c)$. The BIC criterion is chosen as the complete dataset is usually quite large (hundreds of datapoints), and MML and BIC asymptotically converge. These innovations are only used for the multistep prediction scores. The maximum order N_s should be usually around 20 so that the AR model can provide reasonable quality estimates of innovation sequence even if the underlying process is more akin to an ARMA model.

2. Dividing the remaining $(N_c - N_s)$ data points into contiguous training datasets $\mathbf{y}^{\{i\}} \in \mathbb{R}^N$ and testing datasets $\mathbf{y}^{*\{i\}} \in \mathbb{R}^{N_f}$ in the following fashion

$$\mathbf{y}^{\{i\}} = \left[z_{N_d(i)}, \ \ldots, \ z_{(N_d(i)+N)}\right] \tag{9.17}$$

$$\mathbf{y}^{*\{i\}} = \left[z_{(N_d(i)+N+1)}, \ \ldots, \ z_{(N_d(i)+N+N_f)}\right] \tag{9.18}$$

   where $(N_d(i) = N_s + N_g(i-1) + 1)$ is the start of dataset $\mathbf{y}^{\{i\}}$, N is the number of datapoints in each training dataset, N_g is the number of datapoints between the start of each dataset and N_f is the number of datapoints in the testing dataset. The number of datasets is determined by how many complete training/testing blocks can be constructed from the complete dataset $\mathbf{z}$ in this manner.

3. Finally, for each training dataset $\mathbf{y}^{\{i\}}$, a set of candidate ARMA models are fitted and each criterion is required to make nominate a single model as its choice. The predictive performance of this model on unseen data is tested by computing various loss from the test dataset $\mathbf{y}^{*\{i\}}$. The usual scores are the one-step ahead prediction score SPE_1 (evaluated via the Kalman filter), the multi-step prediction score SPE_{N_f} and the negative log-likelihood score L_{N_f}.

Once all the tests have been performed on all the datasets, the mean test scores are computed for examination. In all the real data tests considered for both pure AR and mixed ARMA models the AICc/BIC/KICc/CIC criteria used the ML estimates, the MMLDS/MEKL/MEJ criteria used the MML87 estimates in root space, and Fitzgibbon, Dowe and Vahid's MML87 criteria [16], denoted MML87_ϕ, used MML87 estimates in

coefficient space. In Section 4.9.3 a simple test showed that the MML87S estimates may perform worse than the ML estimates if the generating model has many poles of large magnitude. As the form of the generating source of the real datasets is unknown, the real data tests will provide an interesting measure of the performance of the MML87S method in non-ideal conditions (i.e. when the generating sources do not necessarily conform to the prior distributions).

However, to remove the negative influence of the 'piling-up' phenomenon in the case of MA and ARMA models (see Section 5.8.3 for more details), all ML candidate models with zeros very close to the invertibility boundary were pruned from the candidate sets. This led to (sometimes quite significant) improvements in the performance of all the non-MML criteria in the large majority of cases considered.

9.6.1 Autoregressive Experiments on Real Data

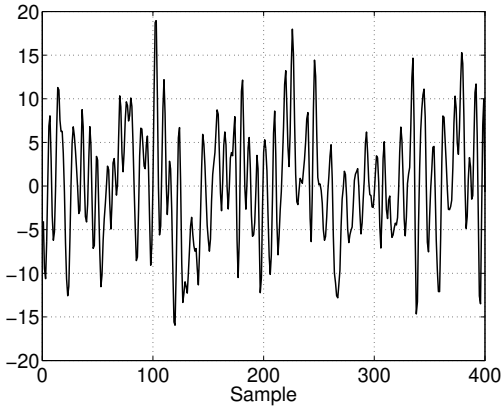
This section tests the MML criteria by examining pure autoregressive modelling of several real data sets. Pure autoregressive models (i.e. with no moving average component) are amongst the most commonly used of all time series models, so it is interesting to determine how the MML based approaches perform on several real datasets.

Earthquake Data

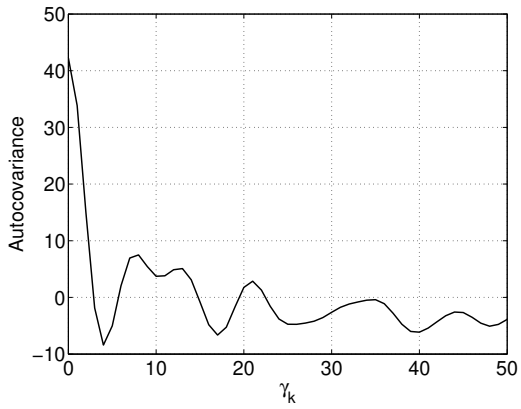
This first experiment tests pure autoregressive modelling on a dataset of ground vibrations during an earthquake in Scandinavia [160]; from the complete dataset an $N = 800$ subset was taken within which the data appears to be approximately stationary. Figure 9.2(a) shows a 400 sample segment of the dataset used, and Figure 9.2(b) shows the observed autocovariances $\hat{\boldsymbol{\gamma}}_{(0:50)}$. The observed autocovariance function drops off quite rapidly but has several oscillations, indicating that the generating source will be a higher order model with poles not too close to the boundary. The data was divided into 100 datasets of sizes $N = \{10, 20, 50\}$, all spaced $N_g = 5$ samples apart, and tests scores were obtained by using the next $N_f = 100$ samples.

Fish Data

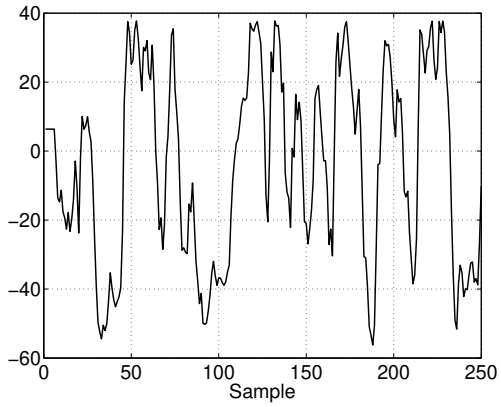
The second dataset examined is a series of the fish population in the Pacific measured at monthly intervals (also analysed in [161]). The series (detrended to have zero mean)



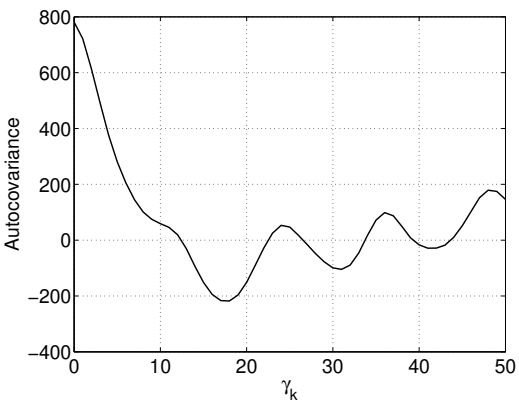
(a) Earthquake Data



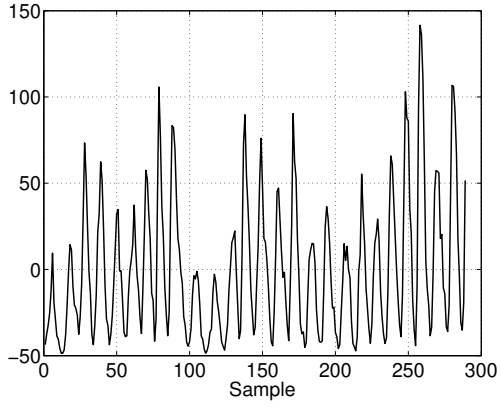
(b) Earthquake Data Observed Autocovariances



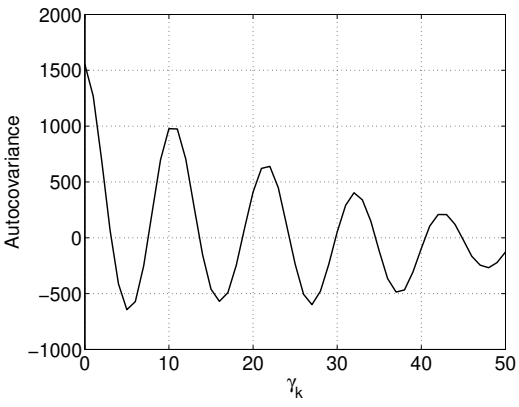
(c) Fish Data



(d) Fish Data Observed Autocovariances



(e) Wolf's Sunspot Data



(f) Sunspot Data Observed Autocovariances

Figure 9.2: Real Datasets used in Autoregressive Tests

Table 9.16: Earthquake Data Test

Score	AICc	BIC	KICc	SC	CIC	MMLDS	MEKL	MEJ	MML87$_\phi$
	$N = 10,\ P \in [0, 4]$								
SPE_{100}	33.819	39.746	33.833	33.941	40.021	18.247	17.411	**17.215**	23.276
SPE_1	7.8382	7.1295	9.3199	7.2470	7.1869	5.5815	5.4193	**5.2222**	6.3941
L_{100}	333.05	450.37	331.16	327.45	451.08	278.43	**264.42**	266.71	361.27
	$N = 20,\ P \in [0, 7]$								
SPE_{100}	21.532	21.613	20.379	21.321	21.779	12.170	11.195	**11.163**	15.558
SPE_1	3.9112	3.8823	4.0051	3.9301	3.8877	3.4587	3.4092	**3.3766**	3.7822
L_{100}	247.02	247.49	238.21	244.04	249.28	228.20	222.09	**221.78**	245.80
	$N = 50,\ P \in [0, 10]$								
SPE_{100}	9.3851	8.3885	8.4072	9.6686	9.4465	6.1642	5.9902	**5.9887**	8.3729
SPE_1	3.1655	3.1241	3.1298	3.1749	3.1610	2.9833	2.9764	**2.9758**	3.1164
L_{100}	210.57	207.24	207.56	207.30	210.56	203.64	203.58	**203.57**	210.13

Table 9.17: Monthly Fish Data Test

Score	AICc	BIC	KICc	SC	CIC	MMLDS	MEKL	MEJ	MML87$_\phi$
	$N = 10,\ P \in [0, 4]$								
SPE_{100}	538.41	708.13	513.02	517.91	696.28	396.56	**353.92**	361.58	453.39
SPE_1	131.62	163.35	143.21	133.31	162.00	115.58	**108.77**	109.96	144.69
L_{100}	453.19	538.51	441.83	454.01	536.18	447.74	**433.15**	439.90	496.04
	$N = 20,\ P \in [0, 7]$								
SPE_{100}	464.83	453.04	449.67	470.21	474.63	231.84	**228.04**	233.19	287.05
SPE_1	120.47	125.08	116.91	123.68	124.63	106.75	**106.03**	106.27	133.36
L_{100}	428.17	430.48	416.82	426.28	432.76	**405.16**	405.34	405.64	452.63
	$N = 50,\ P \in [0, 10]$								
SPE_{100}	125.94	125.35	123.10	124.98	124.93	69.442	**67.895**	68.042	109.63
SPE_1	102.69	104.07	103.55	107.46	102.42	101.15	**101.04**	101.10	104.23
L_{100}	381.40	380.94	380.71	382.51	381.06	**379.81**	379.82	379.85	381.32

is presented in Figure 9.2(c) and the sample autocovariance function is shown in Figure 9.2(d). The sample autocovariance function (ACF) decays at a slower rate than the earthquake data set, indicating that there are poles of the 'generating' process are generally closer to the unit circle boundary than for the earthquake dataset. The data was divided into 60 datasets of sizes $N = \{10, 20, 50\}$, all spaced $N_g = 5$ samples apart, and tests scores were obtained by using the next $N_f = 100$ samples.

Wolf's Sunspot Series

The final dataset examined in this section is the classic Wolf's sunspot numbers series from the years 1700 to 1988. A slightly shorter version of this sequence has been analysed by Tong [162] and Tsay [163] amongst many others who found it to be fitted very well

Table 9.18: Wolf's Sunspots Data Test

Score	AICc	BIC	KICc	SC	CIC	MMLDS	MEKL	MEJ	MML87$_\phi$
	$N = 10,\ P \in [0, 4]$								
SPE$_{50}$	1352.4	1966.6	**1101.9**	1180.5	1976.2	1487.4	1408.1	1366.0	1428.6
SPE$_1$	335.17	339.42	354.44	363.65	340.38	**298.47**	298.51	300.86	349.51
L$_{50}$	252.42	298.96	**248.37**	269.09	299.11	275.46	260.56	262.26	298.25
	$N = 20,\ P \in [0, 7]$								
SPE$_{50}$	1102.0	1126.8	1128.6	1077.9	1088.9	1095.0	1106.5	1085.2	**948.64**
SPE$_1$	264.57	313.82	262.77	283.01	334.54	**262.65**	263.28	263.09	317.88
L$_{50}$	230.98	251.44	230.25	232.75	257.58	229.16	**229.14**	229.24	245.77
	$N = 50,\ P \in [0, 10]$								
SPE$_{100}$	**720.48**	779.25	747.07	782.90	729.45	791.23	792.02	785.63	750.65
SPE$_1$	263.50	**244.63**	252.66	245.71	260.07	258.50	252.58	254.86	259.05
L$_{100}$	215.59	213.35	215.06	**213.48**	215.09	214.98	214.11	214.72	213.90

by a non-linear threshold autoregression. These experiments shall be restricted to linear autoregressive models, and given the non Gaussian nature of the dataset, will provide a good test for the robustness of the all methods. The series (again detrended to have zero mean) is presented in Figure 9.2(e) and the sample autocovariance function is shown in Figure 9.2(f). It is clear that the sample ACF exhibits slowly decaying oscillitary behaviour; the slow rate of decay would indicate that there are roots close to the unit circle, and this is definetely at odds with the prior chosen for the MML schemes. The series is thus deemed to provide a significant challenge for the MML criteria. The data was divided into 30 datasets of sizes $N = \{10, 20, 50\}$, all spaced $N_g = 5$ samples apart, and tests scores were obtained by using the next $N_f = 50$ samples.

Discussion

As would be expected, the MML pole-space criteria perform better for the earthquake data than for the fish data, and better for the fish data than for sunspot data. What is interesting, however, is that even though they are outperformed by several of the other criteria for the sunspot data, the losses are not particularly large. In general, even for this dataset that is clearly at odds with the chosen priors, the MMLD-S and related criteria are outperformed in terms of SPE$_1$ scores only once by the non-Bayesian criteria, and even in this case it is by only a small margin. Overall the MMLD-S criteria have the most trouble on the sunpots data for the $N = 10$, which is not unexpected as for small amounts

of data the priors have a large effect. However, it should still be noted that even in this case the MMLD-S based criteria attain the lowest SPE_1 losses by a considerable amount.

In contrast, the MMLD-S criteria clearly outperform the non-Bayesian criteria on all losses for both the fish data and particularly for the earthquake data. The earthquake data exhibits the most 'normal' behaviour of the three data sets and has a relatively rapidly decaying sample ACF, and thus would appear to suit the chosen pole-space priors quite well. The Fish data is certainly non-normal and has a slower decaying ACF, indicating stronger poles. However, in this case the MMLD-S based criteria again clearly attains the lowest losses. It is interesting to note that in the cases that the MMLD-S criteria lose on the sunspot data (a dataset strongly at odds with the assumed priors), the losses are quite marginal (for example, a 5% loss in terms of SPE_1 for $N = 50$ as compared to BIC on the sunspot data). However, when the MMLD-S criteria outperforms the conventional criteria on the datasets that fit the priors somewhat better, it often does so by a larger amount (for example, a 20% improvement in terms of SPE_1 over BIC for $N = 10$ on the earthquake data). This is a common trend amongst many MML based schemes: when they lose, it is often by small amounts, but when they 'win', it is often by large margins, and indicates the method has some robustness when used on data at odds with the chosen priors.

The performance of the coefficient-space MML criteria MML87_ϕ for the earthquake and fish dataset is uniformly worse than the MMLD-S criteria; for the sunspot dataset it performs better on the SPE_∞ loss for $N = 20$ and $N = 50$, and is generally inferior to MMLD-S in terms of the other losses. Compared against the non-MML criteria, it generally performs better for $N = 10$ and similar for the larger sample sizes. This would suggest that the coefficient-space prior is not as an uninformative a choice as the chosen root-space priors. The difference in performance could also be attributed to the use of the asymptotic Fisher Information Matrix in the MML87 approximation, and the fact that the uniform prior across coefficients induces different priors across pole strength for each different order.

Of course it is not possible to draw sweeping conclusions from three tests; however, it is possible to make the inference that the choice of uniform priors over the pole-space components appears to be a suitably uninformative choice that yields excellent results over datasets exhibiting both strong and weak roots, and non-normality. It should also

be noted, that given the non-normality of the fish and especially the sunspot dataset, and the method used to estimate the innovations (see Section 9.6), the multistep squared prediction errors should be taken with a grain of salt.

9.6.2 Currency Exchange Rate Forecasting Experiment

This experiment examined forecasting of the daily exchange rate of U.S. Dollars and Pound Sterling for the period 1974 to 1994 (as in [6]). This dataset is clearly nonstationary and was detrended by a first difference and the differenced sequence used for the experiments i.e. all fitted models were actually ARIMA(P,1,Q) models [2]. The testing procedure is almost the same as in [6]), the data being divided into segments of 22 datapoints, the first $N = 21$ being used for fitting and the last datapoint for one-step ahead prediction. Additionally, the experiment was extended slightly to test predictions over the next 22 samples to test multi-step ahead forecasting and negative log-likelihood, i.e. $N = 21$, $N_g = 22$, $N_f = 22$. Exactly replicating the experiments was not possible as the parameter estimation method used to fit the ARMA models in [6] was not specified. Thus the decision was made to slightly change the experiment and allow the candidate set of models to be $\mathcal{M} = \{\text{ARMA}(P, Q) : P + Q \leq 7\}$, including the AR(7) model.

Table 9.19: Currency Exchange Rate Data Test

Score	AICc	BIC	KIC	KICc	SC	MMLDS	MEKL	MEJ
SPE_{22}	0.0690	0.0639	0.0675	0.0352	**0.0321**	0.0450	0.0516	0.0669
SPE_1	1.1114	1.1092	1.1090	1.0345	**1.0339**	1.0544	1.0407	1.0895
L_{22}	44.678	44.724	44.700	41.550	**41.535**	42.336	41.977	43.240

The results are presented in Table 9.19. As in [6] the KICc criterion performed well compared to AICc, BIC and KIC, although the SC criterion marginally outperformed KICc in this case. Excluding MEKL and MEJ, the median total order of models (i.e. $P + Q$) chosen by all criteria was zero, and the number of times ARMA(0,0) models were chosen was 187 times for AICc, 210 times for KICc and 211 times for SC. The criteria that selected the ARMA(0,0) model the most frequently performed the best, which indicates strongly that there is little discoverable structure in the data (at least for a sample size of 22). The MMLD-S criterion selected the ARMA(0,0) model 197 times, and outperformed the less conservative criteria (AICc, BIC and KIC) in all three losses while losing somewhat marginally in terms of average losses to KICc and SC. For only one

single dataset did the MMLD grouping of model orders not include the ARMA(0,0) model, indicating that the MMLD-S message lengths rarely rule out the ARMA(0,0) explanation as plausible. The performance of the MEKL and MEJ criteria is quite interesting. The MEJ criterion selected the ARMA(0,0) model only 60 times and had an average total order of approximately 1.4, as opposed to the MMLD-S average total order of approximately 0.2. In this instance, the less conservative nature of the MEJ criterion leads to slightly poorer prediction losses as compared to MMLD-S. In contrast, the MEKL criterion only chose the ARMA(0,0) model 62 times, but had a lower average total model order of 1.1 and managed to outperform the MMLD-S criterion in terms of one-step SPE and log-likelihood loss. This behaviour of MEKL (higher complexity models but better predictive performance) is consistent with the behaviour observed on the synthetic tests.

9.6.3 The Southern Oscillation Index Dataset

This section examines the Southern Oscillation Index (SOI) dataset. This dataset is a measure of the difference between the air pressure at Tahiti and Darwin, and has been analysed by several people over the last decade due to its links to the El Niño phenomenon. The data set used contained monthly measurements from the years 1876 through to 2006. This section undertakes two types of experiments; the first set divide the dataset up into smaller sub-datasets and reports on the results of fitting autoregressive moving average models to these sub-datasets. The second test examines a much larger subset of the sequence.

ARMA Experiments

For the first set of tests the complete dataset was divided into 100 training datasets of size $N = \{11, 20, 50\}$, each $N_g = 10$ samples apart. A set of candidate models were then fitted to the test datasets, and each model selection criteria was required to select a suitable model. Losses were found by predicting over the subsequent $N_f = 50$ samples after each training set. The results are presented in Table 9.20. It is clear that the MML-based criteria perform well, attaining low losses across all sample sizes in all cases but one, and in that case the difference is marginal. In particular, the MEKL criterion has performed very well, uniformly attaining lower losses than all the other criteria, and this appears to be due to the fact that the data does not contain a large amount of structure that can be

Table 9.20: SOI Data Test

Score	AICc	BIC	KIC	KICc	SC	MMLDS	MEKL	MEJ
	$N = 11,\ P \in [0, 4]$							
SPE_{50}	0.3905	0.4458	0.4236	0.3938	0.3636	0.3109	**0.2538**	0.2707
SPE_1	0.8987	0.9274	0.9009	0.9244	0.8886	0.8894	**0.8082**	0.8177
L_{50}	88.290	124.40	100.49	88.946	**88.280**	92.424	88.682	89.144
	$N = 20,\ P \in [0, 7]$							
SPE_{50}	0.2799	0.3176	0.3176	0.2820	0.2692	0.2317	**0.2044**	0.2256
SPE_1	0.8552	0.9058	0.9058	0.8581	0.8465	0.8254	**0.7766**	0.7998
L_{50}	78.466	83.878	83.878	76.662	75.916	76.658	**72.841**	74.378
	$N = 50,\ P \in [0, 10]$							
SPE_{50}	0.3274	0.2236	0.2486	0.2367	0.1966	0.1837	**0.1622**	0.1649
SPE_1	0.8427	0.7520	0.7751	0.7541	0.7537	0.7521	**0.7278**	0.7329
L_{50}	67.616	65.640	68.423	65.636	65.427	65.602	**64.613**	64.891

discovered. The MEJ criterion improves on the MMLD-S losses, but does not perform as well as MEKL, and this is again most likely due to the less conservative nature of MEJ. The SC criterion performs the best of the non-MML methods, which also suggests that there is not a large amount of structure present in the data (as the SC criterion is quite conservative); however, the average total order $(P + Q)$ chosen by MMLD-S criterion for $N = 20$ is 1.07 which is significantly higher than for the exchange rate data (an average total order of 0.2), which indicates that there is more discoverable structure than in the exchange rate data. The average total order chosen by the SC criterion is 0.75, which is lower than the average total order of MMLD-S; this also suggests that the shrinkage of the parameters by the MML87S estimator is beneficial as although the MMLD-S criterion selects on average more complex models, it attains uniformly lower SPE_∞ losses across all samples sizes. The SC criterion performs quite well on this data, uniformly attaining lower SPE_1 and L_{100} losses than the other non-MML criteria, and marginally outperforming MMLD-S in several cases.

Analysis of 1876-1995 Data

An issue examined in previous papers [103, 164] has been whether the long run of predominantly negative values of the SOI during the period of 1990-1995 was inconsistent with the assumptions of stationarity and was unlikely to have arisen in light of the behaviour in previous years. The study in [103] was undertaken using an MCMC approach for exploring the posterior of autoregressive models, using the data from 1950-1990. After

the posterior was explored, it was used to estimate the probability that a run of at least 50 out of a a period of 66 months would be negative (roughly corresponding to the observed El-Niño during that period). In this study, all measurements from 1876 to 1990 of the Southern Oscillation Index were employed in fitting an ARMA model, and the fitted model was used to gauge the likelihood of the El Niño effect. The models selected by the various criteria from the candidate set $\mathcal{M} = \{\text{ARMA}(P,Q) : P \leq 9, Q \leq 9\}$, along with the negative log-likelihood scores and one-step ahead squared prediction errors over the data from 1990-2006 are presented in Table 9.21.

Table 9.21: ARMA Inference of the SOI Dataset, 1876-1990

	AIC	AICc	BIC	KIC	KICc	MMLD-S
Model	(8,9)	(8,9)	(2,4)	(8,9)	(8,9)	(3,3)
L	241.9448	241.9448	238.1786	241.9448	241.9448	**236.6971**
SPE_1	0.7098	0.7098	0.6930	0.7098	0.7098	**0.6843**

Given the large amounts of data involved, the results are very interesting. The AIC/AICc and KIC/KICc criteria all selected the same very high order model; it is well known that these criteria are not consistent and so this behaviour is not particularly surprising. The fact that the corrected and uncorrected variants of AIC and KIC agree is also expected given the asymptotic convergence of the two criteria. The MMLD-S criterion selected a $(3,3)$ structure, but an examination of the posterior probabilities over the model class space revealed that the posterior probabilities were quite diffusely spread across several models; the models in the MMLD grouping are presented in Table 9.22. It is interesting to note that while the BIC criterion selected a different model to MMLD-S, the model it selected possessed the second lowest message length and is considered quite plausible by the MMLD-S criterion. In contrast, the model chosen by AIC/AICc and KIC/KICc had a posterior probability of approximately 3.2×10^{-9} and is therefore considered extremely unlikely. The study in [103] found the posterior probability mass spread

Table 9.22: MMLD Model Group

Model	Message Length	Posterior
(3,3)	1636.6	0.2955
(2,4)	1636.9	0.2229
(2,1)	1637.2	0.1509
(1,1)	1637.3	0.1424

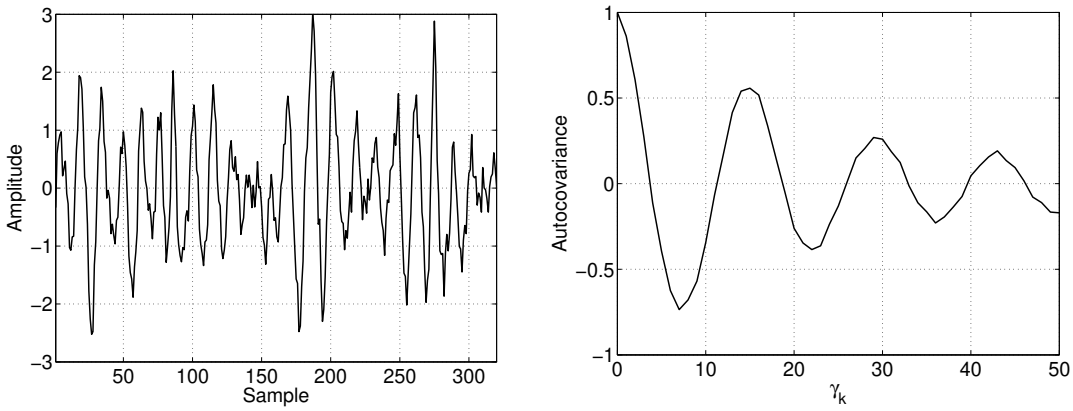
across AR(12) to AR(15) models; this is not inconsistent with our results as such a high order AR model is often indicative of an underlying ARMA-type process.

The synthetic test results showed that the MMLD-S criterion offered a good mix of both predictive ability and ability to select the 'true' model, indicating its decision was not geared towards minimising a single type of predictive criterion at the expense of selecting models that were compact descriptions of the data generating machinery. Given this, it was decided to use the MMLD-S nominated model to explore the SOI series further. The ARMA model chosen by the MMLD-S criterion was

$$\begin{aligned} y_n &= 2.25y_{n-1} - 1.66y_{n-2} + 0.38y_{n-3} \\ &\quad + v_n - 1.78v_{n-1} + 0.93v_{n-2} - 0.06v_{n-3} \\ v_n &\sim \mathcal{N}(0, 0.6) \end{aligned}$$

The parameters with the most effect on the ARMA model's time response are its autoregressive components; these consisted of a single real pole of small magnitude at approximately 0.45, and a complex pole pair of magnitude 0.92 and angular component of 0.2. This translates to a period of approximately 30 months (or 2.5 years), with a half-life of around 6 months. The findings in [103] reported a dominant pole pair with a similar radial strength, but a longer time period of approximately 4 years; however, it should be born in mind that the model found in this thesis used almost 800 points of extra data.

The final test was a brief experiment to determine whether the El Niño observed from the years 1990-1995 was unlikely based on the model estimated from the data up to the year 1990. During this period, the mean SOI level was -0.82 and the variance of the SOI was 0.58; a sequence of 1440 data points was generated 1000 times from the MMLD-S model and the number of years in which one or more periods of 66 months attained both at least 50 samples of the same sign, and an absolute mean (to account for the fact that a period with an equally high average positive level would be considered just as unlikely) at least as large as 0.82. The resulting rate of occurence was approximately 0.51, which is quite high. The conclusion drawn here is that the El Niño event observed from 1990-1995 is consistent with the data (or more correctly, with the model estimated from the data) from the years 1876-1990.



(a) Waves Data

(b) Waves Data Observed Autocovariances

Figure 9.3: Wave Data Set

9.6.4 Wave Experiment

The next real dataset to be tested was the measured force of waves upon a cylinder placed in a watertank [165], and is interesting as a dataset of a physical phenomenon. It also appears to contain considerably more structure than the SOI and exchange rate data and therefore allows all criteria to be tested on a richer dataset. It is assumed there is no measurement noise, so an ARMA structure is applicable. Three different tests were performed by dividing the data into forty datasets of sample sizes $N = \{11, 20, 50\}$ in a similar fashion to the SOI experiments in 9.6.3, with $N_g = 5$, $N_s = 1$ and $N_f = 50$. A segment of the dataset and the observed autocovariances are shown in Figures 9.3(a) and 9.3(b) respectively. It is clear that the observed autocovariances decay reasonably slowly, which is somewhat at odds with the chosen root-space priors. Thus, this dataset is expected to provide a challenging test for the MMLD-S criterion. The results are presented in Table 9.23.

It is apparent that despite the slow decay of the observed autocovariances, the MML-based criteria still perform very well, obtaining the lowest average losses in almost every case. The waves data set appears to contain a significant amount of structure (the average total order for MMLD-S was 3.6 for $N = 20$) and so the risk of overfitting increases as many of the higher order models yield significant decreases in negative log-likelihood. However, by the same token, the risk of underfitting also increases as the bias present in an underparameterised model will be quite significant, and this is evident in squared-prediction error losses of the conservative KICc and SC criteria, which are now amongst the worst for each sample size. Although their conservative nature often leads to lower

Table 9.23: Wave Data Test

Score	AICc	BIC	KIC	KICc	SC	MMLDS	MEKL	MEJ
	$N = 11,\ P \in [0, 4]$							
SPE_{50}	0.7491	0.8340	0.8470	0.8103	0.8807	0.7621	0.7462	**0.7343**
SPE_1	0.3109	0.5043	0.3015	0.3202	0.3199	0.2801	0.2448	**0.2434**
L_{50}	63.825	132.63	82.534	64.874	70.050	57.229	47.389	**47.005**
	$N = 20,\ P \in [0, 7]$							
SPE_{50}	0.7237	0.8624	0.8624	0.6837	0.7652	0.7220	0.6661	**0.6451**
SPE_1	0.1946	**0.1763**	0.1763	0.2293	0.2266	0.1901	0.2069	0.1902
L_{50}	57.762	56.769	56.769	45.312	45.334	**40.191**	40.538	40.535
	$N = 50,\ P \in [0, 10]$							
SPE_{50}	0.4938	0.4893	0.4938	0.4893	0.4906	**0.3628**	0.3932	0.3919
SPE_1	0.1472	0.1504	0.1472	0.1504	0.1536	0.1367	0.1372	**0.1342**
L_{50}	37.761	37.912	37.761	37.912	35.731	**29.047**	29.244	29.362

log-likelihood losses as compared to AICc, BIC and KIC, it is interesting to note that the MML-based criteria all attain clearly lower log-likelihood losses for all sample sizes despite selecting higher average total model orders. The good performance of the MML-based methods on this data is also believed to be due to the large amount of discoverable structure in the data, coupled with the shrinkage present in the MML estimates lowering the selection risk. The MML-methods are less conservative than SC and KICc in general, and when there is a significant amount of structure are more likely to select higher order models that will capture the underlying process behaviour more accurately. This behaviour is also visible in the plots of peformance on synthetic data in Figure 9.1.

9.6.5 Kobe Earthquake Dataset

The final dataset examined is the Kobe earthquake measurement dataset [166]. The observed autocovariances of this dataset decay quite rapidly, but show complex oscilliations indicating that the dataset is rich in structure but does not contain 'slow' poles. This would appear then to be a good dataset for the MML-based criteria, given the choice of pole-space priors. The dataset was divided into 40 training sets of size $N = \{11, 20, 50\}$, each $N_g = 10$ samples apart, with the test data drawn from the next $N_f = 50$ samples. The results are presented in Table 9.24. The results indicate there is a large amount of structure in this dataset, as the average total order selected by MMLD-S is 5.78 (average $P = 2.76$, average $Q = 3.02$), and even KICc selects an average total order 5.24 (average $P = 3.32$, average $Q = 1.92$). The losses in Table 9.24 are *median* scores rather than means, as for $N = 20$

Table 9.24: Kobe Earthquake Data Test (Median Scores)

Score	AICc	BIC	KIC	KICc	SC	MMLDS	MEKL	MEJ
	$N = 11,\ P \in [0, 4]$							
SPE_{50}	0.8455	0.8413	0.8673	0.8702	0.8268	**0.6559**	0.6916	0.6752
SPE_1	0.2197	0.1576	0.1960	0.2253	0.1553	**0.0907**	0.1412	0.1158
L_{50}	163.67	166.92	212.77	177.77	179.34	**46.486**	66.028	62.723
	$N = 20,\ P \in [0, 7]$							
SPE_{50}	0.6886	0.6668	0.6668	0.7013	0.5375	0.4453	0.4271	**0.4181**
SPE_1	0.0807	0.0731	0.0731	0.0971	0.1093	**0.0324**	0.0350	0.0326
L_{50}	128.11	149.15	149.15	168.11	553.20	11.997	8.0690	**-2.461**
	$N = 50,\ P \in [0, 10]$							
SPE_{50}	0.3988	0.3928	0.3988	0.3565	0.3368	0.3214	0.3148	**0.3020**
SPE_1	0.0591	0.0591	0.0591	0.0591	0.0715	0.0197	0.0176	**0.0173**
L_{50}	451.63	378.93	451.63	397.37	378.93	313.11	**100.73**	323.53

and $N = 50$ the non-MML criteria selected several very poor models that inflated the losses significantly. Even with the effects of these models removed through the use of the median, the MML-based methods still clearly outperform the non-MML based methods for all sample sizes and losses. The mean SPE_1 losses (not presented) are drastically affected for $N = 20$ by the selection of an MA(7) model having six out of seven zeros with a radial strength greater than 0.98, and for $N = 50$ by the selection of an MA(10) model have nine out of ten zeros with a radial strength greater than 0.9. Such models are simply not estimated by the MML87 estimators (in comparison, the MA(10) model fitted to the same training dataset had only five zeros with a radial strength exceeding 0.9). Furthmore, such models yield a very high model cost (due to their very small uncertainty region) and will be unlikely to be chosen by any of the MML-based criteria; as an example, when the non-MML criteria chose the MA(7) model with large zeros, the MMLD-S criterion selected an ARMA(2,3) model with the radial strength of all zeros less than 0.9.

Increasing the distance from the unit circle a model's zeros must satisfy to be included as a candidate helps to reduce the chance of selection of such models, but the introduction of such a threshold is quite ad-hoc. Due to the high levels of structure present in the data (for $N = 50$ the mean total order selected by AICc was 8.1) overfitting can be quite dangerous, particularly in terms of log-likelihood losses which is highly sensitive to underestimations of the variance. The squared prediction losses of the MML-based criteria inflate by only around 40% if the mean scores are calculated instead of the medians; in constrast, for $N = 20$ the inflation of the non-MML criteria could be as great as 300%,

suggesting that while the MML87 estimators may suffer some performance degradation if the data is generated by models at odds with the chosen priors, the losses do not seem to be as large as those that can be suffered by the non-MML methods.

9.7 Conclusion

The main findings of this chapter are now summarised:

- For pure AR models using Maximum Likelihood estimates, the MMLD-S criterion achieved similar or superior losses to the MML87S criteria while generally obtaining higher number of correct order selections. When the MML87S estimates were employed, both attained similar losses but the MML87S criteria heavily overestimated the true order.

- For pure MA models using Maximum Likelihood estimates the MML87 criterion based on the asymptotic Fisher Information Matrix was very conservative, often underestimating the model order by a large margin. The MMLD-S criterion performed better in terms of SPE and J divergence losses, and correct order identifications at the expense of increased KL divergence losses.

- The MMLD-S criterion appeared to offer a good mix of predictive performance, often attaining the lowest losses when compared to the non-MML and the MML87/MML87S criteria, as well as performing well in terms of correct order selection and low overestimations of correct order. Even when other criteria attained lower losses, the MMLD-S criteria was rarely outperformed by a large margin.

- In almost all pure AR and mixed ARMA tests the MEKL and MEJ criteria generally attained the lowest scores in terms of all losses, but tended to overestimate the order of the models as compared to MMLD-S.

- Of the non-MML criteria, the AICc, KICc and SC methods appeared perform overall the best, though all three tended to be quite conservative for smaller sample sizes.

- For pure AR, pure MA and mixed ARMA models, the MML-based criteria employing the MML87S estimates generally outperformed the non-MML criteria using the Maximum Likelihood estimates. Even in the case of real data that appeared to be

strongly at odds with the chosen priors, the MML-based criteria remained competetive, achieving similar or slightly worse losses, despite the Bayesian nature of the parameter estimates.

- Tests performed on real data showed that if the data does not exhibit behaviour strongly at odds with the colourless uniform priors on root amplitudes the MML-based criteria using the MML87S estimates can yield very good results. Six real datasets from various disciplines (i.e. financial, physical, metereological) were examined, with varying degrees of structure and varying observed autocovariance decay rates; only the Sunspot dataset that displayed very slow autocovariance decay (i.e. strongly at odds with the chosen priors) and the Exchange Rate data (that possessed very little structure) caused problems for the MML-based criteria, and even in this case the non-MML methods did not outperform the MML-based methods by a large margin.

- The tests on real data suggest that the MML-based methods offer a good compromise between behaviour when their is little discoverable structure and behaviour when the data contains a large amount of structure. When the amount of structure is low, conservative criteria like KICc may outperform the MML methods by small amounts; however, when there is significant structure the MML methods appear to more rapidly discover it and can yield large improvements over the non-MML criteria.

The findings suggest strongly that the MMLD-S and MEKL/MEJ criteria, particularly when equipped with the MML87S parameter estimates offer a very competitive alternative to the conventional ARMA parameter estimation/order selection methods that predominantly feature in the current literature. Even for real data where the true process may be at odds with the chosen priors the MML-based methods remained competitive. In the case of real data generated by processes which appeared to conform to the priors the observed improvements in terms of predictive losses were in many cases quite considerable.

Chapter 10

Conclusion

This thesis has examined the problem of inferring autoregressive (AR), moving average (MA) and mixed autoregressive-moving average (ARMA) models within the Minimum Message Length framework. The MML principle differs from most commonly used approaches in that it explicitly selects *fully specified models*, unifying both parameter estimation and model selection within a single framework. In particular new parameter estimators for AR, MA and ARMA models were derived using Wallace and Freeman's MML87 approximation. Suitable modifications were proposed to the MML87 criterion to help overcome difficulties due to violations of the working assumptions of the quadratic approximation. Additionally, both a new MML criterion derived from the MMLD approximation and a new algorithm for efficiently solving the high dimensional integrals used in this approximation were derived and applied to the ARMA class of models.

10.1 Achievements

The main achievements of this thesis were:

- Derivation of new parameter and order estimation criteria for AR, MA and mixed ARMA models based on the Wallace-Freeman MML87 approximation. In contrast to previous applications of MML to autoregressive models, the models are expressed in *root-space* which allows for a natural and interpretable specification of model priors.

- A new MML approximation, I1F, was introduced as a way to resolve the issue of point estimation within the MMLD framework.

- The development of an algorithm to efficiently approximate the high dimensional integrals in the I1F and MMLD-like equations. This algorithm exploited the properties of MMLD-like uncertainty regions to efficiently evaluate the high dimensional integrals with a time complexity linearly proportional to the number of model parameters.

- Application of the I1F approximation utilising the MMLD-S algorithm to the AR, MA and ARMA model classes. This lead to the introduction of further refinements to the MMLD-S algorithm to efficiently handle redundancies in the parameter space and curved priors based on a sampling approach.

- An empirical examination of the performance of the MML87 and MMLD-S based criteria to the selection of AR, MA and ARMA models. The examination was undertaken on both synthetic and real data and the new methods were compared to traditional estimators and model selection criteria. Additionally, the MMLD approximation was applied to the grouping of model class parameters and criteria based on selecting models with minimum expected losses were also examined.

10.2 Conclusions

The main findings of this thesis were:

- The MML87 based parameter estimation schemes for AR, MA and ARMA models all show excellent performance in terms of squared prediction errors and Kullback-Leibler (KL) divergences in comparison to traditional estimation schemes such as Maximum Likelihood (ML) and Burg estimation.

- For AR/ARMA models, the MML87 estimator tends to 'shrink' poles towards the origin of the unit circle which leads to improvements particularly in capturing long term behaviour of the underlying structure. The MML87 estimators also shrink the zeros in the MA components towards to origin and this appears to result in a complete elimination of the 'piling-up' effect that characterises the ML estimator (see Section 5.8.3 for more details).

- The MML87 approximation has problems for the ARMA class of models when applied to the order selection problem. This is due to singularities in the Fisher Information matrix due to flat spots in the KL divergence surface and redundancies due to the parameterisation in root-space.

- The MMLD-S approximation to the I1F message length formula when applied to the ARMA class of models provides an approximation in which the problems of redundancy and asymmetric uncertainty regions can be explicitly handled. The final MMLD-S criterion produced sensible message lengths for all parameter combinations without the need for any ad-hoc modifications.

- In terms of order selection, the empirical results indicated that the model that minimised the MMLD-S criterion was an excellent choice in terms of both *prediction* capabilities (i.e. ability to predict future data) and *induction* (i.e. selection of 'true' order and dispensation of excess parameters).

- The empirical results further indicate that the Minimum Expected KL and Minimum Expected J divergence criteria based on MMLD grouping of models using MMLD-S message lengths perform very well in terms of prediction while often trading off only a small amount in terms of correct order selection.

- When applied to real data, it was found that despite the Bayesian nature of the MML87 and MMLD-S criteria the choice of colourless uniform priors lead to good performances. The MML based methods were compared against non-Bayesian criteria such as Maximum Likelihood and the AIC, BIC and KIC familes and were found to often outperform the traditional methods. Of course, these benefits came only at the expense of increased computational complexity.

The overall conclusion is that the MML principle of selecting *fully specified* models from data, and of penalising a model based not only on the number of parameters but on the parameter values themselves leads to parameter estimation and order selection criteria that are very competitive with traditional methods. Empirical results in general show that the improved MML based criteria, while not always achieving the lowest test scores, will rarely be outperformed by a large margin while often attaining significantly lower prediction scores. Despite the subjective Bayes nature of the MML criteria, tests on real

data sets strongly suggest that colourless priors over root-space parameters and moving average coefficients are a reasonable general choice if little is known *a priori*. Of course, the improvements exhibited by the MML criteria are all gained only at the expense of extra computational complexity, though this is mediated to a degree by the linear time complexity of the MMLD-S algorithm.

10.3 Future Work

There is scope for future work that builds upon the results of this thesis; these suggestions are all non-trivial and in many cases would consume the better part of an entire thesis themselves and were thus not considered.

- Expanding the MMLD-S approximations to allow the uncertainty region to cross between different root-space structures within models of the same order. This may result in a slightly more efficient coding scheme.

- Application of the MMLD-S to vector autoregressive and vector autoregressive moving-average processes. For the resulting criteria to be efficient clever methods for encoding the structure would have to be considered; within the root-space framework it may be possible to identify shared underlying behaviour by pooling poles.

There are of course many other possible extensions and these are just some of the most obvious directions for further research.

Appendix A

Fisher Information Matrix of the Autoregressive Model

This section summarises the equations required to compute the asymptotic Fisher Information Matrix for an autoregressive model parameterised in root-space. These equations are taken from [135] and [123] and presented here for convenience. Assume the model has poles organised as follows: N_R real poles $\boldsymbol{\alpha} \in (-1,1)^{N_R}$, and N_C complex pole pairs with radii $\mathbf{r} \in (0,1)^{N_C}$ and angles $\boldsymbol{\omega} \in (0,\pi)^{N_R}$. The entries of the asymptotic Fisher Information Matrix are

$$J_{(\alpha_i,\alpha_j)}(\boldsymbol{\beta}) = \frac{1}{1-\alpha_i\alpha_j} \tag{A.1}$$

$$J_{(\alpha_i,r_j)}(\boldsymbol{\beta}) = \frac{2\cos\omega_j - \alpha_i r_j}{1-2\alpha_i r_j\cos\omega_j + (\alpha_i r_j)^2} \tag{A.2}$$

$$J_{(\alpha_i,\omega_j)}(\boldsymbol{\beta}) = -\frac{2r_j\sin\omega_j}{1-2\alpha_i r_j\cos\omega_j + (\alpha_i r_j)^2} \tag{A.3}$$

$$J_{(r_i,r_j)}(\boldsymbol{\beta}) = 4\left(r_i r_j + \cos\omega_i\cdot\cos\omega_j\right)F_{(0,0)} - r_i\cos\omega_j F_{(0,1)} - r_j\cos\omega_i F_{(1,0)} \tag{A.4}$$

$$J_{(r_i,\omega_j)}(\boldsymbol{\beta}) = -4r_j\sin\omega_j\cdot\left[F_{(0,0)}\cos\omega_i - F_{(0,1)}r_i\right] \tag{A.5}$$

$$J_{(\omega_i,\omega_j)}(\boldsymbol{\beta}) = 4r_i r_j\sin\omega_i\cdot\sin\omega_j F_{(0,0)} \tag{A.6}$$

where $F_{(u,v)}$ is

$$F_{(u,v)} = \frac{r_i^u r_j^v}{2 \sin \omega_i \sin \omega_j} \left(G_{(u,v)} - H_{(u,v)} \right) \tag{A.7}$$

and

$$G_{(u,v)} = \frac{\cos\left((u+1)\omega_i - (v+1)\omega_j\right) - r_i r_j \cos(u\omega_i - v\omega_j)}{1 - 2 r_i r_j \cos\left(\omega_i - \omega_j\right) + (r_i r_j)^2} \tag{A.8}$$

$$H_{(u,v)} = \frac{\cos\left((u+1)\omega_i + (v+1)\omega_j\right) - r_i r_j \cos(u\omega_i + v\omega_j)}{1 - 2 r_i r_j \cos\left(\omega_i + \omega_j\right) + (r_i r_j)^2} \tag{A.9}$$

Appendix B

I1F Uncertainty Region of Normal Distribution

B.1 Introduction

This document details the equations required for I1F estimation of the Normal distribution. Consider the likelihood function of the Normal distribution given a mean μ, variance τ and some observed data $\mathbf{y} \in \mathbb{R}^N$

$$L(\mathbf{y}|\mu,\tau) = \frac{N}{2}\log 2\pi + \frac{N}{2}\log\tau + \frac{1}{2\tau}\sum_{i=1}^{N}(y_i-\mu)^2 \tag{B.1}$$

The following priors are chosen for the parameters

$$h(\mu) = \frac{1}{\Upsilon} \tag{B.2}$$

$$h(\tau) = \frac{1}{\tau\Lambda} \tag{B.3}$$

where Υ and Λ are normalising constants. This document details the equations required to solve the penalty cost of I1F message length approximation, to facilitate parameter estimation.

B.2 KL-Divergence

The KL divergence of the Normal distribution of approximating model (μ, τ) from true model (μ^*, τ^*) is

$$\Delta(\mu, \tau || \mu^*, \tau^*) = \frac{N}{2} \log \frac{\tau}{\tau^*} + \frac{N(\mu - \mu^*)^2}{2\tau} + \frac{N\tau^*}{2\tau} - \frac{N}{2} \tag{B.4}$$

The first point to note is that the KL divergence is invariant to shifts in μ; that is

$$\Delta(\mu, \tau || \mu^*, \tau^*) = \Delta(\mu + \varepsilon, \tau || \mu^* + \varepsilon, \tau^*), \ \forall \varepsilon \in \mathbb{R} \tag{B.5}$$

Thus it is convenient to shift μ^* to the origin, resulting in the change of variables ($\tilde{\mu} = \mu - \mu^*$).

B.3 Expected Uncertainty Region

Let $\Omega \equiv \Omega(\mu, \sigma^2)$ denote the *expected uncertainty region* of the Normal distribution for the given parameters. Consider the KL divergence (B.4) as a function of $\tilde{\mu}$ only; it may be re-expressed as

$$\Delta(\tilde{\mu}) = f(\tau)\tilde{\mu}^2 + g(\tau) \tag{B.6}$$

where

$$f(\tau) = \frac{N}{2\tau} \tag{B.7}$$

$$g(\tau) = \frac{N}{2} \log \frac{\tau}{\tau^*} + \frac{N\tau^*}{2\tau} - \frac{N}{2} \tag{B.8}$$

This is clearly a simple quadratic form in which the coefficients depend only on τ. Thus, as the shape of the KL-divergence depends only on τ. The boundary rule states that the boundary of the uncertainty region will lie on a level set of the KL-divergence surface. Let the KL-divergence at the boundary be denoted by ξ. For a given τ and boundary KL ξ, the width of the uncertainty region may be found by solving

$$f(\tau) \left(\frac{w(\tau)}{2} \right)^2 + g(\tau) = \xi \tag{B.9}$$

which yields

$$w(\tau) = \frac{2\sqrt{-f(\tau)(g(\tau)-\xi)}}{f(\tau)} \tag{B.10}$$

Now let the $a, b \in \mathbb{R}^+$ such that $\Delta(0, \tau - a) = \Delta(0, \tau + b) = \xi$, so that $a, b \equiv a(\xi), b(\xi)$. The points $(\tau - a(\xi))$, $(\tau + b(\xi))$ are the points of intersection between the τ axis and the uncertainty region. At any point τ the KL divergence along the μ axis is given by (B.6), and spanned by a width of (B.10); from this, for a given ξ the prior mass integral may be found as

$$\begin{aligned}\int_\Omega h(\theta)d\theta &= \int_{\sigma^2-a(\xi)}^{\sigma^2+b(\xi)} \frac{h(\tau^2)}{\Upsilon} w(\tau^2)d\tau^2 \\ &= \frac{2}{\Upsilon}\int_{\sigma^2-a(\xi)}^{\sigma^2+b(\xi)} h(\tau^2)\sqrt{\left(\tau^2\left(1+\frac{2\xi}{n}-\log\frac{\tau^2}{\sigma^{*2}}\right)-\sigma^{*2}\right)}d\tau^2\end{aligned} \tag{B.11}$$

The round-off integral may be expressed as

$$\int_\Omega h(\theta)\Delta(\theta||\theta^*)d\theta = \int_{\sigma^2-a(\xi)}^{\sigma^2+b(\xi)} \frac{h(\tau^2)}{\Upsilon}\int_{-w(\tau^2)/2}^{w(\tau^2)/2} \left(f(\sigma^2)\mu^2 + g(\sigma^2)\right) d\mu\, d\tau^2 \tag{B.12}$$

The inner integral may be solved exactly to yield

$$\frac{2N}{3\tau}\left(\tau - \tau\log\frac{\tau}{\tau^*} - \tau^* + \frac{2\xi\tau}{N}\right)^{\frac{1}{2}}\left(\tau\log\frac{\tau}{\tau^*} + \tau^* - \tau + \frac{\xi\tau}{N}\right) \tag{B.13}$$

The outer integrals in (B.11) and (B.12) do not admit closed form solutions; however, the problem has been transformed to a single dimensional integral which made be efficiently solved numerically. Given an $a(\xi)$ and $b(\xi)$, there is now a simple way to compute the penalty cost. The issue remains of determining these two points such that they both lie on the same level set.

B.4 Determining $a(\xi)$ and $b(\xi)$

To simplify the determination of the uncertainty region, redefine $\xi \equiv \xi(a)$ and $b \equiv b(a)$, i.e. the uncertainty region is defined by the single point $(\tau - a)$. The function $\xi(a) = \Delta(0, \tau - a)$. Solving

$$\Delta(0, \tau + b) = \Delta(0, \tau - a) \tag{B.14}$$

yields

$$b(a) = \tau^* \left(\exp\left(W_0\left(-\exp\left(-\frac{N+2\xi}{N} \right) \right) \right) + 1 + \frac{2\xi}{N} \right) - 1 \tag{B.15}$$

where $W_0(\cdot)$ is the Lambert W function, restricted to real values. It is now possible to define the uncertainty region completely in terms of b.

B.5 Minimum Circumscribed Rectangle

This section finds the volume of the minimum rectangle that may be circumscribed onto the uncertainty region. Along the τ axis this is trivially given by the $[\tau^* - a, \tau^* + b]$ that minimise the penalty cost. The maximum span across the μ axis may be found analytically; differentiating (B.10) with respect to τ and solving yields the τ for which w is greatest

$$\tau_{MAX} = \exp\left(\frac{2\xi}{N} \right) \tau^* \tag{B.16}$$

The prior over τ remains he same, and a uniform prior over μ is suitable

$$h(\tau) = \frac{1}{\Upsilon} \tag{B.17}$$

where Υ is once again a normalising constant. From this it is clear that as $N \to \infty$, $\tau_{MAX} \to \tau^*$ which is not unexpected given that the uncertainty region will converge to a sphere.

Appendix C

Some Useful Properties of ARMA Models

Several useful and important properties and equations related to ARMA models are summarised in this Appendix, primarily for completeness. Let $\boldsymbol{\theta} = [\mathbf{a}, \mathbf{c}, \sigma^2]$ where $\mathbf{a} \in \mathbb{R}^P$ are the autoregressive coefficients, $\mathbf{c} \in \mathbb{R}^Q$ are the moving average coefficients and σ^2 is the variance of the innovation sequence.

C.1 Autoregressive Autocovariances

A simple procedure to find the first $(P+1)$ autocovariances $\gamma_0, \ldots, \gamma_P$ is to solve the equation [133]

$$(\mathbf{A}_1 + \mathbf{A}_2) \cdot \left[\gamma_P, \ldots, \gamma_1, \frac{1}{2}\gamma_0\right]^T = \left[\mathbf{0}_{(1\times P)}, 1\right] \tag{C.1}$$

with

$$\mathbf{A}_1 = \text{triu}\left(\text{toep}\left([1, \mathbf{a}]\right)\right) \tag{C.2}$$

$$\mathbf{A}_2 = \mathbf{A}_1^T \cdot \tilde{\mathbf{I}}_{(P+1)} \tag{C.3}$$

where triu$(\cdot)$ is an upper-triangular matrix, toep$(\cdot)$ denotes a Toeplitz matrix, and $\tilde{\mathbf{I}}_N$ is the $(N \times N)$ permutation matrix. The higher order autocovariances may then be found from the recurrence relation

$$\gamma_k = \sum_{i=1}^{P} a_i \gamma_{k-i} \tag{C.4}$$

for $k > P$.

C.1.1 The Moving Average Autocovariance Function

As there is no feedback of y_n terms in the pure moving average process, the autocovariance of two observations will be non-zero only in the finite window of the moving average process. All Q autocovariances of a Moving Average process may be computed easily from the expression

$$\boldsymbol{\gamma}_{(0:Q-1)} = \left(\text{triu}\left([1, \mathbf{c}]^T [1, \mathbf{c}]\right) \cdot \mathbf{c}^T\right)^T \tag{C.5}$$

where triu$(\cdot)$ denotes an upper triangular matrix.

C.1.2 Mixed Autoregressive Moving Average Autocovariance Function

Finally, the autocovariance function of the mixed Autoregressive Moving Average model is considered. This may be easily found by exploiting the linearity of the ARMA structure and expressing the model in a cascade decomposition in which the output of the autoregressive component subsequently drives the moving average component. The autocovariances γ'_k of the pure autoregression may be found as per Section C.1. The ARMA autocovariances may then be computed as

$$\gamma_k = \text{Tr}\left(\left([1, \mathbf{c}]^T [1, \mathbf{c}]\right) \cdot \mathbf{G}(k)\right) \tag{C.6}$$

where $\mathbf{G}_k$ is a matrix with elements given by

$$G_{(i,j)}(k) = \gamma'_{(k-i+j)} \tag{C.7}$$

C.2 Evaluation of ARMA Likelihood via the Kalman Filter

Direct evaluation of the ARMA negative log-likelihood function for ARMA models involves the determinants and inversion of $(N \times N)$ matrices. Even if this is done efficiently exploiting the Toeplitz structure of the autocovariance matrices, it still takes roughly order N^2 operations. The Kalman Filter recursions provide an efficient method of evaluating ARMA likelihoods whose time complexity is linear with the number of data points. Originally devised with optimal state estimation as a goal, the Kalman filtering equations have

remarkably deep connections to other problems, including optimal control and matrix factorisation. The algorithm begins by formulating the model in a state-space representation. Define m as

$$m = \max\{P, Q\} \tag{C.8}$$

and then augment the parameter vectors

$$\boldsymbol{\alpha} = \left[\mathbf{a}, \mathbf{0}_{(1\times(m-P))}\right] \tag{C.9}$$

$$\boldsymbol{\beta} = \left[1, \mathbf{c}, \mathbf{0}_{(1\times(m-Q))}\right] \tag{C.10}$$

Then define the matrices

$$\mathbf{F} = \begin{bmatrix} \boldsymbol{\alpha}^T & \mathbf{I}_Q \\ 0 & \mathbf{0}_{(1\times Q)} \end{bmatrix} \tag{C.11}$$

$$\mathbf{Q} = \sigma^2 \boldsymbol{\beta}^T \boldsymbol{\beta} \tag{C.12}$$

$$\mathbf{H} = [1, \mathbf{0}_m] \tag{C.13}$$

where $\mathbf{0}_{(P\times Q)}$ is a $(P \times Q)$ matrix of zeros, $\mathbf{F}$ is the state-transition matrix, $\mathbf{Q}$ is the process noise covariance matrix, and $\mathbf{H}$ is the observation matrix. The likelihood may be computed using the Kalman Filter recursive equations. At step n, first compute the predictions

$$\mathbf{X}_{(n|n-1)} = \mathbf{F} \cdot \mathbf{X}_{(n-1|n-1)} \tag{C.14}$$

$$\mathbf{P}_{(n|n-1)} = \mathbf{F} \cdot \mathbf{P}_{(n-1|n-1)} \cdot \mathbf{F} + \mathbf{Q} \tag{C.15}$$

$$\hat{y}_n = \mathbf{H} \cdot \mathbf{X}_{(n|n-1)} \tag{C.16}$$

From this, the innovations may be estimated

$$\hat{v}_n = y_n - \hat{y}_n \tag{C.17}$$

$$\hat{\sigma}_n^2 = \mathbf{H} \cdot \mathbf{P}_{(n|n-1)} \cdot \mathbf{H}^T \tag{C.18}$$

The update equations are evaluated next:

$$\mathbf{K}_n = \mathbf{P}_{(n|n-1)} \cdot \mathbf{H}^T \cdot \hat{\sigma}_n^{-2} \tag{C.19}$$

$$\mathbf{X}_{(n|n)} = \mathbf{X}_{(n|n-1)} + \mathbf{K}_n \hat{v}_n \tag{C.20}$$

$$\mathbf{P}_{(n|n)} = \mathbf{P}_{(n|n-1)} - \mathbf{K} \cdot \mathbf{H} \cdot \mathbf{P}_{(n|n-1)} \tag{C.21}$$

This process is repeated for all data in $\mathbf{y}$, i.e. for $n = 1 \ldots N$; the negative-log likelihood may then be found as

$$L(\mathbf{y}|\boldsymbol{\theta}) = \frac{N}{2}\log(2\pi) + \frac{1}{2}\sum_{n=1}^{N}\log\hat{\sigma}_n^2 + \frac{1}{2}\sum_{n=1}^{N}\frac{\hat{v}_n^2}{\hat{\sigma}_n^2} \tag{C.22}$$

Clearly, as N grows larger this algorithm becomes much more efficient than direct computation of the likelihood via inversion of the autocovariance matrix. The Kalman filter is so efficient as it transforms the multivariate Normal distribution into a series of univariate Normal distributions, each conditioned on previous data. The only issue that remains is to initialise the algorithm, i.e. appropriate selection of $\mathbf{X}_{(0|0)}$ and $\mathbf{P}_{(0|0)}$. An algorithm is given in [59] to compute these starting values so that the recursions will yield the exact negative log-likelihood. This algorithm selects

$$\mathbf{X}_{(0|0)} = \mathbf{0}_{(m+1)} \tag{C.23}$$

$$\mathbf{S} = \mathbf{I}_{(m^2)} - \mathbf{F} \otimes \mathbf{F} \tag{C.24}$$

$$\mathbf{P}_{(0|0)} = \mathbf{S}^{-1} \cdot \mathrm{vec}(\mathbf{Q}) \tag{C.25}$$

where $\otimes$ denotes the Kronecker product, and $\mathrm{vec}(\mathbf{A})$ builds a column vector by stacking the columns of $\mathbf{A}$. One point to note is that the paper from which the initialisation algorithm was taken [59] seems to possess a small error, and incorrectly transposes the second $\mathbf{F}$ in (C.24) (equation 12 in the cited paper).

Vita

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Schmidt, D. F. and Paplinski, A. P. and Lowe, G. S. (2004), 'On the Design of a Hydraulically Actuated Finger for Dextrous Manipulation.' In *Proceedings of the 2004 IEEE International Conference on Robotics and Automation (IEEE ICRA-2004).* New Orleans, U.S.A.

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